

Physlib Monthly Report

Jun 2026

Generated Mon, 10 Aug 2026 11:28:38 GMT

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Reviewers: Daniel Morrison, Alex Meiburg

Abstract

Auto-generated summary of changes to [leanprover-community/physlib](#) on branch `origin/master` between `d44dd71` and `05930fe`. The full diff is available at [GitHub compare](#).

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1 Introduction

During Jun 2026, 27 contributors submitted 154 commits that changed 46,837 lines (+28,676 / -18,161) across 492 files spanning 47 top-level areas of the repository. This report catalogues 1,496 newly added Lean declarations: 982 lemmas, 317 defs, 109 instances, 43 theorems, 29 abbrevs, 13 structures, 1 inductive, 1 informal definition, and 1 informal lemma.

See the [full compare diff](#) and the [list of commits](#) on GitHub for the raw source.

1.1 Contributors

Ordered by lines changed. People who authored a commit or pull request merged this month.

Contributor	Lines Changed	Commits
Joseph Tooby-Smith	15,350	21
Vasily Ilin	8,962	1
Zhi Kai Pong	5,540	9
juanjfndz	3,103	6
Gregory Loges	2,913	12
Andrea Pari	2,515	4
Bjørn Kjos-Hanssen	1,993	3
Robby Sneiderman	1,736	15
nateabr	1,433	7
Nikolai Kashcheev	1,308	7
Yidi Qi	1,194	13
LibertasSpZ	806	2
Lazar Milikic	742	5
Nicola Bernini	722	20
Justin Findlay	595	3
doxtor6	450	4
Aleksei Bykov	315	2
giuseppe.sorge	309	7
Shibashish	140	3
Yanting Teng	135	1
Christian Krause	131	2
Shlok Vaibhav Singh	61	1
Florian Wiesner	54	2
Tracy McSheery	33	1
Wouter Deconinck	32	1
Ranjan Sharma	16	1
Paul Lezeau	13	1

1.2 Reviewers

People who reviewed a pull request merged this month but did not author one themselves.

Reviewer	PRs Reviewed
Daniel Morrison	11
Alex Meiburg	1

1.3 Modified subfolders

Subfolder	Files	Additions	Deletions
Physlib/Relativity	85	+5,706	-7,342
Physlib/SpaceAndTime	40	+3,364	-1,393
Physlib/Mathematics	35	+2,642	-1,564
Physlib/Particles	44	+2,087	-1,693
PhyslibAlpha/ClassicalFieldTheory	18	+3,083	-0
Physlib/QuantumMechanics	33	+1,916	-791
Physlib/ClassicalMechanics	15	+1,393	-1,014
Physlib/Electromagnetism	31	+1,034	-1,107
scripts	9	+1,683	-10
PhyslibAlpha/QuantumMechanics	2	+1,308	-0
QuantumInfo/ForMathlib	38	+472	-581
Physlib/StringTheory	4	+306	-655
PhyslibAlpha/SpaceAndTime	7	+903	-0
PhyslibAlpha/Mathematics	1	+565	-0
Physlib/QFT	43	+153	-389
Physlib/StatisticalMechanics	6	+116	-365
Physlib/Thermodynamics	3	+81	-330
QuantumInfo/Entropy	4	+150	-173
Physlib/Units	9	+173	-128
Physlib/Cosmology	7	+264	-1
scripts/PhyslibAlpha	5	+242	-0
QuantumInfo/States	6	+73	-102
.github/workflows	3	+118	-54
scripts/MetaPrograms	12	+133	-38
QuantumInfo/ResourceTheory	3	+73	-81
lakefile.lean	1	+0	-125
lakefile.toml	1	+125	-0
PhyslibAlpha/ClassicalMechanics	1	+122	-0
QuantumInfo/Channels	6	+51	-57
Physlib/CondensedMatter	1	+21	-71
AGENTS.md	1	+85	-0
Physlib/FluidDynamics	1	+14	-41
lake-manifest.json	1	+27	-26
Physlib.lean	1	+39	-4
QuantumInfo/ClassicalInfo	3	+18	-15
PhyslibAlpha.lean	1	+32	-0
AI-POLICY.md	1	+31	-0
PhyslibAlpha	1	+30	-0
docs	1	+23	-0
README.md	1	+7	-1
CITATION.cff	1	+7	-0
QuantumInfo/Measurements	1	+1	-4
CONTRIBUTING.md	1	+2	-2
QuantumInfo/Operators	1	+1	-2
lean-toolchain	1	+1	-1
.codespellignore	1	+1	-0
Physlib/Meta	1	+0	-1

1.4 New declarations by kind

Kind	Count
lemma	982
def	317
instance	109
theorem	43
abbrev	29
structure	13
inductive	1
informal_definition	1
informal_lemma	1

2 Details by file

2.1 Physlib.ClassicalMechanics.FreeParticle.Basic

[Physlib/ClassicalMechanics/FreeParticle/Basic.lean](#) on GitHub.

def ClassicalMechanics.FreeParticle.linearMomentum [\[source\]](#)

```
def linearMomentum (s : FreeParticle) (q : Trajectory) (t : Time) : ℝ
```

lemma ClassicalMechanics.FreeParticle.linearMomentum_conserved_of_velocity_const [\[source\]](#)

If a free-particle trajectory has constant velocity, then its linear momentum is constant.

```
lemma linearMomentum_conserved_of_velocity_const (s : FreeParticle) (q : Trajectory)
  (h : ∃ v, ∀ t, s.velocity q t = v) :
  ∃ p, ∀ t, s.linearMomentum q t = p
```

theorem ClassicalMechanics.FreeParticle.linearMomentum_conserved [\[source\]](#)

A free particle satisfying the equation of motion conserves linear momentum.

*Newton's second law implies that the acceleration vanishes, so the velocity is constant.
Since the particle mass is fixed, the linear momentum is constant in time.*

```
theorem linearMomentum_conserved (s : FreeParticle) (q : Trajectory)
  (h : ∀ t, s.NewtonsSecondLaw q t) (hcont : ContDiff ℝ 2 q) :
  ∃ p, ∀ t, s.linearMomentum q t = p
```

2.2 Physlib.ClassicalMechanics.HarmonicOscillator.ConfigurationSpace

[Physlib/ClassicalMechanics/HarmonicOscillator/ConfigurationSpace.lean](#) on GitHub.

lemma ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.coe_apply [\[source\]](#)

```
lemma coe_apply (x : ConfigurationSpace) (i : Fin 1) : x i = x.val i
```

instance ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.«instance@8aef359a» [\[source\]](#)

The topology on configuration space, induced by the chosen coordinate ‘ConfigurationSpace.val’ into ‘EuclideanSpace $\mathbb{R}$ (Fin 1)’.

instance : TopologicalSpace ConfigurationSpace

def ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.valEquiv [\[source\]](#)

The coordinate equivalence between configuration space and its ‘EuclideanSpace $\mathbb{R}$ (Fin 1)’ model, given by ‘ConfigurationSpace.val’ with its wrapper inverse.

def valEquiv : ConfigurationSpace $\approx$ EuclideanSpace $\mathbb{R}$ (Fin 1) **where**

instance ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.«instance@f9bea586» [\[source\]](#)

Configuration space is Hausdorff, transported from ‘EuclideanSpace $\mathbb{R}$ (Fin 1)’ across the coordinate homeomorphism.

instance : T2Space ConfigurationSpace

instance ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.«instance@572511a8» [\[source\]](#)

Configuration space is second countable, transported from ‘EuclideanSpace $\mathbb{R}$ (Fin 1)’ across the coordinate homeomorphism.

instance : SecondCountableTopology ConfigurationSpace

instance ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.«instance@9e9cd011» [\[source\]](#)

The structure of a charted space on ‘ConfigurationSpace’, modeled on its ‘EuclideanSpace $\mathbb{R}$ (Fin 1)’ coordinate via the single global chart ‘valHomeomorphism’.

instance : ChartedSpace (EuclideanSpace $\mathbb{R}$ (Fin 1)) ConfigurationSpace **where**

instance ClassicalMechanics.HarmonicOscillator.ConfigurationSpace.«instance@66ad2370» [\[source\]](#)

The structure of a smooth (indeed analytic) manifold on ‘ConfigurationSpace’. With a single global chart, the only coordinate change is the chart’s self-transition, which is analytic.

instance : IsManifold $\square\mathbb{R}$ (, EuclideanSpace $\mathbb{R}$ (Fin 1)) ω ConfigurationSpace **where**

2.3 Physlib.ClassicalMechanics.HarmonicOscillator.Solution

[Physlib/ClassicalMechanics/HarmonicOscillator/Solution.lean](#) on GitHub.

def ClassicalMechanics.HarmonicOscillator.InitialConditionsFromTwoPositions.toInitialConditions [\[source\]](#)

Convert two-position initial conditions to standard initial conditions at ‘t = 0’.

*Obtained by solving the 2×2 linear system from the trajectory formula at ‘t₁’ and ‘t₂’. See ‘toInitialConditions_trajectory_at_t₁’ and ‘toInitialConditions_trajectory_at_t₂’ in section D.2 for the correctness proofs (valid under ‘sin (S.ω * (t₂ - t₁)) ≠ 0’).*


```

noncomputable def toInitialConditions (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoPositions) : InitialConditions where

```

def ClassicalMechanics.HarmonicOscillator.InitialConditionsFromTwoVelocities.toInitialConditions

[\[source\]](#)

Convert two-velocity initial conditions to standard initial conditions at $t = 0$.

*Obtained by solving the 2×2 linear system from the velocity formula at t_1 and t_2 .
See `toInitialConditions_velocity_at_t1` and `toInitialConditions_velocity_at_t2`
in section D.3 for the correctness proofs (valid under $\sin(S\omega * (t_2 - t_1)) \neq 0$).*

```

noncomputable def toInitialConditions (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoVelocities) : InitialConditions where

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditionsFromTwoPositions.toInitialConditions_trajectory

[\[source\]](#)

*The trajectory from `toInitialConditions` passes through x_{t_1} at time t_1 , provided $\sin(S\omega * (t_2 - t_1)) \neq 0$.*

```

lemma 1toInitialConditions_trajectory_at_t (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoPositions)
  ( $\Delta h : \sin(S\omega * (IC.2t - IC.1t)) \neq 0$ ) :
  (IC.toInitialConditions S).trajectory S IC.1t = IC.1x_t

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditionsFromTwoPositions.toInitialConditions_trajectory

[\[source\]](#)

*The trajectory from `toInitialConditions` passes through x_{t_2} at time t_2 , provided $\sin(S\omega * (t_2 - t_1)) \neq 0$.*

```

lemma 2toInitialConditions_trajectory_at_t (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoPositions)
  ( $\Delta h : \sin(S\omega * (IC.2t - IC.1t)) \neq 0$ ) :
  (IC.toInitialConditions S).trajectory S IC.2t = IC.2x_t

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditionsFromTwoVelocities.toInitialConditions_velocity

[\[source\]](#)

*The trajectory from `toInitialConditions` has velocity v_{t_1} at time t_1 , provided $\sin(S\omega * (t_2 - t_1)) \neq 0$.*

```

lemma 1toInitialConditions_velocity_at_t (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoVelocities)
  ( $\Delta h : \sin(S\omega * (IC.2t - IC.1t)) \neq 0$ ) :
   $\partial_t ((IC.toInitialConditions S).trajectory S) IC.1t = IC.1v_t$ 

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditionsFromTwoVelocities.toInitialConditions_velocity

[\[source\]](#)

*The trajectory from `toInitialConditions` has velocity v_{t_2} at time t_2 , provided $\sin(S\omega * (t_2 - t_1)) \neq 0$.*

```

lemma  $\partial_t$ toInitialConditions_velocity_at_t (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoVelocities)
  ( $\Delta h$  :  $\sin(S\omega \cdot (IC.2t - IC.1t)) \neq 0$ ) :
   $\partial_t ((IC.toInitialConditions S).trajectory S) IC.2t = IC.2v_t$ 

```

def ClassicalMechanics.HarmonicOscillator.AmplitudePhase.toInitialConditions [\[source\]](#)

Convert amplitude-phase initial conditions to standard initial conditions at ' $t = 0$ ', via ' $x_0 = A \cos \varphi$ ' and ' $v_0 = A \omega \sin \varphi$ '.

See 'toInitialConditions_trajectory_eq_cos' and 'toInitialConditions_velocity_eq_sin' in section E.3 for the correctness proofs.

```

noncomputable def toInitialConditions (S : HarmonicOscillator) (IC : AmplitudePhase) :
  InitialConditions where

```

lemma ClassicalMechanics.HarmonicOscillator.AmplitudePhase.toInitialConditions_trajectory_eq_cos

[\[source\]](#)

The trajectory of amplitude-phase initial conditions is the cosine normal form ' $x(t) = A \cos(\omega t - \varphi)$ '.

```

lemma toInitialConditions_trajectory_eq_cos (S : HarmonicOscillator) (IC : AmplitudePhase)
  (t : Time) :
  (IC.toInitialConditions S).trajectory S t
  = EuclideanSpace.single 0 (IC.A *  $\cos(S\omega \cdot t - IC\varphi)$ )

```

lemma ClassicalMechanics.HarmonicOscillator.AmplitudePhase.toInitialConditions_velocity_eq_sin

[\[source\]](#)

The velocity of the amplitude-phase trajectory is ' $v(t) = -A \omega \sin(\omega t - \varphi)$ '.

```

lemma toInitialConditions_velocity_eq_sin (S : HarmonicOscillator) (IC : AmplitudePhase)
  (t : Time) :
   $\partial_t ((IC.toInitialConditions S).trajectory S) t$ 
  = EuclideanSpace.single 0  $(-(IC.A * S\omega \cdot \sin(S\omega \cdot t.val - IC\varphi))$ )

```

def ClassicalMechanics.HarmonicOscillator.AmplitudePhase.fromInitialConditions [\[source\]](#)

Recover amplitude-phase data from standard initial conditions, as the polar coordinates of the phase vector ' $(x_0, v_0 / \omega)$ ' embedded as ' $z = x_0 + (v_0 / \omega) i$ ': the amplitude is ' $\|z\|$ ' and the phase is ' $\text{Complex.arg } z$ '.

See 'toInitialConditions_fromInitialConditions' for the right-inverse identity.

```

noncomputable def fromInitialConditions (S : HarmonicOscillator) (IC : InitialConditions) :
  AmplitudePhase where

```

lemma ClassicalMechanics.HarmonicOscillator.AmplitudePhase.toInitialConditions_fromInitialConditions

[\[source\]](#)

'fromInitialConditions' is a right inverse of 'toInitialConditions': converting initial conditions to amplitude-phase form and back recovers them exactly.

```

lemma toInitialConditions_fromInitialConditions (S : HarmonicOscillator)
  (IC : InitialConditions) :
  (fromInitialConditions S IC).toInitialConditions S = IC

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_eq_cos [\[source\]](#)

Every trajectory of the harmonic oscillator is a single shifted cosine after converting its initial conditions to amplitude-phase form.

```

lemma trajectory_eq_cos (IC : InitialConditions) (t : Time) :
  IC.trajectory S t =
    EuclideanSpace.single 0 ((AmplitudePhase.fromInitialConditions S IC).A *
      cos (Sw. * t - (AmplitudePhase.fromInitialConditions S IC)φ.))

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_velocity_eq_sin [\[source\]](#)

The velocity of every trajectory is the corresponding shifted sine in amplitude-phase form.

```

lemma trajectory_velocity_eq_sin (IC : InitialConditions) (t : Time) :
  ∂t (IC.trajectory S) t =
    EuclideanSpace.single 0 (-((AmplitudePhase.fromInitialConditions S IC).A * Sw. *
      sin (Sw. * t.val - (AmplitudePhase.fromInitialConditions S IC)φ.)))

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_velocity_eq_zero_iff_sin_eq_zero [\[source\]](#)

For nonzero amplitude, the velocity vanishes exactly when the sine factor in amplitude-phase form vanishes.

```

lemma trajectory_velocity_eq_zero_iff_sin_eq_zero (IC : InitialConditions)
  (hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
  ∂t (IC.trajectory S) t = 0 ↔
    sin (Sw. * t.val - (AmplitudePhase.fromInitialConditions S IC)φ.) = 0

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_velocity_eq_zero_iff_exists_int [\[source\]](#)

For nonzero amplitude, the velocity is zero exactly at phase times $\varphi + n\pi$.

```

lemma trajectory_velocity_eq_zero_iff_exists_int (IC : InitialConditions)
  (hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
  ∂t (IC.trajectory S) t = 0 ↔
    ∃ n : ℤ,
      (t : ℝ) =
        ((AmplitudePhase.fromInitialConditions S IC)φ. + n * π) / Sw.

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_velocity_eq_zero_iff_norm_eq [\[source\]](#)

The velocity vanishes exactly when the trajectory has norm equal to the amplitude.

```

lemma trajectory_velocity_eq_zero_iff_norm_eq_amplitude (IC : InitialConditions)
  (t : Time) :
  ∂t (IC.trajectory S) t = 0 ↔
  ||IC.trajectory S ||t = (AmplitudePhase.fromInitialConditions S IC).A

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_eq_zero_iff_cos_eq_zero
[\[source\]](#)

For nonzero amplitude, the trajectory passes through zero exactly when the cosine factor in amplitude-phase form vanishes.

```

lemma trajectory_eq_zero_iff_cos_eq_zero (IC : InitialConditions)
  (hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
  IC.trajectory S t = 0 ↔
  cos (Sω. * t.val - (AmplitudePhase.fromInitialConditions S IC)φ.) = 0

```

lemma ClassicalMechanics.HarmonicOscillator.InitialConditions.trajectory_eq_zero_iff_exists_int
[\[source\]](#)

For nonzero amplitude, the trajectory passes through zero exactly at phase times ‘ $\varphi + (2n + 1)\pi / 2$ ’.

```

lemma trajectory_eq_zero_iff_exists_int (IC : InitialConditions)
  (hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
  IC.trajectory S t = 0 ↔
  ∃ n : ℤ,
    (t : ℝ) =
      ((AmplitudePhase.fromInitialConditions S IC)φ. + (2 * n + 1) * π / 2) / Sω.

```

lemma ClassicalMechanics.HarmonicOscillator.return_time

[\[source\]](#)

Assuming that the initial coordinate and velocity are not simultaneously zero, the time stamps when the harmonic oscillator returns to its initial coordinate and velocity is a multiple of its period

```

lemma return_time (IC : InitialConditions) (non_trivial : IC.ox ≠ 0 ∨ IC.ov ≠ 0)
  (t : Time) (ht : IC.trajectory S t = IC.ox ∧ ∂t (IC.trajectory S) t = IC.ov) :
  ∃ n : ℤ, (n : ℝ) * (T S) = t

```

2.4 Physlib.ClassicalMechanics.Lagrangian.TotalDerivativeEquivalence

[Physlib/ClassicalMechanics/Lagrangian/TotalDerivativeEquivalence.lean](#) on GitHub.

lemma ClassicalMechanics.Lagrangian.isTotalTimeDerivative_explicit

[\[source\]](#)

Explicit reformulation (by the chain rule): $\delta L(t, q, d_t q) = \partial F / \partial t(t, q) + \langle \nabla_r F(t, q), d_t q \rangle$

or

$\delta L(t, q, d_t q) = fderiv \mathbb{R} F(t, q)(1, d_t q)$

```

lemma isTotalTimeDerivative_explicit δ{L : Time → X → X → ℝ} :
  IsTotalTimeDerivative δL ↔ ∃( (F : Time → X → ℝ) (_ : ContDiff ℝ ∞ 1F),
  ∀ t q v, δL t q v = fderiv ℝ 1F (t, q) ((1 : Time), v))

```

2.5 Physlib.Electromagnetism.Distributional.Basic

[Physlib/Electromagnetism/Distributional/Basic.lean](#) on GitHub.

def Electromagnetism.DistElectromagneticPotential.ofStaticScalarPotentialFunction [\[source\]](#)

The creation of an electromagnetic potential from a static scalar-potential function.

If the function is distribution-bounded, it is first promoted to a distribution using ‘Space.distOfFunction’. Otherwise the constructor returns zero.

```
noncomputable def ofStaticScalarPotentialFunction {d} (c : SpeedOfLight)
  φ( : Space d → ℝ) : DistElectromagneticPotential d
```

lemma Electromagnetism.DistElectromagneticPotential.ofStaticScalarPotentialFunction_of_isDistBounded

[\[source\]](#)

```
@[simp]
lemma ofStaticScalarPotentialFunction_of_isDistBounded {d} (c : SpeedOfLight)
  φ( : Space d → ℝ) (φh : Space.IsDistBounded φ) :
  ofStaticScalarPotentialFunction c φ =
  ofStaticScalarPotential c (Space.distOfFunction φ φh)
```

lemma Electromagnetism.DistElectromagneticPotential.ofStaticScalarPotentialFunction_eq_zero_of_not_isD

[\[source\]](#)

```
@[simp]
lemma ofStaticScalarPotentialFunction_eq_zero_of_not_isDistBounded {d}
  (c : SpeedOfLight) φ( : Space d → ℝ) (φh : ¬ Space.IsDistBounded φ) :
  ofStaticScalarPotentialFunction c φ = 0
```

def Electromagnetism.DistElectromagneticPotential.ofVectorPotential

[\[source\]](#)

The creation of an electromagnetic potential from a vector potential.

```
noncomputable def ofVectorPotential {d} (c : SpeedOfLight) :
  ((Time × Space d) → dℝ[]) EuclideanSpace ℝ (Fin d)) → iℝ[]
  DistElectromagneticPotential d where
```

def Electromagnetism.DistElectromagneticPotential.ofStaticVectorPotential

[\[source\]](#)

The creation of an electromagnetic potential from a static vector potential.

```
noncomputable def ofStaticVectorPotential {d} (c : SpeedOfLight) :
  ((Space d) → dℝ[]) EuclideanSpace ℝ (Fin d)) → iℝ[] DistElectromagneticPotential d
```

lemma Electromagnetism.DistElectromagneticPotential.distTensorDeriv_eq_sum_sum [\[source\]](#)

```
lemma distTensorDeriv_eq_sum_sum {d} (A : DistElectromagneticPotential d)
  ε( : □(SpaceTime d, ℝ)) :
  distTensorDeriv A ε ∑= μ, ∑ v, (SpaceTime.distDeriv μ A ε v) •
  Lorentz.CoVector.basis μ ⊗ iℝ[] Lorentz.Vector.basis v
```

lemma Electromagnetism.DistElectromagneticPotential.distTensorDeriv_basis_repr_apply [\[source\]](#)

```
@[simp]
lemma distTensorDeriv_basis_repr_apply {d} μν{ : (Fin 1 ⊗ Fin d) × (Fin 1 ⊗ Fin d)}
  (A : DistElectromagneticPotential d)
  ε( : □(SpaceTime d, ℝ)) :
  (Lorentz.CoVector.basis.tensorProduct Lorentz.Vector.basis).repr (distTensorDeriv A ε) μν =
  distDeriv μν.1 A ε μν.2
```

lemma Electromagnetism.DistElectromagneticPotential.toTensor_distTensorDeriv_basis_repr_apply [\[source\]](#)

```
lemma toTensor_distTensorDeriv_basis_repr_apply {d} (A : DistElectromagneticPotential d)
  ε( : □(SpaceTime d, ℝ)) (b : ComponentIdx (S := realLorentzTensor d)
  (Fin.append ![Color.down] ![Color.up])) :
  (Tensor.basis _).repr (Tensorial.toTensor (distTensorDeriv A ε)) b =
  distDeriv (b 0) A ε (b 1)
```

2.6 Physlib.Electromagnetism.Kinematics.EMPotential

[Physlib/Electromagnetism/Kinematics/EMPotential.lean](#) on GitHub.

instance Electromagnetism.ElectromagneticPotential.«instance@94a1323b» [\[source\]](#)

```
instance {d} : Zero (ElectromagneticPotential d) where
```

lemma Electromagnetism.ElectromagneticPotential.zero_val [\[source\]](#)

```
@[simp]
lemma zero_val {d} : (0 : ElectromagneticPotential d).val = 0
```

lemma Electromagnetism.ElectromagneticPotential.zero_apply [\[source\]](#)

```
@[simp]
lemma zero_apply {d} (x : SpaceTime d) : (0 : ElectromagneticPotential d) x = 0
```

instance Electromagnetism.ElectromagneticPotential.«instance@3f7e7f60» [\[source\]](#)

```
instance {d} : Neg (ElectromagneticPotential d) where
```

lemma Electromagnetism.ElectromagneticPotential.neg_val [\[source\]](#)

```
@[simp]
lemma neg_val {d} (A : ElectromagneticPotential d) :
  (- A).val = - A.val
```

lemma Electromagnetism.ElectromagneticPotential.neg_apply [\[source\]](#)

```
@[simp]
lemma neg_apply {d} (A : ElectromagneticPotential d) (x : SpaceTime d) :
  (- A) x = - A x
```

instance Electromagnetism.ElectromagneticPotential.«instance@6c8e2750»

[\[source\]](#)

```
instance {d} : Sub (ElectromagneticPotential d) where
```

lemma Electromagnetism.ElectromagneticPotential.sub_val

[\[source\]](#)

```
@[simp]
lemma sub_val {d} (A B : ElectromagneticPotential d) :
  (A - B).val = A.val - B.val
```

lemma Electromagnetism.ElectromagneticPotential.sub_apply

[\[source\]](#)

```
@[simp]
lemma sub_apply {d} (A B : ElectromagneticPotential d) (x : SpaceTime d) :
  (A - B) x = A x - B x
```

instance Electromagnetism.ElectromagneticPotential.«instance@2040b1c1»

[\[source\]](#)

```
instance {d} : AddCommGroup (ElectromagneticPotential d) where
```

lemma Electromagnetism.ElectromagneticPotential.action_apply

[\[source\]](#)

```
@[simp]
lemma action_apply {d}  $\Lambda$  ( : LorentzGroup d) (A : ElectromagneticPotential d)
  (x : SpaceTime d) :
 $\Lambda ( \bullet A) x = \Lambda \bullet A \Lambda^{-1} \bullet x$ 
```

instance Electromagnetism.ElectromagneticPotential.«instance@f3d05234»

[\[source\]](#)

```
noncomputable instance {d} :
  DistribMulAction (LorentzGroup d) (ElectromagneticPotential d) where
```

2.7 Physlib.Electromagnetism.Kinematics.FieldStrength

[Physlib/Electromagnetism/Kinematics/FieldStrength.lean](#) on GitHub.

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_eval_eq_basis_repr

[\[source\]](#)

Evaluating both tensor indices of the field strength gives the coefficient in the standard tensor-product basis.

```
lemma toFieldStrength_eval_eq_basis_repr {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d)  $\mu$  (  $v$  : Fin 1  $\otimes$  Fin d) :
  toField {A.toFieldStrength x |  $\mu$ []  $v^T$ []} =
  (Lorentz.CoVector.basis.tensorProduct Lorentz.Vector.basis).repr
  (A.toFieldStrength x)  $\mu$ (,  $v$ )
```

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_eval_apply

[\[source\]](#)

The evaluated components of the field strength tensor in terms of derivatives of the electromagnetic potential.

```

lemma toFieldStrength_eval_apply {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) μ( v : Fin 1 ⊗ Fin d) :
  toField {A.toFieldStrength x | μ[] vT[]} =
  ∑ κ, η( μ κ * ∂_ κ A x v - η v κ * ∂_ κ A x μ)

```

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_eval_apply_eq_single [\[source\]](#)

The evaluated components of the field strength tensor after using diagonal form of the Minkowski metric.

```

lemma toFieldStrength_eval_apply_eq_single {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) μ( v : Fin 1 ⊗ Fin d) :
  toField {A.toFieldStrength x | μ[] vT[]} =
  η μ μ * ∂_ μ A x v - η v v * ∂_ v A x μ

```

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_eval_eq_fieldStrengthMatrix [\[source\]](#)

Index evaluation of the field strength tensor agrees with the corresponding component of the field strength matrix.

```

lemma toFieldStrength_eval_eq_fieldStrengthMatrix {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) μ( v : Fin 1 ⊗ Fin d) :
  toField {A.toFieldStrength x | μ[] vT[]} = A.fieldStrengthMatrix x μ(, v)

```

2.8 Physlib.Electromagnetism.Kinematics.GaugeTransformation

[Physlib/Electromagnetism/Kinematics/GaugeTransformation.lean](#) on GitHub.

def Electromagnetism.ElectromagneticPotential.ofGradient [\[source\]](#)

*The pure-gauge electromagnetic potential $A^\mu = \partial^\mu \chi = \eta^{\mu\nu} \partial_\nu \chi$ built from a gauge function χ . The metric-contracted form $\sum \kappa, \eta \mu \kappa * \partial_\kappa \chi$ is the definition; the diagonal form (which equals $\eta \mu \mu * \partial_\mu \chi$ since η is diagonal) is derived in `ofGradient_apply`.*

```

noncomputable def ofGradient {d} χ( : SpaceTime d → ℝ) : ElectromagneticPotential d where

```

lemma Electromagnetism.ElectromagneticPotential.ofGradient_apply_sum [\[source\]](#)

Unfolding of the summed definition of `ofGradient`.

```

lemma ofGradient_apply_sum {d} χ( : SpaceTime d → ℝ) (x : SpaceTime d) μ( : Fin 1 ⊗ Fin d) :
  ofGradient χ x μ = ∑ κ, η μ κ * ∂_ κ χ x

```

lemma Electromagnetism.ElectromagneticPotential.ofGradient_apply [\[source\]](#)

Evaluation of `ofGradient` in the diagonal form; the off-diagonal entries of η vanish so only the $\kappa = \mu$ term survives.

```

lemma ofGradient_apply {d} χ( : SpaceTime d → ℝ) (x : SpaceTime d) μ( : Fin 1 ⊗ Fin d) :
  ofGradient χ x μ = η μ μ * ∂_ μ χ x

```


lemma Electromagnetism.ElectromagneticPotential.ofGradient_zero[\[source\]](#)

The pure-gauge potential built from the zero gauge function has all components zero.

```
lemma ofGradient_zero {d} (x : SpaceTime d) μ( : Fin 1 ⊗ Fin d) :
  ofGradient (0 : SpaceTime d → ℝ) x μ = 0
```

lemma Electromagnetism.ElectromagneticPotential.ofGradient_add[\[source\]](#)

'ofGradient' is additive in the gauge function (when both summands are differentiable).

```
lemma ofGradient_add {d} χ₁{ χ₂ : SpaceTime d → ℝ}
  (χ₁h : Differentiable ℝ χ₁) (χ₂h : Differentiable ℝ χ₂) :
  ofGradient χ₁( + χ₂) = ofGradient χ₁ + ofGradient χ₂
```

lemma Electromagnetism.ElectromagneticPotential.differentiable_ofGradient[\[source\]](#)

The pure-gauge potential is differentiable when 'χ' is 'C²'.

```
lemma differentiable_ofGradient {d} χ{ : SpaceTime d → ℝ} (χh : ContDiff ℝ 2 χ) :
  Differentiable ℝ (ofGradient χ)
```

lemma Electromagnetism.ElectromagneticPotential.contDiff_ofGradient[\[source\]](#)

The pure-gauge potential is 'Cⁿ' when 'χ' is 'C^{n+1}'.

```
lemma contDiff_ofGradient {n} {d} χ{ : SpaceTime d → ℝ} (χh : ContDiff ℝ (n + 1) χ) :
  ContDiff ℝ n (ofGradient χ)
```

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_ofGradient[\[source\]](#)

A pure-gauge potential has vanishing field strength.

```
lemma toFieldStrength_ofGradient {d} χ{ : SpaceTime d → ℝ} (χh : ContDiff ℝ 2 χ)
  (x : SpaceTime d) : (ofGradient χ).toFieldStrength x = 0
```

lemma Electromagnetism.ElectromagneticPotential.ofGradient_equivariant[\[source\]](#)

'ofGradient' intertwines the Lorentz action on potentials with composition by Λ^{-1} on the gauge function: ' $\Lambda \bullet \text{ofGradient } \chi = \text{ofGradient } (\chi \circ (\Lambda^{-1} \bullet \cdot))$ '.

```
lemma ofGradient_equivariant {d} χ( : SpaceTime d → ℝ) (χh : Differentiable ℝ χ)
  Λ( : LorentzGroup d) :
  Λ • ofGradient χ = ofGradient χ( • Λ-1 • .))
```

def Electromagnetism.ElectromagneticPotential.gaugeTransform[\[source\]](#)

The gauge transformation ' $\hat{A}_\mu \mapsto \hat{A}_\mu + \partial_\mu \chi$ ' of an electromagnetic potential.

```
noncomputable def gaugeTransform {d} χ( : SpaceTime d → ℝ) (A : ElectromagneticPotential d) :
  ElectromagneticPotential d
```

lemma Electromagnetism.ElectromagneticPotential.gaugeTransform_apply[\[source\]](#)*Evaluation of ‘gaugeTransform’.*

lemma gaugeTransform_apply {d} χ (: SpaceTime d $\rightarrow \mathbb{R}$) (A : ElectromagneticPotential d)
 (x : SpaceTime d) : gaugeTransform χ A x = A x + ofGradient χ x

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_gaugeTransform[\[source\]](#)*The field strength tensor is invariant under gauge transformations.*

lemma toFieldStrength_gaugeTransform {d} (A : ElectromagneticPotential d)
 χ (: SpaceTime d $\rightarrow \mathbb{R}$) (hA : Differentiable $\mathbb{R}$ A) (χ h : ContDiff $\mathbb{R}^2 \chi$) (x : SpaceTime d) :
 (gaugeTransform χ A).toFieldStrength x = A.toFieldStrength x

lemma Electromagnetism.ElectromagneticPotential.fieldStrengthMatrix_gaugeTransform[\[source\]](#)*The field strength matrix is invariant under gauge transformations.*

lemma fieldStrengthMatrix_gaugeTransform {d} (A : ElectromagneticPotential d)
 χ (: SpaceTime d $\rightarrow \mathbb{R}$) (hA : Differentiable $\mathbb{R}$ A) (χ h : ContDiff $\mathbb{R}^2 \chi$) (x : SpaceTime d) :
 (gaugeTransform χ A).fieldStrengthMatrix x = A.fieldStrengthMatrix x

lemma Electromagnetism.ElectromagneticPotential.gaugeTransform_zero[\[source\]](#)*Shifting by the zero gauge function is the identity.*

lemma gaugeTransform_zero {d} (A : ElectromagneticPotential d) :
 gaugeTransform (0 : SpaceTime d $\rightarrow \mathbb{R}$) A = A

lemma Electromagnetism.ElectromagneticPotential.gaugeTransform_gaugeTransform[\[source\]](#)*Two successive gauge shifts compose: shifting by χ_2 then χ_1 equals shifting by $\chi_1 + \chi_2$. This upgrades one-step F-invariance to invariance along any finite chain of gauge shifts.*

lemma gaugeTransform_gaugeTransform {d} (A : ElectromagneticPotential d)
 χ_1 (χ_2 : SpaceTime d $\rightarrow \mathbb{R}$) (χ_1 h : Differentiable $\mathbb{R} \chi_1$) (χ_2 h : Differentiable $\mathbb{R} \chi_2$) :
 gaugeTransform χ_1 (gaugeTransform χ_2 A) = gaugeTransform ($\chi_1 + \chi_2$) A

lemma Electromagnetism.ElectromagneticPotential.gaugeTransform_equivariant[\[source\]](#)*Gauge transformations commute with Lorentz transformations: applying Λ and then performing a gauge transformation by ‘ χ ’ equals performing a gauge transformation by ‘ $\chi \circ (\Lambda^{-1} \bullet \cdot)$ ’ and then applying Λ .*

lemma gaugeTransform_equivariant {d} (A : ElectromagneticPotential d)
 χ (: SpaceTime d $\rightarrow \mathbb{R}$) (χ h : Differentiable $\mathbb{R} \chi$) Λ (: LorentzGroup d) :
 $\Lambda \bullet$ gaugeTransform χ A = gaugeTransform ($\chi \circ \Lambda^{-1} \bullet \cdot$) Λ ($\bullet$ A)

lemma Electromagnetism.ElectromagneticPotential.fieldStrengthMatrix_bareGradient_inl_inr[\[source\]](#)

The ‘(inl 0, inr i)’ component of the field strength matrix of the bare-gradient potential ‘ $\hat{B}_\mu := \partial_\mu \chi$ ’ for ‘ $\chi(x) = x^0 \cdot x^i$ ’ equals ‘2’. This witnesses that the bare covariant gradient does not produce a gauge-invariant field strength, so the raised-index contraction ‘ $\eta^{\{\mu\nu\}} \partial_\nu \chi$ ’ in ‘ofGradient’ is necessary (see the module overview).

```
lemma fieldStrengthMatrix_bareGradient_inl_inr {d : ℕ} (i : Fin d)
  (x : SpaceTime d) :
  let  $\chi$  : SpaceTime d → ℝ
```

lemma Electromagnetism.ElectromagneticPotential.toFieldStrength_bareGradient_ne_zero [\[source\]](#)

The field strength of the bare-gradient potential ‘ $\hat{B}_\mu := \partial_\mu \chi$ ’ for ‘ $\chi(x) = x^0 \cdot x^i$ ’ is nonzero (follows from ‘fieldStrengthMatrix_bareGradient_inl_inr’).

```
lemma toFieldStrength_bareGradient_ne_zero {d : ℕ} (i : Fin d)
  (x : SpaceTime d) :
  let  $\chi$  : SpaceTime d → ℝ
```

2.9 Physlib.Electromagnetism.Kinematics.MagneticField

[Physlib/Electromagnetism/Kinematics/MagneticField.lean](#) on GitHub.

lemma Electromagnetism.ElectromagneticPotential.magneticField_coord_eq_fieldStrengthMatrix [\[source\]](#)

```
lemma magneticField_coord_eq_fieldStrengthMatrix {i : Fin 3} {c : SpeedOfLight}
  (A : ElectromagneticPotential) (t : Time)
  (x : Space) (hA : Differentiable ℝ A) :
  A.magneticField c t x i =
  - A.fieldStrengthMatrix ((toTimeAndSpace c).symm (t, x)) (Sum.inr (i+1), Sum.inr (i+2))
```

2.10 Physlib.Mathematics.Calculus.Wirtinger.Basic

[Physlib/Mathematics/Calculus/Wirtinger/Basic.lean](#) on GitHub.

def Physlib.Wirtinger.weightedDirDeriv [\[source\]](#)

The base-point field ‘ $p \mapsto (1/2)(d_{\{b_1\}} f + c \cdot d_{\{b_2\}} f)$ ’, a weighted combination of the real Fréchet derivative of ‘f’ along two directions ‘ b_1 ’, ‘ b_2 ’. The directional Wirtinger operators are its two specializations: ‘dWirtingerDir f v’ at ‘ $c = -i$ ’, ‘(b_1, b_2) = (v, i·v)’, and ‘dWirtingerAntiDir f v’ at ‘ $c = i$ ’. Keeping ‘ b_1 ’, ‘ b_2 ’ free lets §G differentiate it a second time once and reuse the bridge for both operators.

```
def weightedDirDeriv (f : V → ℂ) (c : ℂ) (b₁ b₂ : V) : V → ℂ
```

def Physlib.Wirtinger.dWirtingerDir [\[source\]](#)

The holomorphic directional Wirtinger derivative ‘ $\partial_v f = (1/2)(d_v f - i d_{\{i \cdot v\}} f)$ ’ of ‘ $f : V \rightarrow \mathbb{C}$ ’ along the direction vector ‘ $v : V$ ’, the ‘weightedDirDeriv’ at ‘ $c = -i$ ’.

```
def dWirtingerDir (f : V → ℂ) (v u : V) : ℂ
```

def Physlib.Wirtinger.dWirtingerAntiDir[\[source\]](#)

The anti-holomorphic directional Wirtinger derivative $\partial_{\bar{v}} f = (1/2)(d_v f + i d_{\{i \cdot v\}} f)$ of $f : V \rightarrow \mathbb{C}$ along the direction vector $v : V$, the *weightedDirDeriv* at $c = i$.

```
def dWirtingerAntiDir (f : V → ℂ) (v u : V) : ℂ
```

lemma Physlib.Wirtinger.dWirtingerDir_apply[\[source\]](#)

Definitional unfolding of *dWirtingerDir* to the explicit Wirtinger combination, used to expand the outer operator of a composition without touching the inner one.

```
lemma dWirtingerDir_apply (g : V → ℂ) (v u : V) :
  dWirtingerDir g v u
  = (1 / 2 : ℂ) * (fderiv ℝ g u v - Complex.I * fderiv ℝ g u (Complex.I • v))
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_apply[\[source\]](#)

Definitional unfolding of *dWirtingerAntiDir* to the explicit Wirtinger combination.

```
lemma dWirtingerAntiDir_apply (g : V → ℂ) (v u : V) :
  dWirtingerAntiDir g v u
  = (1 / 2 : ℂ) * (fderiv ℝ g u v + Complex.I * fderiv ℝ g u (Complex.I • v))
```

lemma Physlib.Wirtinger.dWirtingerDir_add[\[source\]](#)

Additivity of *dWirtingerDir*, $\partial_v(g + h) = \partial_v g + \partial_v h$.

```
lemma dWirtingerDir_add {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerDir (g + h) v u = dWirtingerDir g v u + dWirtingerDir h v u
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_add[\[source\]](#)

Additivity of *dWirtingerAntiDir*, $\partial_{\bar{v}}(g + h) = \partial_{\bar{v}} g + \partial_{\bar{v}} h$.

```
lemma dWirtingerAntiDir_add {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerAntiDir (g + h) v u = dWirtingerAntiDir g v u + dWirtingerAntiDir h v u
```

lemma Physlib.Wirtinger.dWirtingerDir_smul[\[source\]](#)

Compatibility of *dWirtingerDir* with complex scalar multiplication, $\partial_v(c \cdot g) = c \cdot \partial_v g$.

```
lemma dWirtingerDir_smul (c : ℂ) {g : V → ℂ} (hg : DifferentiableAt ℝ g u) (v : V) :
  dWirtingerDir (c • g) v u = c • dWirtingerDir g v u
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_smul[\[source\]](#)

Compatibility of *dWirtingerAntiDir* with complex scalar multiplication, $\partial_{\bar{v}}(c \cdot g) = c \cdot \partial_{\bar{v}} g$.

```
lemma dWirtingerAntiDir_smul (c : ℂ) {g : V → ℂ} (hg : DifferentiableAt ℝ g u) (v : V) :
  dWirtingerAntiDir (c • g) v u = c • dWirtingerAntiDir g v u
```

lemma Physlib.Wirtinger.fderiv_mul_apply

[\[source\]](#)

The real Fréchet derivative of a product, evaluated at a tangent 'v'.

```
private lemma fderiv_mul_apply {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  fderiv ℝ (g * h) u v = g u * fderiv ℝ h u v + h u * fderiv ℝ g u v
```

lemma Physlib.Wirtinger.dWirtingerDir_mul

[\[source\]](#)

The Wirtinger Leibniz rule for 'dWirtingerDir', $\partial_v(g \cdot h) = \partial_v g \cdot h + g \cdot \partial_v h$.

```
lemma dWirtingerDir_mul {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerDir (g * h) v u = dWirtingerDir g v u * h u + g u * dWirtingerDir h v u
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_mul

[\[source\]](#)

The Wirtinger Leibniz rule for 'dWirtingerAntiDir', $\bar{\partial}_v(g \cdot h) = \bar{\partial}_v g \cdot h + g \cdot \bar{\partial}_v h$.

```
lemma dWirtingerAntiDir_mul {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerAntiDir (g * h) v u =
    dWirtingerAntiDir g v u * h u + g u * dWirtingerAntiDir h v u
```

lemma Physlib.Wirtinger.dWirtingerDir_fun_sum

[\[source\]](#)

Finite-sum rule for 'dWirtingerDir', $\partial_v(\sum_a F_a) = \sum_a \partial_v F_a$.

```
lemma dWirtingerDir_fun_sum α{ : Type*} {s : Finset α} {F : α → V → ℂ}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (v : V) :
  dWirtingerDir (fun p => ∑ a ∈ s, F a p) v u = ∑ a ∈ s, dWirtingerDir (F a) v u
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_fun_sum

[\[source\]](#)

Finite-sum rule for 'dWirtingerAntiDir', $\bar{\partial}_v(\sum_a F_a) = \sum_a \bar{\partial}_v F_a$.

```
lemma dWirtingerAntiDir_fun_sum α{ : Type*} {s : Finset α} {F : α → V → ℂ}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (v : V) :
  dWirtingerAntiDir (fun p => ∑ a ∈ s, F a p) v u = ∑ a ∈ s, dWirtingerAntiDir (F a) v u
```

lemma Physlib.Wirtinger.fderiv_star_eq

[\[source\]](#)

*Differentiation commutes with conjugation: the real Fréchet derivative of the point-wise conjugate $p \mapsto \text{star}(f p)$ is 'conjCLE' (conjugation on $\mathbb{C}$) composed with 'fderiv ℝ f u'; in physicists' notation, $d f = \text{conj}(d f)$. Conjugation is $\mathbb{R}$ -linear, so it slides through the real derivative unchanged, whereas it does *not* commute with the holomorphic Wirtinger derivative ∂_v . The 'star' conjugates the *output* $f p$, so this is not a derivative in a conjugate variable.*

This is the analytic core of the conjugation lemmas below ('dWirtingerDir_star_comp' and its dual): distributed over the Wirtinger split of 'fderiv ℝ f u' ('realLinear_apply_eq_wirtinger', §D), the outer 'conjCLE' conjugates the two coefficients and swaps the holomorphic and anti-holomorphic parts.

```
lemma fderiv_star_eq {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  {f : E → ℂ} {u : E} (hf : DifferentiableAt ℝ f u) :
  fderiv ℝ (fun p : E => star (f p)) u =
    Complex.conjCLE.toContinuousLinearMap.comp (fderiv ℝ f u)
```

lemma Physlib.Wirtinger.dWirtingerDir_star_comp

[\[source\]](#)

Conjugating the function swaps the operators up to an outer conjugation: ' $\partial_v f = \text{conj}(\partial_v f)$ '.

```
lemma dWirtingerDir_star_comp (hf : DifferentiableAt ℝ f u) (v : V) :
  dWirtingerDir (fun p => star (f p)) v u = star (dWirtingerAntiDir f v u)
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_star_comp

[\[source\]](#)

Conjugating the function swaps the operators up to an outer conjugation: ' $\partial_v f = \text{conj}(\partial_v f)$ '. Dual of 'dWirtingerDir_star_comp'.

```
lemma dWirtingerAntiDir_star_comp (hf : DifferentiableAt ℝ f u) (v : V) :
  dWirtingerAntiDir (fun p => star (f p)) v u = star (dWirtingerDir f v u)
```

lemma Physlib.Wirtinger.realLinear_apply_eq_wirtinger

[\[source\]](#)

*Split a real-linear map ' $\mathbb{C} \rightarrow \mathbb{C}$ ' into its Wirtinger components. Any real-linear ' $L : \mathbb{C} \rightarrow L[\mathbb{R}] \mathbb{C}$ ' splits into a holomorphic and an anti-holomorphic part with the Wirtinger coefficients ' $a = \frac{1}{2}(L 1 - i * L i)$ ', ' $b = \frac{1}{2}(L 1 + i * L i)$ ' as weights:*

*' $L w = a * w + b * \text{star } w$ '.*

This is purely algebraic: ' L ' is an arbitrary real-linear map, no derivative involved. Its use is the Wirtinger chain rule ('dWirtingerDir_comp' below), where the weights of the outer differential ' $L = \text{fderiv } \mathbb{R} g (f u)$ ' are the coefficients ' $\partial g / \partial f$ ', ' $\partial g / \partial \bar{f}$ '.

```
lemma realLinear_apply_eq_wirtinger (L : ℂ → L[ℝ] ℂ) (w : ℂ) :
  L w =
    ((1 / 2 : ℂ) * (L 1 - Complex.I * L Complex.I)) * w
    + ((1 / 2 : ℂ) * (L 1 + Complex.I * L Complex.I)) * star w
```

lemma Physlib.Wirtinger.dWirtingerDir_comp

[\[source\]](#)

The two-term Wirtinger chain rule for 'dWirtingerDir', outer ' $g : \mathbb{C} \rightarrow \mathbb{C}$ ' and inner ' $f : V \rightarrow \mathbb{C}$ ':

' $\partial_v (g \circ f) = (\partial g / \partial f) \cdot \partial_v f + (\partial g / \partial \bar{f}) \cdot \partial_v \bar{f}$ '.

'realLinear_apply_eq_wirtinger' splits the chain rule's outer ' $\mathbb{R}$ '-linear factor into the ' $\partial g / \partial f$ ', ' $\partial g / \partial \bar{f}$ ' coefficients, each multiplying its inner directional derivative ' $\partial_v f$ ', ' $\partial_v \bar{f}$ '.

```
lemma dWirtingerDir_comp {g : ℂ → ℂ} (hg : DifferentiableAt ℝ g (f u))
  (hf : DifferentiableAt ℝ f u) (v : V) :
```

```
dWirtingerDir (fun p => g (f p)) v u =
  dWirtingerDir g 1 (f u) * dWirtingerDir f v u
  + dWirtingerAntiDir g 1 (f u) * dWirtingerDir (fun p => star (f p)) v u
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_comp

[\[source\]](#)

The two-term Wirtinger chain rule for ‘dWirtingerAntiDir’, the anti-holomorphic dual of ‘dWirtingerDir_comp’:

$$\partial_v(g \circ f) = (\partial g / \partial f) \cdot \partial_v f + (\partial g / \partial \bar{f}) \cdot \partial_v \bar{f}.$$

Same outer ‘ $\partial g / \partial f$ ’, ‘ $\partial g / \partial \bar{f}$ ’ coefficients, now each multiplying its anti-holomorphic inner derivative ‘ $\partial_v f$ ’, ‘ $\partial_v \bar{f}$ ’; same proof as ‘dWirtingerDir_comp’.

```
lemma dWirtingerAntiDir_comp {g : ℂ → ℂ} (hg : DifferentiableAt ℝ g (f u))
(hf : DifferentiableAt ℝ f u) (v : V) :
dWirtingerAntiDir (fun p => g (f p)) v u =
  dWirtingerDir g 1 (f u) * dWirtingerAntiDir f v u
  + dWirtingerAntiDir g 1 (f u) * dWirtingerAntiDir (fun p => star (f p)) v u
```

lemma Physlib.Wirtinger.fderiv_comp_clm_apply

[\[source\]](#)

Chain rule for an inner continuous linear map ‘L’. Because the derivative of a linear map is the map itself, the real Fréchet derivative of ‘g ∘ L’ at ‘u’, applied to ‘x’, equals the derivative of ‘g’ at ‘L u’ applied to ‘L x’.

```
private lemma fderiv_comp_clm_apply {g : V' → ℂ} {L : V → Lℝ[] V'} {u : V}
(hg : DifferentiableAt ℝ g (L u)) (x : V) :
fderiv ℝ (fun p => g (L p)) u x = fderiv ℝ g (L u) (L x)
```

lemma Physlib.Wirtinger.dWirtingerDir_comp_conjLinear

[\[source\]](#)

Domain conjugation swaps the operators: precomposing with a conjugate-‘ℂ’-linear ‘L’ turns the holomorphic derivative of ‘g ∘ L’ at ‘u’ into the anti-holomorphic derivative of ‘g’ at the mapped point ‘L u’, in the mapped direction ‘L v’: ‘ $\partial_v(g \circ L)$ ’ at ‘u’ equals ‘ $\partial_{\{L v\}} g$ ’ at ‘L u’.

```
lemma dWirtingerDir_comp_conjLinear {g : V' → ℂ} {L : V → Lℝ[] V'} {u : V}
(hL : ∀ x : V, L (Complex.I • x) = -(Complex.I • L x))
(hg : DifferentiableAt ℝ g (L u)) (v : V) :
dWirtingerDir (fun p => g (L p)) v u = dWirtingerAntiDir g (L v) (L u)
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_comp_conjLinear

[\[source\]](#)

Dual of ‘dWirtingerDir_comp_conjLinear’: the anti-holomorphic derivative of ‘g ∘ L’ at ‘u’ is the holomorphic derivative of ‘g’ at the mapped point ‘L u’, in the mapped direction ‘L v’: ‘ $\partial_v(g \circ L)$ ’ at ‘u’ equals ‘ $\partial_{\{L v\}} g$ ’ at ‘L u’.

```
lemma dWirtingerAntiDir_comp_conjLinear {g : V' → ℂ} {L : V → Lℝ[] V'} {u : V}
(hL : ∀ x : V, L (Complex.I • x) = -(Complex.I • L x))
(hg : DifferentiableAt ℝ g (L u)) (v : V) :
dWirtingerAntiDir (fun p => g (L p)) v u = dWirtingerDir g (L v) (L u)
```

lemma Physlib.Wirtinger.dWirtingerDir_eq_of_clinear[\[source\]](#)

Holomorphic collapse: along a direction where the real derivative is $\mathbb{C}$ -linear, the holomorphic derivative is the full real derivative, $\partial_v f = d_v f$.

```
lemma dWirtingerDir_eq_of_clinear {v : V}
  (h : fderiv ℝ f u (Complex.I • v) = Complex.I • fderiv ℝ f u v) :
  dWirtingerDir f v u = fderiv ℝ f u v
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_eq_zero_of_clinear[\[source\]](#)

Holomorphic collapse: the anti-holomorphic derivative vanishes along a direction of $\mathbb{C}$ -linearity, $\bar{\partial}_v f = 0$.

```
lemma dWirtingerAntiDir_eq_zero_of_clinear {v : V}
  (h : fderiv ℝ f u (Complex.I • v) = Complex.I • fderiv ℝ f u v) :
  dWirtingerAntiDir f v u = 0
```

lemma Physlib.Wirtinger.dWirtingerDir_eq_zero_of_antilinear[\[source\]](#)

Anti-holomorphic collapse: a direction of conjugate- $\mathbb{C}$ -linearity kills the holomorphic derivative, $\partial_v f = 0$.

```
lemma dWirtingerDir_eq_zero_of_antilinear {v : V}
  (h : fderiv ℝ f u (Complex.I • v) = -(Complex.I • fderiv ℝ f u v)) :
  dWirtingerDir f v u = 0
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_eq_of_antilinear[\[source\]](#)

Anti-holomorphic collapse: the anti-holomorphic derivative is the full real derivative along a direction of conjugate- $\mathbb{C}$ -linearity, $\bar{\partial}_v f = d_v f$.

```
lemma dWirtingerAntiDir_eq_of_antilinear {v : V}
  (h : fderiv ℝ f u (Complex.I • v) = -(Complex.I • fderiv ℝ f u v)) :
  dWirtingerAntiDir f v u = fderiv ℝ f u v
```

lemma Physlib.Wirtinger.hasFDerivAt_fderiv_apply[\[source\]](#)

The field $p \mapsto d_p f$ is the evaluation map b composed with $fderiv ℝ f$, so when $fderiv ℝ f$ is differentiable its derivative is $fderiv ℝ (fderiv ℝ f) u$ post-composed with that evaluation.

```
private lemma hasFDerivAt_fderiv_apply (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (b : V) :
  HasFDerivAt (fun p => fderiv ℝ f p b)
    ((ContinuousLinearMap.apply ℝ ℂ b).comp (fderiv ℝ (fderiv ℝ f) u)) u
```

lemma Physlib.Wirtinger.hasFDerivAt_weightedDirDeriv[\[source\]](#)

The weightedDirDeriv is differentiable wherever $fderiv ℝ f$ is.

```
private lemma hasFDerivAt_weightedDirDeriv (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (c : ℂ) (ib zb : V) :
  HasFDerivAt (weightedDirDeriv f c ib zb)
    ((1 / 2 : ℂ) • ((ContinuousLinearMap.apply ℝ ℂ ib).comp (fderiv ℝ (fderiv ℝ f) u)
      + c • (ContinuousLinearMap.apply ℝ ℂ zb).comp (fderiv ℝ (fderiv ℝ f) u))) u
```


lemma Physlib.Wirtinger.fderiv_weightedDirDeriv[\[source\]](#)

The bridge: differentiating a ‘weightedDirDeriv’ along a third direction ‘a’ lands on the second real Fréchet derivative ‘fderiv $\mathbb{R}$ (fderiv $\mathbb{R}$ f) u a b’ in the two slots.

```
private lemma fderiv_weightedDirDeriv (hf' : DifferentiableAt  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u)
  (c :  $\mathbb{C}$ ) (1b 2b a : V) :
  fderiv  $\mathbb{R}$  (weightedDirDeriv f c 1b 2b) u a
  = (1 / 2 :  $\mathbb{C}$ ) * (fderiv  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u a 1b
    + c * fderiv  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u a 2b)
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_eq_weightedDirDeriv[\[source\]](#)

A directional derivative is a ‘weightedDirDeriv’: anti-holomorphic with ‘c = i’.

```
private lemma dWirtingerAntiDir_eq_weightedDirDeriv (w : V) :
  (fun p => dWirtingerAntiDir f w p) = weightedDirDeriv f Complex.I w (Complex.I • w)
```

lemma Physlib.Wirtinger.dWirtingerDir_eq_weightedDirDeriv[\[source\]](#)

A directional derivative is a ‘weightedDirDeriv’: holomorphic with ‘c = -i’.

```
private lemma dWirtingerDir_eq_weightedDirDeriv (v : V) :
  (fun p => dWirtingerDir f v p) = weightedDirDeriv f (-Complex.I) v (Complex.I • v)
```

lemma Physlib.Wirtinger.fderiv_dWirtingerAntiDir[\[source\]](#)

Differentiating the anti-holomorphic directional derivative lands on the second real Fréchet derivative in the two slots.

```
private lemma fderiv_dWirtingerAntiDir (hf' : DifferentiableAt  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u)
  (w a : V) :
  fderiv  $\mathbb{R}$  (fun p => dWirtingerAntiDir f w p) u a
  = (1 / 2 :  $\mathbb{C}$ ) * (fderiv  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u a w
    + Complex.I * fderiv  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u a (Complex.I • w))
```

lemma Physlib.Wirtinger.fderiv_dWirtingerDir[\[source\]](#)

Differentiating the holomorphic directional derivative lands on the second real Fréchet derivative in the two slots.

```
private lemma fderiv_dWirtingerDir (hf' : DifferentiableAt  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u)
  (v a : V) :
  fderiv  $\mathbb{R}$  (fun p => dWirtingerDir f v p) u a
  = (1 / 2 :  $\mathbb{C}$ ) * (fderiv  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u a v
    - Complex.I * fderiv  $\mathbb{R}$  (fderiv  $\mathbb{R}$  f) u a (Complex.I • v))
```

lemma Physlib.Wirtinger.differentiableAt_dWirtingerDir[\[source\]](#)

On a ‘ C^2 ’ field the holomorphic directional derivative is itself real-differentiable.

```
lemma differentiableAt_dWirtingerDir (hf2 : ContDiffAt  $\mathbb{R}$  2 f u) (v : V) :
  DifferentiableAt  $\mathbb{R}$  (fun p => dWirtingerDir f v p) u
```

lemma Physlib.Wirtinger.differentiableAt_dWirtingerAntiDir[\[source\]](#)

On a ' C^2 ' field the anti-holomorphic directional derivative is itself real-differentiable.

```
lemma differentiableAt_dWirtingerAntiDir (hf2 : ContDiffAt ℝ 2 f u) (w : V) :
  DifferentiableAt ℝ (fun p => dWirtingerAntiDir f w p) u
```

lemma Physlib.Wirtinger.dWirtingerDir_congr_of_eventuallyEq[\[source\]](#)

The holomorphic directional derivative depends only on the field near the point: fields agreeing on a neighbourhood have equal derivative.

```
lemma dWirtingerDir_congr_of_eventuallyEq {ι f z f : V → ℂ} {u : V}
  (h : ι f f=[nhds u] z f) (v : V) :
  dWirtingerDir ι f v u = dWirtingerDir z f v u
```

lemma Physlib.Wirtinger.dWirtingerAntiDir_congr_of_eventuallyEq[\[source\]](#)

The anti-holomorphic directional derivative depends only on the field near the point; dual of 'dWirtingerDir_congr_of_eventuallyEq'.

```
lemma dWirtingerAntiDir_congr_of_eventuallyEq {ι f z f : V → ℂ} {u : V}
  (h : ι f f=[nhds u] z f) (v : V) :
  dWirtingerAntiDir ι f v u = dWirtingerAntiDir z f v u
```

theorem Physlib.Wirtinger.dWirtingerDir_dWirtingerAntiDir_comm[\[source\]](#)

***Schwarz's theorem** for the directional Wirtinger operators: on a ' C^2 ' field ' f ' the holomorphic and anti-holomorphic directional derivatives commute in any two directions, ' $\partial_v \partial_w f = \partial_w \partial_v f$ '.*

The commutation adds no analytic input. By the §G bridge each order expands into a real-linear combination of the second real Fréchet derivative ' $fderiv ℝ (fderiv ℝ f) u$ ' on the four directions ' v ', ' $i \cdot v$ ', ' w ', ' $i \cdot w$ '. The two orders give the same combination up to transposing the two slots of that second derivative, and ' $ContDiffAt.isSymmSndFDerivAt$ ', the symmetry of ordinary mixed second partials, equates them.

```
theorem dWirtingerDir_dWirtingerAntiDir_comm (hf2 : ContDiffAt ℝ 2 f u) (v w : V) :
  dWirtingerDir (fun p => dWirtingerAntiDir f w p) v u
  = dWirtingerAntiDir (fun p => dWirtingerDir f v p) w u
```

2.11 Physlib.Mathematics.Calculus.Wirtinger.Coordinate

[Physlib/Mathematics/Calculus/Wirtinger/Coordinate.lean](#) on GitHub.

def Physlib.Wirtinger.dWirtingerCoord[\[source\]](#)

Holomorphic Wirtinger derivative along the I -th coordinate of ' $\iota \rightarrow \mathbb{C}$ '.

```
def dWirtingerCoord (f : ι( → ℂ) → ℂ) (I : ι) : ι( → ℂ) → ℂ
```

def Physlib.Wirtinger.dWirtingerAntiCoord[\[source\]](#)

Anti-holomorphic Wirtinger derivative along the I -th coordinate of $\iota \rightarrow \mathbb{C}$.

```
def dWirtingerAntiCoord (f :  $\iota \rightarrow \mathbb{C}$ )  $\rightarrow \mathbb{C}$  (I :  $\iota$ ) :  $\iota \rightarrow \mathbb{C} \rightarrow \mathbb{C}$ 
```

lemma Physlib.Wirtinger.dWirtingerCoord_apply[\[source\]](#)

Real-Fréchet form of ‘dWirtingerCoord’: ‘dWirtingerCoord f I u = (1/2)($\partial_x - i \cdot \partial_y$) f’, the derivatives along the slot- I real and imaginary coordinate directions. Unconditional — the directional definition makes it definitional (‘Complex.I • Pi.single I 1 = Pi.single I Complex.I’); no differentiability hypothesis is needed.

```
lemma dWirtingerCoord_apply {f :  $\iota \rightarrow \mathbb{C}$   $\rightarrow \mathbb{C}$ }
  {u :  $\iota \rightarrow \mathbb{C}$ } (I :  $\iota$ ) :
  dWirtingerCoord f I u = (1 / 2 :  $\mathbb{C}$ ) * (fderiv  $\mathbb{R}$  f u (Pi.single I 1)
    - Complex.I * fderiv  $\mathbb{R}$  f u (Pi.single I Complex.I))
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_apply[\[source\]](#)

Real-Fréchet form of ‘dWirtingerAntiCoord’: ‘ $\partial_I f = (1/2)(\partial_x + i \cdot \partial_y)$ f’, mirror of ‘dWirtingerCoord_apply’ with the sign flip on the imaginary-direction term. Unconditional, as for ‘dWirtingerCoord_apply’.

```
lemma dWirtingerAntiCoord_apply {f :  $\iota \rightarrow \mathbb{C}$   $\rightarrow \mathbb{C}$ }
  {u :  $\iota \rightarrow \mathbb{C}$ } (I :  $\iota$ ) :
  dWirtingerAntiCoord f I u = (1 / 2 :  $\mathbb{C}$ ) * (fderiv  $\mathbb{R}$  f u (Pi.single I 1)
    + Complex.I * fderiv  $\mathbb{R}$  f u (Pi.single I Complex.I))
```

lemma Physlib.Wirtinger.clinear_of_holomorphic[\[source\]](#)

The real derivative of a holomorphic ‘ $f : E \rightarrow \mathbb{C}$ ’ is ‘ $\mathbb{C}$ ’-linear along every direction — the hypothesis the foundation collapse ‘dWirtingerDir_eq_of_clinear’ consumes. A holomorphic ‘ f ’ has ‘fderiv $\mathbb{R}$ f u = (fderiv $\mathbb{C}$ f u).restrictScalars $\mathbb{R}$ ’ (‘DifferentiableAt.fderiv_restrictScalars’), so its real derivative agrees with the ‘ $\mathbb{C}$ ’-linear complex derivative on every direction. Used at ‘ $E = \iota \rightarrow \mathbb{C}$ ’ for the coordinate collapse and at ‘ $E = \mathbb{C}$ ’ for the outer-‘ g ’ chain rule (§D).

```
private lemma clinear_of_holomorphic {E : Type*} [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
  [NormedSpace  $\mathbb{C}$  E] [IsScalarTower  $\mathbb{R}$   $\mathbb{C}$  E] {f : E  $\rightarrow \mathbb{C}$ } {u : E}
  (hf : DifferentiableAt  $\mathbb{C}$  f u) (d : E) :
  fderiv  $\mathbb{R}$  f u (Complex.I • d) = Complex.I • fderiv  $\mathbb{R}$  f u d
```

lemma Physlib.Wirtinger.dWirtingerCoord_congr_of_eventuallyEq_apply[\[source\]](#)

‘dWirtingerCoord’ is local: functions agreeing on a neighbourhood of ‘ u ’ have equal holomorphic Wirtinger derivative at ‘ u ’ (‘ $f_1 \stackrel{f}{=} [nhds\ u] f_2 \implies \partial_I f_1\ u = \partial_I f_2\ u$ ’).

```
lemma dWirtingerCoord_congr_of_eventuallyEq_apply {f f' :  $\iota \rightarrow \mathbb{C}$   $\rightarrow \mathbb{C}$ }
  {u :  $\iota \rightarrow \mathbb{C}$ } (h : f  $\stackrel{f}{=} [nhds\ u] f'$ ) (I :  $\iota$ ) :
  dWirtingerCoord f I u = dWirtingerCoord f' I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_neg_apply[\[source\]](#)

Pointwise negation rule for the holomorphic coordinate derivative at ‘u’: $\partial_I (-f) = -\partial_I f$. Used with ‘dWirtingerCoord_add_apply’ to assemble the coordinate-difference rule ‘dWirtingerCoord_coordDiff’ (§C).

```
lemma dWirtingerCoord_neg_apply {u : ι( → ℂ)} (I : ι) :
  dWirtingerCoord (fun v => -(f v)) I u = -(dWirtingerCoord f I u)
```

lemma Physlib.Wirtinger.fderiv_coordProj[\[source\]](#)

The real Fréchet derivative of the J -th coordinate projection $v \mapsto v J$. Consumed by the Kronecker coordinate-value lemmas ‘dWirtingerCoord_coordProj’ / ‘dWirtingerCoord_AntiCoord_coordProj’, which feed it the slot- I real and imaginary directions.

```
private lemma fderiv_coordProj (J : ι) (u d : ι( → ℂ)) :
  fderiv ℝ (fun v : ι( → ℂ) => v J) u d = d J
```

lemma Physlib.Wirtinger.dWirtingerCoord_add_apply[\[source\]](#)

Pointwise additivity of the holomorphic coordinate derivative at ‘u’: $\partial_I (f + g) = \partial_I f + \partial_I g$. Used to assemble the coordinate-difference rule ‘dWirtingerCoord_coordDiff’ (§C).

```
lemma dWirtingerCoord_add_apply {u : ι( → ℂ)}
  (hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
  dWirtingerCoord (f + g) I u = dWirtingerCoord f I u + dWirtingerCoord g I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_smul_apply[\[source\]](#)

Pointwise compatibility with complex scalar multiplication at ‘u’: $\partial_I (c \bullet f) = c \bullet \partial_I f$.

```
lemma dWirtingerCoord_smul_apply {u : ι( → ℂ)}
  (c : ℂ) (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerCoord (c • f) I u = c • dWirtingerCoord f I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_mul_apply[\[source\]](#)

Pointwise Leibniz rule for the holomorphic coordinate derivative at ‘u’: $\partial_I (f \cdot g) = \partial_I f \cdot g + f \cdot \partial_I g$.

```
lemma dWirtingerCoord_mul_apply {u : ι( → ℂ)}
  (hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
  dWirtingerCoord (f * g) I u =
    dWirtingerCoord f I u * g u + f u * dWirtingerCoord g I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_fun_sum_apply[\[source\]](#)

Pointwise finite-sum rule for holomorphic coordinate derivatives at ‘u’: $\partial_I (\sum_{a \in s, F a}) = \sum_{a \in s, \partial_I (F a)}$.

```
lemma dWirtingerCoord_fun_sum_apply α{ : Type*} {s : Finset α}
  {F : α → ι( → ℂ) → ℂ} {u : ι( → ℂ)}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (I : ι) :
```

```
dWirtingerCoord (fun v => ∑ a ∈ s, F a v) I u =
  ∑ a ∈ s, dWirtingerCoord (F a) I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_eq_complex_fderiv_apply

[\[source\]](#)

For a holomorphic function, ‘dWirtingerCoord’ is the complex Fréchet derivative in the corresponding coordinate direction: ‘ $\partial_I f = \text{fderiv } \mathbb{C} f u \text{ (Pi.single } I \text{ 1)}$ ’.

```
lemma dWirtingerCoord_eq_complex_fderiv_apply {u : ι( → ℂ)}
  (hf : DifferentiableAt ℂ f u) (I : ι) :
  dWirtingerCoord f I u = fderiv ℂ f u (Pi.single I 1)
```

def Physlib.Wirtinger.conjCLM

[\[source\]](#)

Pointwise conjugation bundled as an $\mathbb{R}$ -linear CLM (conjugate- $\mathbb{C}$ -linear): the I -th component is conjugation of the I -th coordinate ‘Complex.conjCLE $\circ$ proj I ’. Its underlying function is the ‘star’ of ‘ $\iota \rightarrow \mathbb{C}$ ’ (‘conjCLM_apply’). Bundling ‘star’ as a ‘ $\rightarrow L/\mathbb{R}$ ’ is what lets the anti-holomorphic lemmas below feed it to the foundation ‘dWirtingerDir_comp_conjLinear’, which requires its domain map as a continuous linear map.

```
private def conjCLM : ι( → ℂ) → LR[] ι( → ℂ)
```

lemma Physlib.Wirtinger.conjCLM_smul_I

[\[source\]](#)

‘conjCLM’ is conjugate- $\mathbb{C}$ -linear: ‘conj ($i \cdot d$) = $-(i \cdot \text{conj } d)$ ’. The ‘hL’ hypothesis the foundation ‘dWirtingerDir_comp_conjLinear’ / ‘dWirtingerAntiDir_comp_conjLinear’ consume.

```
private lemma conjCLM_smul_I (d : ι → ℂ) :
  conjCLM (Complex.I • d) = -(Complex.I • conjCLM ι( := ι) d)
```

lemma Physlib.Wirtinger.dWirtingerCoord_eq_zero_of_antiHolomorphic_apply

[\[source\]](#)

An *anti-holomorphic* function ‘ $v \mapsto g \text{ (star } v)$ ’ (a holomorphic ‘ g ’ precomposed with conjugation) depends on the coordinates only through their conjugates ‘ $\bar{z}$ ’. Its holomorphic derivative therefore vanishes, ‘ $\partial_I (g \circ \text{star}) = 0$ ’, since ‘ ∂_I ’ differentiates w.r.t. ‘ z ’ and treats ‘ $\bar{z}$ ’ as constant. Dual to ‘ $\partial_I f = 0$ ’ for holomorphic ‘ f ’.

```
lemma dWirtingerCoord_eq_zero_of_antiHolomorphic_apply {u : ι( → ℂ)}
  (hg : DifferentiableAt ℂ g (star u)) (I : ι) :
  dWirtingerCoord (fun v : ι( → ℂ) => g (star v)) I u = 0
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_congr_of_eventuallyEq_apply

[\[source\]](#)

‘dWirtingerAntiCoord’ is local (‘ $f_1 \stackrel{f}{=} [nhds \ u] f_2 \implies \partial_I f_1 \ u = \partial_I f_2 \ u$ ’).

```
lemma dWirtingerAntiCoord_congr_of_eventuallyEq_apply {ιf zf : ι( → ℂ) → ℂ}
  {u : ι( → ℂ)} (h : ιf  $\stackrel{f}{=} [nhds \ u] zf$ ) (I : ι) :
  dWirtingerAntiCoord ιf I u = dWirtingerAntiCoord zf I u
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_neg_apply[\[source\]](#)

Pointwise negation rule for the anti-holomorphic coordinate derivative at 'u': $\partial_I (-f) = -\partial_I f$. Used with 'dWirtingerAntiCoord_add_apply' to assemble the coordinate-difference rule 'dWirtingerAntiCoord_coordDiff' (§C).

```
lemma dWirtingerAntiCoord_neg_apply {u : ι( → ℂ)} (I : ι) :
  dWirtingerAntiCoord (fun v => -(f v)) I u = -(dWirtingerAntiCoord f I u)
```

lemma Physlib.Wirtinger.dWirtingerCoord_conjCoord[\[source\]](#)

$\partial_I \bar{z}^J = 0$. The conjugate of 'dWirtingerAntiCoord_coordProj' ($\bar{z}^J = \text{star } z^J$), read off through 'dWirtingerDir_star_comp' rather than recomputed. Used to assemble the coordinate-difference rule 'dWirtingerCoord_coordDiff' (§C).

```
lemma dWirtingerCoord_conjCoord (I J : ι) :
  dWirtingerCoord (fun u : ι( → ℂ) => star (u J)) I = 0
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_conjCoord[\[source\]](#)

$\partial_I \bar{z}^J = \delta_{IJ}$. The conjugate of 'dWirtingerCoord_coordProj', read off through 'dWirtingerAntiDir_star_comp' rather than recomputed. Used to assemble the coordinate-difference rule 'dWirtingerAntiCoord_coordDiff' (§C).

```
lemma dWirtingerAntiCoord_conjCoord (I J : ι) :
  dWirtingerAntiCoord (fun u : ι( → ℂ) => star (u J)) I =
    fun _ => if I = J then 1 else 0
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_add_apply[\[source\]](#)

Pointwise additivity of the anti-holomorphic coordinate derivative at 'u': $\partial_I (f + g) = \partial_I f + \partial_I g$. Used to assemble the coordinate-difference rule 'dWirtingerAntiCoord_coordDiff' (§C).

```
lemma dWirtingerAntiCoord_add_apply {u : ι( → ℂ)}
  (hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
  dWirtingerAntiCoord (f + g) I u = dWirtingerAntiCoord f I u + dWirtingerAntiCoord g I u
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_smul_apply[\[source\]](#)

Pointwise compatibility with complex scalar multiplication at 'u': $\partial_I (c \bullet f) = c \bullet \partial_I f$.

```
lemma dWirtingerAntiCoord_smul_apply {u : ι( → ℂ)}
  (c : ℂ) (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerAntiCoord (c • f) I u = c • dWirtingerAntiCoord f I u
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_mul_apply[\[source\]](#)

Pointwise Leibniz rule for the anti-holomorphic coordinate derivative at 'u': $\partial_I (f \cdot g) = \partial_I f \cdot g + f \cdot \partial_I g$.

```
lemma dWirtingerAntiCoord_mul_apply {u : ι( → ℂ)}
  (hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
  dWirtingerAntiCoord (f * g) I u =
```

```
dWirtingerAntiCoord f I u * g u + f u * dWirtingerAntiCoord g I u
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_fun_sum_apply
[\[source\]](#)

Pointwise finite-sum rule for anti-holomorphic coordinate derivatives at ‘u’: ‘ $\partial_I (\sum a \in s, F a) = \sum a \in s, \partial_I (F a)$ ’.

```
lemma dWirtingerAntiCoord_fun_sum_apply α { : Type*} {s : Finset α}
  {F : α → ι( → ℂ) → ℂ} {u : ι( → ℂ)}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (I : ι) :
  dWirtingerAntiCoord (fun v => ∑ a ∈ s, F a v) I u =
    ∑ a ∈ s, dWirtingerAntiCoord (F a) I u
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_eq_complex_fderiv_apply
[\[source\]](#)

For an anti-holomorphic function the anti-holomorphic Wirtinger derivative equals the complex Fréchet derivative of ‘g’ at ‘star u’ along the slot-I real coordinate direction, ‘ $\partial_I (g \circ \text{star}) = \text{fderiv } \mathbb{C} g (\text{star } u)$ (Pi.single I 1)’. Dual of ‘dWirtingerCoord_eq_complex_fderiv_apply’.

```
lemma dWirtingerAntiCoord_eq_complex_fderiv_apply {u : ι( → ℂ)}
  (hg : DifferentiableAt ℂ g (star u)) (I : ι) :
  dWirtingerAntiCoord (fun v : ι( → ℂ) => g (star v)) I u =
    fderiv ℂ g (star u) (Pi.single I 1)
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_eq_zero_of_holomorphic_apply
[\[source\]](#)

Holomorphic functions have zero anti-holomorphic coordinate derivative: ‘ $\partial_I f = 0$ ’.

```
lemma dWirtingerAntiCoord_eq_zero_of_holomorphic_apply {u : ι( → ℂ)}
  (hf : DifferentiableAt ℂ f u) (I : ι) :
  dWirtingerAntiCoord f I u = 0
```

lemma Physlib.Wirtinger.coordDiff_eq_add_neg
[\[source\]](#)

The coordinate difference ‘ $z^J - \bar{z}^J$ ’ as a sum of the holomorphic coordinate and the negated conjugate coordinate — the form on which the ‘dWirtingerCoord’ / ‘dWirtingerAntiCoord’ additivity rules apply.

```
private lemma coordDiff_eq_add_neg (J : ι) :
  (fun v : ι( → ℂ) => v J - star (v J))
  = (fun v => v J) + (fun v => -(star (v J)))
```

lemma Physlib.Wirtinger.dWirtingerCoord_coordDiff
[\[source\]](#)

‘ $\partial_I (z^J - \bar{z}^J) = \delta_{IJ}$ ’, from ‘ $\partial_I z^J = \delta_{IJ}$ ’ and ‘ $\partial_I \bar{z}^J = 0$ ’.

```
lemma dWirtingerCoord_coordDiff (I J : ι) :
  dWirtingerCoord (fun v : ι( → ℂ) => v J - star (v J)) I
  = fun _ => if I = J then 1 else 0
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_coordDiff[\[source\]](#)

$\partial_I (z^J - \bar{z}^J) = -\delta_{IJ}$, from $\partial_I z^J = 0$ and $\partial_I \bar{z}^J = \delta_{IJ}$.

```
lemma dWirtingerAntiCoord_coordDiff (I J : ι) :
  dWirtingerAntiCoord (fun v : ι(→ℂ) => v J - star (v J)) I
    = fun _ => if I = J then -1 else 0
```

lemma Physlib.Wirtinger.dWirtingerCoord_comp_apply[\[source\]](#)

The two-term coordinate chain rule for a real-differentiable outer g , pointwise at u : $\partial_I (g \circ f) = (\partial g / \partial f) \cdot \partial_I f + (\partial g / \partial \bar{f}) \cdot \partial_I \bar{f}$, with coefficients $\partial g / \partial f = dWirtingerDir\ g\ 1\ (f\ u)$ and $\partial g / \partial \bar{f} = dWirtingerAntiDir\ g\ 1\ (f\ u)$. The $d = \text{Pi.single } I\ 1$ case of the foundation $dWirtingerDir_comp$. The single-term holomorphic specialization $dWirtingerCoord_comp_holomorphic_apply$ is proved from this.

```
lemma dWirtingerCoord_comp_apply {u : ι(→ℂ)}
  (hg : DifferentiableAt ℝ g (f u)) (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerCoord (fun v => g (f v)) I u =
    dWirtingerDir g 1 (f u) * dWirtingerCoord f I u
    + dWirtingerAntiDir g 1 (f u) *
      dWirtingerCoord (fun v : ι(→ℂ) => star (f v)) I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_comp_holomorphic_apply[\[source\]](#)

The single-term coordinate chain rule for a holomorphic outer g , pointwise at u : $\partial_I (g \circ f) = \text{deriv } g\ (f\ u) \cdot \partial_I f$. From the two-term $dWirtingerCoord_comp_apply$: for holomorphic g the anti-holomorphic coefficient $dWirtingerAntiDir\ g\ 1\ (f\ u)$ vanishes and $dWirtingerDir\ g\ 1\ (f\ u)$ collapses to $\text{deriv } g\ (f\ u)$, both off the $\mathbb{C}$ -linearity $\text{clinear_of_holomorphic}$ (§B).

```
lemma dWirtingerCoord_comp_holomorphic_apply {u : ι(→ℂ)}
  (hg : DifferentiableAt ℂ g (f u)) (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerCoord (fun v => g (f v)) I u =
    deriv g (f u) * dWirtingerCoord f I u
```

lemma Physlib.Wirtinger.dWirtingerCoord_star_comp_apply[\[source\]](#)

Conjugating the inner field swaps the two operators: $\partial_I \bar{f} = \text{conj } (\partial_I f)$. The $d = \text{Pi.single } I\ 1$ case of the foundation $dWirtingerDir_star_comp$.

```
lemma dWirtingerCoord_star_comp_apply {u : ι(→ℂ)}
  (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerCoord (fun v : ι(→ℂ) => star (f v)) I u =
    star (dWirtingerAntiCoord f I u)
```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_comp_apply[\[source\]](#)

The two-term coordinate chain rule for a real-differentiable outer g , anti-holomorphic version, pointwise at u : $\partial_I (g \circ f) = (\partial g / \partial f) \cdot \partial_I f + (\partial g / \partial \bar{f}) \cdot \partial_I \bar{f}$. Same outer coefficients as $dWirtingerCoord_comp_apply$, now multiplying the anti-holomorphic inner derivatives. The $d = \text{Pi.single } I\ 1$ case of the foundation $dWirtingerAntiDir_comp$. The single-term holomorphic specialization $dWirtingerAntiCoord_comp_holomorphic_apply$ is proved from this.


```

lemma dWirtingerAntiCoord_comp_apply {u :  $\iota(\rightarrow \mathbb{C})$ }
  (hg : DifferentiableAt  $\mathbb{R}$  g (f u)) (hf : DifferentiableAt  $\mathbb{R}$  f u) (I :  $\iota$ ) :
  dWirtingerAntiCoord (fun v => g (f v)) I u =
    dWirtingerDir g 1 (f u) * dWirtingerAntiCoord f I u
    + dWirtingerAntiDir g 1 (f u) *
      dWirtingerAntiCoord (fun v :  $\iota(\rightarrow \mathbb{C})$  => star (f v)) I u

```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_comp_holomorphic_apply

[\[source\]](#)

The single-term coordinate chain rule for a holomorphic outer ‘g’, anti-holomorphic version, pointwise at ‘u’: ‘ $\partial_I (g \circ f) = \text{deriv } g (f u) \cdot \partial_I f$ ’. As in ‘dWirtingerCoord_comp_holomorphic_apply’, the ‘ $\partial g / \partial \bar{f}$ ’ channel vanishes and ‘ $\partial g / \partial f$ ’ collapses to ‘ $\text{deriv } g (f u)$ ’.

```

lemma dWirtingerAntiCoord_comp_holomorphic_apply {u :  $\iota(\rightarrow \mathbb{C})$ }
  (hg : DifferentiableAt  $\mathbb{C}$  g (f u)) (hf : DifferentiableAt  $\mathbb{R}$  f u) (I :  $\iota$ ) :
  dWirtingerAntiCoord (fun v => g (f v)) I u =
    deriv g (f u) * dWirtingerAntiCoord f I u

```

lemma Physlib.Wirtinger.dWirtingerAntiCoord_star_comp_apply

[\[source\]](#)

Conjugating the inner field swaps the two operators: ‘ $\partial_I f = \text{conj } (\partial_I f)$ ’. Dual of ‘dWirtingerCoord_star_comp_apply’, the ‘ $d = \text{Pi.single } I \ 1$ ’ case of the foundation ‘dWirtingerAntiDir_star_comp’.

```

lemma dWirtingerAntiCoord_star_comp_apply {u :  $\iota(\rightarrow \mathbb{C})$ }
  (hf : DifferentiableAt  $\mathbb{R}$  f u) (I :  $\iota$ ) :
  dWirtingerAntiCoord (fun v :  $\iota(\rightarrow \mathbb{C})$  => star (f v)) I u =
    star (dWirtingerCoord f I u)

```

lemma Physlib.Wirtinger.differentiableAt_dWirtingerCoord

[\[source\]](#)

On a ‘ C^2 ’ function the holomorphic coordinate Wirtinger derivative is itself real-differentiable (‘ $\text{DifferentiableAt } \mathbb{R} (\partial_I f) u$ ’) — the ‘ $d = \text{Pi.single } I \ 1$ ’ case of ‘differentiableAt_dWirtingerDir’.

```

lemma differentiableAt_dWirtingerCoord (hf2 : ContDiffAt  $\mathbb{R} \ 2$  f u) (I :  $\iota$ ) :
  DifferentiableAt  $\mathbb{R}$  (fun v => dWirtingerCoord f I v) u

```

lemma Physlib.Wirtinger.differentiableAt_dWirtingerAntiCoord

[\[source\]](#)

On a ‘ C^2 ’ function the anti-holomorphic coordinate Wirtinger derivative is itself real-differentiable (‘ $\text{DifferentiableAt } \mathbb{R} (\partial_I f) u$ ’) — the ‘ $d = \text{Pi.single } J \ 1$ ’ case of ‘differentiableAt_dWirtingerAntiDir’.

```

lemma differentiableAt_dWirtingerAntiCoord (hf2 : ContDiffAt  $\mathbb{R} \ 2$  f u) (J :  $\iota$ ) :
  DifferentiableAt  $\mathbb{R}$  (fun v => dWirtingerAntiCoord f J v) u

```

theorem Physlib.Wirtinger.dWirtingerCoord_dWirtingerAntiCoord_comm

[\[source\]](#)

Schwarz’s theorem for the coordinate Wirtinger operators: on a ‘ C^2 ’ function the holomorphic and anti-holomorphic derivatives commute, ‘ $\partial_I \partial_J f =$

$\partial_J \partial_I f$: The ‘ $d = \text{Pi.single } I \ 1$ ’, ‘ $e = \text{Pi.single } J \ 1$ ’ case of the general ‘ $d\text{WirtingerDir_}d\text{WirtingerAntiDir_comm}$ ’.

```
theorem dWirtingerCoord_dWirtingerAntiCoord_comm (hf2 : ContDiffAt ℝ 2 f u) (I J : ι) :
  dWirtingerCoord (fun v => dWirtingerAntiCoord f J v) I u
  = dWirtingerAntiCoord (fun v => dWirtingerCoord f I v) J u
```

2.12 Physlib.Mathematics.ConjModule

[Physlib/Mathematics/ConjModule.lean](#) on GitHub.

def ConjModule

[\[source\]](#)

The conjugate module of ‘ M ’: the same additive group with the scalar action twisted by conjugation, ‘ $r \bullet v = \text{star } r \bullet v$ ’. A type synonym so the twisted action stays off ‘ M ’.

```
def ConjModule (M : Type*)
```

instance ConjModule.«instance@88e8497f»

[\[source\]](#)

```
instance : AddCommGroup (ConjModule M)
```

instance ConjModule.instModule

[\[source\]](#)

The twisted action ‘ $r \bullet v = \text{star } r \bullet v$ ’, obtained by restricting scalars along the conjugation ring endomorphism ‘ $\text{starRingEnd } k$ ’.

```
instance instModule : Module k (ConjModule M)
```

def conjEquiv

[\[source\]](#)

The canonical conjugate-linear equivalence ‘ $M \simeq_{s\iota[\text{starRingEnd } k]} \text{ConjModule } M$ ’, the identity on the underlying additive group.

```
def conjEquiv : M  $\simeq_{s\iota[\text{starRingEnd } k]}$  ConjModule M where
```

def ConjModule.involution

[\[source\]](#)

Conjugating twice returns the original module: the ‘ k ’-linear isomorphism ‘ $\text{ConjModule } (\text{ConjModule } M) \simeq_{\iota[k]} M$ ’. It is ‘ k ’-linear, not merely semilinear, because ‘ $\text{starRingEnd } k$ ’ composed with itself is the identity.

```
def involution : ConjModule (ConjModule M)  $\simeq_{\iota[k]}$  M
```

def ConjModule.starFinsupp

[\[source\]](#)

Coordinate-wise conjugation on ‘ $\iota \rightarrow_0 k$ ’, a conjugate-linear self-equivalence.

```
noncomputable def starFinsupp :  $\iota \rightarrow_0 k \simeq_{s\iota[\text{starRingEnd } k]} \iota \rightarrow_0 k$  where
```

def Basis.conj[\[source\]](#)

A basis of 'M' transported to a basis of 'ConjModule M': the same basis vectors, with coordinates conjugated ('(Basis.conj b).repr v = star ∘ b.repr v').

```
noncomputable def _root_.Basis.conj (b : Basis ι k M) : Basis ι k (ConjModule M)
```

2.13 Physlib.Mathematics.CrossProductMatrix

[Physlib/Mathematics/CrossProductMatrix.lean](#) on GitHub.

def Matrix.crossProductMatrix[\[source\]](#)

*The hat map $[\omega]_x$: the skew-symmetric 3×3 matrix acting as the cross product with ω , i.e. $[\omega]_x *_v v = \omega \times_3 v$.*

```
def crossProductMatrix ω( : Fin 3 → ℝ) : Matrix (Fin 3) (Fin 3) ℝ
```

lemma Matrix.crossProductMatrix_mulVec[\[source\]](#)

The hat matrix acts on a vector as the cross product.

```
@[simp]
lemma crossProductMatrix_mulVec ω( v : Fin 3 → ℝ) :
  crossProductMatrix ω *_ v = ω ×₃ v
```

lemma Matrix.crossProductMatrix_transpose[\[source\]](#)

The hat map produces skew-symmetric matrices.

```
@[simp]
lemma crossProductMatrix_transpose ω( : Fin 3 → ℝ) :
  (crossProductMatrix ω)ᵀ = - crossProductMatrix ω
```

def Matrix.crossProductVee[\[source\]](#)

The vee map: reads the vector off a 3×3 matrix. It is a left inverse of the hat map.

```
def crossProductVee (A : Matrix (Fin 3) (Fin 3) ℝ) : Fin 3 → ℝ
```

lemma Matrix.crossProductVee_crossProductMatrix[\[source\]](#)

The vee map is a left inverse of the hat map.

```
@[simp]
lemma crossProductVee_crossProductMatrix ω( : Fin 3 → ℝ) :
  crossProductVee (crossProductMatrix ω) = ω
```

2.14 Physlib.Mathematics.KroneckerDelta

[Physlib/Mathematics/KroneckerDelta.lean](#) on GitHub.

def KroneckerDelta.generalizedKroneckerDelta[\[source\]](#)

Generalized Kronecker delta: $\delta^{\mu_1 \dots \mu_n}_{\nu_1 \dots \nu_n} = \det(\delta[\mu_i, \nu_j])$.

This is defined for any finite type α with decidable equality.

```
def generalizedKroneckerDelta α{ ι : Type} [DecidableEq α]
  [DecidableEq ι] [Fintype ι]
  μ( : ι → α) ν( : ι → α) : ℤ
```

lemma KroneckerDelta.generalizedKroneckerDelta_swap[\[source\]](#)

Swapping two of the upper indices of the generalized Kronecker delta negates it. This is one row transposition of the underlying determinant.

```
lemma generalizedKroneckerDelta_swap α{ ι : Type} [DecidableEq α] [DecidableEq ι] [Fintype ι]
  μ( ν : ι → α) {i j : ι} (hij : i ≠ j) :
  generalizedKroneckerDelta μ( ◦ Equiv.swap i j) ν = - generalizedKroneckerDelta μ ν
```

2.15 Physlib.Mathematics.LinearPMap

[Physlib/Mathematics/LinearPMap.lean](#) on GitHub.

lemma LinearPMap.sub_le_zero[\[source\]](#)

```
lemma sub_le_zero : f - f ≤ 0
```

lemma LinearPMap.neg_add_le_zero[\[source\]](#)

```
lemma neg_add_le_zero : -f + f ≤ 0
```

lemma LinearPMap.le_iff_neg_le_neg[\[source\]](#)

```
lemma le_iff_neg_le_neg : 1g ≤ 2g ↔ -1g ≤ -2g
```

lemma LinearPMap.le_neg_iff_neg_le[\[source\]](#)

```
lemma le_neg_iff_neg_le : 1g ≤ -2g ↔ -1g ≤ 2g
```

lemma LinearPMap.add_sub_le_cancel[\[source\]](#)

```
lemma add_sub_le_cancel : 1f + (2f - 1f) ≤ 2f
```

lemma LinearPMap.add_sub_le_cancel_left[\[source\]](#)

```
lemma add_sub_le_cancel_left : 1f + 2f - 1f ≤ 2f
```

lemma LinearPMap.add_sub_le_cancel_right[\[source\]](#)

```
lemma add_sub_le_cancel_right : 1f + 2f - 2f ≤ 1f
```

lemma LinearPMap.add_add_sub_le_cancel[\[source\]](#)

```
lemma add_add_sub_le_cancel : 1f + 2f + (3f - 2f) ≤ 1f + 3f
```

lemma LinearPMap.add_sub_sub_le_cancel[\[source\]](#)

```
lemma add_sub_sub_le_cancel : 1f + 2f - (1f - 3f) ≤ 2f + 3f
```

lemma LinearPMap.sub_sub_sub_le_cancel_right[\[source\]](#)

```
lemma sub_sub_sub_le_cancel_right : 1f - 2f - (3f - 2f) ≤ 1f - 3f
```

lemma LinearPMap.sub_sub_sub_le_cancel_left[\[source\]](#)

```
lemma sub_sub_sub_le_cancel_left : 1f - 2f - (1f - 3f) ≤ 3f - 2f
```

lemma LinearPMap.sub_le_of_le_add[\[source\]](#)

```
lemma sub_le_of_le_add (h : g ≤ 1g + 2g) : g - 2g ≤ 1g
```

lemma LinearPMap.sub_add_le_cancel[\[source\]](#)

```
lemma sub_add_le_cancel : 1f - 2f + 2f ≤ 1f
```

lemma LinearPMap.add_le_of_le_sub[\[source\]](#)

```
lemma add_le_of_le_sub (h : g ≤ 1g - 2g) : g + 2g ≤ 1g
```

lemma LinearPMap.add_left_le_of_le[\[source\]](#)

```
lemma add_left_le_of_le (h : 1g ≤ 2g) : f + 1g ≤ f + 2g
```

lemma LinearPMap.add_right_le_of_le[\[source\]](#)

```
lemma add_right_le_of_le (h : 1g ≤ 2g) : 1g + f ≤ 2g + f
```

lemma LinearPMap.sub_right_le_of_le[\[source\]](#)

```
lemma sub_right_le_of_le (h : 1g ≤ 2g) : 1g - f ≤ 2g - f
```

lemma LinearPMap.sub_left_le_of_le[\[source\]](#)

```
lemma sub_left_le_of_le (h : 1g ≤ 2g) : f - 1g ≤ f - 2g
```

lemma LinearPMap.zero_smul_le[\[source\]](#)

```
lemma zero_smul_le (f : E →1 [] F) : (0 : []) • f ≤ 0
```

lemma LinearPMap.zero_smul_eq[\[source\]](#)`@[simp]``lemma zero_smul_eq {f : E →ℓ.[] F} (h : f.domain = τ) : (0 : []) • f = 0`**def LinearPMap.sum**[\[source\]](#)*A finite sum of partial linear maps.**‘sum f’ and ‘ $\sum a, f a$ ’ are equal, but not by definition. With ‘sum f’ both ‘domain’ and ‘toFun’ are made explicit.*`def sum : E →ℓ.[R] F where`**lemma LinearPMap.sum\_domain**[\[source\]](#)`lemma sum_domain : (sum f).domain = [] a, (f a).domain`**lemma LinearPMap.sum\_domain\_le**[\[source\]](#)`lemma sum_domain_le (a : α) : (sum f).domain ≤ (f a).domain`**lemma LinearPMap.sum\_apply**[\[source\]](#)`@[simp]``lemma sum_apply ψ(: (sum f).domain) : sum f ψ = $\sum a, f a$ (ψ, sum_domain_le f a ψ).2`**def LinearPMap.compRestricted**[\[source\]](#)*‘ $g \circ_r f$ ’ is the composition of ‘g’ with ‘f’ restricted to a domain consisting of exactly those ‘ $x : f.domain$ ’ for which ‘ $f x \in g.domain$ ’.*`def compRestricted : E →ℓ.[R] G`**lemma LinearPMap.compRestricted\_domain\_le**[\[source\]](#)`lemma compRestricted_domain_le : (g ∘r f).domain ≤ f.domain`**lemma LinearPMap.compRestricted\_domain**[\[source\]](#)`lemma compRestricted_domain : (g ∘r f).domain = (g.domain.comap f.toFun).map f.domain.subtype`**lemma LinearPMap.mem\_compRestricted\_domain\_iff**[\[source\]](#)`lemma mem_compRestricted_domain_iff {x : E} :``x ∈ (v ∘r u).domain ↔ ∃ h : x ∈ u.domain, u (x, ) h ∈ v.domain`**lemma LinearPMap.mem\_compRestricted\_domain\_iff'**[\[source\]](#)`lemma mem_compRestricted_domain_iff' {x : E} :``x ∈ (v ∘r u).domain ↔ ∃ y : u.domain, x = y ∧ ∃ y' : v.domain, u y = y'`

lemma LinearPMap.mem_domain_of_mem_compRestricted_domain[\[source\]](#)

```
lemma mem_domain_of_mem_compRestricted_domain (x : (v ◦r u).domain) : u ⟨x, x⟩.2.2 ∈ v.domain
```

lemma LinearPMap.compRestricted_apply[\[source\]](#)

```
@[simp]
```

```
lemma compRestricted_apply (x : (v ◦r u).domain) :  
  (v ◦r u) x = v ⟨u ⟨x, x⟩.2.2, mem_domain_of_mem_compRestricted_domain x⟩
```

lemma LinearPMap.compRestricted_zero[\[source\]](#)

The zero map is right-absorbing.

```
@[simp]
```

```
lemma compRestricted_zero : g ◦r (0 : E →1.[R] F) = 0
```

lemma LinearPMap.compRestricted_assoc[\[source\]](#)

```
lemma compRestricted_assoc {H : Type*} [AddCommGroup H] [Module R H]  
  (1f : G →1.[R] H) (2f : F →1.[R] G) (3f : E →1.[R] F) :  
  (1f ◦r 2f) ◦r 3f = 1f ◦r 2f ◦r 3f
```

lemma LinearPMap.compRestricted_eq_comp[\[source\]](#)

‘compRestricted’ is the same as ‘comp’ when the range of ‘u’ is contained in ‘v.domain’.

```
lemma compRestricted_eq_comp (h : ∀ x : u.domain, u x ∈ v.domain) :  
  v ◦r u = v.comp u h
```

lemma LinearPMap.comp_le_compRestricted[\[source\]](#)

‘compRestricted’ is maximal amongst compositions of ‘v’ with domain restrictions of ‘u’.

```
lemma comp_le_compRestricted  
  {S : Submodule R E} (h : ∀ x : (u.domRestrict S).domain, u ⟨x, x⟩.2.2 ∈ v.domain) :  
  v.comp (u.domRestrict S) h ≤ v ◦r u
```

lemma LinearPMap.compRestricted_mono_left[\[source\]](#)

```
lemma compRestricted_mono_left {g g' : F →1.[R] G} (h : g ≤ g') (f : E →1.[R] F) :  
  g ◦r f ≤ g' ◦r f
```

lemma LinearPMap.compRestricted_mono_right[\[source\]](#)

```
lemma compRestricted_mono_right (g : F →1.[R] G) {f f' : E →1.[R] F} (h : f ≤ f') :  
  g ◦r f ≤ g ◦r f'
```

lemma LinearPMap.neg_compRestricted[\[source\]](#)

```
@[simp]
```

```
lemma neg_compRestricted : (-g) ◦r f = -g ◦r f
```

lemma LinearPMap.compRestricted_neg[\[source\]](#)

@[simp]

lemma compRestricted_neg : $g \circ_r (-f) = -g \circ_r f$ **lemma LinearPMap.add_compRestricted**[\[source\]](#)**lemma** add_compRestricted : $(1g + 2g) \circ_r f = 1g \circ_r f + 2g \circ_r f$ **lemma LinearPMap.sub_compRestricted**[\[source\]](#)**lemma** sub_compRestricted : $(1g - 2g) \circ_r f = 1g \circ_r f - 2g \circ_r f$ **lemma LinearPMap.compRestricted_add_ge**[\[source\]](#)**lemma** compRestricted_add_ge : $g \circ_r 1f + g \circ_r 2f \leq g \circ_r (1f + 2f)$ **lemma LinearPMap.compRestricted_sub_ge**[\[source\]](#)**lemma** compRestricted_sub_ge : $g \circ_r 1f - g \circ_r 2f \leq g \circ_r (1f - 2f)$ **lemma LinearPMap.compRestricted_smul**[\[source\]](#)

lemma compRestricted_smul {S : Type*} [DivisionRing S]
 [Module S E] [Module S F] [Module S G] [SMulCommClass S S F] [SMulCommClass S S G]
 {c : S} (hc : c ≠ 0) (g : F →_ℓ.[S] G) (f : E →_ℓ.[S] F) :
 $g \circ_r (c \bullet f) = c \bullet (g \circ_r f)$

lemma LinearPMap.smul_compRestricted[\[source\]](#)

@[simp]

lemma smul_compRestricted {M : Type*} [Monoid M] [DistribMulAction M G] [SMulCommClass R M G]
 (c : M) (g : F →_ℓ.[R] G) (f : E →_ℓ.[R] F) :
 $(c \bullet g) \circ_r f = c \bullet (g \circ_r f)$

instance LinearPMap.instMonoid[\[source\]](#)**instance** instMonoid : Monoid (E →_ℓ.[R] E) **where****lemma LinearPMap.mul_def**[\[source\]](#)**lemma** mul_def (1f 2f : E →_ℓ.[R] E) : $1f * 2f = 1f \circ_r 2f$ **lemma LinearPMap.one_domain**[\[source\]](#)

@[simp]

lemma one_domain : $(1 : E \rightarrow_{\ell} [R] E).domain = \top$

lemma LinearPMap.one_toFun[\[source\]](#)

```
@[simp]
lemma one_toFun : (l : E →l.[R] E).toFun = topEquiv.toLinearMap
```

lemma LinearPMap.one_coe[\[source\]](#)

```
@[simp]
lemma one_coe : (l : E →l.[R] E).toFun' = ↑topEquiv.toLinearMap
```

lemma LinearPMap.inverse_ker[\[source\]](#)

```
lemma inverse_ker : f.inverse.toFun.ker = ⊥
```

lemma LinearPMap.inverse_inverse[\[source\]](#)

```
lemma inverse_inverse : f.inverse.inverse = f
```

lemma LinearPMap.inverse_compRestricted_eq[\[source\]](#)

```
lemma inverse_compRestricted_eq : f.inverse ∘r f = domRestrict 1 f.domain
```

lemma LinearPMap.compRestricted_inverse_eq[\[source\]](#)

```
lemma compRestricted_inverse_eq : f ∘r f.inverse = domRestrict 1 f.inverse.domain
```

2.16 Physlib.Particles.BeyondTheStandardModel.RHN.AnomalyCancellation.Basic

[Physlib/Particles/BeyondTheStandardModel/RHN/AnomalyCancellation/Basic.lean](#) on GitHub.

lemma SMCharges.sum_one[\[source\]](#)

```
lemma sum_one [AddCommMonoid M] (f : Fin (vSMSpecies 1).numberCharges → M) :
  ∑ i, f i = f {0, by }simp
```

2.17 Physlib.Particles.StandardModel.AnomalyCancellation.Basic

[Physlib/Particles/StandardModel/AnomalyCancellation/Basic.lean](#) on GitHub.

lemma SMCharges.sum_SMSpecies_numberCharges_one[\[source\]](#)

```
lemma sum_SMSpecies_numberCharges_one {M} [AddCommMonoid M]
  (f : Fin (SMSpecies 1).numberCharges → M) :
  ∑ i, f i = f {0, by }simp
```

lemma SMCharges.toSpecies_apply_eq[\[source\]](#)

```
lemma toSpecies_apply_eq (i : Fin 5) (S : (SMCharges n).Charges) :
  toSpecies i S = fun j => toSpeciesEquiv S i j
```

2.18 Physlib.Particles.StandardModel.AnomalyCancellation.NoGrav.One.LinearParameterization

[Physlib/Particles/StandardModel/AnomalyCancellation/NoGrav/One/LinearParameterization.lean](#) on GitHub.

lemma SM.SMNoGrav.One.linearParameters.toSpecies_apply_asCharges [\[source\]](#)

```
lemma toSpecies_apply_asCharges (S : linearParameters) (i : Fin 5) :
  toSpecies i S.asCharges = fun _ => S.asCharges i
```

2.19 Physlib.Particles.StandardModel.Basic

[Physlib/Particles/StandardModel/Basic.lean](#) on GitHub.

def StandardModel.gaugeGroup $\mathbb{Z}_6$ Unitary0fRoot [\[source\]](#)

The unitary complex number associated to a sixth root of unity.

```
noncomputable def  $\mathbb{Z}_6$ gaugeGroupUnitary0fRoot  $\alpha$ ( : rootsOfUnity 6  $\mathbb{C}$ ) : unitary  $\mathbb{C}$ 
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ Unitary0fRoot_coe [\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_6$ gaugeGroupUnitary0fRoot_coe  $\alpha$ ( : rootsOfUnity 6  $\mathbb{C}$ ) :
  ( $\mathbb{Z}_6$ gaugeGroupUnitary0fRoot  $\alpha$  :  $\mathbb{C}$ ) =  $\alpha$ (( :  $\mathbb{C}^\times$ ) :  $\mathbb{C}$ )
```

def StandardModel.gaugeGroup $\mathbb{Z}_6$ SU30fRoot [\[source\]](#)

The ‘SU(3)’ scalar matrix associated to a sixth root of unity.

```
noncomputable def  $\mathbb{Z}_6$ gaugeGroupSU30fRoot  $\alpha$ ( : rootsOfUnity 6  $\mathbb{C}$ ) :
  specialUnitaryGroup (Fin 3)  $\mathbb{C}$ 
```

def StandardModel.gaugeGroup $\mathbb{Z}_6$ SU20fRoot [\[source\]](#)

The ‘SU(2)’ scalar matrix associated to a sixth root of unity.

```
noncomputable def  $\mathbb{Z}_6$ gaugeGroupSU20fRoot  $\alpha$ ( : rootsOfUnity 6  $\mathbb{C}$ ) :
  specialUnitaryGroup (Fin 2)  $\mathbb{C}$ 
```

def StandardModel.gaugeGroup $\mathbb{Z}_6$ 0fRoot [\[source\]](#)

The element of ‘GaugeGroupI’ associated to a sixth root of unity.

```
noncomputable def  $\mathbb{Z}_6$ gaugeGroup0fRoot  $\alpha$ ( : rootsOfUnity 6  $\mathbb{C}$ ) : GaugeGroupI
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ 0fRoot_toSU3 [\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_6$ gaugeGroup0fRoot_toSU3  $\alpha$ ( : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toSU3 ( $\mathbb{Z}_6$ gaugeGroup0fRoot  $\alpha$ ) =  $\mathbb{Z}_6$ gaugeGroupSU30fRoot  $\alpha$ 
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ OfRoot_toSU2[\[source\]](#)

@[simp]

lemma $\mathbb{Z}_6$ gaugeGroupOfRoot_toSU2 α (: rootsOfUnity 6 $\mathbb{C}$) :
 GaugeGroupI.toSU2 ($\mathbb{Z}_6$ gaugeGroupOfRoot α) = $\mathbb{Z}_6$ gaugeGroupSU20fRoot α

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ OfRoot_toU1[\[source\]](#)

@[simp]

lemma $\mathbb{Z}_6$ gaugeGroupOfRoot_toU1 α (: rootsOfUnity 6 $\mathbb{C}$) :
 GaugeGroupI.toU1 ($\mathbb{Z}_6$ gaugeGroupOfRoot α) = $\mathbb{Z}_6$ gaugeGroupUnitaryOfRoot α

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ OfRoot_mem_center[\[source\]](#)

lemma $\mathbb{Z}_6$ gaugeGroupOfRoot_mem_center α (: rootsOfUnity 6 $\mathbb{C}$) :
 $\mathbb{Z}_6$ gaugeGroupOfRoot $\alpha \in \text{Subgroup.center GaugeGroupI}$

def StandardModel.gaugeGroup $\mathbb{Z}_6$ Hom[\[source\]](#)

The homomorphism from sixth roots of unity to ‘GaugeGroupI’.

noncomputable def $\mathbb{Z}_6$ gaugeGroupHom : rootsOfUnity 6 $\mathbb{C} \rightarrow^*$ GaugeGroupI **where**

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ Hom_apply[\[source\]](#)

@[simp]

lemma $\mathbb{Z}_6$ gaugeGroupHom_apply α (: rootsOfUnity 6 $\mathbb{C}$) :
 $\mathbb{Z}_6$ gaugeGroupHom α = $\mathbb{Z}_6$ gaugeGroupOfRoot α

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ Hom_toSU3[\[source\]](#)

@[simp]

lemma $\mathbb{Z}_6$ gaugeGroupHom_toSU3 α (: rootsOfUnity 6 $\mathbb{C}$) :
 GaugeGroupI.toSU3 ($\mathbb{Z}_6$ gaugeGroupHom α) = $\mathbb{Z}_6$ gaugeGroupSU30fRoot α

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ Hom_toSU2[\[source\]](#)

@[simp]

lemma $\mathbb{Z}_6$ gaugeGroupHom_toSU2 α (: rootsOfUnity 6 $\mathbb{C}$) :
 GaugeGroupI.toSU2 ($\mathbb{Z}_6$ gaugeGroupHom α) = $\mathbb{Z}_6$ gaugeGroupSU20fRoot α

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ Hom_toU1[\[source\]](#)

@[simp]

lemma $\mathbb{Z}_6$ gaugeGroupHom_toU1 α (: rootsOfUnity 6 $\mathbb{C}$) :
 GaugeGroupI.toU1 ($\mathbb{Z}_6$ gaugeGroupHom α) = $\mathbb{Z}_6$ gaugeGroupUnitaryOfRoot α

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ OfRoot_mem[\[source\]](#)

lemma $\mathbb{Z}_6$ gaugeGroupOfRoot_mem α (: rootsOfUnity 6 $\mathbb{C}$) :
 $\mathbb{Z}_6$ gaugeGroupOfRoot $\alpha \in \mathbb{Z}_6$ gaugeGroupSubGroup

lemma StandardModel.mem_gaugeGroup $\mathbb{Z}_6$ SubGroup_iff[\[source\]](#)

lemma $\mathbb{Z}_6$ mem_gaugeGroupSubGroup_iff (g : GaugeGroupI) :
 g $\in \mathbb{Z}_6$ gaugeGroupSubGroup $\leftrightarrow \exists \alpha : \text{rootsOfUnity } 6 \mathbb{C}, \mathbb{Z}_6$ gaugeGroupOfRoot $\alpha = g$

lemma StandardModel.gaugeGroup $\mathbb{Z}_6$ SubGroup_le_center[\[source\]](#)

lemma $\mathbb{Z}_6$ gaugeGroupSubGroup_le_center :
 $\mathbb{Z}_6$ gaugeGroupSubGroup $\leq$ Subgroup.center GaugeGroupI

instance StandardModel.gaugeGroup $\mathbb{Z}_6$ SubGroup_normal[\[source\]](#)

instance $\mathbb{Z}_6$ gaugeGroupSubGroup_normal : $\mathbb{Z}_6$ gaugeGroupSubGroup.Normal **where**

instance StandardModel.«instance@25db9a02»[\[source\]](#)

noncomputable instance : Group $\mathbb{Z}_6$ GaugeGroup

def StandardModel.GaugeGroup $\mathbb{Z}_6$.mk[\[source\]](#)

The quotient map from ‘GaugeGroupI’ to ‘GaugeGroup $\mathbb{Z}_6$ ’.

noncomputable def mk : GaugeGroupI $\rightarrow^*$ $\mathbb{Z}_6$ GaugeGroup

lemma StandardModel.GaugeGroup $\mathbb{Z}_6$.mk_gaugeGroup $\mathbb{Z}_6$ OfRoot[\[source\]](#)

@[simp]
lemma $\mathbb{Z}_6$ mk_gaugeGroupOfRoot $\alpha(: \text{rootsOfUnity } 6 \mathbb{C})$:
 mk ($\mathbb{Z}_6$ gaugeGroupOfRoot α) = 1

def StandardModel.gaugeGroup $\mathbb{Z}_2$ RootTo $\mathbb{Z}_6$ Root[\[source\]](#)

The inclusion of second roots of unity into sixth roots of unity.

noncomputable def $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot : rootsOfUnity 2 $\mathbb{C} \rightarrow^*$ rootsOfUnity 6 $\mathbb{C}$

def StandardModel.gaugeGroup $\mathbb{Z}_2$ OfRoot[\[source\]](#)

The element of ‘GaugeGroupI’ associated to a second root of unity.

noncomputable def $\mathbb{Z}_2$ gaugeGroupOfRoot $\alpha(: \text{rootsOfUnity } 2 \mathbb{C})$: GaugeGroupI

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ OfRoot_toSU3[\[source\]](#)

@[simp]
lemma $\mathbb{Z}_2$ gaugeGroupOfRoot_toSU3 $\alpha(: \text{rootsOfUnity } 2 \mathbb{C})$:
 GaugeGroupI.toSU3 ($\mathbb{Z}_2$ gaugeGroupOfRoot α) =
 $\mathbb{Z}_6$ gaugeGroupSU3OfRoot ($\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot α)

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ OfRoot_toSU2[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_2$ gaugeGroupOfRoot_toSU2  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 ( $\mathbb{Z}_2$ gaugeGroupOfRoot  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU2OfRoot ( $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ OfRoot_toU1[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_2$ gaugeGroupOfRoot_toU1  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 ( $\mathbb{Z}_2$ gaugeGroupOfRoot  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupUnitaryOfRoot ( $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ OfRoot_mem_center[\[source\]](#)

```
lemma  $\mathbb{Z}_2$ gaugeGroupOfRoot_mem_center  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
   $\mathbb{Z}_2$ gaugeGroupOfRoot  $\alpha \in \text{Subgroup.center GaugeGroupI}$ 
```

def StandardModel.gaugeGroup $\mathbb{Z}_2$ Hom[\[source\]](#)

The homomorphism from second roots of unity to ‘GaugeGroupI’.

```
noncomputable def  $\mathbb{Z}_2$ gaugeGroupHom : rootsOfUnity 2  $\mathbb{C} \rightarrow^*$  GaugeGroupI
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ Hom_apply[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_2$ gaugeGroupHom_apply  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
   $\mathbb{Z}_2$ gaugeGroupHom  $\alpha = \mathbb{Z}_2$ gaugeGroupOfRoot  $\alpha$ 
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ Hom_toSU3[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_2$ gaugeGroupHom_toSU3  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toSU3 ( $\mathbb{Z}_2$ gaugeGroupHom  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU3OfRoot ( $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ Hom_toSU2[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_2$ gaugeGroupHom_toSU2  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 ( $\mathbb{Z}_2$ gaugeGroupHom  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU2OfRoot ( $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ Hom_toU1[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_2$ gaugeGroupHom_toU1  $\alpha$ ( : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 ( $\mathbb{Z}_2$ gaugeGroupHom  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupUnitaryOfRoot ( $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ OfRoot_mem[\[source\]](#)

lemma $\mathbb{Z}_2$ gaugeGroupOfRoot_mem α (: rootsOfUnity 2 $\mathbb{C}$) :
 $\mathbb{Z}_2$ gaugeGroupOfRoot $\alpha \in \mathbb{Z}_2$ gaugeGroupSubGroup

lemma StandardModel.mem_gaugeGroup $\mathbb{Z}_2$ SubGroup_iff[\[source\]](#)

lemma $\mathbb{Z}_2$ mem_gaugeGroupSubGroup_iff (g : GaugeGroupI) :
 $g \in \mathbb{Z}_2$ gaugeGroupSubGroup $\leftrightarrow \exists \alpha : \text{rootsOfUnity } 2 \mathbb{C}, \mathbb{Z}_2$ gaugeGroupOfRoot $\alpha = g$

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ SubGroup_le_gaugeGroup $\mathbb{Z}_6$ SubGroup[\[source\]](#)

lemma $\mathbb{Z}_2\mathbb{Z}_6$ gaugeGroupSubGroup_le_gaugeGroupSubGroup :
 $\mathbb{Z}_2$ gaugeGroupSubGroup $\leq \mathbb{Z}_6$ gaugeGroupSubGroup

lemma StandardModel.gaugeGroup $\mathbb{Z}_2$ SubGroup_le_center[\[source\]](#)

lemma $\mathbb{Z}_2$ gaugeGroupSubGroup_le_center :
 $\mathbb{Z}_2$ gaugeGroupSubGroup $\leq$ Subgroup.center GaugeGroupI

instance StandardModel.gaugeGroup $\mathbb{Z}_2$ SubGroup_normal[\[source\]](#)

instance $\mathbb{Z}_2$ gaugeGroupSubGroup_normal : $\mathbb{Z}_2$ gaugeGroupSubGroup.Normal **where**

instance StandardModel.«instance@6ca2807b»[\[source\]](#)

noncomputable instance : Group $\mathbb{Z}_2$ GaugeGroup

def StandardModel.GaugeGroup $\mathbb{Z}_2$.mk[\[source\]](#)

The quotient map from ‘GaugeGroupI’ to ‘GaugeGroup $\mathbb{Z}_2$ ’.

noncomputable def mk : GaugeGroupI $\rightarrow^*$ $\mathbb{Z}_2$ GaugeGroup

lemma StandardModel.GaugeGroup $\mathbb{Z}_2$.mk_gaugeGroup $\mathbb{Z}_2$ OfRoot[\[source\]](#)

@[simp]
lemma $\mathbb{Z}_2$ mk_gaugeGroupOfRoot α (: rootsOfUnity 2 $\mathbb{C}$) :
mk ($\mathbb{Z}_2$ gaugeGroupOfRoot α) = 1

def StandardModel.gaugeGroup $\mathbb{Z}_3$ RootTo $\mathbb{Z}_6$ Root[\[source\]](#)

The inclusion of third roots of unity into sixth roots of unity.

noncomputable def $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot : rootsOfUnity 3 $\mathbb{C} \rightarrow^*$ rootsOfUnity 6 $\mathbb{C}$

def StandardModel.gaugeGroup $\mathbb{Z}_3$ OfRoot[\[source\]](#)

The element of ‘GaugeGroupI’ associated to a third root of unity.

noncomputable def $\mathbb{Z}_3$ gaugeGroupOfRoot α (: rootsOfUnity 3 $\mathbb{C}$) : GaugeGroupI

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ OfRoot_toSU3[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_3$ gaugeGroupOfRoot_toSU3  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  GaugeGroupI.toSU3 ( $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU3OfRoot ( $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ OfRoot_toSU2[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_3$ gaugeGroupOfRoot_toSU2  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 ( $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU2OfRoot ( $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ OfRoot_toU1[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_3$ gaugeGroupOfRoot_toU1  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 ( $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupUnitaryOfRoot ( $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ OfRoot_mem_center[\[source\]](#)

```
lemma  $\mathbb{Z}_3$ gaugeGroupOfRoot_mem_center  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
   $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha \in \text{Subgroup.center GaugeGroupI}$ 
```

def StandardModel.gaugeGroup $\mathbb{Z}_3$ Hom[\[source\]](#)

The homomorphism from third roots of unity to ‘GaugeGroupI’.

```
noncomputable def  $\mathbb{Z}_3$ gaugeGroupHom : rootsOfUnity 3  $\mathbb{C} \rightarrow^*$  GaugeGroupI
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ Hom_apply[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_3$ gaugeGroupHom_apply  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
   $\mathbb{Z}_3$ gaugeGroupHom  $\alpha = \mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha$ 
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ Hom_toSU3[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_3$ gaugeGroupHom_toSU3  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  GaugeGroupI.toSU3 ( $\mathbb{Z}_3$ gaugeGroupHom  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU3OfRoot ( $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ Hom_toSU2[\[source\]](#)

```
@[simp]
lemma  $\mathbb{Z}_3$ gaugeGroupHom_toSU2  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 ( $\mathbb{Z}_3$ gaugeGroupHom  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupSU2OfRoot ( $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ Hom_toU1[\[source\]](#)

@[simp]

```
lemma  $\mathbb{Z}_3$ gaugeGroupHom_toU1  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 ( $\mathbb{Z}_3$ gaugeGroupHom  $\alpha$ ) =
     $\mathbb{Z}_6$ gaugeGroupUnitaryOfRoot ( $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupRootToRoot  $\alpha$ )
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ OfRoot_mem[\[source\]](#)

```
lemma  $\mathbb{Z}_3$ gaugeGroupOfRoot_mem  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
   $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha \in \mathbb{Z}_3$ gaugeGroupSubGroup
```

lemma StandardModel.mem_gaugeGroup $\mathbb{Z}_3$ SubGroup_iff[\[source\]](#)

```
lemma  $\mathbb{Z}_3$ mem_gaugeGroupSubGroup_iff (g : GaugeGroupI) :
  g  $\in \mathbb{Z}_3$ gaugeGroupSubGroup  $\leftrightarrow \exists \alpha$  : rootsOfUnity 3  $\mathbb{C}$ ,  $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha = g$ 
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ SubGroup_le_gaugeGroup $\mathbb{Z}_6$ SubGroup[\[source\]](#)

```
lemma  $\mathbb{Z}_3\mathbb{Z}_6$ gaugeGroupSubGroup_le_gaugeGroupSubGroup :
   $\mathbb{Z}_3$ gaugeGroupSubGroup  $\leq \mathbb{Z}_6$ gaugeGroupSubGroup
```

lemma StandardModel.gaugeGroup $\mathbb{Z}_3$ SubGroup_le_center[\[source\]](#)

```
lemma  $\mathbb{Z}_3$ gaugeGroupSubGroup_le_center :
   $\mathbb{Z}_3$ gaugeGroupSubGroup  $\leq$  Subgroup.center GaugeGroupI
```

instance StandardModel.gaugeGroup $\mathbb{Z}_3$ SubGroup_normal[\[source\]](#)

```
instance  $\mathbb{Z}_3$ gaugeGroupSubGroup_normal :  $\mathbb{Z}_3$ gaugeGroupSubGroup.Normal where
```

instance StandardModel.«instance@2ccf6670»[\[source\]](#)

```
noncomputable instance : Group  $\mathbb{Z}_3$ GaugeGroup
```

def StandardModel.GaugeGroup $\mathbb{Z}_3$.mk[\[source\]](#)

The quotient map from ‘GaugeGroupI’ to ‘GaugeGroup $\mathbb{Z}_3$ ’.

```
noncomputable def mk : GaugeGroupI  $\rightarrow^*$   $\mathbb{Z}_3$ GaugeGroup
```

lemma StandardModel.GaugeGroup $\mathbb{Z}_3$.mk_gaugeGroup $\mathbb{Z}_3$ OfRoot[\[source\]](#)

@[simp]

```
lemma  $\mathbb{Z}_3$ mk_gaugeGroupOfRoot  $\alpha$ ( : rootsOfUnity 3  $\mathbb{C}$ ) :
  mk ( $\mathbb{Z}_3$ gaugeGroupOfRoot  $\alpha$ ) = 1
```

instance StandardModel.«instance@e87cb8a7»[\[source\]](#)

```
noncomputable instance (q : GaugeGroupQuot) : Group (GaugeGroup q)
```


def StandardModel.GaugeGroupQuot.subgroup[\[source\]](#)

The central subgroup of ‘GaugeGroupI’ quotiented by a gauge-group quotient choice.

```
noncomputable def subgroup : GaugeGroupQuot → Subgroup GaugeGroupI
| Z₆. => Z₆gaugeGroupSubGroup
| Z₂. => Z₂gaugeGroupSubGroup
| Z₃. => Z₃gaugeGroupSubGroup
| .I => 1
```

lemma StandardModel.GaugeGroupQuot.subgroup_le_center[\[source\]](#)

The subgroup attached to a gauge-group quotient choice lies in the center of ‘GaugeGroupI’.

```
lemma subgroup_le_center (q : GaugeGroupQuot) :
  subgroup q ≤ Subgroup.center GaugeGroupI
```

instance StandardModel.GaugeGroupQuot.subgroup_normal[\[source\]](#)

The subgroup attached to a gauge-group quotient choice is normal in ‘GaugeGroupI’.

```
instance subgroup_normal (q : GaugeGroupQuot) : (subgroup q).Normal
```

def StandardModel.GaugeGroupQuot.quotientMap[\[source\]](#)

The quotient map from ‘GaugeGroupI’ to the gauge group selected by a quotient choice.

```
noncomputable def quotientMap (q : GaugeGroupQuot) : GaugeGroupI →* GaugeGroup q
```

lemma StandardModel.GaugeGroupQuot.quotientMap_I_apply[\[source\]](#)

```
@[simp]
lemma quotientMap_I_apply (g : GaugeGroupI) :
  quotientMap .I g = g
```

lemma StandardModel.GaugeGroupQuot.quotientMap_Z₆_gaugeGroupZ₆OfRoot[\[source\]](#)

```
@[simp]
lemma Z₆Z₆quotientMap__gaugeGroupOfRoot α( : rootsOfUnity 6 ℂ) :
  quotientMap Z₆. (Z₆gaugeGroupOfRoot α) = 1
```

lemma StandardModel.GaugeGroupQuot.quotientMap_Z₂_gaugeGroupZ₂OfRoot[\[source\]](#)

```
@[simp]
lemma Z₂Z₂quotientMap__gaugeGroupOfRoot α( : rootsOfUnity 2 ℂ) :
  quotientMap Z₂. (Z₂gaugeGroupOfRoot α) = 1
```

lemma StandardModel.GaugeGroupQuot.quotientMap_Z₃_gaugeGroupZ₃OfRoot[\[source\]](#)

```
@[simp]
lemma Z₃Z₃quotientMap__gaugeGroupOfRoot α( : rootsOfUnity 3 ℂ) :
  quotientMap Z₃. (Z₃gaugeGroupOfRoot α) = 1
```

lemma StandardModel.GaugeGroupQuot.mem_subgroup_iff_quotientMap_eq_one [\[source\]](#)

The kernel of the quotient map is the subgroup selected by the quotient choice.

```
lemma mem_subgroup_iff_quotientMap_eq_one (q : GaugeGroupQuot) (g : GaugeGroupI) :
  g ∈ subgroup q ↔ quotientMap q g = 1
```

lemma StandardModel.GaugeGroupQuot.quotientMap_eq_iff [\[source\]](#)

Two representatives have the same image under the selected quotient map exactly when their quotient lies in the subgroup selected by the quotient choice.

```
lemma quotientMap_eq_iff (q : GaugeGroupQuot) (g h : GaugeGroupI) :
  quotientMap q g = quotientMap q h ↔ g / h ∈ subgroup q
```

2.20 Physlib.Particles.StandardModel.HiggsBoson.Basic

[Physlib/Particles/StandardModel/HiggsBoson/Basic.lean](#) on GitHub.

instance StandardModel.HiggsVec.«instance@d58bd5cc» [\[source\]](#)

```
instance : SMulCommClass ℂ GaugeGroupI HiggsVec where
```

instance StandardModel.HiggsVec.«instance@34481c4b» [\[source\]](#)

```
instance : SMulCommClass ℝ GaugeGroupI HiggsVec where
```

def StandardModel.HiggsVec.toRealScalars [\[source\]](#)

The underlying real values of the Higgs vector.

```
def toRealScalars : HiggsVec → ℝ[] (Fin 4 → ℝ) where
```

lemma StandardModel.HiggsVec.toRealScalars_smul_real [\[source\]](#)

```
lemma toRealScalars_smul_real (a : ℝ) φ( : HiggsVec) :
  toRealScalars (a • φ) = a • toRealScalars φ
```

lemma StandardModel.HiggsVec.ofReal_toRealScalars [\[source\]](#)

```
lemma ofReal_toRealScalars (a : ℝ) :
  toRealScalars (ofReal a) = 4![Real.sqrt a, 0, 0, 0]
```

lemma StandardModel.HiggsVec.ofReal_toRealScalars_norm [\[source\]](#)

```
lemma ofReal_toRealScalars_norm φ( : HiggsVec) :
  toRealScalars (ofReal ||φ|| ( ^ 2)) = 4||φ||![, 0, 0, 0]
```

2.21 Physlib.Particles.StandardModel.HiggsBoson.EffectivePotential

[Physlib/Particles/StandardModel/HiggsBoson/EffectivePotential.lean](#) on GitHub.

abbrev StandardModel.HiggsField.EffectivePotential[\[source\]](#)*A general potential of the Higgs field.***abbrev** EffectivePotential : Type**def StandardModel.HiggsField.EffectivePotential.IsInvariant**[\[source\]](#)*The proposition that the general potential is invariant under the global action of the gauge group.***def** IsInvariant (V : EffectivePotential) : Prop**lemma StandardModel.HiggsField.EffectivePotential.IsInvariant.eq_on_orbits**[\[source\]](#)*An invariant potential is equal on gauge orbits.*

lemma eq_on_orbits $\phi_1 \phi_2 : \text{HiggsVec}$ {V : EffectivePotential} (h : IsInvariant V)
 ($\phi h : \phi_1 \in \text{MulAction.orbit GaugeGroupI } \phi_2$) :
 $V \phi_1 = V \phi_2$

lemma StandardModel.HiggsField.EffectivePotential.IsInvariant.eq_of_norm_eq[\[source\]](#)*An invariant potential is equal on Higgs vectors with identical norms.*

lemma eq_of_norm_eq $\phi_1 \phi_2 : \text{HiggsVec}$ {V : EffectivePotential} (h : IsInvariant V)
 ($\phi h : \|\phi_1\| = \|\phi_2\|$) :
 $V \phi_1 = V \phi_2$

lemma StandardModel.HiggsField.EffectivePotential.IsInvariant.factors_through_norm[\[source\]](#)

lemma factors_through_norm {V : EffectivePotential} (h : IsInvariant V) :
 $\exists (f : \mathbb{R} \rightarrow \mathbb{R}), V = f \circ \text{norm}$

def StandardModel.HiggsField.EffectivePotential.HasMaxMassDimLE[\[source\]](#)*The proposition that the potential ‘V’ has a maximum mass dimension less then or equal to ‘n’ - also implying it is a polynomial.***def** HasMaxMassDimLE (V : EffectivePotential) (n : $\mathbb{N}$) : Prop**def StandardModel.HiggsField.EffectivePotential.polynomial**[\[source\]](#)*The polynomial associated to a potential ‘V’ with a maximum mass dimension less than or equal to ‘n’.*

def polynomial (V : EffectivePotential) {n : $\mathbb{N}$ } (h : HasMaxMassDimLE V n) :
 MvPolynomial (Fin 4) $\mathbb{R}$

lemma StandardModel.HiggsField.EffectivePotential.polynomial_totalDegree[\[source\]](#)

lemma polynomial_totalDegree {V : EffectivePotential} {n : $\mathbb{N}$ } (h : HasMaxMassDimLE V n) :
 (polynomial V h).totalDegree $\leq n$

lemma StandardModel.HiggsField.EffectivePotential.apply_eq_polynomial[\[source\]](#)

```
lemma apply_eq_polynomial {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  φ( : HiggsVec) : V φ = (polynomial V h).eval φ.toRealScalars
```

def StandardModel.HiggsField.EffectivePotential.termOfMassDim[\[source\]](#)

The part of a potential at a given mass-dimension.

```
def termOfMassDim (V : EffectivePotential) {n : ℕ} (h : HasMaxMassDimLE V n) (m : ℕ) :
  HiggsVec → ℝ
```

lemma StandardModel.HiggsField.EffectivePotential.termOfMassDim_eq_zero_of_max_lt [\[source\]](#)

```
lemma termOfMassDim_eq_zero_of_max_lt {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  {m : ℕ} (hm : n < m) φ( : HiggsVec) :
  termOfMassDim V h m φ = 0
```

lemma StandardModel.HiggsField.EffectivePotential.termOfMassDim_homogeneity[\[source\]](#)

```
lemma termOfMassDim_homogeneity {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n) (m :
  ℕ)
  φ( : HiggsVec) (t : ℝ) : termOfMassDim V h m (t • φ) = t ^ m * termOfMassDim V h m φ
```

lemma StandardModel.HiggsField.EffectivePotential.apply_eq_sum_termOfMassDim[\[source\]](#)

```
lemma apply_eq_sum_termOfMassDim {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  φ( : HiggsVec) :
  V φ = ∑ m ∈ Finset.range (n + 1), termOfMassDim V h m φ
```

lemma StandardModel.HiggsField.EffectivePotential.apply_smul_eq_sum_termOfMassDim [\[source\]](#)

```
lemma apply_smul_eq_sum_termOfMassDim {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  φ( : HiggsVec) (t : ℝ) :
  V (t • φ) = ∑ m ∈ Finset.range (n + 1), t ^ m * termOfMassDim V h m φ
```

lemma StandardModel.HiggsField.EffectivePotential.termOfMassDim_isInvariant[\[source\]](#)

```
lemma termOfMassDim_isInvariant {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  (m : ℕ) (hV : IsInvariant V) : IsInvariant (termOfMassDim V h m)
```

lemma StandardModel.HiggsField.EffectivePotential.termOfMassDim_eq_mul_norm[\[source\]](#)

```
lemma termOfMassDim_eq_mul_norm {V : EffectivePotential} {n : ℕ}
  (h : HasMaxMassDimLE V n) (m : ℕ) (hV : IsInvariant V) φ( : HiggsVec) :
  ∃ c, termOfMassDim V h m φ = c * ||φ|| ^ m
```

lemma StandardModel.HiggsField.EffectivePotential.termOfMassDim_zero_of_odd[\[source\]](#)

```
lemma termOfMassDim_zero_of_odd {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n) (m :
  ℕ)
  (hV : IsInvariant V) ϕ( : HiggsVec) (hodd : Odd m) :
  termOfMassDim V h m ϕ = 0
```

lemma `StandardModel.HiggsField.EffectivePotential.apply_eq_sum_even_termOfMassDim` [\[source\]](#)

```
lemma apply_eq_sum_even_termOfMassDim {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  )
  (hV : IsInvariant V) ϕ( : HiggsVec) :
  V ϕ = ∑ m ∈ Finset.range (n / 2 + 1), termOfMassDim V h (2 * m) ϕ
```

lemma `StandardModel.HiggsField.EffectivePotential.apply_eq_sum_even_termOfMassDim_fin` [\[source\]](#)

```
lemma apply_eq_sum_even_termOfMassDim_fin {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE
  V n)
  (hV : IsInvariant V) ϕ( : HiggsVec) :
  V ϕ = ∑ m : Fin (n/2 + 1), termOfMassDim V h (2 * m) ϕ
```

lemma `StandardModel.HiggsField.EffectivePotential.apply_eq_sum_norm_pow` [\[source\]](#)

The potential is equal to the sum of norms to even powers.

```
lemma apply_eq_sum_norm_pow {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  (hV : IsInvariant V) ϕ( : HiggsVec) :
  ∃ c : Fin (n/2 + 1) → ℝ, V ϕ = ∑ m, c m • ||ϕ|| ^ (2 * m.1)
```

2.22 Physlib.Particles.SuperSymmetry.MSSMNu.AnomalyCancellation.Basic

[Physlib/Particles/SuperSymmetry/MSSMNu/AnomalyCancellation/Basic.lean](#) on GitHub.

lemma `MSSMCharges.sum_MSSMSpecies_numberCharges_eq_expand` [\[source\]](#)

```
lemma sum_MSSMSpecies_numberCharges_eq_expand [AddCommMonoid M]
  (f : Fin MSSMSpecies.numberCharges → M) :
  ∑ i, f i = f ⟨0, by⟩simp + f ⟨1, by⟩simp + f ⟨2, by⟩simp
```

2.23 Physlib.Particles.SuperSymmetry.N1.Basic

[Physlib/Particles/SuperSymmetry/N1/Basic.lean](#) on GitHub.

abbrev `SUSY.N1.ChiralScalarConfiguration` [\[source\]](#)

The chiral scalar configuration: a complex value for each chiral label. This is the sector's only field data. Declared as an 'abbrev' so that unification sees through it to 'ChiralIndexingType → ℂ' and applies Mathlib's function-space calculus lemmas directly.

```
abbrev ChiralScalarConfiguration (ChiralIndexingType : Type*)
```

inductive SUSY.N1.ChiralColor[\[source\]](#)

The four colours carried by a chiral-sector index: holomorphy ('chiral' versus 'anti', a scalar versus its complex conjugate) crossed with variance ('up' versus 'down', contravariant versus covariant). Carrying both axes here lets the single index type ' ι ' label the scalars.

```
inductive ChiralColor | chiralUp | chiralDown | antiUp | antiDown
```

def SUSY.N1.ChiralColor.tau[\[source\]](#)

The dual colour: flips variance and preserves holomorphy. Two indices may contract exactly when their colours are ' τ '-related, so ' $V^\wedge I$ ' pairs only with ' $V_\wedge I$ ' (same holomorphy, opposite variance) and never with a conjugate index.

```
def tau : ChiralColor → ChiralColor
| chiralUp => chiralDown
| chiralDown => chiralUp
| antiUp => antiDown
| antiDown => antiUp
```

def SUSY.N1.ChiralColor.bar[\[source\]](#)

The conjugate colour: flips holomorphy ('chiral'↔'anti') and preserves variance. Complex conjugation sends an index to its conjugate carrier, so 'bar' swaps 'chiral*' with 'anti*'. Distinct from the variance dual 'tau'; the two commute ('bar_tau').

```
def bar : ChiralColor → ChiralColor
| chiralUp => antiUp
| antiUp => chiralUp
| chiralDown => antiDown
| antiDown => chiralDown
```

abbrev SUSY.N1.chiralModule[\[source\]](#)

The carrier module of each colour, distinct for all four: the holomorphic vectors ' $\iota \rightarrow \mathbb{C}$ ' and their dual ' $\text{Module.Dual } \mathbb{C} (\iota \rightarrow \mathbb{C})$ ' on the chiral side, and the conjugate module ' $\text{ConjModule } \dots$ ' of each (where ' i ' acts as ' $-i$ ' on the anti side. Variance is the vector/dual axis, holomorphy the conjugate-module axis; both are genuine carrier data, not labels tracked separately.

```
abbrev chiralModule : ChiralColor → Type
| .chiralUp   =>  $\iota \rightarrow \mathbb{C}$ 
| .chiralDown =>  $\text{Module.Dual } \mathbb{C} (\iota \rightarrow \mathbb{C})$ 
| .antiUp     =>  $\text{ConjModule } (\iota \rightarrow \mathbb{C})$ 
| .antiDown   =>  $\text{ConjModule } (\text{Module.Dual } \mathbb{C} (\iota \rightarrow \mathbb{C}))$ 
```

instance SUSY.N1.instAddCommGroupChiralModule[\[source\]](#)

```
instance instAddCommGroupChiralModule :  $\forall c, \text{AddCommGroup } (\text{chiralModule } \iota (:= \iota) c)$ 
| .chiralUp | .chiralDown | .antiUp | .antiDown => inferInstance
```

instance SUSY.N1.instModuleChiralModule[\[source\]](#)

```
noncomputable instance instModuleChiralModule : ∀ c, Module ℂ (chiralModule ι( := ι) c)
| .chiralUp | .chiralDown | .antiUp | .antiDown => inferInstance
```

def SUSY.N1.chiralRep

[\[source\]](#)

The representation on each colour, taken trivial over the trivial group ‘Unit’: the chiral scalars carry no charge in this sector.

```
def chiralRep : (c : ChiralColor) → Representation ℂ Unit (chiralModule ι( := ι) c)
```

def SUSY.N1.piBasis

[\[source\]](#)

The standard basis of the vector carrier ‘ $\iota \rightarrow \mathbb{C}$ ’ (the indicator functions); the basis of ‘chiralUp’ and the reference basis for the δ pairing.

```
def piBasis : Basis ι ℂ ι( → ℂ)
```

def SUSY.N1.chiralBasis

[\[source\]](#)

The basis of each colour’s carrier, all indexed by ‘ ι ’: ‘piBasis’ on the holomorphic vectors, its dual ‘piBasis.dualBasis’ on the holomorphic covectors, and the ‘Basis.conj’ of each on the anti-holomorphic side (coordinates ‘star’-ed).

```
noncomputable def chiralBasis : (c : ChiralColor) → Basis ι ℂ (chiralModule ι( := ι) c)
| .chiralUp    => piBasis
| .chiralDown  => piBasis.dualBasis
| .antiUp      => Basis.conj piBasis
| .antiDown    => Basis.conj piBasis.dualBasis
```

def SUSY.N1.deltaCap

[\[source\]](#)

The δ cap ‘ $\sum_I b_I \otimes b_I$ ’: the rank-2 tensor in ‘ $M \otimes M$ ’ with two upper indices, whose components in the basis ‘ b ’ are ‘ δ^{ij} ’. It is an element of ‘ $M \otimes M$ ’ (the inverse-metric “cap”), not a linear map, and serves as the ‘metric’ field of the species (whose two slots share a colour).

```
def deltaCap (b : Basis ι ℂ M) : M ⊗ ℂ[] M
```

def SUSY.N1.deltaBil₂

[\[source\]](#)

The δ pairing between two based modules sharing the index ‘ ι ’: the dot product of coordinate vectors ‘ $(x, y) \mapsto \sum_I (b x)_I (b' y)_I$ ’. Built from Mathlib’s ‘Basis.toDual b' ’ (the canonical δ map ‘ $M \rightarrow \text{Module.Dual } M$ ’, sending ‘ b ’ to its dual basis) precomposed on the second slot with the basis transport ‘ $b' \simeq b$ ’.

```
def _2deltaBil (b : Basis ι ℂ M) (b' : Basis ι ℂ N) : M →_ι ℂ[] N →_ι ℂ[] ℂ
```

lemma SUSY.N1.deltaBil₂_apply

[\[source\]](#)

‘deltaBil₂ $b b' x y = \sum_I (b x)_I (b' y)_I$ ’.

```
lemma _2deltaBil_apply (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (x : M) (y : N) :
  _2deltaBil b b' x y = ∑ I, b.equivFun x I * b'.equivFun y I
```

def SUSY.N1.deltaContr₂[\[source\]](#)

The two-module δ contraction ‘ $M \otimes N \rightarrow \mathbb{C}$ ’.

```
def  $\mathbb{Z}$ deltaContr (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) : M  $\otimes$   $\mathbb{C}$ [ ] N  $\rightarrow$   $\iota$   $\mathbb{C}$ [ ]  $\mathbb{C}$ 
```

lemma SUSY.N1.deltaContr₂_tmul[\[source\]](#)

‘ $\text{deltaContr}_2 \ b \ b' \ (x \otimes_t y) = \sum_I I \ (b \ x)_I \ (b' \ y)_I$ ’.

```
lemma  $\mathbb{Z}$ deltaContr_tmul (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) (x : M) (y : N) :  
 $\mathbb{Z}$ deltaContr b b' (x  $\otimes_t$   $\mathbb{C}$ [ ] y) =  $\sum I$ , b.equivFun x I * b'.equivFun y I
```

lemma SUSY.N1.deltaContr₂_tmul_basis[\[source\]](#)

‘ $\text{deltaContr}_2 \ b \ b' \ (x \otimes_t b' \ J) = x_{J'}$ ’: pairing with the second basis reads off a coordinate.

```
lemma  $\mathbb{Z}$ deltaContr_tmul_basis (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) (x : M) (J :  $\iota$ ) :  
 $\mathbb{Z}$ deltaContr b b' (x  $\otimes_t$   $\mathbb{C}$ [ ] b' J) = b.equivFun x J
```

lemma SUSY.N1.deltaContr₂_basis_basis[\[source\]](#)

‘ $\text{deltaContr}_2 \ b \ b' \ (b \ I \otimes_t b' \ J) = \delta_{IJ}$ ’: the two bases are δ -dual.

```
lemma  $\mathbb{Z}$ deltaContr_basis_basis (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) (I J :  $\iota$ ) :  
 $\mathbb{Z}$ deltaContr b b' (b I  $\otimes_t$   $\mathbb{C}$ [ ] b' J) = if I = J then 1 else 0
```

lemma SUSY.N1.deltaContr₂_comm[\[source\]](#)

‘ $\text{deltaContr}_2 \ b \ b' \ (x \otimes_t y) = \text{deltaContr}_2 \ b' \ b \ (y \otimes_t x)$ ’: swapping slots swaps the two bases.

```
lemma  $\mathbb{Z}$ deltaContr_comm (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) (x : M) (y : N) :  
 $\mathbb{Z}$ deltaContr b b' (x  $\otimes_t$   $\mathbb{C}$ [ ] y) =  $\mathbb{Z}$ deltaContr b' b (y  $\otimes_t$   $\mathbb{C}$ [ ] x)
```

def SUSY.N1.deltaCap₂[\[source\]](#)

The two-module δ cap ‘ $\sum_I I \ b_I \otimes \ b'_I \in M \otimes N$ ’.

```
def  $\mathbb{Z}$ deltaCap (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) : M  $\otimes$   $\mathbb{C}$ [ ] N
```

lemma SUSY.N1.deltaCap₂_comm[\[source\]](#)

‘ $\text{comm} \ (\text{deltaCap}_2 \ b \ b') = \text{deltaCap}_2 \ b' \ b$ ’: swapping the two factors swaps the two bases.

```
lemma  $\mathbb{Z}$ deltaCap_comm (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) :  
TensorProduct.comm  $\mathbb{C}$  M N ( $\mathbb{Z}$ deltaCap b b') =  $\mathbb{Z}$ deltaCap b' b
```

lemma SUSY.N1.deltaUnit₂_symm[\[source\]](#)

The ‘ unit_symm ’ law (two-module, ‘ toSpanSingleton ’ form): ‘ $\text{deltaCap}_2 \ b' \ b$ ’ is the swap of ‘ $\text{deltaCap}_2 \ b \ b'$ ’.


```

lemma  $\mathbb{Z}$ deltaUnit_symm (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) :
  LinearMap.toSpanSingleton  $\mathbb{C}$  _ ( $\mathbb{Z}$ deltaCap b' b) 1 =
    LinearMap.lTensor N (LinearEquiv.refl  $\mathbb{C}$  M).toLinearMap
      (TensorProduct.comm  $\mathbb{C}$  M N (LinearMap.toSpanSingleton  $\mathbb{C}$  _ ( $\mathbb{Z}$ deltaCap b b') 1))

```

lemma SUSY.N1.deltaContr₂_unit

[\[source\]](#)

The snake identity (two-module, ‘contr_unit’ law): contracting ‘ $x \in M$ ’ into the ‘ M ’-leg of ‘deltaCap₂ b’ $b \in N \otimes M$ ’ returns ‘ x ’.

```

lemma  $\mathbb{Z}$ deltaContr_unit (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) (x : M) :
  (TensorProduct.lid  $\mathbb{C}$  M) (( $\mathbb{Z}$ deltaContr b b').rTensor M
    ((TensorProduct.assoc  $\mathbb{C}$  M N M).symm
      (x  $\otimes$   $\mathbb{C}$ []) LinearMap.toSpanSingleton  $\mathbb{C}$  _ ( $\mathbb{Z}$ deltaCap b' b) 1))) = x

```

lemma SUSY.N1.deltaContr₂_metric

[\[source\]](#)

The ‘contr_metric’ law (two-module): contracting the inner ‘ M ’/‘ N ’ legs of ‘deltaCap b $\otimes$ deltaCap b’ yields ‘deltaCap₂ b’ b’.

```

lemma  $\mathbb{Z}$ deltaContr_metric (b : Basis  $\iota$   $\mathbb{C}$  M) (b' : Basis  $\iota$   $\mathbb{C}$  N) :
  (TensorProduct.comm  $\mathbb{C}$  M N ((TensorProduct.lid  $\mathbb{C}$  N).lTensor M
    ((( $\mathbb{Z}$ deltaContr b b').rTensor N).lTensor M
      (((TensorProduct.assoc  $\mathbb{C}$  M N N).symm.toLinearMap.lTensor M)
        ((TensorProduct.assoc  $\mathbb{C}$  M M (N  $\otimes$   $\mathbb{C}$ []) N))
          (LinearMap.toSpanSingleton  $\mathbb{C}$  _ (deltaCap b) 1  $\otimes$   $\mathbb{C}$ [])
            LinearMap.toSpanSingleton  $\mathbb{C}$  _ (deltaCap b') 1)))))) =
    LinearMap.toSpanSingleton  $\mathbb{C}$  _ ( $\mathbb{Z}$ deltaCap b' b) 1

```

def SUSY.N1.chiralTensor

[\[source\]](#)

The chiral-index tensor species, bundled with its conjugation. Its four colours ‘chiral’/‘anti’ $\times$ ‘up’/‘down’ carry the four distinct carriers of $\mathcal{SC}$. ‘contr c’ is the two-module δ pairing of a colour against its variance dual ‘ τ c’ (‘ V c $\otimes$ V (τ c) $\rightarrow$ $\mathbb{C}$ ’); ‘unit c’ is the δ cap across those two carriers; ‘metric c’ is the single-colour δ cap ‘ $\sum_I b_I \otimes b_I$ ’. Each ‘TensorSpecies’ coherence law reduces, by case analysis on the colour, to the corresponding abstract two-module δ lemma of $\mathcal{SD}$. The conjugation flips holomorphy (‘ChiralColor.bar’) while preserving variance; every basis is indexed by ‘ ι ’, so ‘barIdx_eq’ is ‘rfl’, and ‘conj_contrComm’ is ‘star $\delta = \delta$ ’. Instantiating ‘ConjTensorSpecies’ this way gives the chiral sector both the framework’s generic tensor API and its conjugation API (‘conjT’ and its laws) on one object.

```

def chiralTensor : ConjTensorSpecies  $\mathbb{C}$  ChiralColor Unit (chiralModule  $\iota$  (:=  $\iota$ )) (fun _ =>  $\iota$ )
  (chiralRep  $\iota$  (:=  $\iota$ )) (chiralBasis  $\iota$  (:=  $\iota$ )) where

```

def SUSY.N1.conjScalar

[\[source\]](#)

Conjugation of a scalar tensor, normalized back to the scalar colour list ‘![]’.

```

def conjScalar (t : (chiralTensor  $\iota$  (:=  $\iota$ )).Tensor ![]) :
  (chiralTensor  $\iota$  (:=  $\iota$ )).Tensor ![]

```

def SUSY.N1.conjChiralCovector[\[source\]](#)

Conjugation of a holomorphic covector, normalized to the anti-holomorphic covector colour list ‘![antiDown]’.

```
def conjChiralCovector
  (t : (chiralTensor ι( := ι)).Tensor ![chiralDown]) :
  (chiralTensor ι( := ι)).Tensor ![antiDown]
```

lemma SUSY.N1.toField_conjScalar[\[source\]](#)

For scalar tensors, ‘toField’ of the normalized tensor conjugate is the complex conjugate of ‘toField’.

```
lemma toField_conjScalar (t : (chiralTensor ι( := ι)).Tensor ![]) :
  (conjScalar t).toField = star t.toField
```

lemma SUSY.N1.repr_conjChiralCovector[\[source\]](#)

Component formula for the holomorphic covector conjugate: the ‘![I]’ basis component of ‘conjChiralCovector t’ is the complex conjugate of the ‘![I]’ component of ‘t’.

```
lemma repr_conjChiralCovector
  (t : (chiralTensor ι( := ι)).Tensor ![chiralDown]) (I : ι) :
  (basis (S := (chiralTensor ι( := ι)).toTensorSpecies) ![antiDown]).repr
    (conjChiralCovector t) ![I] =
    star ((basis (S := (chiralTensor ι( := ι)).toTensorSpecies) ![chiralDown]).repr t ![I])
```

lemma SUSY.N1.conjChiralCovector_add[\[source\]](#)

Conjugation of a holomorphic covector is additive.

```
@[simp]
lemma conjChiralCovector_add
  (ι1 ι2 : (chiralTensor ι( := ι)).Tensor ![chiralDown]) :
  conjChiralCovector (ι1 + ι2) = conjChiralCovector ι1 + conjChiralCovector ι2
```

lemma SUSY.N1.conjChiralCovector_smul[\[source\]](#)

Conjugation of a holomorphic covector is conjugate-linear: a scalar ‘r’ pulls out as ‘star r’.

```
@[simp]
lemma conjChiralCovector_smul (r : ℂ)
  (t : (chiralTensor ι( := ι)).Tensor ![chiralDown]) :
  conjChiralCovector (r • t) = star r • conjChiralCovector t
```

2.24 Physlib.Particles.SuperSymmetry.SU5.ChargeSpectrum.AllowsTerm

[Physlib/Particles/SuperSymmetry/SU5/ChargeSpectrum/AllowsTerm.lean](#) on GitHub.

instance SuperSymmetry.SU5.ChargeSpectrum.«instance@7b86df89»[\[source\]](#)

```
instance (x : ChargeSpectrum []) (q5 : []) (T : PotentialTerm) :
  Decidable (AllowsTermQ5 x q5 T)
```

instance SuperSymmetry.SU5.ChargeSpectrum.«instance@35144a57»

[\[source\]](#)

```
instance (x : ChargeSpectrum []) (q10 : []) (T : PotentialTerm) :
  Decidable (AllowsTermQ10 x q10 T)
```

2.25 Physlib.Particles.SuperSymmetry.SU5.ChargeSpectrum.PhenoClosed

[Physlib/Particles/SuperSymmetry/SU5/ChargeSpectrum/PhenoClosed.lean](#) on GitHub.

instance SuperSymmetry.SU5.ChargeSpectrum.«instance@1aa8df08»

[\[source\]](#)

```
instance {S5 S10 : Finset []} {charges : Multiset (ChargeSpectrum [])} :
  Decidable (ContainsPhenoCompletionsOfMinimallyAllows S5 S10 charges)
```

2.26 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.Basic

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/Basic.lean](#) on GitHub.

abbrev LinearPMap.resolvent

[\[source\]](#)

The resolvent, $(T - z \bullet 1)^{-1}$.

```
abbrev resolvent (T : H →1ℂ.[ ] H) (z : ℂ) : H →1ℂ.[ ] H
```

def LinearPMap.IsLowerBound

[\[source\]](#)

*'IsLowerBound T z c' is the property that ' $c * \|x\| \leq \|T x - z \bullet x\|$ ' for all ' $x : T.domain$ '.*

```
def IsLowerBound (T : H →1ℂ.[ ] H) (z : ℂ) (c : ℝ) : Prop
```

lemma LinearPMap.isLowerBound_neg

[\[source\]](#)

```
lemma isLowerBound_neg {T : H →1ℂ.[ ] H} {z : ℂ} {c : ℝ} (h : IsLowerBound T z c) :
  IsLowerBound (-T) (-z) c
```

lemma LinearPMap.isLowerBound_of_right_le

[\[source\]](#)

```
lemma isLowerBound_of_right_le
  {T : H →1ℂ.[ ] H} {z : ℂ} {z1 z2 : ℝ} (hle : z1 ≤ z2) (h : IsLowerBound T z z2) :
  IsLowerBound T z z1
```

lemma LinearPMap.isLowerBound_of_left_le

[\[source\]](#)

```
lemma isLowerBound_of_left_le
  {z1 z2 : H →1ℂ.[ ] H} (hle : z1 ≤ z2) {z : ℂ} {c : ℝ} (h : IsLowerBound z2 z c) :
  IsLowerBound z1 z c
```

lemma LinearPMap.isLowerBound_closure[\[source\]](#)**lemma** isLowerBound_closure $\{T : H \rightarrow \mathbb{C} \cdot []\ H\} \{z : \mathbb{C}\} \{c : \mathbb{R}\} (h : \text{IsLowerBound } T \ z \ c) : \text{IsLowerBound } T.\text{closure } z \ c$ **def LinearPMap.regularityDomain**[\[source\]](#)*The regular points for 'T'.**'z : $\mathbb{C}$ ' is a regular point for 'T' iff there exists a constant ' $c > 0$ ' such that ' $c * \|x\| \leq \|(T - z \bullet 1) x\|$ ' for all ' $x \in T.\text{domain}$ '.***def** regularityDomain (T : H $\rightarrow$ $\mathbb{C} \cdot []$ H) : Set $\mathbb{C}$ **lemma LinearPMap.regularityDomain_neg**[\[source\]](#)

@[simp]

lemma regularityDomain_neg (T : H $\rightarrow$ $\mathbb{C} \cdot []$ H) : (-T).regularityDomain = -T.regularityDomain**lemma LinearPMap.regularityDomain_smul**[\[source\]](#)

@[simp]

lemma regularityDomain_smul (T : H $\rightarrow$ $\mathbb{C} \cdot []$ H) {w : $\mathbb{C}$ } (hw : w $\neq$ 0) :
(w • T).regularityDomain = w • T.regularityDomain**lemma LinearPMap.regularityDomain_antitone**[\[source\]](#)*'T $\leq$ T'' implies 'T'.regularityDomain $\subseteq$ T.regularityDomain'.***lemma** regularityDomain_antitone : Antitone (regularityDomain (H := H))**lemma LinearPMap.mem_regularityDomain_iff**[\[source\]](#)*'z' is a regular point for 'T' iff 'T - z • 1' has a continuous (equivalently, bounded) inverse.***lemma** mem_regularityDomain_iff {T : H $\rightarrow$ $\mathbb{C} \cdot []$ H} {z : $\mathbb{C}$ } :
z $\in$ T.regularityDomain $\leftrightarrow$ (T - z • 1).toFun.ker = $\perp$ $\wedge$ Continuous $\square$ (T z)**lemma LinearPMap.ball_subset_regularityDomain**[\[source\]](#)*The regularity domain of 'T' contains open balls with radii controlled by the lower bounds.***lemma** ball_subset_regularityDomain $\{T : H \rightarrow \mathbb{C} \cdot []\ H\} \{z : \mathbb{C}\} \{c : \mathbb{R}\} (h : \text{IsLowerBound } T \ z \ c) : \text{ball } z \ c \subseteq T.\text{regularityDomain}$ **lemma LinearPMap.regularityDomain_isOpen**[\[source\]](#)*The regularity domain is an open set.***lemma** regularityDomain_isOpen (T : H $\rightarrow$ $\mathbb{C} \cdot []$ H) : IsOpen T.regularityDomain

lemma LinearPMap.regularityDomain_closure[\[source\]](#)

‘ T ’ and ‘ $T.closure$ ’ have the same regularity domain.

```
lemma regularityDomain_closure (T : H →ℂ H) :
  T.closure.regularityDomain = T.regularityDomain
```

lemma LinearPMap.IsClosable.closure_range_sub_eq_range_closure_sub[\[source\]](#)

```
lemma IsClosable.closure_range_sub_eq_range_closure_sub [CompleteSpace H]
  {T : H →ℂ H} (hT : T.IsClosable) {z : ℂ} (hz : z ∈ T.regularityDomain) :
  (T - z • 1).toFun.range.closure = (T.closure - z • 1).toFun.range
```

lemma LinearPMap.IsClosed.sub_range_isClosed[\[source\]](#)

```
lemma IsClosed.sub_range_isClosed [CompleteSpace H]
  {T : H →ℂ H} (hT : T.IsClosed) {z : ℂ} (hz : z ∈ T.regularityDomain) :
  _root_.IsClosed ((T - z • 1).toFun.range : Set H)
```

lemma LinearPMap.IsUnbounded.orthogonal_closure_sub_range[\[source\]](#)

‘ $(T.closure - z • 1).range^\perp = (T^\dagger - \text{conj } z • 1).ker$ ’

```
lemma IsUnbounded.orthogonal_closure_sub_range [CompleteSpace H]
  {T : H →ℂ H} (hT : T.IsUnbounded) (z : ℂ) :
  (T.closure - z • 1).toFun.□range
  = (†T - conj z • 1).toFun.ker.map (†T - conj z • 1).domain.subtype
```

lemma LinearPMap.IsUnbounded.orthogonal_adjoint_sub_ker[\[source\]](#)

‘ $(T^\dagger - \text{conj } z • 1).ker^\perp = (T.closure - z • 1).range$ ’

```
lemma IsUnbounded.orthogonal_adjoint_sub_ker [CompleteSpace H]
  {T : H →ℂ H} (hT : T.IsUnbounded) {z : ℂ} (hz : z ∈ T.regularityDomain) :
  ((†T - conj z • 1).toFun.ker.map (†T - conj z • 1).domain.subtype)□
  = (T.closure - z • 1).toFun.range
```

def LinearPMap.deficiencySubspace[\[source\]](#)

For a partial linear map ‘ T ’ and any complex number ‘ z ’, the closed submodule which is orthogonal to the range of ‘ $T - z • 1$ ’.

‘ $T.defectNumber z$ ’ is defined as the rank of this subspace.

```
def deficiencySubspace (T : H →ℂ H) (z : ℂ) : ClosedSubmodule ℂ H
```

lemma LinearPMap.deficiencySubspace_coe[\[source\]](#)

```
@[simp]
lemma deficiencySubspace_coe (T : H →ℂ H) (z : ℂ) :
  T.deficiencySubspace z = (T - z • 1).toFun.□range
```

def LinearPMap.defectNumber[\[source\]](#)

The rank of ‘ T .deficiencySubspace $z = (T - z \bullet 1).range^\perp$ ’.

```
def defectNumber (T : H →1ℂ.[ ] H) (z : ℂ) : Cardinal
```

lemma LinearPMap.defectNumber_eq[\[source\]](#)

```
lemma defectNumber_eq (T : H →1ℂ.[ ] H) (z : ℂ) :
  T.defectNumber z = Module.rank ℂ (T.deficiencySubspace z)
```

lemma LinearPMap.IsClosed.defectNumber_eq_zero_iff[\[source\]](#)

```
lemma IsClosed.defectNumber_eq_zero_iff [CompleteSpace H]
  {T : H →1ℂ.[ ] H} (hT : T.IsClosed) {z : ℂ} (hz : z ∈ T.regularityDomain) :
  T.defectNumber z = 0 ↔ (T - z • 1).toFun.range = T
```

lemma LinearPMap.defectNumber_closure[\[source\]](#)

‘ T ’ and ‘ T .closure’ have the same defect number at points in their regularity domain.

```
lemma defectNumber_closure [CompleteSpace H]
  {T : H →1ℂ.[ ] H} {z : ℂ} (hz : z ∈ T.regularityDomain) :
  T.closure.defectNumber z = T.defectNumber z
```

lemma Submodule.inf_ne_bot_of_rank_lt[\[source\]](#)

```
lemma _root_.Submodule.inf_ne_bot_of_rank_lt
  {E F : Submodule ℂ H} [E.HasOrthogonalProjection] (h_rank : Module.rank ℂ E < Module.rank
  ℂ F) :
  ⊥ E ⊂ F ≠ ⊥
```

lemma LinearPMap.IsClosed.exists_inner_eq_zero_of_defectNumber_lt[\[source\]](#)

```
lemma IsClosed.exists_inner_eq_zero_of_defectNumber_lt [CompleteSpace H]
  {T : H →1ℂ.[ ] H} (hT : T.IsClosed)
  {z₁ z₂ : ℂ} (hz₁ : z₁ ∈ T.regularityDomain) (h : T.defectNumber z₁ < T.defectNumber z₂) :
  ∃ x : T.domain, x ≠ 0 ∧ ⊥ T x - z₁ • x, T x - z₂ • ⊥ Cx_ = 0
```

lemma LinearPMap.IsClosable.defectNumber_eq_of_mem_ball[\[source\]](#)

```
lemma IsClosable.defectNumber_eq_of_mem_ball [CompleteSpace H] {T : H →1ℂ.[ ] H} (hT : T.
  IsClosable)
  {z₁ z₂ : ℂ} {c : ℝ} (h : IsLowerBound T z₁ c) (h_ball : z₂ ∈ ball z₁ c) :
  T.defectNumber z₁ = T.defectNumber z₂
```

lemma LinearPMap.IsClosable.defectNumber_const[\[source\]](#)

The defect number is constant on each connected component of the regularity domain.

```
lemma IsClosable.defectNumber_const [CompleteSpace H]
  {T : H →1ℂ.[ ] H} (hT : T.IsClosable)
  {z₁ z₂ : ℂ} (hz : z₂ ∈ connectedComponentIn T.regularityDomain z₁) :
  T.defectNumber z₁ = T.defectNumber z₂
```

def LinearPMap.numericalRange[\[source\]](#)

The set $\{\langle x, T x \rangle_{\mathbb{C}} \mid x \in T.\text{domain} \wedge \|x\| = 1\} \subseteq \mathbb{C}$.

```
def numericalRange (T : H →ℂ.[.] H) : Set ℂ
```

lemma LinearPMap.numericalRange_eq[\[source\]](#)

```
lemma numericalRange_eq (T : H →ℂ.[.] H) : 0 T = (fun x ↦ ‖x‖, T ‖Cx‖) ' ' {x | ‖x‖ = 1}
```

lemma LinearPMap.mem_numericalRange[\[source\]](#)

```
lemma mem_numericalRange {T : H →ℂ.[.] H} {x : T.domain} (hx : x ≠ 0) :
  ‖(‖x‖ ^ 2)-1 * ‖x‖, T ‖Cx‖ ∈ 0 T
```

lemma LinearPMap.numericalRange_nonempty[\[source\]](#)

```
lemma numericalRange_nonempty {T : H →ℂ.[.] H} (hT : T.domain ≠ ⊥) : 0 (T).Nonempty
```

lemma LinearPMap.numericalRange_neg[\[source\]](#)

```
@[simp]
```

```
lemma numericalRange_neg (T : H →ℂ.[.] H) : 0 (-T) = 0- T
```

lemma LinearPMap.numericalRange_smul[\[source\]](#)

```
@[simp]
```

```
lemma numericalRange_smul (T : H →ℂ.[.] H) (c : ℂ) : 0 (c • T) = c • 0 T
```

lemma LinearPMap.numericalRange_sub_const[\[source\]](#)

```
lemma numericalRange_sub_const (T : H →ℂ.[.] H) (c : ℂ) : 0 (T - c • 1) = 0 T - {c}
```

lemma LinearPMap.compl_closure_numericalRange_subset_regularityDomain[\[source\]](#)

The regularity domain contains the exterior of the numerical range.

```
lemma compl_closure_numericalRange_subset_regularityDomain (T : H →ℂ.[.] H) :
  (_root_.closure 0 (T))c ⊆ T.regularityDomain
```

lemma LinearPMap.exists_phase_add_im_eq_zero[\[source\]](#)

```
private lemma exists_phase_add_im_eq_zero (z z : ℂ) :
  ∃ θ : ℝ, (exp (I * θ) * z + exp (-I * θ) * z).im = 0
```

theorem LinearPMap.numericalRange_convex[\[source\]](#)

The Toeplitz-Hausdorff theorem.

```
theorem numericalRange_convex (T : H →ℂ.[.] H) : Convex ℝ 0 (T)
```

def LinearPMap.resolventSet[\[source\]](#)

The resolvent set, ' ρ ', of a partial linear map.

A complex number ' z ' is in ' ρT ' iff the linear map ' $T - z \bullet 1$ ' from ' $T.domain$ ' to ' H ' is a bijection with continuous (equivalently, bounded) inverse.

```
def resolventSet (T : H →ℂ.[] H) : Set ℂ
```

lemma LinearPMap.resolventSet_eq[\[source\]](#)

```
lemma resolventSet_eq (T : H →ℂ.[] H) :
  ρ T = {z | (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range = T ∧ Continuous (λ z)}
```

lemma LinearPMap.mem_resolventSet_iff[\[source\]](#)

```
lemma mem_resolventSet_iff {T : H →ℂ.[] H} {z : ℂ} :
  z ∈ ρ T ↔ (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range = T ∧ Continuous (λ z)
```

lemma LinearPMap.resolventSet_eq_empty[\[source\]](#)

If an operator is not closed then its resolvent set is empty.

```
lemma resolventSet_eq_empty [CompleteSpace H] {T : H →ℂ.[] H} (h : ¬T.IsClosed) : ρ T = ∅
```

lemma LinearPMap.resolventSet_subset_regularityDomain[\[source\]](#)

```
lemma resolventSet_subset_regularityDomain (T : H →ℂ.[] H) : ρ T ⊆ T.regularityDomain
```

lemma LinearPMap.IsClosed.resolventSet_eq[\[source\]](#)

For a closed operator the continuity of the resolvent is redundant in the definition of the resolvent set.

```
lemma IsClosed.resolventSet_eq [CompleteSpace H] {T : H →ℂ.[] H} (hT : T.IsClosed) :
  ρ T = {z : ℂ | (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range = T}
```

lemma LinearPMap.IsClosed.resolventSet_eq'[\[source\]](#)

For a closed operator the resolvent set consists of those regular points for which the defect number is zero.

```
lemma IsClosed.resolventSet_eq' [CompleteSpace H] {T : H →ℂ.[] H} (hT : T.IsClosed) :
  ρ T = T.regularityDomain ∩ T.defectNumber-1 {0}
```

lemma LinearPMap.resolventSet_isOpen[\[source\]](#)

The resolvent set is an open subset of $\mathbb{C}$.

```
lemma resolventSet_isOpen [CompleteSpace H] (T : H →ℂ.[] H) : IsOpen ρ(T)
```


def LinearPMap.spectrum[\[source\]](#)

The spectrum, ' σ ', of a partial linear map.

' σ T ' is the complement of ' ρ T '. A complex number ' z ' is in ' σ T ' iff the linear map ' $T - z \bullet 1$ ' from ' $T.\text{domain}$ ' to ' H ' fails to be bijective or ' $(T - z \bullet 1)^{-1}$ ' is not continuous (equivalently, is not bounded).

```
def spectrum (T : H →ℂ.[] H) : Set ℂ
```

lemma LinearPMap.spectrum_eq[\[source\]](#)

```
lemma spectrum_eq (T : H →ℂ.[] H) : σ T = ρ( T)c
```

lemma LinearPMap.mem_spectrum_iff[\[source\]](#)

```
lemma mem_spectrum_iff {T : H →ℂ.[] H} {z : ℂ} :
  z ∈ σ T ↔ (T - z • 1).toFun.ker ≠ ⊥ ∨ (T - z • 1).toFun.range ≠ T ∨ ¬Continuous ( T z)
```

lemma LinearPMap.spectrum_eq_univ[\[source\]](#)

If an operator is not closed then its spectrum is all of $\mathbb{C}$.

```
lemma spectrum_eq_univ [CompleteSpace H] {T : H →ℂ.[] H} (h : ¬T.IsClosed) : σ T = univ
```

lemma LinearPMap.spectrum_isClosed[\[source\]](#)

The spectrum is a closed subset of $\mathbb{C}$.

```
lemma spectrum_isClosed [CompleteSpace H] (T : H →ℂ.[] H) : _root_.IsClosed σ( T)
```

def LinearPMap.pointSpectrum[\[source\]](#)

The point spectrum, ' σ^p ', of a partial linear map.

A complex number ' z ' is in ' σ^p T ' iff ' $T - z \bullet 1$ ' is not injective.

```
def pointSpectrum (T : H →ℂ.[] H) : Set ℂ
```

lemma LinearPMap.pointSpectrum_eq[\[source\]](#)

```
lemma pointSpectrum_eq (T : H →ℂ.[] H) : σp T = {z | (T - z • 1).toFun.ker ≠ ⊥}
```

lemma LinearPMap.mem_pointSpectrum_iff[\[source\]](#)

```
lemma mem_pointSpectrum_iff {T : H →ℂ.[] H} {z : ℂ} : z ∈ σp T ↔ (T - z • 1).toFun.ker ≠ ⊥
```

lemma LinearPMap.pointSpectrum_subset_spectrum[\[source\]](#)

```
lemma pointSpectrum_subset_spectrum (T : H →ℂ.[] H) : σp T ⊆ σ T
```

def LinearPMap.residualSpectrum[\[source\]](#)

The residual spectrum, σ^r , of a partial linear map.

A complex number z is in $\sigma^r T$ iff $T - z \bullet 1$ is injective but not surjective and $(T - z \bullet 1)^{-1}$ is continuous (equivalently, bounded).

```
def residualSpectrum (T : H →ℂ H) : Set ℂ
```

lemma LinearPMap.residualSpectrum_eq[\[source\]](#)

```
lemma residualSpectrum_eq (T : H →ℂ H) :
```

```
σr T = {z | (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range ≠ T ∧ Continuous (λ (T z))}
```

lemma LinearPMap.mem_residualSpectrum_iff[\[source\]](#)

```
lemma mem_residualSpectrum_iff {T : H →ℂ H} {z : ℂ} :
```

```
z ∈ σr T ↔ (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range ≠ T ∧ Continuous (λ (T z))
```

lemma LinearPMap.residualSpectrum_subset_spectrum[\[source\]](#)

```
lemma residualSpectrum_subset_spectrum (T : H →ℂ H) : σr T ⊆ σ T
```

lemma LinearPMap.residualSpectrum_subset_regularityDomain[\[source\]](#)

```
lemma residualSpectrum_subset_regularityDomain (T : H →ℂ H) : σr T ⊆ T.regularityDomain
```

def LinearPMap.continuousSpectrum[\[source\]](#)

The continuous spectrum, σ^c , of a partial linear map.

A complex number z is in $\sigma^c T$ iff the range of $T - z \bullet 1$ is not closed.

```
def continuousSpectrum (T : H →ℂ H) : Set ℂ
```

lemma LinearPMap.continuousSpectrum_eq[\[source\]](#)

```
lemma continuousSpectrum_eq (T : H →ℂ H) :
```

```
σc T = {z | ¬_root_.IsClosed ((T - z • 1).toFun.range : Set H)}
```

lemma LinearPMap.mem_continuousSpectrum_iff[\[source\]](#)

```
lemma mem_continuousSpectrum_iff {T : H →ℂ H} {z : ℂ} :
```

```
z ∈ σc T ↔ ¬_root_.IsClosed ((T - z • 1).toFun.range : Set H)
```

lemma LinearPMap.continuousSpectrum_subset_spectrum[\[source\]](#)

```
lemma continuousSpectrum_subset_spectrum (T : H →ℂ H) : σc T ⊆ σ T
```

lemma LinearPMap.IsClosed.spectrum_eq[\[source\]](#)

```
lemma IsClosed.spectrum_eq [CompleteSpace H] {T : H →ℂ H} (hT : T.IsClosed) :
  σ T = σp T ∪ σr T ∪ σc T
```

lemma LinearPMap.pointSpectrum_inter_residualSpectrum[\[source\]](#)

```
lemma pointSpectrum_inter_residualSpectrum (T : H →ℂ H) : σp T ∩ σr T = ∅
```

lemma LinearPMap.resolvent_sub[\[source\]](#)

```
lemma resolvent_sub
  {1T 2T : H →ℂ H} (hT : 2T.domain ≤ 1T.domain) {z : ℂ} (h1z : z ∈ ρ 1T) (h2z : z ∈ ρ 2T) :
  :
  □ 1T z - □ 2T z = □ 1T z * (2T - 1T) * □ 2T z
```

lemma LinearPMap.resolvent_sub'[\[source\]](#)

```
lemma resolvent_sub' {T : H →ℂ H} (z1 z2 : ℂ) (h1z : z1 ∈ ρ T) (h2z : z2 ∈ ρ T) :
  □ T z1 - □ T z2 = (z1 - z2) • □ (T z1 * □ T z2)
```

2.27 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.SelfAdjoint

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/SelfAdjoint.lean](#) on GitHub.

lemma LinearPMap.IsSelfAdjoint.resolventSet_eq_regularityDomain[\[source\]](#)

```
lemma resolventSet_eq_regularityDomain : ρ T = T.regularityDomain
```

lemma LinearPMap.IsSelfAdjoint.mem_resolventSet_of_range_eq_top[\[source\]](#)

‘(T - z • 1).range = ⊤’ is a sufficient condition for ‘z ∈ ρ T’ (and it is a necessary condition by definition of ‘ρ’).

```
lemma mem_resolventSet_of_range_eq_top {z : ℂ} (h : (T - z • 1).toFun.range = ⊤) : z ∈ ρ T
```

lemma LinearPMap.IsSelfAdjoint.spectrum_real[\[source\]](#)

The spectrum of a self-adjoint operator is real.

```
lemma spectrum_real : σ T ⊆ range ofReal
```

lemma LinearPMap.IsSelfAdjoint.residualSpectrum_eq_empty[\[source\]](#)

The residual spectrum of a self-adjoint operator is empty.

```
lemma residualSpectrum_eq_empty : σr T = ∅
```

2.28 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.SpectralMeasure

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/SpectralMeasure.lean](#) on GitHub.

instance «instance@38b2a7ba»[\[source\]](#)

```
instance (H : Type*) [SeminormedAddCommGroup H] [InnerProductSpace ℂ H] :
  IsAddTorsionFree (H →L[ℂ] H) where
```

structure SpectralMeasure[\[source\]](#)

A spectral measure on a measurable space ‘ α ’ is a σ -additive function ‘Set $\alpha \rightarrow H \rightarrow L[\mathbb{C}] H$ ’ such that each set is mapped to a star projection on ‘ H ’, the empty set and non-measurable sets are mapped to zero, and ‘univ’ is mapped to the identity.

```
structure SpectralMeasure
   $\alpha$ ( : Type*) [MeasurableSpace  $\alpha$ ]
  (H : Type*) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
  extends VectorMeasure  $\alpha$  (H →L[ℂ] H) where
  isStarProjection' :  $\forall$  A, IsStarProjection (measureOf' A)
  univ' : measureOf' univ = 1
```

instance SpectralMeasure.instCoeVectorMeasure[\[source\]](#)

```
instance instCoeVectorMeasure : Coe (SpectralMeasure  $\alpha$  H) (VectorMeasure  $\alpha$  (H →L[ℂ] H))
```

instance SpectralMeasure.instCoeFun[\[source\]](#)

```
instance instCoeFun : CoeFun (SpectralMeasure  $\alpha$  H) fun _  $\mapsto$  Set  $\alpha \rightarrow H \rightarrow L[\mathbb{C}] H$ 
```

lemma SpectralMeasure.isStarProjection[\[source\]](#)

```
lemma isStarProjection (A : Set  $\alpha$ ) : IsStarProjection  $\mu$ (S A)
```

lemma SpectralMeasure.univ[\[source\]](#)

```
@[simp]
lemma univ :  $\mu$ S univ = 1
```

lemma SpectralMeasure.comp_self[\[source\]](#)

```
@[simp]
lemma comp_self (A : Set  $\alpha$ ) :  $\mu$ S A  $\circ$ L  $\mu$ S A =  $\mu$ S A
```

lemma SpectralMeasure.comp_of_disjoint[\[source\]](#)

```
lemma comp_of_disjoint
  {A B : Set  $\alpha$ } (h : Disjoint A B) (hA : MeasurableSet A) (hB : MeasurableSet B) :
   $\mu$ S A  $\circ$ L  $\mu$ S B = 0
```

lemma SpectralMeasure.comp_eq_of_inter[\[source\]](#)

```
lemma comp_eq_of_inter {A B : Set  $\alpha$ } (hA : MeasurableSet A) (hB : MeasurableSet B) :
   $\mu$ S A  $\circ$ L  $\mu$ S B =  $\mu$ S (A  $\cap$  B)
```

lemma SpectralMeasure.commute[\[source\]](#)

```
lemma commute (A B : Set  $\alpha$ ) : Commute  $\mu(S A)$   $\mu(S B)$ 
```

2.29 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.Symmetric

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/Symmetric.lean](#) on GitHub.

def LinearPMap.IsSymmetric.realNumericalRange[\[source\]](#)

The projection of the numerical range onto the real axis.

```
def realNumericalRange (T : H  $\rightarrow$   $\mathbb{C}$ .[] H) : Set  $\mathbb{R}$ 
```

lemma LinearPMap.IsSymmetric.realNumericalRange_eq[\[source\]](#)

```
lemma realNumericalRange_eq (T : H  $\rightarrow$   $\mathbb{C}$ .[] H) :  $\theta_{re} T = re \text{ '' } \theta T$ 
```

lemma LinearPMap.IsSymmetric.realNumericalRange_neg[\[source\]](#)

[@\[simp\]](#)

```
lemma realNumericalRange_neg (T : H  $\rightarrow$   $\mathbb{C}$ .[] H) :  $\theta_{re} (-T) = \theta_{re} T$ 
```

lemma LinearPMap.IsSymmetric.im_eq_zero_of_mem_numericalRange[\[source\]](#)

The numerical range of a symmetric operator is contained in the real axis.

```
lemma im_eq_zero_of_mem_numericalRange {z :  $\mathbb{C}$ } (hz : z  $\in$   $\theta T$ ) : z.im = 0
```

lemma LinearPMap.IsSymmetric.numericalRange_subset[\[source\]](#)

The numerical range of a symmetric operator is contained in the real axis.

```
lemma numericalRange_subset :  $\theta T \subseteq \text{range ofReal}$ 
```

lemma LinearPMap.IsSymmetric.numericalRange_eq[\[source\]](#)

The numerical range of a symmetric operator is equal to its projection onto the real axis.

```
lemma numericalRange_eq :  $\theta T = \text{ofReal '' } \theta_{re} T$ 
```

lemma LinearPMap.IsSymmetric.closure_numericalRange_subset[\[source\]](#)

```
lemma closure_numericalRange_subset :  $\_root\_closure \theta(T) \subseteq \text{range ofReal}$ 
```

lemma LinearPMap.IsSymmetric.mem_regularityDomain_of_im_ne_zero[\[source\]](#)

The regularity domain of a symmetric operator contains all complex numbers with non-zero imaginary part.

```
lemma mem_regularityDomain_of_im_ne_zero {z :  $\mathbb{C}$ } (hz : z.im  $\neq$  0) : z  $\in$  T.regularityDomain
```

lemma LinearPMap.IsSymmetric.compl_ofReal_subset_regularityDomain[\[source\]](#)

The regularity domain of a symmetric operator contains all complex numbers with non-zero imaginary part.

```
lemma compl_ofReal_subset_regularityDomain : (range ofReal)c ⊆ T.regularityDomain
```

lemma LinearPMap.IsSymmetric.Iio_subset_regularityDomain[\[source\]](#)

If ‘m’ is a lower bound on the numerical range then the regularity domain contains ‘(-∞,m)’.

```
lemma Iio_subset_regularityDomain {m : ℝ} (h : m ∈ lowerBounds θre( T)) :  
  ofReal 'Iio m ⊆ T.regularityDomain
```

lemma LinearPMap.IsSymmetric.Ioi_subset_regularityDomain[\[source\]](#)

If ‘m’ is an upper bound on the numerical range then the regularity domain contains ‘(m,∞)’.

```
lemma Ioi_subset_regularityDomain {m : ℝ} (h : m ∈ upperBounds θre( T)) :  
  ofReal 'Ioi m ⊆ T.regularityDomain
```

lemma LinearPMap.IsSymmetric.regularityDomain_isConnected_iff[\[source\]](#)

The regularity domain of a symmetric operator is connected iff it contains a real number.

```
lemma regularityDomain_isConnected_iff :  
  IsConnected T.regularityDomain ↔ (ofReal -1 T.regularityDomain).Nonempty
```

lemma LinearPMap.IsSymmetric.regularityDomain_isConnected_of_bddBelow[\[source\]](#)

The regularity domain of a lower semibounded symmetric operator is connected.

```
lemma regularityDomain_isConnected_of_bddBelow (h : BddBelow θre( T)) :  
  IsConnected T.regularityDomain
```

lemma LinearPMap.IsSymmetric.regularityDomain_isConnected_of_bddAbove[\[source\]](#)

The regularity domain of an upper semibounded symmetric operator is connected.

```
lemma regularityDomain_isConnected_of_bddAbove (h : BddAbove θre( T)) :  
  IsConnected T.regularityDomain
```

lemma LinearPMap.IsSymmetric.pointSpectrum_real[\[source\]](#)

Eigenvalues of a symmetric unbounded operator are real.

```
lemma pointSpectrum_real : σp T ⊆ range ofReal
```

2.30 Physlib.QuantumMechanics.DDimensions.Operators.Unbounded

[Physlib/QuantumMechanics/DDimensions/Operators/Unbounded.lean](#) on GitHub.

lemma LinearPMap.HasDenseDomain.orthogonal_range[\[source\]](#)

$$'U.range^\perp = U\dagger.ker'$$

c.f. 'LinearMap.orthogonal_range' and 'ContinuousLinearMap.orthogonal_range'

```
lemma HasDenseDomain.orthogonal_range [CompleteSpace H] (h : U.HasDenseDomain) :
  U.toFun.[]range = †U.toFun.ker.map †U.domain.subtype
```

lemma LinearPMap.HasDenseDomain.orthogonal_adjoint_ker[\[source\]](#)

$$'U\dagger.ker^\perp = U.range.closure'$$

```
lemma HasDenseDomain.orthogonal_adjoint_ker [CompleteSpace H] [CompleteSpace H']
  (h : U.HasDenseDomain) :
  (†U.toFun.ker.map †U.domain.subtype)[] = U.toFun.range.closure
```

lemma LinearPMap.IsClosed.closure_eq[\[source\]](#)

```
lemma IsClosed.closure_eq (h : U.IsClosed) : U.closure = U
```

lemma LinearPMap.adjoint_zero[\[source\]](#)

@[simp]

```
lemma adjoint_zero [CompleteSpace H] : (0 : H →1 ℂ.[ ] H')† = 0
```

lemma LinearPMap.adjoint_sub_le_sub_adjoint[\[source\]](#)

```
lemma adjoint_sub_le_sub_adjoint [CompleteSpace H]
  (₁U ₂U : H →1 ℂ.[ ] H') (₁₂h : (₁U - ₂U).HasDenseDomain) : ₁†U - ₂†U ≤ (₁U - ₂U)†
```

lemma LinearMap.continuous_iff_bounded[\[source\]](#)

*'f : E →₁ ℂ.[] F' is continuous iff there exists 'M > 0' s.t. '||f x|| ≤ M * ||x||' for all 'x : E'.*

This is a (convenient) immediate consequence of 'IsBoundedLinearMap.isLinearMap_and_continuous_iff'.

```
lemma _root_.LinearMap.continuous_iff_bounded [] {E F : Type*} [NontriviallyNormedField []]
  [SeminormedAddCommGroup E] [NormedSpace [] E] [SeminormedAddCommGroup F] [NormedSpace [] F]
  {f : E →1 [] F} : Continuous f ↔ ∃ M, 0 < M ∧ ∀ x : E, ||f x|| ≤ M * ||x||
```

lemma LinearPMap.isClosable_of_continuous[\[source\]](#)

Continuous operators are closable.

```
lemma isClosable_of_continuous (h : Continuous U) : U.IsClosable
```

lemma LinearPMap.closure_domain_eq_domain_closure_of_continuous[\[source\]](#)

A strengthening of 'closure_domain_le_domain_closure' for continuous operators.

```
lemma closure_domain_eq_domain_closure_of_continuous [CompleteSpace H'] (h : Continuous U) :
  U.closure.domain = U.domain.closure
```

lemma LinearPMap.isClosed_iff_isClosed_domain_of_continuous[\[source\]](#)

A continuous operator is closed iff its domain is closed.

```
lemma isClosed_iff_isClosed_domain_of_continuous [CompleteSpace H'] (h : Continuous U) :
  U.IsClosed ↔ _root_.IsClosed (U.domain : Set H)
```

lemma LinearPMap.IsClosed.isClosed_toFun_graph[\[source\]](#)

```
lemma IsClosed.isClosed_toFun_graph (hU : U.IsClosed) :
  _root_.IsClosed (U.toFun.graph : Set (U.domain × H'))
```

lemma LinearPMap.IsClosed.continuous_of_isClosed_domain[\[source\]](#)

*The ****closed graph theorem**** for partial linear maps: a closed operator with closed domain is continuous.*

This follows immediately from ‘LinearMap.continuous_of_isClosed_graph’ and the completeness of ‘H’ and ‘H’.

```
lemma IsClosed.continuous_of_isClosed_domain [CompleteSpace H] [CompleteSpace H']
  (hU : U.IsClosed) (h : _root_.IsClosed (U.domain : Set H)) :
  Continuous U
```

lemma LinearPMap.IsClosable.add_continuous[\[source\]](#)

Closability is preserved upon adding a continuous operator.

```
lemma IsClosable.add_continuous
  (₁h : ₁U.IsClosable) (₂h : Continuous ₂U) (h : ₁U.domain ≤ ₂U.domain) :
  (₁U + ₂U).IsClosable
```

lemma LinearPMap.IsClosable.sub_continuous[\[source\]](#)

Closability is preserved upon subtracting a continuous operator.

```
lemma IsClosable.sub_continuous
  (₁h : ₁U.IsClosable) (₂h : Continuous ₂U) (h : ₁U.domain ≤ ₂U.domain) : (₁U - ₂U).
  IsClosable
```

lemma LinearPMap.IsClosed.add_continuous[\[source\]](#)

Closedness is preserved upon adding a continuous operator.

```
lemma IsClosed.add_continuous [CompleteSpace H']
  (₁h : ₁U.IsClosed) (₂h : Continuous ₂U) (h : ₁U.domain ≤ ₂U.domain) : (₁U + ₂U).IsClosed
```

lemma LinearPMap.IsClosed.sub_continuous[\[source\]](#)

Closedness is preserved upon subtracting a continuous operator.

```
lemma IsClosed.sub_continuous [CompleteSpace H']
  (₁h : ₁U.IsClosed) (₂h : Continuous ₂U) (h : ₁U.domain ≤ ₂U.domain) : (₁U - ₂U).IsClosed
```


lemma LinearPMap.IsUnbounded.orthogonal_adjoint_range

[\[source\]](#)

$$‘U\dagger.range^\perp = U.closure.ker’$$

lemma IsUnbounded.orthogonal_adjoint_range [CompleteSpace H] [CompleteSpace H']
(h : U.IsUnbounded) : $\dagger U.toFun.\square range = U.closure.toFun.ker.map\ U.closure.domain.subtype$

lemma LinearPMap.IsUnbounded.orthogonal_closure_ker

[\[source\]](#)

$$‘U.closure.ker^\perp = U\dagger.range’$$

lemma IsUnbounded.orthogonal_closure_ker [CompleteSpace H] [CompleteSpace H'] (h : U.
IsUnbounded) :
(U.closure.toFun.ker.map U.closure.domain.subtype) $\square = \dagger U.toFun.range.closure$

2.31 Physlib.QuantumMechanics.FiniteTarget.HilbertSpace

[Physlib/QuantumMechanics/FiniteTarget/HilbertSpace.lean](#) on GitHub.

def QuantumMechanics.FiniteHilbertSpace.equivEuclidean

[\[source\]](#)

The equivalence between ‘FiniteHilbertSpace d’ and ‘EuclideanSpace $\mathbb{C}$ d’ given by ‘val’.

def equivEuclidean : FiniteHilbertSpace d $\approx$ EuclideanSpace $\mathbb{C}$ d **where**

instance QuantumMechanics.FiniteHilbertSpace.«instance@89dd6b4e»

[\[source\]](#)

noncomputable instance : AddCommGroup (FiniteHilbertSpace d)

instance QuantumMechanics.FiniteHilbertSpace.«instance@ba152d22»

[\[source\]](#)

noncomputable instance : Module $\mathbb{C}$ (FiniteHilbertSpace d)

lemma QuantumMechanics.FiniteHilbertSpace.val_add

[\[source\]](#)

@[simp]

lemma val_add ψ (ϕ : FiniteHilbertSpace d) : $\psi(+ \phi).val = \psi.val + \phi.val$

lemma QuantumMechanics.FiniteHilbertSpace.val_smul

[\[source\]](#)

@[simp]

lemma val_smul (c : $\mathbb{C}$) ψ (: FiniteHilbertSpace d) : $(c \cdot \psi).val = c \cdot \psi.val$

lemma QuantumMechanics.FiniteHilbertSpace.val_zero

[\[source\]](#)

@[simp]

lemma val_zero : $(0 : FiniteHilbertSpace d).val = 0$

def QuantumMechanics.FiniteHilbertSpace.linearEquivEuclidean

[\[source\]](#)

The equivalence between ‘FiniteHilbertSpace d’ and ‘EuclideanSpace $\mathbb{C}$ d’ as a $\mathbb{C}$ ’-linear equivalence, upgrading ‘equivEuclidean’.

```
noncomputable def linearEquivEuclidean : FiniteHilbertSpace d  $\approx_{\mathbb{C}}$  EuclideanSpace  $\mathbb{C}$  d
```

instance QuantumMechanics.FiniteHilbertSpace.«instance@a3aa9745»

[\[source\]](#)

```
noncomputable instance : NormedAddCommGroup (FiniteHilbertSpace d)
```

lemma QuantumMechanics.FiniteHilbertSpace.norm_eq_val

[\[source\]](#)

@[simp]

```
lemma norm_eq_val  $\psi$  ( : FiniteHilbertSpace d) :  $\|\psi\| = \|\psi\|.val$ 
```

instance QuantumMechanics.FiniteHilbertSpace.«instance@d6d3ba74»

[\[source\]](#)

```
noncomputable instance : InnerProductSpace  $\mathbb{C}$  (FiniteHilbertSpace d)
```

lemma QuantumMechanics.FiniteHilbertSpace.inner_eq_val

[\[source\]](#)

@[simp]

```
lemma inner_eq_val  $\psi$  (  $\phi$  : FiniteHilbertSpace d) :  $\text{inner } \mathbb{C} \psi \phi = \text{inner } \mathbb{C} \psi.val \phi.val$ 
```

instance QuantumMechanics.FiniteHilbertSpace.«instance@df6e240e»

[\[source\]](#)

```
instance : FiniteDimensional  $\mathbb{C}$  (FiniteHilbertSpace d)
```

instance QuantumMechanics.FiniteHilbertSpace.«instance@25da29e8»

[\[source\]](#)

```
instance : CompleteSpace (FiniteHilbertSpace d)
```

def QuantumMechanics.FiniteHilbertSpace.isometryEquivEuclidean

[\[source\]](#)

The equivalence between ‘FiniteHilbertSpace d’ and ‘EuclideanSpace $\mathbb{C}$ d’ as a linear isometry equivalence, upgrading ‘linearEquivEuclidean’.

```
noncomputable def isometryEquivEuclidean : FiniteHilbertSpace d  $\approx_{\mathbb{C}}$  EuclideanSpace  $\mathbb{C}$  d where
```

def QuantumMechanics.FiniteHilbertSpace.basisFun

[\[source\]](#)

The standard orthonormal basis of ‘FiniteHilbertSpace d’, indexed by ‘d’.

```
noncomputable def basisFun (d : Type*) [Fintype d] [DecidableEq d] :  
  OrthonormalBasis d  $\mathbb{C}$  (FiniteHilbertSpace d)
```

lemma QuantumMechanics.FiniteHilbertSpace.basisFun_apply

[\[source\]](#)

```
lemma basisFun_apply (i : d) : basisFun d i = (EuclideanSpace.single i )1
```

2.32 Physlib.Relativity.LorentzGroup.Basic

[Physlib/Relativity/LorentzGroup/Basic.lean](#) on GitHub.

lemma LorentzGroup.continuous_self_mul_dual

[\[source\]](#)

```
lemma continuous_self_mul_dual {d : ℕ} :
  Continuous (fun Λ : Matrix (Fin 1 ⊕ Fin d) (Fin 1 ⊕ Fin d) ℝ => Λ * dual Λ)
```

lemma LorentzGroup.isClosed

[\[source\]](#)

The Lorentz group is closed in the ambient matrix space.

```
lemma isClosed (d : ℕ) :
  IsClosed (LorentzGroup d : Set (Matrix (Fin 1 ⊕ Fin d) (Fin 1 ⊕ Fin d) ℝ))
```

lemma LorentzGroup.isClosedEmbedding_val

[\[source\]](#)

The coercion from the Lorentz group to its ambient matrix space is a closed embedding.

```
lemma isClosedEmbedding_val {d : ℕ} :
  Topology.IsClosedEmbedding
  (Subtype.val : LorentzGroup d → Matrix (Fin 1 ⊕ Fin d) (Fin 1 ⊕ Fin d) ℝ)
```

2.33 Physlib.Relativity.Tensors.Basic

[Physlib/Relativity/Tensors/Basic.lean](#) on GitHub.

lemma TensorSpecies.Tensor.Pure.update_eq_function_update

[\[source\]](#)

```
lemma update_eq_function_update {n : ℕ} {c : Fin n → C} [inst : DecidableEq (Fin n)]
  (p : Pure S c) (i : Fin n)
  (x : V (c i)) : update p i x = Function.update p i x
```

lemma TensorSpecies.Tensor.Pure.update_diff

[\[source\]](#)

```
lemma update_diff {n : ℕ} {c : Fin n → C} [inst : DecidableEq (Fin n)] (p : Pure S c) (i j :
  Fin n)
  (x : V (c i)) (hij : i ≠ j) : (update p i x) j = p j
```

lemma TensorSpecies.Tensor.componentMap_eq_repr

[\[source\]](#)

‘componentMap’ is the basis representation ‘(basis c).repr’ definitionally; this bridges the two notations wherever a component computation meets a ‘repr’ rewrite.

```
lemma componentMap_eq_repr {n : ℕ} (c : Fin n → C) (t : S.Tensor c)
  ψ( : ComponentIdx (S := S) c) : componentMap c t ψ = (basis c).repr t ψ
```

lemma TensorSpecies.Tensor.componentMap_ext

[\[source\]](#)

Two tensors with the same colour sequence are equal when all their components agree.

```
lemma componentMap_ext {n : ℕ} {c : Fin n → C} {t₁ t₂ : S.Tensor c}
  (h : ∀ b, componentMap c t₁ b = componentMap c t₂ b) : t₁ = t₂
```

instance TensorSpecies.Tensor.«instance@180d6a48»[\[source\]](#)

```

instance {k : Type} [Field k] {C : Type} {G : Type} [Group G]
  {V : C → Type} ∀[ c, AddCommGroup (V c)] ∀[ c, Module k (V c)]
  {basisIdx : C → Type} ∀[ c, Fintype (basisIdx c)] ∀[ c, DecidableEq (basisIdx c)]
  {rep : (c : C) → Representation k G (V c)} {b : (c : C) → Basis (basisIdx c) k (V c)}
  (S : TensorSpecies k C G V basisIdx rep b)
  {c : Fin n → C} : FiniteDimensional k (S.Tensor c)

```

instance TensorSpecies.Tensor.«instance@357a4078»[\[source\]](#)

```

noncomputable instance {k : Type} [RCLike k] {C : Type} {G : Type} [Group G]
  {V : C → Type} ∀[ c, AddCommGroup (V c)] ∀[ c, Module k (V c)]
  {basisIdx : C → Type} ∀[ c, Fintype (basisIdx c)] ∀[ c, DecidableEq (basisIdx c)]
  {rep : (c : C) → Representation k G (V c)} {b : (c : C) → Basis (basisIdx c) k (V c)}
  (S : TensorSpecies k C G V basisIdx rep b)
  {c : Fin n → C} : TopologicalSpace (S.Tensor c)

```

instance TensorSpecies.Tensor.«instance@958788b8»[\[source\]](#)

```

instance {k : Type} [RCLike k] {C : Type} {G : Type} [Group G]
  {V : C → Type} ∀[ c, AddCommGroup (V c)] ∀[ c, Module k (V c)]
  {basisIdx : C → Type} ∀[ c, Fintype (basisIdx c)] ∀[ c, DecidableEq (basisIdx c)]
  {rep : (c : C) → Representation k G (V c)} {b : (c : C) → Basis (basisIdx c) k (V c)}
  (S : TensorSpecies k C G V basisIdx rep b)
  {c : Fin n → C} : IsTopologicalAddGroup (S.Tensor c)

```

lemma TensorSpecies.Tensor.Pure.rep_cast[\[source\]](#)

```

lemma rep_cast {g : G} {c c1 : C} (p : V c) (h : c = c1) :
  LinearEquiv.cast (R := k) h (rep c g p) = rep c1 g (LinearEquiv.cast (R := k) h p)

```

lemma TensorSpecies.Tensor.Pure.actionP_cast[\[source\]](#)

```

lemma actionP_cast {g : G} {p : Pure S c} (h : c = c1) :
  LinearEquiv.cast (R := k) h (g • p) = g • LinearEquiv.cast (R := k) h p

```

instance TensorSpecies.Tensor.«instance@713c1a5c»[\[source\]](#)

```

noncomputable instance : SMul G (S.Tensor c) where

```

instance TensorSpecies.Tensor.«instance@62e8adcb»[\[source\]](#)

```

noncomputable instance : DistribMulAction G (S.Tensor c) where

```

lemma TensorSpecies.Tensor.Pure.permP_equivariant[\[source\]](#)

```

lemma Pure.permP_equivariant {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ{ : Fin m → Fin n} (h : IsReindexing c c1 σ) (g : G) (p : Pure S c) :
  Pure.permP σ h (g • p) = g • Pure.permP σ h p

```

lemma TensorSpecies.Tensor.toField_injective[\[source\]](#)

```
lemma toField_injective {c : Fin 0 → C} :
  Function.Injective (toField : S.Tensor c → k)
```

lemma TensorSpecies.Tensor.toField_basis[\[source\]](#)

```
lemma toField_basis {c : Fin 0 → C} (b : ComponentIdx (S := S) c) :
  toField (basis c b) = 1
```

lemma TensorSpecies.Tensor.eq_smul_toField[\[source\]](#)

```
lemma eq_smul_toField {c : Fin 0 → C} (t : Tensor S c) :
  t = toField t • (basis c (@default (ComponentIdx (S := S) c) Unique.instInhabited))
```

2.34 Physlib.Relativity.Tensors.ComplexTensor.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Basic.lean](#) on GitHub.

abbrev complexLorentzTensor.modules[\[source\]](#)

The modules associated with each of the different types of complex Lorentz vector space.

```
abbrev modules : Color → Type
| Color.upL => Fermion.LeftHandedWeyl
| Color.downL => Fermion.DualLeftHandedWeyl
| Color.upR => Fermion.RightHandedWeyl
| Color.downR => Fermion.DualRightHandedWeyl
| Color.up => Lorentz.CContrModule
| Color.down => Lorentz.CCoModule
```

instance complexLorentzTensor.modulesAddCommGroup[\[source\]](#)

```
instance modulesAddCommGroup : ∀ c, AddCommGroup (modules c)
| Color.upL => inferInstance
| Color.downL => inferInstance
| Color.upR => inferInstance
| Color.downR => inferInstance
| Color.up => inferInstance
| Color.down => inferInstance
```

instance complexLorentzTensor.modulesModule[\[source\]](#)

```
noncomputable instance modulesModule : ∀ c, Module ℂ (modules c)
| Color.upL => inferInstance
| Color.downL => inferInstance
| Color.upR => inferInstance
| Color.downR => inferInstance
| Color.up => inferInstance
| Color.down => inferInstance
```

abbrev complexLorentzTensor.basis[\[source\]](#)

The basis associated with each of the different types of complex Lorentz vector space.

```
abbrev basis (c : Color) : Module.Basis (Fin (repDim c)) ℂ (modules c)
```

abbrev complexLorentzTensor.rep[\[source\]](#)

The reps associated with each of the different types of complex Lorentz vector space.

```
abbrev rep (c : Color) : Representation ℂ SL(2, ℂ) (modules c)
```

lemma complexLorentzTensor.contrPCoeff_basis[\[source\]](#)

```
lemma contrPCoeff_basis {n : ℕ} {c : Fin n → complexLorentzTensor.Color} (i j : Fin n)
  (hij : i ≠ j ∧ (complexLorentzTensor. (c i) = c j))
  (b : ComponentIdx (S := complexLorentzTensor) c) :
  Pure.contrPCoeff i j hij (Pure.basisVector c b) = if b i =
    Fin.cast (by simp <| hij.2, repDim_tau]) (b j)
  then 1 else 0
```

2.35 Physlib.Relativity.Tensors.ComplexTensor.Metrics.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Metrics/Basic.lean](#) on GitHub.

abbrev complexLorentzTensor.dualLeftMetric[\[source\]](#)

The metric ' ε_{aa} ' as a complex Lorentz tensor.

```
abbrev dualLeftMetric : ℂT[.downL, .downL]
```

abbrev complexLorentzTensor.dualRightMetric[\[source\]](#)

The metric ' $\varepsilon_{\dot{a}\dot{a}}$ ' as a complex Lorentz tensor.

```
abbrev dualRightMetric : ℂT[.downR, .downR]
```

lemma complexLorentzTensor.dualLeftMetric_eq_fromConstPair[\[source\]](#)

```
lemma dualLeftMetric_eq_fromConstPair : εL' = fromConstPair Fermion.dualLeftMetric
```

lemma complexLorentzTensor.dualRightMetric_eq_fromConstPair[\[source\]](#)

```
lemma dualRightMetric_eq_fromConstPair : εR' = fromConstPair Fermion.dualRightMetric
```

lemma complexLorentzTensor.dualLeftMetric_eq_fromPairT[\[source\]](#)

```
lemma dualLeftMetric_eq_fromPairT : εL' = fromPairT (Fermion.dualLeftMetricVal)
```

lemma complexLorentzTensor.dualRightMetric_eq_fromPairT[\[source\]](#)

```
lemma dualRightMetric_eq_fromPairT : εR' = fromPairT (Fermion.dualRightMetricVal)
```

lemma complexLorentzTensor.dualLeftMetric_eq_dualLeftBasis[\[source\]](#)

```

lemma dualLeftMetric_eq_dualLeftBasis :  $\varepsilon L'$  =
  fromPairT (dualLeftBasis 0  $\otimes_{\mathbb{C}}$  dualLeftBasis 1)
  - fromPairT (dualLeftBasis 1  $\otimes_{\mathbb{C}}$  dualLeftBasis 0)

```

lemma complexLorentzTensor.dualRightMetric_eq_dualRightBasis[\[source\]](#)

```

lemma dualRightMetric_eq_dualRightBasis :  $\varepsilon R'$  =
  fromPairT (dualRightBasis 0  $\otimes_{\mathbb{C}}$  dualRightBasis 1)
  - fromPairT (dualRightBasis 1  $\otimes_{\mathbb{C}}$  dualRightBasis 0)

```

lemma complexLorentzTensor.dualLeftMetric_eq_basis[\[source\]](#)

```

lemma dualLeftMetric_eq_basis :  $\varepsilon L'$  =
  (Tensor.basis (S := complexLorentzTensor) ![Color.downL, Color.downL]
    (fun | 0 => (0 : Fin 2) | 1 => (1 : Fin 2)))
  - (Tensor.basis (S := complexLorentzTensor)
    ![Color.downL, Color.downL] (fun | 0 => (1 : Fin 2) | 1 => (0 : Fin 2)))

```

lemma complexLorentzTensor.dualRightMetric_eq_basis[\[source\]](#)

```

lemma dualRightMetric_eq_basis :  $\varepsilon R'$  =
  (Tensor.basis (S := complexLorentzTensor)
    ![Color.downR, Color.downR] (fun | 0 => (0 : Fin 2) | 1 => (1 : Fin 2)))
  - (Tensor.basis (S := complexLorentzTensor)
    ![Color.downR, Color.downR] (fun | 0 => (1 : Fin 2) | 1 => (0 : Fin 2)))

```

lemma complexLorentzTensor.dualLeftMetric_eq_ofRat[\[source\]](#)

```

lemma dualLeftMetric_eq_ofRat :  $\varepsilon L'$  = ofRat fun f =>
  if f 0 = Fin.cast (by rfl) (0 : Fin 2)  $\wedge$  f 1 = Fin.cast (by rfl) (1 : Fin 2) then 1 else
  if f 1 = Fin.cast (by rfl) (0 : Fin 2)  $\wedge$  f 0 = Fin.cast (by rfl) (1 : Fin 2) then
  - 1 else 0

```

lemma complexLorentzTensor.dualRightMetric_eq_ofRat[\[source\]](#)

```

lemma dualRightMetric_eq_ofRat :  $\varepsilon R'$  = ofRat fun f =>
  if f 0 = Fin.cast (by rfl) (0 : Fin 2)  $\wedge$  f 1 = Fin.cast (by rfl) (1 : Fin 2) then 1 else
  if f 1 = Fin.cast (by rfl) (0 : Fin 2)  $\wedge$  f 0 = Fin.cast (by rfl) (1 : Fin 2) then
  - 1 else 0

```

lemma complexLorentzTensor.actionT_dualLeftMetric[\[source\]](#)

The tensor ‘dualLeftMetric’ is invariant under the action of ‘ $SL(2, \mathbb{C})$ ’.

```

lemma actionT_dualLeftMetric (g : SLC(2,)) : g •  $\varepsilon L'$  =  $\varepsilon L'$ 

```

lemma complexLorentzTensor.actionT_dualRightMetric[\[source\]](#)

The tensor ‘dualRightMetric’ is invariant under the action of ‘ $SL(2, \mathbb{C})$ ’.

```

lemma actionT_dualRightMetric (g : SLC(2,)) : g •  $\varepsilon R'$  =  $\varepsilon R'$ 

```

2.36 Physlib.Relativity.Tensors.ComplexTensor.Metrics.Lemmas

[Physlib/Relativity/Tensors/ComplexTensor/Metrics/Lemmas.lean](#) on GitHub.

lemma complexLorentzTensor.dualLeftMetric_antisymm [\[source\]](#)

The dual-left metric is antisymmetric $\{\varepsilon L' \mid \alpha \alpha' = -\varepsilon L' \mid \alpha' \alpha\}^T$:

lemma dualLeftMetric_antisymm : $\{\varepsilon L' \mid \alpha \alpha' = -\varepsilon L' \mid \alpha' \alpha\}^T$

lemma complexLorentzTensor.dualRightMetric_antisymm [\[source\]](#)

The dual-right metric is antisymmetric $\{\varepsilon R' \mid \beta \beta' = -\varepsilon R' \mid \beta' \beta\}^T$:

lemma dualRightMetric_antisymm : $\{\varepsilon R' \mid \alpha \alpha' = -\varepsilon R' \mid \alpha' \alpha\}^T$

lemma complexLorentzTensor.leftMetric_contr_dualLeftMetric [\[source\]](#)

The contraction of the left metric with the dual-left metric is the unit $\{\varepsilon L \mid \alpha \beta \otimes \varepsilon L' \mid \beta \gamma = \delta L \mid \alpha \gamma\}^T$:

lemma leftMetric_contr_dualLeftMetric : $\{\varepsilon L \mid \alpha \beta \otimes \varepsilon L' \mid \beta \gamma = \delta L \mid \alpha \gamma\}^T$

lemma complexLorentzTensor.rightMetric_contr_dualRightMetric [\[source\]](#)

The contraction of the right metric with the dual-right metric is the unit $\{\varepsilon R \mid \alpha \beta \otimes \varepsilon R' \mid \beta \gamma = \delta R \mid \alpha \gamma\}^T$:

lemma rightMetric_contr_dualRightMetric : $\{\varepsilon R \mid \alpha \beta \otimes \varepsilon R' \mid \beta \gamma = \delta R \mid \alpha \gamma\}^T$

lemma complexLorentzTensor.dualLeftMetric_contr_leftMetric [\[source\]](#)

The contraction of the dual-left metric with the left metric is the unit $\{\varepsilon L' \mid \alpha \beta \otimes \varepsilon L \mid \beta \gamma = \delta L' \mid \alpha \gamma\}^T$:

lemma dualLeftMetric_contr_leftMetric : $\{\varepsilon L' \mid \alpha \beta \otimes \varepsilon L \mid \beta \gamma = \delta L' \mid \alpha \gamma\}^T$

lemma complexLorentzTensor.dualRightMetric_contr_rightMetric [\[source\]](#)

The contraction of the dual-right metric with the right metric is the unit $\{\varepsilon R' \mid \alpha \beta \otimes \varepsilon R \mid \beta \gamma = \delta R' \mid \alpha \gamma\}^T$:

lemma dualRightMetric_contr_rightMetric : $\{\varepsilon R' \mid \alpha \beta \otimes \varepsilon R \mid \beta \gamma = \delta R' \mid \alpha \gamma\}^T$

2.37 Physlib.Relativity.Tensors.ComplexTensor.Units.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Units/Basic.lean](#) on GitHub.

abbrev complexLorentzTensor.dualLeftLeftUnit [\[source\]](#)

The unit δ_a^a as a complex Lorentz tensor.

abbrev dualLeftLeftUnit : $\mathbb{C}T[\text{.downL}, \text{.upL}]$

abbrev complexLorentzTensor.leftDualLeftUnit

[\[source\]](#)

The unit ' δ_a^a ' as a complex Lorentz tensor.

abbrev leftDualLeftUnit : $\mathbb{C}T[.upL, .downL]$

abbrev complexLorentzTensor.dualRightRightUnit

[\[source\]](#)

The unit ' $\delta_{\dot{a}}^{\dot{a}}$ ' as a complex Lorentz tensor.

abbrev dualRightRightUnit : $\mathbb{C}T[.downR, .upR]$

abbrev complexLorentzTensor.rightDualRightUnit

[\[source\]](#)

The unit ' $\delta^{\dot{a}}_{\dot{a}}$ ' as a complex Lorentz tensor.

abbrev rightDualRightUnit : $\mathbb{C}T[.upR, .downR]$

lemma complexLorentzTensor.dualLeftLeftUnit_eq_fromConstPair

[\[source\]](#)

lemma dualLeftLeftUnit_eq_fromConstPair : $\delta L' = \text{fromConstPair Fermion.dualLeftLeftUnit}$

lemma complexLorentzTensor.leftDualLeftUnit_eq_fromConstPair

[\[source\]](#)

lemma leftDualLeftUnit_eq_fromConstPair : $\delta L = \text{fromConstPair Fermion.leftDualLeftUnit}$

lemma complexLorentzTensor.dualRightRightUnit_eq_fromConstPair

[\[source\]](#)

lemma dualRightRightUnit_eq_fromConstPair : $\delta R' = \text{fromConstPair Fermion.dualRightRightUnit}$

lemma complexLorentzTensor.rightDualRightUnit_eq_fromConstPair

[\[source\]](#)

lemma rightDualRightUnit_eq_fromConstPair : $\delta R = \text{fromConstPair Fermion.rightDualRightUnit}$

lemma complexLorentzTensor.dualLeftLeftUnit_eq_fromPairT

[\[source\]](#)

lemma dualLeftLeftUnit_eq_fromPairT : $\delta L' = \text{fromPairT (Fermion.dualLeftLeftUnitVal)}$

lemma complexLorentzTensor.leftDualLeftUnit_eq_fromPairT

[\[source\]](#)

lemma leftDualLeftUnit_eq_fromPairT : $\delta L = \text{fromPairT (Fermion.leftDualLeftUnitVal)}$

lemma complexLorentzTensor.dualRightRightUnit_eq_fromPairT

[\[source\]](#)

lemma dualRightRightUnit_eq_fromPairT : $\delta R' = \text{fromPairT (Fermion.dualRightRightUnitVal)}$

lemma complexLorentzTensor.rightDualRightUnit_eq_fromPairT

[\[source\]](#)

lemma rightDualRightUnit_eq_fromPairT : $\delta R = \text{fromPairT (Fermion.rightDualRightUnitVal)}$

lemma complexLorentzTensor.dualLeftLeftUnit_eq_dualLeftBasis_leftBasis[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_dualLeftBasis_leftBasis : δL' =
  ∑ i, fromPairT (dualLeftBasis i @_tC[] leftBasis i)
```

lemma complexLorentzTensor.leftDualLeftUnit_eq_leftBasis_dualLeftBasis[\[source\]](#)

```
lemma leftDualLeftUnit_eq_leftBasis_dualLeftBasis : δL =
  ∑ i, fromPairT (leftBasis i @_tC[] dualLeftBasis i)
```

lemma complexLorentzTensor.dualRightRightUnit_eq_dualRightBasis_rightBasis[\[source\]](#)

```
lemma dualRightRightUnit_eq_dualRightBasis_rightBasis : δR' =
  ∑ i, fromPairT (dualRightBasis i @_tC[] rightBasis i)
```

lemma complexLorentzTensor.rightDualRightUnit_eq_rightBasis_dualRightBasis[\[source\]](#)

```
lemma rightDualRightUnit_eq_rightBasis_dualRightBasis : δR =
  ∑ i, fromPairT (rightBasis i @_tC[] dualRightBasis i)
```

lemma complexLorentzTensor.dualLeftLeftUnit_eq_basis[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_basis : δL' =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.downL, Color.upL] (fun | 0 => i | 1 => i)
```

lemma complexLorentzTensor.leftDualLeftUnit_eq_basis[\[source\]](#)

```
lemma leftDualLeftUnit_eq_basis : δL =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.upL, Color.downL] (fun | 0 => i | 1 => i)
```

lemma complexLorentzTensor.dualRightRightUnit_eq_basis[\[source\]](#)

```
lemma dualRightRightUnit_eq_basis : δR' =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.downR, Color.upR] (fun | 0 => i | 1 => i)
```

lemma complexLorentzTensor.rightDualRightUnit_eq_basis[\[source\]](#)

```
lemma rightDualRightUnit_eq_basis : δR =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.upR, Color.downR] (fun | 0 => i | 1 => i)
```

lemma complexLorentzTensor.dualLeftLeftUnit_eq_ofRat[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_ofRat : δL' = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma complexLorentzTensor.leftDualLeftUnit_eq_ofRat[\[source\]](#)

```
lemma leftDualLeftUnit_eq_ofRat : δL = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma complexLorentzTensor.dualRightRightUnit_eq_ofRat[\[source\]](#)

```
lemma dualRightRightUnit_eq_ofRat : δR' = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma complexLorentzTensor.rightDualRightUnit_eq_ofRat[\[source\]](#)

```
lemma rightDualRightUnit_eq_ofRat : δR = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma complexLorentzTensor.actionT_dualLeftLeftUnit[\[source\]](#)

The tensor ‘dualLeftLeftUnit’ is invariant under the action of ‘ $SL(2, \mathbb{C})$ ’.

```
lemma actionT_dualLeftLeftUnit (g : SLC(2,)) : g • δL' = δL'
```

lemma complexLorentzTensor.actionT_leftDualLeftUnit[\[source\]](#)

The tensor ‘leftDualLeftUnit’ is invariant under the action of ‘ $SL(2, \mathbb{C})$ ’.

```
lemma actionT_leftDualLeftUnit (g : SLC(2,)) : g • δL = δL
```

lemma complexLorentzTensor.actionT_dualRightRightUnit[\[source\]](#)

The tensor ‘dualRightRightUnit’ is invariant under the action of ‘ $SL(2, \mathbb{C})$ ’.

```
lemma actionT_dualRightRightUnit (g : SLC(2,)) : g • δR' = δR'
```

lemma complexLorentzTensor.actionT_rightDualRightUnit[\[source\]](#)

The tensor ‘rightDualRightUnit’ is invariant under the action of ‘ $SL(2, \mathbb{C})$ ’.

```
lemma actionT_rightDualRightUnit (g : SLC(2,)) : g • δR = δR
```

2.38 Physlib.Relativity.Tensors.ComplexTensor.Units.Symm

[Physlib/Relativity/Tensors/ComplexTensor/Units/Symm.lean](#) on GitHub.

lemma complexLorentzTensor.dualLeftLeftUnit_symm[\[source\]](#)

Swapping indices of ‘dualLeftLeftUnit’ returns ‘leftDualLeftUnit’: ‘ $\{\delta L' \mid \alpha \alpha' = \delta L \mid \alpha' \alpha\}^T$ ’.

```
lemma dualLeftLeftUnit_symm : δ{L' | α α' = δL | α' α}^T
```

lemma complexLorentzTensor.leftDualLeftUnit_symm[\[source\]](#)

Swapping indices of 'leftDualLeftUnit' returns 'dualLeftLeftUnit': $\{\delta L \mid \alpha \alpha' = \delta L' \mid \alpha' \alpha\}^T$.

lemma leftDualLeftUnit_symm : $\delta\{L \mid \alpha \alpha' = \delta L' \mid \alpha' \alpha^T\}$

lemma complexLorentzTensor.dualRightRightUnit_symm[\[source\]](#)

Swapping indices of 'dualRightRightUnit' returns 'rightDualRightUnit': $\{\delta R \mid \beta \beta' = \delta R' \mid \beta' \beta\}^T$.

lemma dualRightRightUnit_symm : $\delta\{R' \mid \beta \beta' = \delta R \mid \beta' \beta^T\}$

lemma complexLorentzTensor.rightDualRightUnit_symm[\[source\]](#)

Swapping indices of 'rightDualRightUnit' returns 'dualRightRightUnit': $\{\delta R \mid \beta \beta' = \delta R' \mid \beta' \beta\}^T$.

lemma rightDualRightUnit_symm : $\delta\{R \mid \beta \beta' = \delta R' \mid \beta' \beta^T\}$

2.39 Physlib.Relativity.Tensors.ComplexTensor.Vector.Pre.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Vector/Pre/Basic.lean](#) on GitHub.

lemma Lorentz.CoModule.SL2CRep_val[\[source\]](#)

lemma CoModule.SL2CRep_val (M : SL(2, ℂ)) (v : CoModule) :
 ((CoModule.SL2CRep M) v).val =
 (LorentzGroup.toComplex (SL2C.toLorentzGroup M))^{-T} v * v.val

def Lorentz.inclCoRealLorentz[\[source\]](#)

The semilinear map including real Lorentz co-vectors into complex covariant Lorentz vectors.

def inclCoRealLorentz : CoMod 3 →_{SL} [Complex.ofRealHom] CoModule **where**

lemma Lorentz.inclCoRealLorentz_val[\[source\]](#)

lemma inclCoRealLorentz_val (v : CoMod 3) :
 (inclCoRealLorentz v).val = ofRealHom ∘ v.ℝtoFinId

lemma Lorentz.complexCoBasis_of_real[\[source\]](#)

lemma complexCoBasis_of_real (i : Fin 1 ⊕ Fin 3) :
 (complexCoBasis i) = inclCoRealLorentz (CoMod.stdBasis i)

lemma Lorentz.inclCoRealLorentz_ρ[\[source\]](#)

lemma pinclCoRealLorentz_ (M : SL(2, ℂ)) (v : CoMod 3) :
 (CoModule.SL2CRep M) (inclCoRealLorentz v) =
 inclCoRealLorentz ((Co 3)ρ. (SL2C.toLorentzGroup M) v)

2.40 Physlib.Relativity.Tensors.ComplexTensor.Vector.Pre.Modules

[Physlib/Relativity/Tensors/ComplexTensor/Vector/Pre/Modules.lean](#) on GitHub.

lemma Lorentz.CoCModule.ext

[\[source\]](#)

```
@[ext]
lemma ext (ψ' : CCModule) (h : ψ.val = ψ'.val) : ψ = ψ'
```

lemma Lorentz.CoCModule.val_add

[\[source\]](#)

```
@[simp]
lemma val_add (ψ' : CCModule) : (ψ + ψ').val = ψ.val + ψ'.val
```

lemma Lorentz.CoCModule.val_smul

[\[source\]](#)

```
@[simp]
lemma val_smul (r : C) (ψ : CCModule) : (r • ψ).val = r • ψ.val
```

2.41 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Basic.lean](#) on GitHub.

def Fermion.leftHandedRep

[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the fundamental representation of $SL(2, \mathbb{C})$. In index notation corresponds to a Weyl fermion with indices ψ^a .

```
def leftHandedRep : Representation C SLC(2,) LeftHandedWeyl where
```

def Fermion.dualLeftHandedRep

[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the representation of $SL(2, \mathbb{C})$ given by $M \rightarrow (M^{-1})^T$. In index notation corresponds to a Weyl fermion with indices ψ_a .

```
def dualLeftHandedRep : Representation C SLC(2,) DualLeftHandedWeyl where
```

def Fermion.dualLeftBasis

[\[source\]](#)

The standard basis on dual-left-handed Weyl fermions.

```
def dualLeftBasis : Basis (Fin 2) C DualLeftHandedWeyl
```

lemma Fermion.dualLeftBasis_toFin2C

[\[source\]](#)

```
@[simp]
lemma C dualLeftBasis_toFin2 (i : Fin 2) : (dualLeftBasis i).CtoFin2 = Pi.single i 1
```

lemma Fermion.dualLeftBasis_ρ_apply

[\[source\]](#)

```
@[simp]
lemma pdualLeftBasis__apply (M : SLC(2,)) (i j : Fin 2) :
  (LinearMap.toMatrix dualLeftBasis dualLeftBasis) (dualLeftHandedRep M) i j = (M⁻¹.1¹)ᵀ i j
```

def Fermion.rightHandedRep[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the conjugate representation of $SL(2, \mathbb{C})$. In index notation corresponds to a Weyl fermion with indices $\psi^{\dot{a}}$.

```
def rightHandedRep : Representation ℂ SLC(2,) RightHandedWeyl where
```

def Fermion.dualRightHandedRep[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the representation of $SL(2, \mathbb{C})$ given by $M \rightarrow (M^{-1})^t$. In index notation this corresponds to a Weyl fermion with index $\psi_{\dot{a}}$.

```
def dualRightHandedRep : Representation ℂ SLC(2,) DualRightHandedWeyl where
```

def Fermion.dualRightBasis[\[source\]](#)

The standard basis on dual-right-handed Weyl fermions.

```
def dualRightBasis : Basis (Fin 2) ℂ DualRightHandedWeyl
```

lemma Fermion.dualRightBasis_toFin2ℂ[\[source\]](#)

```
@[simp]
```

```
lemma C dualRightBasis_toFin2 (i : Fin 2) : (dualRightBasis i).CtoFin2 = Pi.single i 1
```

lemma Fermion.dualRightBasis_ρ_apply[\[source\]](#)

```
@[simp]
```

```
lemma pdualRightBasis__apply (M : SLC(2,)) (i j : Fin 2) :
  (LinearMap.toMatrix dualRightBasis dualRightBasis) (dualRightHandedRep M) i j =
  ((M⁻¹).1¹).conjTranspose i j
```

def Fermion.leftHandedToDual[\[source\]](#)

The morphism between the representation ‘leftHanded’ and the representation ‘dualLeftHanded’ defined by multiplying an element of ‘leftHanded’ by the matrix $\varepsilon^{a0a1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

```
def leftHandedToDual : leftHandedRep.IntertwiningMap dualLeftHandedRep where
```

lemma Fermion.leftHandedToDual_hom_apply[\[source\]](#)

```
lemma leftHandedToDual_hom_apply ψ( : LeftHandedWeyl) :
  leftHandedToDual ψ =
  DualLeftHandedWeyl.CtoFin2Equiv.symm (!! [0, 1; -1, 0] v * ψ.CtoFin2)
```

def Fermion.leftHandedDualTo[\[source\]](#)

The morphism from ‘dualLeftHanded’ to ‘leftHanded’ defined by multiplying an element of DualLeftHandedWeyl by the matrix $\varepsilon_{a1a2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

```
def leftHandedDualTo : dualLeftHandedRep.IntertwiningMap leftHandedRep where
```

lemma Fermion.leftHandedDualTo_hom_apply[\[source\]](#)

```
lemma leftHandedDualTo_hom_apply ψ( : DualLeftHandedWeyl) :
  leftHandedDualTo ψ =
  LeftHandedWeyl.ḲtoFin2Equiv.symm (![0, -1; 1, 0] v* ψ.ḲtoFin2)
```

def Fermion.leftHandedDualEquiv[\[source\]](#)

The equivalence between the representation ‘leftHanded’ and the representation ‘dualLeftHanded’ defined by multiplying an element of ‘leftHanded’ by the matrix ‘ $\varepsilon^{a_0 a_1} = ![0, 1; -1, 0]$ ’.

```
def leftHandedDualEquiv : leftHandedRep.Equiv dualLeftHandedRep
```

lemma Fermion.leftHandedDualEquiv_hom_hom_apply[\[source\]](#)

‘leftHandedDualEquiv’ acting on an element ‘ $\psi : \text{leftHanded}$ ’ corresponds to multiplying ‘ ψ ’ by the matrix ‘ $![0, 1; -1, 0]$ ’.

```
lemma leftHandedDualEquiv_hom_hom_apply ψ( : LeftHandedWeyl) :
  leftHandedDualEquiv ψ =
  DualLeftHandedWeyl.ḲtoFin2Equiv.symm (![0, 1; -1, 0] v* ψ.ḲtoFin2)
```

lemma Fermion.leftHandedDualEquiv_inv_hom_apply[\[source\]](#)

The inverse of ‘leftHandedDualEquiv’ acting on an element ‘ $\psi : \text{dualLeftHanded}$ ’ corresponds to multiplying ‘ ψ ’ by the matrix ‘ $![0, -1; 1, 0]$ ’.

```
lemma leftHandedDualEquiv_inv_hom_apply ψ( : DualLeftHandedWeyl) :
  leftHandedDualEquiv.symm ψ =
  LeftHandedWeyl.ḲtoFin2Equiv.symm (![0, -1; 1, 0] v* ψ.ḲtoFin2)
```

informal_definition Fermion.rightHandedWeylDualEquiv[\[source\]](#)

The linear equivalence between ‘rightHandedWeyl’ and ‘DualRightHandedWeyl’ given by multiplying an element of ‘rightHandedWeyl’ by the matrix ‘ $\varepsilon^{a_0 a_1} = ![0, 1; -1, 0]$ ’.

```
informal_definition rightHandedWeylDualEquiv where
  deps := [``rightHandedRep, ``dualRightHandedRep]
  tag := "6VZR4"
```

informal_lemma Fermion.rightHandedWeylDualEquiv_equivariant[\[source\]](#)

The linear equivalence ‘rightHandedWeylDualEquiv’ is equivariant with respect to the action of ‘ $SL(2, C)$ ’ on ‘rightHandedWeyl’ and ‘DualRightHandedWeyl’.

```
informal_lemma rightHandedWeylDualEquiv_equivariant where
  deps := [``rightHandedWeylDualEquiv]
  tag := "6VZSG"
```

2.42 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Contraction

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Contraction.lean](#) on GitHub.

def Fermion.leftDualBi[\[source\]](#)

The bi-linear map corresponding to contraction of a left-handed Weyl fermion with a dual-left-handed Weyl fermion.

```
def leftDualBi : LeftHandedWeyl →1ℂ[] DualLeftHandedWeyl →1ℂ[] ℂ where
```

def Fermion.dualLeftBi[\[source\]](#)

The bi-linear map corresponding to contraction of a dual-left-handed Weyl fermion with a left-handed Weyl fermion.

```
def dualLeftBi : DualLeftHandedWeyl →1ℂ[] LeftHandedWeyl →1ℂ[] ℂ where
```

def Fermion.rightDualBi[\[source\]](#)

The bi-linear map corresponding to contraction of a right-handed Weyl fermion with a dual-right-handed Weyl fermion.

```
def rightDualBi : RightHandedWeyl →1ℂ[] DualRightHandedWeyl →1ℂ[] ℂ where
```

def Fermion.dualRightBi[\[source\]](#)

The bi-linear map corresponding to contraction of a dual-right-handed Weyl fermion with a right-handed Weyl fermion.

```
def dualRightBi : DualRightHandedWeyl →1ℂ[] RightHandedWeyl →1ℂ[] ℂ where
```

def Fermion.leftDualContraction[\[source\]](#)

The linear map from $\text{leftHandedWeyl} \otimes \text{DualLeftHandedWeyl}$ to $\mathbb{C}$ given by summing over components of leftHandedWeyl and $\text{DualLeftHandedWeyl}$ in the standard basis (i.e. the dot product). Physically, the contraction of a left-handed Weyl fermion with a dual-left-handed Weyl fermion. In index notation this is $\psi^a \varphi_a$.

```
def leftDualContraction : (leftHandedRep.tprod dualLeftHandedRep).IntertwiningMap  
(Representation.trivial ℂ SL(2,) ℂ) where
```

lemma Fermion.leftDualContraction_hom_tmul[\[source\]](#)

```
lemma leftDualContraction_hom_tmul ψ( : LeftHandedWeyl)  
  φ( : DualLeftHandedWeyl) :  
  leftDualContraction ψ( ⊗t φ) = ψ.CtoFin2 □v φ.CtoFin2
```

lemma Fermion.leftDualContraction_basis[\[source\]](#)

```
lemma leftDualContraction_basis (i j : Fin 2) :  
  leftDualContraction (leftBasis i ⊗t dualLeftBasis j) = if i.1 = j.1 then (1 : ℂ) else 0
```

def Fermion.dualLeftContraction[\[source\]](#)

The linear map from $\text{DualLeftHandedWeyl} \otimes \text{leftHandedWeyl}$ to $\mathbb{C}$ given by summing over components of $\text{DualLeftHandedWeyl}$ and leftHandedWeyl in the standard basis

(i.e. the dot product). Physically, the contraction of a dual-left-handed Weyl fermion with a left-handed Weyl fermion. In index notation this is $\varphi_a \psi^a$.

```
def dualLeftContraction : (dualLeftHandedRep.tprod leftHandedRep).IntertwiningMap
  (Representation.trivial  $\mathbb{C}$  SL(2,)  $\mathbb{C}$ ) where
```

lemma Fermion.dualLeftContraction_hom_tmul

[\[source\]](#)

```
lemma dualLeftContraction_hom_tmul  $\phi$  ( : DualLeftHandedWeyl)  $\psi$  ( : LeftHandedWeyl) :
  dualLeftContraction  $\phi$  ( $\otimes_t \psi$ ) =  $\phi$ .CtoFin2  $\sqcap_v \psi$ .CtoFin2
```

lemma Fermion.dualLeftContraction_basis

[\[source\]](#)

```
lemma dualLeftContraction_basis (i j : Fin 2) :
  dualLeftContraction (dualLeftBasis i  $\otimes_t$  leftBasis j) = if i.1 = j.1 then (1 :  $\mathbb{C}$ ) else 0
```

def Fermion.rightDualContraction

[\[source\]](#)

The linear map from ‘rightHandedWeyl $\otimes$ DualRightHandedWeyl’ to ‘ $\mathbb{C}$ ’ given by summing over components of ‘rightHandedWeyl’ and ‘DualRightHandedWeyl’ in the standard basis (i.e. the dot product). The contraction of a right-handed Weyl fermion with a left-handed Weyl fermion. In index notation this is $\psi^{\dot{a}} \varphi_{\dot{a}}$.

```
def rightDualContraction : (rightHandedRep.tprod dualRightHandedRep).IntertwiningMap
  (Representation.trivial  $\mathbb{C}$  SL(2,)  $\mathbb{C}$ ) where
```

lemma Fermion.rightDualContraction_hom_tmul

[\[source\]](#)

```
lemma rightDualContraction_hom_tmul  $\psi$  ( : RightHandedWeyl)
   $\phi$  ( : DualRightHandedWeyl) :
  rightDualContraction  $\psi$  ( $\otimes_t \phi$ ) =  $\psi$ .CtoFin2  $\sqcap_v \phi$ .CtoFin2
```

lemma Fermion.rightDualContraction_basis

[\[source\]](#)

```
lemma rightDualContraction_basis (i j : Fin 2) :
  rightDualContraction (rightBasis i  $\otimes_t$  dualRightBasis j) =
  if i.1 = j.1 then (1 :  $\mathbb{C}$ ) else 0
```

def Fermion.dualRightContraction

[\[source\]](#)

The linear map from DualRightHandedWeyl $\otimes$ rightHandedWeyl to $\mathbb{C}$ given by summing over components of DualRightHandedWeyl and rightHandedWeyl in the standard basis (i.e. the dot product). The contraction of a right-handed Weyl fermion with a left-handed Weyl fermion. In index notation this is $\varphi_{\dot{a}} \psi^{\dot{a}}$.

```
def dualRightContraction : (dualRightHandedRep.tprod rightHandedRep).IntertwiningMap
  (Representation.trivial  $\mathbb{C}$  SL(2,)  $\mathbb{C}$ ) where
```

lemma Fermion.dualRightContraction_hom_tmul

[\[source\]](#)

```
lemma dualRightContraction_hom_tmul  $\phi$  ( : DualRightHandedWeyl)
   $\psi$  ( : RightHandedWeyl) :
  dualRightContraction  $\phi$  ( $\otimes_t \psi$ ) =  $\phi$ .CtoFin2  $\sqcap_v \psi$ .CtoFin2
```

lemma Fermion.dualRightContraction_basis[\[source\]](#)

```
lemma dualRightContraction_basis (i j : Fin 2) :
  dualRightContraction (dualRightBasis i ⊗ₜ rightBasis j) =
    if i.1 = j.1 then (1 : ℂ) else 0
```

lemma Fermion.leftDualContraction_tmul_symm[\[source\]](#)

```
lemma leftDualContraction_tmul_symm ψ( : LeftHandedWeyl) φ( : DualLeftHandedWeyl) :
  leftDualContraction ψ( ⊗ₜ ℂ[] φ) = dualLeftContraction φ( ⊗ₜ ℂ[] ψ)
```

lemma Fermion.dualLeftContraction_tmul_symm[\[source\]](#)

```
lemma dualLeftContraction_tmul_symm φ( : DualLeftHandedWeyl) ψ( : LeftHandedWeyl) :
  dualLeftContraction φ( ⊗ₜ ℂ[] ψ) = leftDualContraction ψ( ⊗ₜ ℂ[] φ)
```

lemma Fermion.rightDualContraction_tmul_symm[\[source\]](#)

```
lemma rightDualContraction_tmul_symm ψ( : RightHandedWeyl) φ( : DualRightHandedWeyl) :
  rightDualContraction ψ( ⊗ₜ ℂ[] φ) = dualRightContraction φ( ⊗ₜ ℂ[] ψ)
```

lemma Fermion.dualRightContraction_tmul_symm[\[source\]](#)

```
lemma dualRightContraction_tmul_symm φ( : DualRightHandedWeyl) ψ( : RightHandedWeyl) :
  dualRightContraction φ( ⊗ₜ ℂ[] ψ) = rightDualContraction ψ( ⊗ₜ ℂ[] φ)
```

2.43 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Metric

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Metric.lean](#) on GitHub.

lemma Fermion.leftMetricVal_expand_tmul'[\[source\]](#)

```
lemma leftMetricVal_expand_tmul' : leftMetricVal = leftBasis 1 ⊗ₜ ℂ[] leftBasis 0
  - leftBasis 0 ⊗ₜ ℂ[] leftBasis 1
```

def Fermion.dualLeftMetricVal[\[source\]](#)

The metric ϵ_{aa} as an element of $(\text{dualLeftHanded} \otimes \text{dualLeftHanded}).V$.

```
def dualLeftMetricVal : (DualLeftHandedWeyl ⊗ₜ ℂ[] DualLeftHandedWeyl)
```

lemma Fermion.dualLeftMetricVal_expand_tmul[\[source\]](#)

Expansion of dualLeftMetricVal into the left basis.

```
lemma dualLeftMetricVal_expand_tmul : dualLeftMetricVal =
  dualLeftBasis 0 ⊗ₜ ℂ[] dualLeftBasis 1 - dualLeftBasis 1 ⊗ₜ ℂ[] dualLeftBasis 0
```

def Fermion.dualLeftMetric[\[source\]](#)

The metric ϵ_{aa} as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \longrightarrow \text{dualLeftHanded} \otimes \text{dualLeftHanded}$, making manifest its invariance under the action of $SL(2, \mathbb{C})$.

```
def dualLeftMetric : (Representation.trivial ℂ SL(2, ℂ).IntertwiningMap
  (dualLeftHandedRep.tprod dualLeftHandedRep) where
```

lemma Fermion.dualLeftMetric_apply_one [\[source\]](#)

```
lemma dualLeftMetric_apply_one : dualLeftMetric (1 : ℂ) = dualLeftMetricVal
```

lemma Fermion.rightMetricVal_expand_tmul' [\[source\]](#)

```
lemma rightMetricVal_expand_tmul' : rightMetricVal = rightBasis 1 ⊗_t ℂ[] rightBasis 0
  - rightBasis 0 ⊗_t ℂ[] rightBasis 1
```

def Fermion.dualRightMetricVal [\[source\]](#)

The metric $\varepsilon_{\{ \dot{a} \} \{ \dot{a} \}}$ as an element of $(\text{dualRightHanded} \otimes \text{dualRightHanded}).V$.

```
def dualRightMetricVal : DualRightHandedWeyl ⊗ ℂ[] DualRightHandedWeyl
```

lemma Fermion.dualRightMetricVal_expand_tmul [\[source\]](#)

Expansion of rightMetricVal into the left basis.

```
lemma dualRightMetricVal_expand_tmul : dualRightMetricVal =
  dualRightBasis 0 ⊗_t ℂ[] dualRightBasis 1 - dualRightBasis 1 ⊗_t ℂ[] dualRightBasis 0
```

def Fermion.dualRightMetric [\[source\]](#)

The metric $\varepsilon_{\{ \dot{a} \} \{ \dot{a} \}}$ as a morphism $\mathcal{K}_{\text{Rep } \mathbb{C} SL(2, \mathbb{C})} \rightarrow \text{dualRightHanded} \otimes \text{dualRightHanded}$, making manifest its invariance under the action of $SL(2, \mathbb{C})$.

```
def dualRightMetric : (Representation.trivial ℂ SL(2, ℂ).IntertwiningMap
  (dualRightHandedRep.tprod dualRightHandedRep) where
```

lemma Fermion.dualRightMetric_apply_one [\[source\]](#)

```
lemma dualRightMetric_apply_one : dualRightMetric (1 : ℂ) = dualRightMetricVal
```

lemma Fermion.leftDualContraction_apply_metric [\[source\]](#)

```
lemma leftDualContraction_apply_metric :
  (TensorProduct.comm ℂ _ _ <|
    (TensorProduct.lid ℂ _).lTensor _ <|
    (leftDualContraction.toLinearMap.rTensor _).lTensor _ <|
    (TensorProduct.assoc ℂ _ _).symm.toLinearMap.lTensor _ <|
    TensorProduct.assoc ℂ _ _ (_ ⊗ ℂ[] _) <|
    (leftMetric 1) ⊗_t ℂ[] (dualLeftMetric 1)) = dualLeftLeftUnit (1 : ℂ)
```

lemma Fermion.dualLeftContraction_apply_metric [\[source\]](#)

```

lemma dualLeftContraction_apply_metric :
  (TensorProduct.comm C _ _ <|
   (TensorProduct.lid C _).lTensor _ <|
   (dualLeftContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc C _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc C _ _ (_ @C[] _) <|
   (dualLeftMetric 1) @C[] (leftMetric 1)) = leftDualLeftUnit (1 : C)

```

lemma Fermion.rightDualContraction_apply_metric

[\[source\]](#)

```

lemma rightDualContraction_apply_metric :
  (TensorProduct.comm C _ _ <|
   (TensorProduct.lid C _).lTensor _ <|
   (rightDualContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc C _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc C _ _ (_ @C[] _) <|
   (rightMetric 1) @C[] (dualRightMetric 1)) = dualRightRightUnit (1 : C)

```

lemma Fermion.dualRightContraction_apply_metric

[\[source\]](#)

```

lemma dualRightContraction_apply_metric :
  (TensorProduct.comm C _ _ <|
   (TensorProduct.lid C _).lTensor _ <|
   (dualRightContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc C _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc C _ _ (_ @C[] _) <|
   (dualRightMetric 1) @C[] (rightMetric 1)) = rightDualRightUnit (1 : C)

```

2.44 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Modules

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Modules.lean](#) on GitHub.

structure Fermion.LeftHandedWeyl

[\[source\]](#)

The module in which left handed fermions live. This is equivalent to ‘ $\text{Fin } 2 \rightarrow \mathbb{C}$ ’.

```

structure LeftHandedWeyl where
  val : Fin 2 → C

```

def Fermion.LeftHandedWeyl.toFin2CFun

[\[source\]](#)

The equivalence between ‘LeftHandedWeyl’ and ‘ $\text{Fin } 2 \rightarrow \mathbb{C}$ ’.

```

def CtoFin2Fun : LeftHandedWeyl ≈ (Fin 2 → C) where

```

instance Fermion.LeftHandedWeyl.«instance@c5c29288»

[\[source\]](#)

The instance of ‘AddCommMonoid’ on ‘LeftHandedWeyl’ defined via its equivalence with ‘ $\text{Fin } 2 \rightarrow \mathbb{C}$ ’.

```

instance : AddCommMonoid LeftHandedWeyl

```

instance Fermion.LeftHandedWeyl.«instance@c3de7b53»

[\[source\]](#)

The instance of ‘AddCommGroup’ on ‘LeftHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommGroup LeftHandedWeyl
```

instance Fermion.LeftHandedWeyl.«instance@db8c5fc4»

[\[source\]](#)

The instance of ‘Module’ on ‘LeftHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : Module ℂ LeftHandedWeyl
```

def Fermion.LeftHandedWeyl.toFin2CEquiv

[\[source\]](#)

The linear equivalence between ‘LeftHandedWeyl’ and ‘(Fin 2 → ℂ)’.

```
@[simps!]
def CtoFin2Equiv : LeftHandedWeyl ≈ℂ (Fin 2 → ℂ) where
```

abbrev Fermion.LeftHandedWeyl.toFin2ℂ

[\[source\]](#)

The underlying element of ‘Fin 2 → ℂ’ of a element in ‘LeftHandedWeyl’ defined through the linear equivalence ‘toFin2CEquiv’.

```
abbrev CtoFin2 ψ( : LeftHandedWeyl)
```

structure Fermion.DualLeftHandedWeyl

[\[source\]](#)

The module in which dual-left handed fermions live. This is equivalent to ‘Fin 2 → ℂ’.

```
structure DualLeftHandedWeyl where
  val : Fin 2 → ℂ
```

def Fermion.DualLeftHandedWeyl.toFin2CFun

[\[source\]](#)

The equivalence between ‘DualLeftHandedWeyl’ and ‘Fin 2 → ℂ’.

```
def CtoFin2Fun : DualLeftHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance Fermion.DualLeftHandedWeyl.«instance@ed87b73f»

[\[source\]](#)

The instance of ‘AddCommMonoid’ on ‘DualLeftHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommMonoid DualLeftHandedWeyl
```

instance Fermion.DualLeftHandedWeyl.«instance@61470bec»

[\[source\]](#)

The instance of ‘AddCommGroup’ on ‘DualLeftHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommGroup DualLeftHandedWeyl
```

instance Fermion.DualLeftHandedWeyl.«instance@c5aed84d»

[\[source\]](#)

The instance of ‘Module’ on ‘DualLeftHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : Module ℂ DualLeftHandedWeyl
```

def Fermion.DualLeftHandedWeyl.toFin2ℂEquiv

[\[source\]](#)

The linear equivalence between ‘DualLeftHandedWeyl’ and ‘(Fin 2 → ℂ)’.

```
@[simps!]
```

```
def CtoFin2Equiv : DualLeftHandedWeyl ≈ℂ [] (Fin 2 → ℂ) where
```

abbrev Fermion.DualLeftHandedWeyl.toFin2ℂ

[\[source\]](#)

The underlying element of ‘Fin 2 → ℂ’ of a element in ‘DualLeftHandedWeyl’ defined through the linear equivalence ‘toFin2ℂEquiv’.

```
abbrev CtoFin2 ψ( : DualLeftHandedWeyl)
```

structure Fermion.RightHandedWeyl

[\[source\]](#)

The module in which right handed fermions live. This is equivalent to ‘Fin 2 → ℂ’.

```
structure RightHandedWeyl where
```

```
val : Fin 2 → ℂ
```

def Fermion.RightHandedWeyl.toFin2ℂFun

[\[source\]](#)

The equivalence between ‘RightHandedWeyl’ and ‘Fin 2 → ℂ’.

```
def CtoFin2Fun : RightHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance Fermion.RightHandedWeyl.«instance@97ccc5aa»

[\[source\]](#)

The instance of ‘AddCommMonoid’ on ‘RightHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommMonoid RightHandedWeyl
```

instance Fermion.RightHandedWeyl.«instance@4a60c666»

[\[source\]](#)

The instance of ‘AddCommGroup’ on ‘RightHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommGroup RightHandedWeyl
```

instance Fermion.RightHandedWeyl.«instance@753cf914»

[\[source\]](#)

The instance of ‘Module’ on ‘RightHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : Module ℂ RightHandedWeyl
```

def Fermion.RightHandedWeyl.toFin2CEquiv

[\[source\]](#)

The linear equivalence between ‘RightHandedWeyl’ and ‘(Fin 2 → ℂ)’.

```
@[simps!]
```

```
def CtoFin2Equiv : RightHandedWeyl ≈ ℂ[] (Fin 2 → ℂ) where
```

abbrev Fermion.RightHandedWeyl.toFin2C

[\[source\]](#)

The underlying element of ‘Fin 2 → ℂ’ of a element in ‘RightHandedWeyl’ defined through the linear equivalence ‘toFin2CEquiv’.

```
abbrev CtoFin2 ψ ( : RightHandedWeyl)
```

structure Fermion.DualRightHandedWeyl

[\[source\]](#)

The module in which dual-right handed fermions live. This is equivalent to ‘Fin 2 → ℂ’.

```
structure DualRightHandedWeyl where
```

```
val : Fin 2 → ℂ
```

def Fermion.DualRightHandedWeyl.toFin2CFun

[\[source\]](#)

The equivalence between ‘DualRightHandedWeyl’ and ‘Fin 2 → ℂ’.

```
def CtoFin2Fun : DualRightHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance Fermion.DualRightHandedWeyl.«instance@c7559440»

[\[source\]](#)

The instance of ‘AddCommMonoid’ on ‘DualRightHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommMonoid DualRightHandedWeyl
```

instance Fermion.DualRightHandedWeyl.«instance@a8cda771»

[\[source\]](#)

The instance of ‘AddCommGroup’ on ‘DualRightHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : AddCommGroup DualRightHandedWeyl
```

instance Fermion.DualRightHandedWeyl.«instance@b26a3288»

[\[source\]](#)

The instance of ‘Module’ on ‘DualRightHandedWeyl’ defined via its equivalence with ‘Fin 2 → ℂ’.

```
instance : Module ℂ DualRightHandedWeyl
```

```
def Fermion.DualRightHandedWeyl.toFin2CEquiv
```

[\[source\]](#)

The linear equivalence between ‘DualRightHandedWeyl’ and ‘(Fin 2 → ℂ)’.

```
@[simps!]
```

```
def CtoFin2Equiv : DualRightHandedWeyl ≈ℂ[] (Fin 2 → ℂ) where
```

```
abbrev Fermion.DualRightHandedWeyl.toFin2C
```

[\[source\]](#)

The underlying element of ‘Fin 2 → ℂ’ of a element in ‘DualRightHandedWeyl’ defined through the linear equivalence ‘toFin2CEquiv’.

```
abbrev CtoFin2 ψ( : DualRightHandedWeyl)
```

2.45 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Two

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Two.lean](#) on GitHub.

```
def Fermion.dualLeftdualLeftToMatrix
```

[\[source\]](#)

Equivalence of ‘dualLeftHanded ⊗ dualLeftHanded’ to ‘2 x 2’ complex matrices.

```
def dualLeftdualLeftToMatrix : (DualLeftHandedWeyl ⊗ℂ[] DualLeftHandedWeyl) ≈ℂ[]  
Matrix (Fin 2) (Fin 2) ℂ
```

```
lemma Fermion.dualLeftdualLeftToMatrix_symm_expand_tmul
```

[\[source\]](#)

Expanding ‘dualLeftdualLeftToMatrix’ in terms of the standard basis.

```
lemma dualLeftdualLeftToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) ℂ) :  
dualLeftdualLeftToMatrix.symm M = ∑ i, ∑ j, M i j •  
(dualLeftBasis i ⊗ℂ[] dualLeftBasis j)
```

```
def Fermion.leftDualLeftToMatrix
```

[\[source\]](#)

Equivalence of ‘leftHanded ⊗ dualLeftHanded’ to ‘2 x 2’ complex matrices.

```
def leftDualLeftToMatrix : (LeftHandedWeyl ⊗ℂ[] DualLeftHandedWeyl) ≈ℂ[]  
Matrix (Fin 2) (Fin 2) ℂ
```

```
lemma Fermion.leftDualLeftToMatrix_symm_expand_tmul
```

[\[source\]](#)

Expanding ‘leftDualLeftToMatrix’ in terms of the standard basis.

```
lemma leftDualLeftToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) ℂ) :  
leftDualLeftToMatrix.symm M = ∑ i, ∑ j, M i j • (leftBasis i ⊗ℂ[] dualLeftBasis j)
```

```
def Fermion.dualLeftLeftToMatrix
```

[\[source\]](#)

Equivalence of ‘dualLeftHanded ⊗ leftHanded’ to ‘2 x 2’ complex matrices.


```
def dualLeftLeftToMatrix : (DualLeftHandedWeyl @C[] LeftHandedWeyl) ≈1C[]
  Matrix (Fin 2) (Fin 2) C
```

lemma Fermion.dualLeftLeftToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding ‘dualLeftLeftToMatrix’ in terms of the standard basis.

```
lemma dualLeftLeftToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  dualLeftLeftToMatrix.symm M = ∑ i, ∑ j, M i j • (dualLeftBasis i @1C[] leftBasis j)
```

def Fermion.dualRightDualRightToMatrix

[\[source\]](#)

Equivalence of ‘dualRightHanded ⊗ dualRightHanded’ to ‘2 x 2’ complex matrices.

```
def dualRightDualRightToMatrix : (DualRightHandedWeyl @C[] DualRightHandedWeyl) ≈1C[]
  Matrix (Fin 2) (Fin 2) C
```

lemma Fermion.dualRightDualRightToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding ‘dualRightDualRightToMatrix’ in terms of the standard basis.

```
lemma dualRightDualRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  dualRightDualRightToMatrix.symm M =
  ∑ i, ∑ j, M i j • (dualRightBasis i @1C[] dualRightBasis j)
```

def Fermion.rightDualRightToMatrix

[\[source\]](#)

Equivalence of ‘rightHanded ⊗ dualRightHanded’ to ‘2 x 2’ complex matrices.

```
def rightDualRightToMatrix : (RightHandedWeyl @C[] DualRightHandedWeyl) ≈1C[]
  Matrix (Fin 2) (Fin 2) C
```

lemma Fermion.rightDualRightToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding ‘rightDualRightToMatrix’ in terms of the standard basis.

```
lemma rightDualRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  rightDualRightToMatrix.symm M = ∑ i, ∑ j, M i j • (rightBasis i @1C[] dualRightBasis j)
```

def Fermion.dualRightRightToMatrix

[\[source\]](#)

Equivalence of ‘dualRightHanded ⊗ rightHanded’ to ‘2 x 2’ complex matrices.

```
def dualRightRightToMatrix : (DualRightHandedWeyl @C[] RightHandedWeyl) ≈1C[]
  Matrix (Fin 2) (Fin 2) C
```

lemma Fermion.dualRightRightToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding ‘dualRightRightToMatrix’ in terms of the standard basis.

```
lemma dualRightRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  dualRightRightToMatrix.symm M = ∑ i, ∑ j, M i j • (dualRightBasis i @1C[] rightBasis j)
```

def Fermion.dualLeftDualRightToMatrix[\[source\]](#)

Equivalence of ‘dualLeftHanded $\otimes$ dualRightHanded’ to ‘ 2×2 ’ complex matrices.

```
def dualLeftDualRightToMatrix : (DualLeftHandedWeyl @C[] DualRightHandedWeyl) → C[]
Matrix (Fin 2) (Fin 2) C
```

lemma Fermion.dualLeftDualRightToMatrix_symm_expand_tm1[\[source\]](#)

Expanding ‘dualLeftDualRightToMatrix’ in terms of the standard basis.

```
lemma dualLeftDualRightToMatrix_symm_expand_tm1 (M : Matrix (Fin 2) (Fin 2) C) :
dualLeftDualRightToMatrix.symm M = ∑ i, ∑ j, M i j •
(dualLeftBasis i @C[] dualRightBasis j)
```

lemma Fermion.coe_linearEquivFunOnFinite[\[source\]](#)

The coercion of ‘Finsupp.linearEquivFunOnFinite’ to a function is the underlying finitely-supported function, used to bridge it with ‘Matrix.mulVec’.

```
private lemma coe_linearEquivFunOnFinite (g : (Fin 2 × Fin 2) → C) :
Finsupp.linearEquivFunOnFinite C C (Fin 2 × Fin 2) g = ↑g
```

lemma Fermion.dualLeftdualLeftToMatrix_ρ[\[source\]](#)

*The group action of ‘ $SL(2, \mathbb{C})$ ’ on ‘dualLeftHanded $\otimes$ dualLeftHanded’ is equivalent to ‘ $(M \cdot 1^{-1})^T * \text{leftLeftToMatrix } v * (M \cdot 1^{-1})$ ’.*

```
lemma pdualLeftdualLeftToMatrix_ (v : (DualLeftHandedWeyl @C[] DualLeftHandedWeyl)) (M : SL
C(2,)) :
dualLeftdualLeftToMatrix (TensorProduct.map (dualLeftHandedRep M) (dualLeftHandedRep M) v) =
=
(M⁻.1¹)ᵀ * dualLeftdualLeftToMatrix v * (M⁻.1¹)
```

lemma Fermion.leftDualLeftToMatrix_ρ[\[source\]](#)

*The group action of ‘ $SL(2, \mathbb{C})$ ’ on ‘leftHanded $\otimes$ dualLeftHanded’ is equivalent to ‘ $M \cdot 1 * \text{leftDualLeftToMatrix } v * (M \cdot 1^{-1})$ ’.*

```
lemma pleftDualLeftToMatrix_ (v : (LeftHandedWeyl @C[] DualLeftHandedWeyl)) (M : SLC(2,)) :
leftDualLeftToMatrix (TensorProduct.map (leftHandedRep M) (dualLeftHandedRep M) v) =
M.1 * leftDualLeftToMatrix v * (M⁻.1¹)
```

lemma Fermion.dualLeftLeftToMatrix_ρ[\[source\]](#)

*The group action of ‘ $SL(2, \mathbb{C})$ ’ on ‘dualLeftHanded $\otimes$ leftHanded’ is equivalent to ‘ $(M \cdot 1^{-1})^T * \text{leftDualLeftToMatrix } v * (M \cdot 1)^T$ ’.*

```
lemma pdualLeftLeftToMatrix_ (v : (DualLeftHandedWeyl @C[] LeftHandedWeyl)) (M : SLC(2,)) :
dualLeftLeftToMatrix (TensorProduct.map (dualLeftHandedRep M) (leftHandedRep M) v) =
(M⁻.1¹)ᵀ * dualLeftLeftToMatrix v * (M.1)ᵀ
```

lemma Fermion.dualRightDualRightToMatrix_ρ[\[source\]](#)

*The group action of ‘ $SL(2, \mathbb{C})$ ’ on ‘ $dualRightHanded \otimes dualRightHanded$ ’ is equivalent to ‘ $((M.1^{-1}).conjTranspose * rightRightToMatrix v * ((M.1^{-1}).conjTranspose)^T$ ’.*

```
lemma pdualRightDualRightToMatrix_ (v : (DualRightHandedWeyl @C[] DualRightHandedWeyl))
(M : SLC(2,)) :
dualRightDualRightToMatrix (TensorProduct.map (dualRightHandedRep M) (dualRightHandedRep M)
v) =
((M⁻.1¹).conjTranspose) * dualRightDualRightToMatrix v * ((M⁻.1¹).conjTranspose)ᵀ
```

lemma Fermion.rightDualRightToMatrix_ρ

[\[source\]](#)

*The group action of ‘ $SL(2, \mathbb{C})$ ’ on ‘ $rightHanded \otimes dualRightHanded$ ’ is equivalent to ‘ $(M.1.map star) * rightDualRightToMatrix v * ((M.1^{-1}).conjTranspose)^T$ ’.*

```
lemma prightDualRightToMatrix_ (v : (RightHandedWeyl @C[] DualRightHandedWeyl)) (M : SLC(2,)) :
rightDualRightToMatrix (TensorProduct.map (rightHandedRep M) (dualRightHandedRep M) v) =
(M.1.map star) * rightDualRightToMatrix v * ((M⁻.1¹).conjTranspose)ᵀ
```

lemma Fermion.dualRightRightToMatrix_ρ

[\[source\]](#)

*The group action of ‘ $SL(2, \mathbb{C})$ ’ on ‘ $dualRightHanded \otimes rightHanded$ ’ is equivalent to ‘ $((M.1^{-1}).conjTranspose * rightDualRightToMatrix v * (M.1.map star))^T$ ’.*

```
lemma pdualRightRightToMatrix_ (v : (DualRightHandedWeyl @C[] RightHandedWeyl)) (M : SLC(2,)) :
dualRightRightToMatrix (TensorProduct.map (dualRightHandedRep M) (rightHandedRep M) v) =
((M⁻.1¹).conjTranspose) * dualRightRightToMatrix v * (M.1.map star)ᵀ
```

lemma Fermion.dualLeftDualRightToMatrix_ρ

[\[source\]](#)

```
lemma pdualLeftDualRightToMatrix_ (v : (DualLeftHandedWeyl @C[] DualRightHandedWeyl))
(M : SLC(2,)) :
dualLeftDualRightToMatrix (TensorProduct.map (dualLeftHandedRep M) (dualRightHandedRep M) v) =
(M⁻.1¹)ᵀ * dualLeftDualRightToMatrix v * ((M⁻.1¹).conjTranspose)ᵀ
```

lemma Fermion.dualLeftdualLeftToMatrix_ρ_symm

[\[source\]](#)

```
lemma pdualLeftdualLeftToMatrix__symm (v : Matrix (Fin 2) (Fin 2) C) (M : SLC(2,)) :
TensorProduct.map (dualLeftHandedRep M) (dualLeftHandedRep M)
(dualLeftdualLeftToMatrix.symm v) =
dualLeftdualLeftToMatrix.symm ((M⁻.1¹)ᵀ * v * (M⁻.1¹))
```

lemma Fermion.leftDualLeftToMatrix_ρ_symm

[\[source\]](#)

```
lemma pleftDualLeftToMatrix__symm (v : Matrix (Fin 2) (Fin 2) C) (M : SLC(2,)) :
TensorProduct.map (leftHandedRep M) (dualLeftHandedRep M) (leftDualLeftToMatrix.symm v) =
leftDualLeftToMatrix.symm (M.1 * v * (M⁻.1¹))
```

lemma Fermion.dualLeftLeftToMatrix_ρ_symm

[\[source\]](#)

```
lemma pdualLeftLeftToMatrix__symm (v : Matrix (Fin 2) (Fin 2) C) (M : SLC(2,)) :
TensorProduct.map (dualLeftHandedRep M) (leftHandedRep M) (dualLeftLeftToMatrix.symm v) =
dualLeftLeftToMatrix.symm ((M⁻.1¹)ᵀ * v * (M.1)ᵀ)
```

lemma Fermion.dualRightDualRightToMatrix_ρ_symm[\[source\]](#)

```
lemma pdualRightDualRightToMatrix__symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SLℂ(2,)) :
  TensorProduct.map (dualRightHandedRep M) (dualRightHandedRep M)
    (dualRightDualRightToMatrix.symm v) =
  dualRightDualRightToMatrix.symm (((M⁻¹).conjTranspose) * v * ((M⁻¹).conjTranspose)ᵀ)
```

lemma Fermion.rightDualRightToMatrix_ρ_symm[\[source\]](#)

```
lemma prightDualRightToMatrix__symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SLℂ(2,)) :
  TensorProduct.map (rightHandedRep M) (dualRightHandedRep M) (rightDualRightToMatrix.symm v)
  =
  rightDualRightToMatrix.symm ((M.l.map star) * v * (((M⁻¹).conjTranspose)ᵀ))
```

lemma Fermion.dualRightRightToMatrix_ρ_symm[\[source\]](#)

```
lemma pdualRightRightToMatrix__symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SLℂ(2,)) :
  TensorProduct.map (dualRightHandedRep M) (rightHandedRep M) (dualRightRightToMatrix.symm v)
  =
  dualRightRightToMatrix.symm (((M⁻¹).conjTranspose) * v * (M.l.map star)ᵀ)
```

lemma Fermion.dualLeftDualRightToMatrix_ρ_symm[\[source\]](#)

```
lemma pdualLeftDualRightToMatrix__symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SLℂ(2,)) :
  TensorProduct.map (dualLeftHandedRep M) (dualRightHandedRep M)
    (dualLeftDualRightToMatrix.symm v) =
  dualLeftDualRightToMatrix.symm ((M⁻¹)ᵀ * v * ((M⁻¹).conjTranspose)ᵀ)
```

lemma Fermion.dualLeftDualRightToMatrix_ρ_symm_selfAdjoint[\[source\]](#)

```
lemma pdualLeftDualRightToMatrix__symm_selfAdjoint (v : Matrix (Fin 2) (Fin 2) ℂ)
  (hv : IsSelfAdjoint v) (M : SLℂ(2,)) :
  TensorProduct.map (dualLeftHandedRep M) (dualRightHandedRep M)
    (dualLeftDualRightToMatrix.symm v) =
  dualLeftDualRightToMatrix.symm (SL2C.toSelfAdjointMap (M.ᵀ.transpose¹) (v, ) hv)
```

2.46 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Unit

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Unit.lean](#) on GitHub.

def Fermion.leftDualLeftUnitVal[\[source\]](#)

The left-dual-left unit ‘ δ_a^a ’ as an element of ‘ $(\text{leftHanded} \otimes \text{dualLeftHanded}).V$ ’.

```
def leftDualLeftUnitVal : (LeftHandedWeyl ⊗ ℂ[] DualLeftHandedWeyl)
```

lemma Fermion.leftDualLeftUnitVal_expand_tmul[\[source\]](#)

Expansion of ‘leftDualLeftUnitVal’ into the basis.

```
lemma leftDualLeftUnitVal_expand_tmul : leftDualLeftUnitVal =
  leftBasis 0 ⊗_t ℂ[] dualLeftBasis 0 + leftBasis 1 ⊗_t ℂ[] dualLeftBasis 1
```

def Fermion.leftDualLeftUnit[\[source\]](#)

The left-dual-left unit δ_a^a as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} SL(2, \mathbb{C})) \rightarrow \text{leftHanded} \otimes \text{dualLeftHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def leftDualLeftUnit : (Representation.trivial  $\mathbb{C} SL(2, \mathbb{C})$ ).IntertwiningMap
  (leftHandedRep.tprod dualLeftHandedRep) where
```

lemma Fermion.leftDualLeftUnit_apply_one[\[source\]](#)

```
lemma leftDualLeftUnit_apply_one : leftDualLeftUnit (1 :  $\mathbb{C}$ ) = leftDualLeftUnitVal
```

def Fermion.dualLeftLeftUnitVal[\[source\]](#)

The dual-left-left unit δ_a^a as an element of $(\text{dualLeftHanded} \otimes \text{leftHanded}).V$.

```
def dualLeftLeftUnitVal : (DualLeftHandedWeyl  $\otimes$   $\mathbb{C}$ []) LeftHandedWeyl
```

lemma Fermion.dualLeftLeftUnitVal_expand_tmul[\[source\]](#)

Expansion of $\text{dualLeftLeftUnitVal}$ into the basis.

```
lemma dualLeftLeftUnitVal_expand_tmul : dualLeftLeftUnitVal =
  dualLeftBasis 0  $\otimes$   $\mathbb{C}$ [] leftBasis 0 + dualLeftBasis 1  $\otimes$   $\mathbb{C}$ [] leftBasis 1
```

def Fermion.dualLeftLeftUnit[\[source\]](#)

The dual-left-left unit δ_a^a as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} SL(2, \mathbb{C})) \rightarrow \text{dualLeftHanded} \otimes \text{leftHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def dualLeftLeftUnit :
  (Representation.trivial  $\mathbb{C} SL(2, \mathbb{C})$ ).IntertwiningMap
  (dualLeftHandedRep.tprod leftHandedRep) where
```

lemma Fermion.dualLeftLeftUnit_apply_one[\[source\]](#)

Applying the morphism dualLeftLeftUnit to 1 returns $\text{dualLeftLeftUnitVal}$.

```
lemma dualLeftLeftUnit_apply_one : dualLeftLeftUnit (1 :  $\mathbb{C}$ ) = dualLeftLeftUnitVal
```

def Fermion.rightDualRightUnitVal[\[source\]](#)

The right-dual-right unit $\delta^{\dot{a}}_{\dot{a}}$ as an element of $(\text{rightHanded} \otimes \text{dualRightHanded}).V$.

```
def rightDualRightUnitVal : RightHandedWeyl  $\otimes$   $\mathbb{C}$ [] DualRightHandedWeyl
```

lemma Fermion.rightDualRightUnitVal_expand_tmul[\[source\]](#)

Expansion of $\text{rightDualRightUnitVal}$ into the basis.

```
lemma rightDualRightUnitVal_expand_tmul : rightDualRightUnitVal =
  rightBasis 0  $\otimes$   $\mathbb{C}$ [] dualRightBasis 0 + rightBasis 1  $\otimes$   $\mathbb{C}$ [] dualRightBasis 1
```

def Fermion.rightDualRightUnit[\[source\]](#)

The right-dual-right unit $\delta_{\{ \dot{a} \} \{ \dot{a} \}}$ as a morphism $\mathcal{K}_{\text{--}}(\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{rightHanded} \otimes \text{dualRightHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def rightDualRightUnit : (Representation.trivial  $\mathbb{C}$  SLC(2,)  $\mathbb{C}$ ).IntertwiningMap
  (rightHandedRep.tprod dualRightHandedRep) where
```

lemma Fermion.rightDualRightUnit_apply_one[\[source\]](#)

```
lemma rightDualRightUnit_apply_one : rightDualRightUnit (1 :  $\mathbb{C}$ ) = rightDualRightUnitVal
```

def Fermion.dualRightRightUnitVal[\[source\]](#)

The dual-right-right unit $\delta_{\{ \dot{a} \} \{ \dot{a} \}}$ as an element of $(\text{rightHanded} \otimes \text{dualRightHanded}).V$.

```
def dualRightRightUnitVal : (DualRightHandedWeyl  $\otimes$   $\mathbb{C}$ []) RightHandedWeyl)
```

lemma Fermion.dualRightRightUnitVal_expand_tmul[\[source\]](#)

Expansion of $\text{dualRightRightUnitVal}$ into the basis.

```
lemma dualRightRightUnitVal_expand_tmul : dualRightRightUnitVal =
  dualRightBasis 0  $\otimes$   $\mathbb{C}$ [] rightBasis 0 + dualRightBasis 1  $\otimes$   $\mathbb{C}$ [] rightBasis 1
```

def Fermion.dualRightRightUnit[\[source\]](#)

The dual-right-right unit $\delta_{\{ \dot{a} \} \{ \dot{a} \}}$ as a morphism $\mathcal{K}_{\text{--}}(\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{dualRightHanded} \otimes \text{rightHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def dualRightRightUnit : (Representation.trivial  $\mathbb{C}$  SLC(2,)  $\mathbb{C}$ ).IntertwiningMap
  (dualRightHandedRep.tprod rightHandedRep) where
```

lemma Fermion.dualRightRightUnit_apply_one[\[source\]](#)

```
lemma dualRightRightUnit_apply_one : dualRightRightUnit (1 :  $\mathbb{C}$ ) = dualRightRightUnitVal
```

lemma Fermion.contr_dualLeftLeftUnit[\[source\]](#)

Contraction on the right with dualLeftLeftUnit does nothing.

```
lemma contr_dualLeftLeftUnit (x : LeftHandedWeyl) :
  (TensorProduct.lid  $\mathbb{C}$  _ <|
   leftDualContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc  $\mathbb{C}$  _ _).symm <|
   x  $\otimes$   $\mathbb{C}$ []) (dualLeftLeftUnit (1 :  $\mathbb{C}$ )) = x
```

lemma Fermion.contr_leftDualLeftUnit[\[source\]](#)

Contraction on the right with leftDualLeftUnit does nothing.

```

lemma contr_leftDualLeftUnit (x : DualLeftHandedWeyl) :
  (TensorProduct.lid C _ <|
   dualLeftContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc C _ _).symm <|
   x ⊗ C[] (leftDualLeftUnit (1 : C))) = x

```

lemma Fermion.contr_dualRightRightUnit

[\[source\]](#)

Contraction on the right with ‘dualRightRightUnit’ does nothing.

```

lemma contr_dualRightRightUnit (x : RightHandedWeyl) :
  (TensorProduct.lid C _ <|
   rightDualContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc C _ _).symm <|
   x ⊗ C[] (dualRightRightUnit (1 : C))) = x

```

lemma Fermion.contr_rightDualRightUnit

[\[source\]](#)

Contraction on the right with ‘rightDualRightUnit’ does nothing.

```

lemma contr_rightDualRightUnit (x : DualRightHandedWeyl) :
  (TensorProduct.lid C _ <|
   dualRightContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc C _ _).symm <|
   x ⊗ C[] (rightDualRightUnit (1 : C))) = x

```

lemma Fermion.dualLeftLeftUnit_symm

[\[source\]](#)

```

lemma dualLeftLeftUnit_symm :
  dualLeftLeftUnit (1 : C) = LinearMap.lTensor _ (LinearEquiv.refl _ _).toLinearMap
  (TensorProduct.comm C _ _ (leftDualLeftUnit (1 : C)))

```

lemma Fermion.leftDualLeftUnit_symm

[\[source\]](#)

```

lemma leftDualLeftUnit_symm :
  leftDualLeftUnit (1 : C) = LinearMap.lTensor _ (LinearEquiv.refl _ _).toLinearMap
  (TensorProduct.comm C _ _ (dualLeftLeftUnit (1 : C)))

```

lemma Fermion.dualRightRightUnit_symm

[\[source\]](#)

```

lemma dualRightRightUnit_symm :
  dualRightRightUnit (1 : C) = LinearMap.lTensor _ (LinearEquiv.refl _ _).toLinearMap
  (TensorProduct.comm C _ _ (rightDualRightUnit (1 : C)))

```

lemma Fermion.rightDualRightUnit_symm

[\[source\]](#)

```

lemma rightDualRightUnit_symm :
  rightDualRightUnit (1 : C) = LinearMap.lTensor _ (LinearEquiv.refl _ _).toLinearMap
  (TensorProduct.comm C _ _ (dualRightRightUnit (1 : C)))

```

2.47 Physlib.Relativity.Tensors.ComponentIdx.Basic

[Physlib/Relativity/Tensors/ComponentIdx/Basic.lean](#) on GitHub.

abbrev TensorSpecies.Tensor.ComponentIdx[\[source\]](#)

Given a list of indices $c : \text{Fin } n \rightarrow C$, the type $\text{ComponentIdx } c$ is the type of component indices of a tensor with those colors. For instance, if $c = ![\text{.up}, \text{.down}]$, then an element of $\text{ComponentIdx } c$ specifies one basis index for .up and one for .down .

```
@[nolint unusedArguments]
abbrev ComponentIdx {n : ℕ} {S : TensorSpecies k C G V basisIdx rep b}
  (c : Fin n → C) : Type
```

lemma TensorSpecies.Tensor.ComponentIdx.congr_right[\[source\]](#)

```
lemma ComponentIdx.congr_right {n : ℕ} {c : Fin n → C} (b : ComponentIdx (S := S) c)
  (i j : Fin n) (h : i = j) : b i = basisIdxCongr (by simp [h]) (b j)
```

def TensorSpecies.Tensor.ComponentIdx.cast[\[source\]](#)

Casting of a ComponentIdx through equivalent color maps.

```
def ComponentIdx.cast {n m : ℕ} {c : Fin n → C} {cm : Fin m → C}
  (h : n = m) (hc : c = cm ∘ Fin.cast h) (b : ComponentIdx (S := S) c) :
  ComponentIdx (S := S) cm
```

2.48 Physlib.Relativity.Tensors.ComponentIdx.Contraction

[Physlib/Relativity/Tensors/ComponentIdx/Contraction.lean](#) on GitHub.

def TensorSpecies.Tensor.ComponentIdx.dropPair[\[source\]](#)

The ComponentIdx obtained by dropping two components.

```
def dropPair {n : ℕ} {c : Fin (n + 1 + 1) → C}
  (i j : Fin (n + 1 + 1)) (b : ComponentIdx (S := S) c) :
  ComponentIdx (S := S) (c ∘ Fin.succSuccAbove i j)
```

def TensorSpecies.Tensor.ComponentIdx.DropPairSection[\[source\]](#)

Given a coordinate parameter $b : \prod k, \text{Fin } (S.\text{repDim } ((c \circ i.\text{succAbove} \circ j.\text{succAbove}) k))$, the coordinate parameter $\prod k, \text{Fin } (S.\text{repDim } (c k))$ which map down to b .

```
def DropPairSection {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i : Fin (n + 1 + 1)} {j : Fin (n + 1 + 1)}
  (b : ComponentIdx (S := S) (c ∘ Fin.succSuccAbove i j)) :
  Finset (ComponentIdx (S := S) c)
```

lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.mem_iff_apply_succSuccAbove_eq[\[source\]](#)

```
lemma mem_iff_apply_succSuccAbove_eq {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)}
  (b : ComponentIdx (c ∘ Fin.succSuccAbove i j))
  (b' : ComponentIdx c) :
  b' ∈ DropPairSection (S := S) b ↔
  ∀ m, b' (Fin.succSuccAbove i j m) = b m
```


lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.mem_self_of_dropPair [\[source\]](#)

```
@[simp]
lemma mem_self_of_dropPair {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)}
  (b : ComponentIdx (c)) :
  b ∈ DropPairSection (S := S) (b.dropPair i j)
```

def TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFin [\[source\]](#)

Given a ‘b’ in ‘ComponentIdx (c ∘ Fin.succSuccAbove i j)’ and an ‘x’ in ‘Fin (S.repDim (c i)) × Fin (S.repDim (c j))’, the corresponding coordinate parameter in ‘ComponentIdx c’.

```
def ofFin {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j) (b : ComponentIdx (S := S) (c ∘ Fin.succSuccAbove i j))
  (x : basisIdx (c i) × basisIdx (c j)) :
  ComponentIdx (S := S) c
```

lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFin_apply_fst [\[source\]](#)

```
@[simp]
lemma ofFin_apply_fst {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j) (b : ComponentIdx (c ∘ Fin.succSuccAbove i j))
  (x : basisIdx (c i) × basisIdx (c j)) :
  ofFin (S := S) hij b x i = x.1
```

lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFin_apply_snd [\[source\]](#)

```
@[simp]
lemma ofFin_apply_snd {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j) (b : ComponentIdx (c ∘ Fin.succSuccAbove i j))
  (x : basisIdx (c i) × basisIdx (c j)) :
  ofFin (S := S) hij b x j = x.2
```

lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFin_mem_succSuccAboveSection [\[source\]](#)

```
lemma ofFin_mem_succSuccAboveSection {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j) (b : ComponentIdx (c ∘ Fin.succSuccAbove i j))
  (x : basisIdx (c i) × basisIdx (c j)) :
  ofFin (S := S) hij b x ∈ DropPairSection b
```

def TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFinEquiv [\[source\]](#)

The equivalence between ‘ContrSection b’ and ‘basisIdx (c i) × basisIdx (c j)’.

```
def ofFinEquiv {n : ℕ} {c : Fin n.succ.succ → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j)
  (b : ComponentIdx (c ∘ Fin.succSuccAbove i j)) :
  basisIdx (c i) × basisIdx (c j) ≈ DropPairSection (S := S) b where
```

lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFinEquiv_apply_fst [\[source\]](#)

```
@[simp]
lemma ofFinEquiv_apply_fst {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j) (b : ComponentIdx (c ∘ Fin.succSuccAbove i j))
  (x : basisIdx (c i) × basisIdx (c j)) :
  (ofFinEquiv (S := S) hij b x).1 i = x.1
```

lemma TensorSpecies.Tensor.ComponentIdx.DropPairSection.ofFinEquiv_apply_snd [\[source\]](#)

```
@[simp]
lemma ofFinEquiv_apply_snd {n : ℕ} {c : Fin (n + 1 + 1) → C}
  {i j : Fin (n + 1 + 1)} (hij : i ≠ j) (b : ComponentIdx (c ∘ Fin.succSuccAbove i j))
  (x : basisIdx (c i) × basisIdx (c j)) :
  (ofFinEquiv (S := S) hij b x).1 j = x.2
```

2.49 Physlib.Relativity.Tensors.ComponentIdx.Product

[Physlib/Relativity/Tensors/ComponentIdx/Product.lean](#) on GitHub.

def TensorSpecies.Tensor.ComponentIdx.prod [\[source\]](#)

The equivalence between ‘ComponentIdx (Fin.append c c1)’ and ‘ComponentIdx c × ComponentIdx c1’ formed by products.

```
def ComponentIdx.prod {n1 n2 : ℕ} {c : Fin n1 → C} {c1 : Fin n2 → C} :
  ComponentIdx (S := S) (Fin.append c c1) ≈
  ComponentIdx (S := S) c × ComponentIdx (S := S) c1 where
```

lemma TensorSpecies.Tensor.ComponentIdx.prod_symm_natAdd [\[source\]](#)

```
@[simp]
lemma ComponentIdx.prod_symm_natAdd {n1 n2 : ℕ} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p : ComponentIdx (S := S) c) (q : ComponentIdx (S := S) c1) (i : Fin n2) :
  ComponentIdx.prod.symm (p, q) (Fin.natAdd n1 i) =
  basisIdxCongr (by simp) (q i)
```

lemma TensorSpecies.Tensor.ComponentIdx.prod_symm_castAdd [\[source\]](#)

```
@[simp]
lemma ComponentIdx.prod_symm_castAdd {n1 n2 : ℕ} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p : ComponentIdx (S := S) c) (q : ComponentIdx (S := S) c1) (i : Fin n1) :
  ComponentIdx.prod.symm (p, q) (Fin.castAdd n2 i) =
  basisIdxCongr (by simp) (p i)
```

2.50 Physlib.Relativity.Tensors.ComponentIdx.Single

[Physlib/Relativity/Tensors/ComponentIdx/Single.lean](#) on GitHub.

def TensorSpecies.Tensor.ComponentIdx.single [\[source\]](#)

The equivalence between component indices for a single color and the basis indices of that color.

```
def ComponentIdx.single {c : C} :
  ComponentIdx (S := S) ![c] ≈ basisIdx c where
```

lemma TensorSpecies.Tensor.ComponentIdx.single_apply

[\[source\]](#)

```
@[simp]
lemma ComponentIdx.single_apply {c : C} (b : ComponentIdx (S := S) ![c]) :
  ComponentIdx.single (S := S) b = basisIdxCongr (by simp) (b 0)
```

lemma TensorSpecies.Tensor.ComponentIdx.single_symm_apply

[\[source\]](#)

```
@[simp]
lemma ComponentIdx.single_symm_apply {c : C} (b : basisIdx c) (i : Fin 1) :
  (ComponentIdx.single (S := S) (c := c)).symm b i = basisIdxCongr (by simp) b
```

2.51 Physlib.Relativity.Tensors.Conjugation.Basic

[Physlib/Relativity/Tensors/Conjugation/Basic.lean](#) on GitHub.

structure ConjTensorSpecies

[\[source\]](#)

A tensor species bundled with a conjugation. Beyond the ‘TensorSpecies’ it ‘extends’, it carries the conjugate-colour involution ‘bar’, the index-set identification ‘barIdx_eq’, and the contraction-coherence ‘conj_contrComm’. Also carries ‘bar_involution’ and ‘bar_tau’ (that ‘bar’ is involutive and commutes with the variance dual).

```
structure ConjTensorSpecies (k : Type) [CommRing k] [StarRing k] (C : Type) (G : Type) [Group G]
]
(V : C → Type) ∀[ c, AddCommGroup (V c)] ∀[ c, Module k (V c)]
(basisIdx : C → Type) ∀[ c, Fintype (basisIdx c)] ∀[ c, DecidableEq (basisIdx c)]
(rep : (c : C) → Representation k G (V c)) (b : (c : C) → Basis (basisIdx c) k (V c))
extends TensorSpecies k C G V basisIdx rep b where
bar : C → C
bar_involution : Function.Involutive bar
bar_tau : ∀ c, bar (toTensorSpeciesT. c) = toTensorSpeciesT. (bar c)
barIdx_eq : ∀ c, basisIdx (bar c) = basisIdx c
conj_contrComm : ∀ (d : C) (x : basisIdx d) (y : basisIdx (toTensorSpeciesT. d)),
  star (toTensorSpecies.contr d (b d x ⊗k b (toTensorSpeciesT. d) y))
    = toTensorSpecies.contr (bar d) (b (bar d) ((Equiv.cast (barIdx_eq d)).symm x) ⊗k
      b (toTensorSpeciesT. (bar d)) (basisIdxCongr (bar_tau d)
        ((Equiv.cast (barIdx_eq (toTensorSpeciesT. d)).symm y))))
```

def ConjTensorSpecies.componentReindex

[\[source\]](#)

Reindex component labels of ‘bar ∘ c’ back to ‘c’, slotwise via the cast ‘barIdx_eq’.

```
def componentReindex {n : ℕ} (c : Fin n → C) :
  ComponentIdx (S := S.toTensorSpecies) (S.bar ∘ c) ≈ ComponentIdx (S := S.toTensorSpecies) c
```

def ConjTensorSpecies.conjT

[\[source\]](#)

Conjugation of a tensor: conjugate the components and reindex the basis to the conjugate colours. ‘conj’-semilinear by construction (see ‘conjT_smul’).

```
def conjT {n : ℕ} {c : Fin n → C} (t : S.Tensor c) : S.Tensor (S.bar ∘ c)
```

lemma ConjTensorSpecies.conjT_eq_permT_iff

[\[source\]](#)

Componentwise criterion for ‘ $\text{conjT } t = \text{permT } \sigma \text{ h } t'$ ’. The conjugate of ‘ t ’ equals the recolouring ‘ $\text{permT } \sigma \text{ h } t'$ ’ exactly when, at every component, the ‘star’-conjugated reindexed component of ‘ t ’ matches the corresponding component of ‘ $\text{permT } \sigma \text{ h } t'$ ’. This packages the ‘ $\text{componentMap_conjT}$ ’ expansion and the ‘ repr ’/‘ componentMap ’ bridge that the reality and Hermiticity proofs downstream would otherwise repeat by hand; the caller is left only with the species-specific permutation bookkeeping on the right-hand side.

```
lemma conjT_eq_permT_iff {n m : ℕ} {c : Fin n → C} {c' : Fin m → C}
σ { : Fin n → Fin m} (h : IsReindexing c' (S.bar ∘ c) σ)
(t : S.Tensor c) (t' : S.Tensor c') :
S.conjT t = permT σ h t' ↔
  ∀ φ : ComponentIdx (S := S.toTensorSpecies) (S.bar ∘ c),
    star (componentMap c t (S.componentReindex c φ))
      = (Tensor.basis (S.bar ∘ c)).repr (permT σ h t') φ
```

lemma ConjTensorSpecies.barIdx_involutive_symm

[\[source\]](#)

Undoing the two cast reindexings (at ‘ $\text{bar } c$ ’ then at ‘ c ’) on a label of the doubly-conjugated colour returns the ‘ bar_involution ’ cast. Free from ‘ barIdx_eq ’ by ‘ cast_cast ’ + proof irrelevance, with no coherence field needed.

```
private lemma barIdx_involutive_symm (c : C) (y : basisIdx (S.bar (S.bar c))) :
Equiv.cast (S.barIdx_eq c) (Equiv.cast (S.barIdx_eq (S.bar c)) y)
= basisIdxCongr (S.bar_involution c) y
```

lemma ConjTensorSpecies.isReindexing_bar_bar

[\[source\]](#)

The identity permutation satisfies ‘ IsReindexing ’ from ‘ c ’ to ‘ $\text{bar} \circ \text{bar} \circ c$ ’, as ‘ bar ’ is an involution.

```
lemma isReindexing_bar_bar {n : ℕ} (c : Fin n → C) :
IsReindexing c (S.bar ∘ S.bar ∘ c) (id : Fin n → Fin n)
```

lemma ConjTensorSpecies.conjT_conjT

[\[source\]](#)

Conjugation is an involution: conjugating twice returns the original tensor, up to the ‘ bar_involution ’ recolouring (the identity permutation ‘ permT ’).

```
lemma conjT_conjT {n : ℕ} {c : Fin n → C} (t : S.Tensor c) :
S.conjT (S.conjT t)
= permT id (S.isReindexing_bar_bar c) t
```

lemma ConjTensorSpecies.contrCond_bar

[\[source\]](#)

Slots with dual colour in ‘ c ’ have dual colour in ‘ $\text{bar} \circ c$ ’, as ‘ bar ’ commutes with ‘ τ ’.

```
lemma contrCond_bar {n : ℕ} {c : Fin (n + 1 + 1) → C} {i j : Fin (n + 1 + 1)}
(h : i ≠ j ∧ S.τ. (c i) = c j) :
i ≠ j ∧ S.τ. ((S.bar ∘ c) i) = (S.bar ∘ c) j
```

lemma ConjTensorSpecies.conjT_contrT[\[source\]](#)

Conjugation commutes with contraction: conjugating a contracted tensor equals contracting the conjugate on the ‘bar’-images of the same two slots.

```
lemma conjT_contrT {n : ℕ} {c : Fin (n + 1 + 1) → C} (i j : Fin (n + 1 + 1))
  (h : i ≠ j ∧ Sτ. (c i) = c j) (t : S.Tensor c) :
  S.conjT (contrT n i j h t)
    = contrT n i j (S.contrCond_bar h) (S.conjT t)
```

def ConjTensorSpecies.conjEquiv[\[source\]](#)

The conjugate-linear slot isomorphism ‘ $V c \simeq_{sl} [starRingEnd\ k] V (\bar{c})$ ’: conjugate the coordinates in the species basis and relabel to the conjugate colour via ‘barIdx_eq’.

```
def conjEquiv {c : C} : V c ≈sl [starRingEnd k] V (S.bar c)
```

def ConjTensorSpecies.IsHermitian[\[source\]](#)

A pairing ‘ $g : V c \otimes V (\bar{c}) \rightarrow k$ ’ is Hermitian when conjugating and swapping its two slots via ‘conjEquiv’ returns its ‘star’: ‘ $g (x \otimes y) = star (g (conjEquiv.symm\ y \otimes conjEquiv\ x))$ ’.

```
def IsHermitian {c : C} (g : V c ⊗[k] V (S.bar c) →[k] k) : Prop
```

2.52 Physlib.Relativity.Tensors.Constructors

[Physlib/Relativity/Tensors/Constructors.lean](#) on GitHub.

lemma TensorSpecies.Tensor.fromPairT_eq_pure[\[source\]](#)

```
lemma fromPairT_eq_pure {c1 c2 : C} (x : V c1) (y : V c2) :
  fromPairT (S := S) (x ⊗[k] y) = Pure.toTensor (fun | 0 => x | 1 => y)
```

2.53 Physlib.Relativity.Tensors.Contraction.SuccSuccAbove

[Physlib/Relativity/Tensors/Contraction/SuccSuccAbove.lean](#) on GitHub.

lemma Fin.apply_succSuccAbove_symm[\[source\]](#)

```
@[simp, nolint simpVarHead]
lemma apply_succSuccAbove_symm {c : Fin (n + 1 + 1) → C} (i j : Fin (n + 1 + 1))
  (k : Fin n) : c (succSuccAbove i j k) = c (succSuccAbove j i k)
```

2.54 Physlib.Relativity.Tensors.Evaluation

[Physlib/Relativity/Tensors/Evaluation.lean](#) on GitHub.

lemma TensorSpecies.Tensor.evalT_basis_single[\[source\]](#)

Evaluating the single-index basis tensor ‘basis ![c] (single.symm b)’ at the index ‘x’ yields the field element ‘1’ if ‘b = x’ (transported across ‘![c] 0 = c’) and ‘0’ otherwise: evaluation of a one-index basis tensor is the Kronecker delta.

```

lemma evalT_basis_single {c : C} (b : basisIdx c) (x : basisIdx (![c] 0)) :
  (evalT 0 x (basis (S := S) ![c] (ComponentIdx.single.symm b))).toField =
  if basisIdxCongr (by simp) b = x then 1 else 0

```

lemma TensorSpecies.Tensor.eq_sum_evalT_of_single_tensor_basis

[source]

Basis expansion of a one-index tensor: every $t : \text{Tensor } S \text{ } ![c]$ is the sum over basis indices i of its evaluation coefficient $\text{toField } (\text{evalT } 0 \ i \ t)$ times the corresponding basis tensor.

```

lemma eq_sum_evalT_of_single_tensor_basis {c : C} (t : Tensor S ![c]) :
  t =  $\sum i$ , toField (evalT 0 i t) • basis ![c] (ComponentIdx.single.symm
    (basisIdxCongr (by simp) i))

```

lemma TensorSpecies.Tensor.eq_sum_evalT

[source]

Reconstruction of a tensor from the evaluations of its last index: every $t : \text{Tensor } S \ c$ is the sum over basis indices i of the evaluation $\text{evalT } (\text{Fin.last } n) \ i \ t$ tensored with the basis covector $\text{basis } ![c \ (\text{Fin.last } n)] \ (\text{single.symm } i)$, with the appended index permuted back into the last slot.

```

lemma eq_sum_evalT {n : ℕ} {c : Fin (n + 1) → C} (t : Tensor S c) :
  t =  $\sum i$ , permT id (IsReindexing.append_succ_last c) (prodT (evalT (Fin.last n) i t)
    (basis ![c \ (\text{Fin.last } n)] (ComponentIdx.single.symm i)))

```

lemma TensorSpecies.Tensor.eq_sum_evalT_zero

[source]

Reconstruction of a tensor from the evaluations of its first index: every $t : \text{Tensor } S \ c$ is the sum over basis indices i of the basis covector $\text{basis } ![c \ 0] \ (\text{single.symm } i)$ tensored with the evaluation $\text{evalT } 0 \ i \ t$, with the prepended index permuted back into the first slot. This is the first-index analogue of eq_sum_evalT .

```

lemma eq_sum_evalT_zero {n : ℕ} {c : Fin (n + 1) → C} (t : Tensor S c) :
  t =  $\sum i$ , permT _ (IsReindexing.append_of_first c)
    (prodT (basis ![c \ 0] (ComponentIdx.single.symm i)) (evalT 0 i t))

```

2.55 Physlib.Relativity.Tensors.LeviCivita.Basic

Physlib/Relativity/Tensors/LeviCivita/Basic.lean on GitHub.

def realLorentzTensor.leviCivita

[source]

The Levi-Civita tensor $\epsilon^{uv\delta}$ as a real Lorentz tensor in $d = 3$, with $\epsilon^{0123} = 1$.

The component on a multi-index f is the generalized Kronecker delta of f against the identity, i.e. the sign of f when f is a permutation and 0 otherwise.

```

noncomputable def leviCivita : ℝT[3, .up, .up, .up, .up]

```

lemma realLorentzTensor.leviCivita_eq_ofInt

[source]

The $\text{TensorInt.toTensor}$ form of the Levi-Civita tensor.

```

lemma leviCivita_eq_ofInt :  $\varepsilon_4 =$ 
  TensorInt.toTensor (S := realLorentzTensor 3)
  (c := ![Color.up, Color.up, Color.up, Color.up]) fun f =>
  generalizedKroneckerDelta (fun i => finSumFinEquiv (f i)) (id : Fin 4 → Fin 4)

```

lemma realLorentzTensor.leviCivita_basis_repr_apply

[\[source\]](#)

The components of the Levi-Civita tensor in the standard basis are the generalized Kronecker delta of the multi-index against the identity.

```

lemma leviCivita_basis_repr_apply
  (b : ComponentIdx (S := realLorentzTensor 3) ![Color.up, Color.up, Color.up, Color.up]) :
  (Tensor.basis _).repr  $\varepsilon_4$  b
  = (generalizedKroneckerDelta (fun i => finSumFinEquiv (b i)) (id : Fin 4 → Fin 4) :  $\mathbb{R}$ )

```

lemma realLorentzTensor.leviCivita_basis_repr_eq_zero_of_eq

[\[source\]](#)

The Levi-Civita tensor vanishes on any multi-index with a repeated value: if two distinct index positions ' $i \neq j$ ' carry the same basis index, the component is zero.

```

lemma leviCivita_basis_repr_eq_zero_of_eq
  {b : ComponentIdx (S := realLorentzTensor 3) ![Color.up, Color.up, Color.up, Color.up]}
  {i j : Fin 4} (hij : i ≠ j) (h : b i = b j) :
  (Tensor.basis _).repr  $\varepsilon_4$  b = 0

```

lemma realLorentzTensor.leviCivita_antisymm

[\[source\]](#)

The Levi-Civita tensor is antisymmetric in its first two indices ' $\{\varepsilon_4 \mid \mu \nu \rho \sigma = -\varepsilon_4 \mid \nu \mu \rho \sigma\}^T$ '.

```

lemma leviCivita_antisymm :  $\varepsilon\{4 \mid \mu \nu \rho \sigma = -\varepsilon(4 \mid \nu \mu \rho \sigma)^T\}$ 

```

lemma realLorentzTensor.leviCivita_antisymm_mid

[\[source\]](#)

The Levi-Civita tensor is antisymmetric in its middle two indices ' $\{\varepsilon_4 \mid \mu \nu \rho \sigma = -\varepsilon_4 \mid \mu \rho \nu \sigma\}^T$ '.

```

lemma leviCivita_antisymm_mid :  $\varepsilon\{4 \mid \mu \nu \rho \sigma = -\varepsilon(4 \mid \mu \rho \nu \sigma)^T\}$ 

```

lemma realLorentzTensor.leviCivita_antisymm_last

[\[source\]](#)

The Levi-Civita tensor is antisymmetric in its last two indices ' $\{\varepsilon_4 \mid \mu \nu \rho \sigma = -\varepsilon_4 \mid \mu \nu \sigma \rho\}^T$ '.

```

lemma leviCivita_antisymm_last :  $\varepsilon\{4 \mid \mu \nu \rho \sigma = -\varepsilon(4 \mid \mu \nu \sigma \rho)^T\}$ 

```

2.56 Physlib.Relativity.Tensors.Product

[Physlib/Relativity/Tensors/Product.lean](#) on GitHub.

lemma TensorSpecies.Tensor.Pure.prodP_update_left

[\[source\]](#)

@[simp]

```
lemma Pure.prodP_update_left {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p1 : Pure S c) (p2 : Pure S c1) (i : Fin n1) (x : V (c i)) :
  Pure.prodP (Pure.update p1 i x) p2
= Pure.update (Pure.prodP p1 p2) (Fin.castAdd n2 i)
  (LinearEquiv.cast (R := k) (by simp) x)
```

lemma TensorSpecies.Tensor.Pure.prodP_update_right

[\[source\]](#)

@[simp]

```
lemma Pure.prodP_update_right {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p1 : Pure S c) (p2 : Pure S c1) (i : Fin n2) (x : V (c1 i)) :
  Pure.prodP p1 (Pure.update p2 i x)
= Pure.update (Pure.prodP p1 p2) (Fin.natAdd n1 i)
  (LinearEquiv.cast (R := k) (by simp) x)
```

lemma TensorSpecies.Tensor.Pure.prodP_update_add_toTensor_left

[\[source\]](#)

```
lemma Pure.prodP_update_add_toTensor_left {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p1 : Pure S c) (p2 : Pure S c1) (i : Fin n2) (x y : V (c1 i)) :
  (Pure.prodP p1 (Pure.update p2 i (x + y))).toTensor
= (Pure.prodP p1 (Pure.update p2 i x)).toTensor
+ (Pure.prodP p1 (Pure.update p2 i y)).toTensor
```

lemma TensorSpecies.Tensor.prodT_basis'

[\[source\]](#)

```
lemma prodT_basis' {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (b : ComponentIdx c) (b1 : ComponentIdx (S := S) c1) :
  (basis c b).prodT (basis c1 b1) =
  basis (Fin.append c c1) (ComponentIdx.prod.symm (b, b1))
```

lemma TensorSpecies.Tensor.prodT_zero_right

[\[source\]](#)

```
lemma prodT_zero_right {n} {c : Fin n → C}
  {c1 : Fin 0 → C} (t : S.Tensor c) (t1 : S.Tensor c1) :
  prodT t t1 = (toField t1) • permT id (IsReindexing.append_zero_right) t
```

2.57 Physlib.Relativity.Tensors.RealTensor.Basic

[Physlib/Relativity/Tensors/RealTensor/Basic.lean](#) on GitHub.

abbrev realLorentzTensor.modules

[\[source\]](#)

The modules associated with each of the different types of real Lorentz vector space.

```
abbrev modules (d : ℕ) : Color → Type
| Color.up => Lorentz.ContrMod d
| Color.down => Lorentz.CoMod d
```

instance realLorentzTensor.modulesAddCommGroup

[\[source\]](#)

```
instance modulesAddCommGroup (d : ℕ) : ∀ c, AddCommGroup (modules d c)
| Color.up => inferInstance
```



```
| Color.down => inferInstance
```

instance realLorentzTensor.modulesModule

[\[source\]](#)

```
instance modulesModule (d : ℕ) : ∀ c, Module ℝ (modules d c)
| Color.up => inferInstance
| Color.down => inferInstance
```

2.58 Physlib.Relativity.Tensors.RealTensor.CoVector.Basic

[Physlib/Relativity/Tensors/RealTensor/CoVector/Basic.lean](#) on GitHub.

lemma Lorentz.CoVector.eq_of_apply_eq

[\[source\]](#)

```
@[ext]
lemma eq_of_apply_eq {d : ℕ} {v w : CoVector d} (h : ∀ i, v i = w i) : v = w
```

lemma Lorentz.CoVector.apply_sum

[\[source\]](#)

```
@[simp]
lemma apply_sum {d : ℕ} ι{ : Type} [Fintype ι] (f : ι → CoVector d) (i : Fin 1 ⊕ Fin d) :
  Σ( j, f j) i = Σ j, f j i
```

2.59 Physlib.Relativity.Tensors.RealTensor.CoVector.Representation

[Physlib/Relativity/Tensors/RealTensor/CoVector/Representation.lean](#) on GitHub.

def Lorentz.CoVector.rep

[\[source\]](#)

The representation of the Lorentz group on Lorentz vectors.

```
def rep {d : ℕ} : Representation ℝ (LorentzGroup d) (CoVector d) where
```

lemma Lorentz.CoVector.rep_apply_eq_mulVec

[\[source\]](#)

```
lemma rep_apply_eq_mulVec (d : ℕ) Λ( : LorentzGroup d) (v : CoVector d) :
  rep Λ v = (LorentzGroup.transpose Λ-1) v * v
```

lemma Lorentz.CoVector.rep_apply_eq_sum

[\[source\]](#)

```
lemma rep_apply_eq_sum (d : ℕ) Λ( : LorentzGroup d) (v : CoVector d) (k : Fin 1 ⊕ Fin d) :
  rep Λ v k = Σ j, Λ-1.1 j k • v j
```

lemma Lorentz.CoVector.rep_apply_eq_sum_coe

[\[source\]](#)

```
lemma rep_apply_eq_sum_coe (d : ℕ) Λ( : LorentzGroup d) (v : CoVector d) (k : Fin 1 ⊕ Fin d) :
  rep Λ v k = Σ j, Λ-1.11 j k • v j
```

lemma Lorentz.CoVector.rep_apply_basis[\[source\]](#)

```
lemma rep_apply_basis {d} μ( : Fin 1 ⊕ Fin d) Λ( : LorentzGroup d) :
  rep Λ (basis μ) = ∑ j, Λ⁻¹ j μ j • basis j
```

lemma Lorentz.CoVector.rep_toMatrix[\[source\]](#)

```
lemma rep_toMatrix (d : ℕ) Λ( : LorentzGroup d) :
  LinearMap.toMatrix basis basis (rep Λ) = Λ⁻¹ᵀ
```

lemma Lorentz.CoVector.rep_injective[\[source\]](#)

```
lemma rep_injective (d : ℕ) Λ( : LorentzGroup d) : Function.Injective (rep Λ)
```

lemma Lorentz.CoVector.rep_surjective[\[source\]](#)

```
lemma rep_surjective (d : ℕ) Λ( : LorentzGroup d) : Function.Surjective (rep Λ)
```

lemma Lorentz.CoVector.rep_bijective[\[source\]](#)

```
lemma rep_bijective (d : ℕ) Λ( : LorentzGroup d) : Function.Bijective (rep Λ)
```

2.60 Physlib.Relativity.Tensors.RealTensor.CoVector.Tensorial

[Physlib/Relativity/Tensors/RealTensor/CoVector/Tensorial.lean](#) on GitHub.

def Lorentz.CoVector.indexEquiv[\[source\]](#)

The equivalence between the type of indices of a Lorentz vector and ‘Fin 1 ⊕ Fin d’.

```
def indexEquiv {d : ℕ} :
  ComponentIdx (S := (realLorentzTensor d)) ![Color.down] ≈ Fin 1 ⊕ Fin d
```

instance Lorentz.CoVector.tensorial[\[source\]](#)

```
instance tensorial {d : ℕ} : Tensorial (realLorentzTensor d) ! [.down] (CoVector d) where
```

lemma Lorentz.CoVector.toTensor_symm_apply[\[source\]](#)

```
lemma toTensor_symm_apply {d : ℕ} (p : ℝT[d, .down]) :
  (toTensor (self := tensorial)).symm p =
  (Equiv.piCongrLeft' _ indexEquiv <|
  Finsupp.equivFunOnFinite <|
  (Tensor.basis (S := (realLorentzTensor d)) _).repr p)
```

lemma Lorentz.CoVector.toTensor_symm_pure[\[source\]](#)

```
lemma toTensor_symm_pure {d : ℕ} (p : Pure (realLorentzTensor d) ! [.down]) (i : Fin 1 ⊕ Fin d) :
  (toTensor (self := tensorial)).symm p.toTensor i =
  ((Lorentz.coBasis d).repr (p 0)) (indexEquiv.symm i 0)
```

lemma Lorentz.CoVector.toTensor_symm_basis[\[source\]](#)

```
lemma toTensor_symm_basis {d : ℕ} μ( : Fin 1 ⊗ Fin d) :
  (toTensor (self := tensorial)).symm (Tensor.basis ![Color.down] (indexEquiv.symm μ)) =
  basis μ
```

lemma Lorentz.CoVector.toTensor_basis_eq_tensor_basis[\[source\]](#)

```
lemma toTensor_basis_eq_tensor_basis {d : ℕ} μ( : Fin 1 ⊗ Fin d) :
  toTensor (basis μ) = Tensor.basis ![Color.down] (indexEquiv.symm μ)
```

lemma Lorentz.CoVector.basis_eq_map_tensor_basis[\[source\]](#)

```
lemma basis_eq_map_tensor_basis {d} : basis =
  ((Tensor.basis (S := realLorentzTensor d) ![Color.down])).map
  toTensor.symm).reindex indexEquiv
```

lemma Lorentz.CoVector.tensor_basis_map_eq_basis_reindex[\[source\]](#)

```
lemma tensor_basis_map_eq_basis_reindex {d} :
  (Tensor.basis (S := realLorentzTensor d) ![Color.down]).map toTensor.symm =
  basis.reindex indexEquiv.symm
```

lemma Lorentz.CoVector.tensor_basis_repr_toTensor_apply[\[source\]](#)

```
lemma tensor_basis_repr_toTensor_apply {d : ℕ} (p : CoVector d) μ( : ComponentIdx ![Color.down]
) :
  (Tensor.basis ![Color.down]).repr (toTensor p) μ =
  p (indexEquiv μ)
```

lemma Lorentz.CoVector.smul_eq_sum[\[source\]](#)

```
lemma smul_eq_sum {d : ℕ} (i : Fin 1 ⊗ Fin d) Λ( : LorentzGroup d) (p : CoVector d) :
  Λ • p i = ∑ j, Λ⁻¹.1 j i * p j
```

lemma Lorentz.CoVector.smul_eq_mulVec[\[source\]](#)

```
lemma smul_eq_mulVec {d} Λ( : LorentzGroup d) (p : CoVector d) :
  Λ • p = (LorentzGroup.transpose Λ⁻¹).1 v* p
```

lemma Lorentz.CoVector.smul_add[\[source\]](#)

```
lemma smul_add {d : ℕ} Λ( : LorentzGroup d) (p q : CoVector d) :
  Λ • (p + q) = Λ • p + Λ • q
```

lemma Lorentz.CoVector.smul_sub[\[source\]](#)

```
@[simp]
lemma smul_sub {d : ℕ} Λ( : LorentzGroup d) (p q : CoVector d) :
  Λ • (p - q) = Λ • p - Λ • q
```

lemma Lorentz.CoVector.smul_zero[\[source\]](#)

```
lemma smul_zero {d : ℕ} Λ( : LorentzGroup d) :
  Λ • (0 : CoVector d) = 0
```

lemma Lorentz.CoVector.smul_neg[\[source\]](#)

```
lemma smul_neg {d : ℕ} Λ( : LorentzGroup d) (p : CoVector d) :
  Λ • (-p) = - Λ( • p)
```

def Lorentz.CoVector.actionCLM[\[source\]](#)

The Lorentz action on vectors as a continuous linear map.

```
def actionCLM {d : ℕ} Λ( : LorentzGroup d) :
  CoVector d → ℝ[] CoVector d
```

lemma Lorentz.CoVector.actionCLM_apply[\[source\]](#)

```
lemma actionCLM_apply {d : ℕ} Λ( : LorentzGroup d) (p : CoVector d) :
  actionCLM Λ p = Λ • p
```

lemma Lorentz.CoVector.smul_basis[\[source\]](#)

```
lemma smul_basis {d : ℕ} Λ( : LorentzGroup d) μ( : Fin 1 ⊕ Fin d) :
  Λ • basis μ = ∑ v, Λ⁻¹.1 μ v • basis v
```

2.61 Physlib.Relativity.Tensors.RealTensor.ToComplex

[Physlib/Relativity/Tensors/RealTensor/ToComplex.lean](#) on GitHub.

def realLorentzTensor.toComplexVector[\[source\]](#)

For a given color, the map turning a real Lorentz vector into a complex one.

```
noncomputable def toComplexVector (c : realLorentzTensor.Color) :
  realLorentzTensor.modules 3 c →ℝ [Complex.ofRealHom] complexLorentzTensor.modules
  (colorToComplex c) where
```

lemma realLorentzTensor.toComplexVector_up_eq_inclCongrRealLorentz[\[source\]](#)

```
lemma toComplexVector_up_eq_inclCongrRealLorentz (v : Lorentz.ContrMod 3) :
  toComplexVector Color.up v = Lorentz.inclCongrRealLorentz v
```

lemma realLorentzTensor.toComplexVector_down_eq_inclCoRealLorentz[\[source\]](#)

```
lemma toComplexVector_down_eq_inclCoRealLorentz (v : Lorentz.CoMod 3) :
  toComplexVector Color.down v = Lorentz.inclCoRealLorentz v
```

def realLorentzTensor.toComplexPure[\[source\]](#)

The function which turns a real pure tensor into a complex one.

```
noncomputable def toComplexPure {c : Fin n → Color} (p : Pure realLorentzTensor c) :
  Pure complexLorentzTensor (colorToComplex ∘ c)
```

lemma realLorentzTensor.toComplexPure_component[\[source\]](#)

```
lemma toComplexPure_component {c : Fin n → Color} (p : Pure realLorentzTensor c)
  (φ : ComponentIdx c) : (toComplexPure p).component (ComponentIdx.complexify φ) =
  p.component φ
```

lemma realLorentzTensor.actionP_toComplexPure[\[source\]](#)

```
lemma actionP_toComplexPure {n : ℕ} (c : Fin n → Color) (p : Pure realLorentzTensor c)
  (Λ : SL(2, ℂ)) :
  Λ • toComplexPure p = toComplexPure (toLorentzGroup Λ • p)
```

lemma realLorentzTensor.toComplex_pure[\[source\]](#)

```
lemma toComplex_pure {n : ℕ} (c : Fin n → Color) (p : Pure realLorentzTensor c) :
  toComplex p.toTensor = (toComplexPure p).toTensor
```

lemma realLorentzTensor.isReindexing_prodTColorToComplex[\[source\]](#)

```
lemma isReindexing_prodTColorToComplex {n m : ℕ}
  {c : Fin n → realLorentzTensor.Color} {c1 : Fin m → realLorentzTensor.Color} :
  IsReindexing (Fin.append (colorToComplex ∘ c) (colorToComplex ∘ c1))
    (colorToComplex ∘ Fin.append c c1)
    (id : Fin (n + m) → Fin (n + m))
```

2.62 Physlib.Relativity.Tensors.RealTensor.Vector.Basic

[Physlib/Relativity/Tensors/RealTensor/Vector/Basic.lean](#) on GitHub.

lemma Lorentz.Vector.eq_of_apply_eq[\[source\]](#)

@[ext]

```
lemma eq_of_apply_eq {d : ℕ} {v w : Vector d} (h : ∀ i, v i = w i) : v = w
```

def Lorentz.Vector.ofSpatialComponent[\[source\]](#)

The continuous linear map corresponding to the creation of a Lorentz Vector with only non-zero spatial components.

```
def ofSpatialComponent {d : ℕ} : EuclideanSpace ℝ (Fin d) → LR[] Vector d where
```

2.63 Physlib.Relativity.Tensors.RealTensor.Vector.Representation

[Physlib/Relativity/Tensors/RealTensor/Vector/Representation.lean](#) on GitHub.

def Lorentz.Vector.rep[\[source\]](#)*The representation of the Lorentz group on Lorentz vectors.***def** rep {d : ℕ} : Representation ℝ (LorentzGroup d) (Vector d) **where****lemma Lorentz.Vector.rep_apply_eq_mulVec**[\[source\]](#)**lemma** rep_apply_eq_mulVec (d : ℕ) Λ(: LorentzGroup d) (v : Vector d) :
rep Λ v = Λ v * v**lemma Lorentz.Vector.rep_apply_eq_sum**[\[source\]](#)**lemma** rep_apply_eq_sum (d : ℕ) Λ(: LorentzGroup d) (v : Vector d) (k : Fin 1 ⊕ Fin d) :
rep Λ v k = ∑ j, Λ.1 k j • v j**lemma Lorentz.Vector.rep_apply_basis**[\[source\]](#)**lemma** rep_apply_basis {d} μ(: Fin 1 ⊕ Fin d) Λ(: LorentzGroup d) :
rep Λ (basis μ) = ∑ j, Λ.1 j μ • basis j**lemma Lorentz.Vector.rep_toMatrix**[\[source\]](#)**lemma** rep_toMatrix (d : ℕ) Λ(: LorentzGroup d) :
LinearMap.toMatrix basis basis (rep Λ) = Λ.1**lemma Lorentz.Vector.rep_injective**[\[source\]](#)**lemma** rep_injective (d : ℕ) Λ(: LorentzGroup d) : Function.Injective (rep Λ)**lemma Lorentz.Vector.rep_surjective**[\[source\]](#)**lemma** rep_surjective (d : ℕ) Λ(: LorentzGroup d) : Function.Surjective (rep Λ)**lemma Lorentz.Vector.rep_bijective**[\[source\]](#)**lemma** rep_bijective (d : ℕ) Λ(: LorentzGroup d) : Function.Bijective (rep Λ)**lemma Lorentz.Vector.rep_contDiff**[\[source\]](#)

@[fun_prop]

lemma rep_contDiff (d : ℕ) {n} Λ(: LorentzGroup d) : ContDiff ℝ n (rep Λ)**lemma Lorentz.Vector.rep_left_injective**[\[source\]](#)**lemma** rep_left_injective (d : ℕ) : Function.Injective (rep (d := d))**2.64 Physlib.Relativity.Tensors.RealTensor.Vector.Tensorial**[Physlib/Relativity/Tensors/RealTensor/Vector/Tensorial.lean](#) on GitHub.

def Lorentz.Vector.indexEquiv[\[source\]](#)

The equivalence between the type of indices of a Lorentz vector and $\text{Fin } 1 \oplus \text{Fin } d$.

```
def indexEquiv {d : ℕ} :
  ComponentIdx (S := (realLorentzTensor d)) ![Color.up] ≈ Fin 1 ⊕ Fin d
```

instance Lorentz.Vector.tensorial[\[source\]](#)

```
instance tensorial {d : ℕ} : Tensorial (realLorentzTensor d) ! [.up] (Vector d) where
```

lemma Lorentz.Vector.toTensor_symm_apply[\[source\]](#)

```
lemma toTensor_symm_apply {d : ℕ} (p : ℝT[d, .up]) :
  (toTensor (self := tensorial)).symm p =
  (Equiv.piCongrLeft' _ indexEquiv <|
  Finsupp.equivFunOnFinite <|
  (Tensor.basis (S := (realLorentzTensor d)) _).repr p)
```

lemma Lorentz.Vector.toTensor_symm_pure[\[source\]](#)

```
lemma toTensor_symm_pure {d : ℕ} (p : Pure (realLorentzTensor d) ! [.up]) (i : Fin 1 ⊕ Fin d) :
  (toTensor (self := tensorial)).symm p.toTensor i =
  ((Lorentz.contrBasis d).repr (p 0)) (indexEquiv.symm i 0)
```

lemma Lorentz.Vector.toTensor_symm_basis[\[source\]](#)

```
lemma toTensor_symm_basis {d : ℕ} μ( : Fin 1 ⊕ Fin d) :
  (toTensor (self := tensorial)).symm (Tensor.basis ! [Color.up] (indexEquiv.symm μ)) =
  basis μ
```

lemma Lorentz.Vector.toTensor_basis_eq_tensor_basis[\[source\]](#)

```
lemma toTensor_basis_eq_tensor_basis {d : ℕ} μ( : Fin 1 ⊕ Fin d) :
  toTensor (basis μ) = Tensor.basis ! [Color.up] (indexEquiv.symm μ)
```

lemma Lorentz.Vector.basis_eq_map_tensor_basis[\[source\]](#)

```
lemma basis_eq_map_tensor_basis {d} : basis =
  ((Tensor.basis
  (S := realLorentzTensor d) ! [Color.up])).map toTensor.symm).reindex indexEquiv
```

lemma Lorentz.Vector.tensor_basis_map_eq_basis_reindex[\[source\]](#)

```
lemma tensor_basis_map_eq_basis_reindex {d} :
  (Tensor.basis (S := realLorentzTensor d) ! [Color.up]).map toTensor.symm =
  basis.reindex indexEquiv.symm
```

lemma Lorentz.Vector.tensor_basis_repr_toTensor_apply[\[source\]](#)

```
lemma tensor_basis_repr_toTensor_apply {d : ℕ} (p : Vector d) μ( : ComponentIdx ! [Color.up]) :
  (Tensor.basis ! [Color.up]).repr (toTensor p) μ =
  p (indexEquiv μ)
```

lemma Lorentz.Vector.smul_eq_sum[\[source\]](#)

```
lemma smul_eq_sum {d : ℕ} (i : Fin 1 ⊕ Fin d) (Λ : LorentzGroup d) (p : Vector d) :
  Λ • p i = ∑ j, Λ.1 i j * p j
```

lemma Lorentz.Vector.smul_eq_mulVec[\[source\]](#)

```
lemma smul_eq_mulVec {d} (Λ : LorentzGroup d) (p : Vector d) :
  Λ • p = Λ.1 v* p
```

lemma Lorentz.Vector.smul_add[\[source\]](#)

```
lemma smul_add {d : ℕ} (Λ : LorentzGroup d) (p q : Vector d) :
  Λ • (p + q) = Λ • p + Λ • q
```

lemma Lorentz.Vector.smul_sub[\[source\]](#)

```
@[simp]
lemma smul_sub {d : ℕ} (Λ : LorentzGroup d) (p q : Vector d) :
  Λ • (p - q) = Λ • p - Λ • q
```

lemma Lorentz.Vector.smul_zero[\[source\]](#)

```
lemma smul_zero {d : ℕ} (Λ : LorentzGroup d) :
  Λ • (0 : Vector d) = 0
```

lemma Lorentz.Vector.smul_neg[\[source\]](#)

```
lemma smul_neg {d : ℕ} (Λ : LorentzGroup d) (p : Vector d) :
  Λ • (-p) = - Λ • p
```

lemma Lorentz.Vector.neg_smul[\[source\]](#)

```
lemma neg_smul {d} (Λ : LorentzGroup d) (p : Vector d) :
  Λ(-) • p = - Λ • p
```

lemma LorentzGroup.eq_of_action_vector_eq[\[source\]](#)

```
lemma _root_.LorentzGroup.eq_of_action_vector_eq {d : ℕ}
  Λ{ Λ' : LorentzGroup d} (h : ∀ p : Vector d, Λ • p = Λ' • p) :
  Λ = Λ'
```

def Lorentz.Vector.actionCLM[\[source\]](#)

The Lorentz action on vectors as a continuous linear map.

```
def actionCLM {d : ℕ} (Λ : LorentzGroup d) :
  Vector d → LR[] Vector d
```


lemma Lorentz.Vector.actionCLM_apply[\[source\]](#)

```
lemma actionCLM_apply {d : ℕ} (Λ : LorentzGroup d) (p : Vector d) :
  actionCLM Λ p = Λ • p
```

lemma Lorentz.Vector.actionCLM_injective[\[source\]](#)

```
lemma actionCLM_injective {d : ℕ} (Λ : LorentzGroup d) :
  Function.Injective (actionCLM Λ)
```

lemma Lorentz.Vector.actionCLM_surjective[\[source\]](#)

```
lemma actionCLM_surjective {d : ℕ} (Λ : LorentzGroup d) :
  Function.Surjective (actionCLM Λ)
```

lemma Lorentz.Vector.smul_basis[\[source\]](#)

```
lemma smul_basis {d : ℕ} (Λ : LorentzGroup d) (μ : Fin 1 ⊗ Fin d) :
  Λ • basis μ = ∑ v, Λ.1 v μ • basis v
```

2.65 Physlib.Relativity.Tensors.Reindexing

[Physlib/Relativity/Tensors/Reindexing.lean](#) on GitHub.

def TensorSpecies.Tensor.IsReindexing[\[source\]](#)

Given two lists of indices 'c : Fin n → C' and 'c1 : Fin m → C' a map 'σ : Fin m → Fin n' satisfies the condition 'IsReindexing c c1 σ' if it is: - A bijection - Forms a commutative triangle with 'c' and 'c1'.

```
def IsReindexing {n m : ℕ} (c : Fin n → C) (c1 : Fin m → C)
  σ ( : Fin m → Fin n) : Prop
```

lemma TensorSpecies.Tensor.IsReindexing.injective[\[source\]](#)

```
lemma injective {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ { : Fin m → Fin n} (h : IsReindexing c c1 σ) : Function.Injective σ
```

lemma TensorSpecies.Tensor.IsReindexing.surjective[\[source\]](#)

```
lemma surjective {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ { : Fin m → Fin n} (h : IsReindexing c c1 σ) : Function.Surjective σ
```

lemma TensorSpecies.Tensor.IsReindexing.auto[\[source\]](#)

```
lemma auto {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ { : Fin m → Fin n} (h : IsReindexing c c1 σ := by {simp [IsReindexing]; try decide}) :
  IsReindexing c c1 σ
```

lemma TensorSpecies.Tensor.IsReindexing.on_id[\[source\]](#)

@[simp]

```
lemma on_id {n : ℕ} {c c1 : Fin n → C} :
  IsReindexing c c1 (id : Fin n → Fin n) ↔ ∀ i, c i = c1 i
```

lemma TensorSpecies.Tensor.IsReindexing.on_id_symm[\[source\]](#)

```
lemma on_id_symm {n : ℕ} {c c1 : Fin n → C} (h : IsReindexing c1 c id) :
  IsReindexing c c1 (id : Fin n → Fin n)
```

def TensorSpecies.Tensor.IsReindexing.inv[\[source\]](#)

For a map ' σ ' satisfying ' $\text{IsReindexing } c \text{ c1 } \sigma$ ', the inverse of that map.

```
def inv {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ( : Fin m → Fin n) (h : IsReindexing c c1 σ) : Fin n → Fin m
```

def TensorSpecies.Tensor.IsReindexing.toEquiv[\[source\]](#)

For a map ' $\sigma : \text{Fin } m \rightarrow \text{Fin } n$ ' satisfying ' $\text{IsReindexing } c \text{ c1 } \sigma$ ', that map lifted to an equivalence between ' $\text{Fin } n$ ' and ' $\text{Fin } m$ '.

```
def toEquiv {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ{ : Fin m → Fin n} (h : IsReindexing c c1 σ) :
  Fin n ≈ Fin m where
```

lemma TensorSpecies.Tensor.IsReindexing.apply_inv_apply[\[source\]](#)

```
lemma apply_inv_apply {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ( : Fin m → Fin n) (h : IsReindexing c c1 σ) (x : Fin m) :
  h.inv σ σ( x) = x
```

lemma TensorSpecies.Tensor.IsReindexing.inv_apply_apply[\[source\]](#)

```
lemma inv_apply_apply {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ( : Fin m → Fin n) (h : IsReindexing c c1 σ) (x : Fin n) :
  σ (h.inv σ x) = x
```

lemma TensorSpecies.Tensor.IsReindexing.preserve_color[\[source\]](#)

```
lemma preserve_color {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ{ : Fin m → Fin n} (h : IsReindexing c c1 σ) :
  ∀ (x : Fin m), c1 x = (c ∘ σ) x
```

lemma TensorSpecies.Tensor.IsReindexing.inv_perserve_color[\[source\]](#)

@[simp, noint simpVarHead]

```
lemma inv_perserve_color {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ{ : Fin m → Fin n} (h : IsReindexing c c1 σ) (x : Fin n) :
  c1 (h.inv σ x) = c x
```

lemma TensorSpecies.Tensor.IsReindexing.toEquiv_symm_perserve_color[\[source\]](#)`@[simp, nolint simpVarHead]`

```

lemma toEquiv_symm_perserve_color {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ{ : Fin m → Fin n} (h : IsReindexing c c1 σ) (x : Fin m) :
  c (h.toEquiv.symm x) = c1 x

```

lemma TensorSpecies.Tensor.IsReindexing.symm[\[source\]](#)

The inverse of a map satisfying ‘IsReindexing c c1 σ’ is a reindexing of ‘c1’ by ‘c’.

```

lemma symm {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  σ{ : Fin m → Fin n} (h : IsReindexing c c1 σ) : IsReindexing c1 c (h.inv σ)

```

lemma TensorSpecies.Tensor.IsReindexing.symm_of_id[\[source\]](#)

```

lemma symm_of_id {n : ℕ} {c c1 : Fin n → C} (h : IsReindexing c c1 id) :
  IsReindexing c1 c id

```

lemma TensorSpecies.Tensor.IsReindexing.comp[\[source\]](#)

The composition of two maps satisfying ‘IsReindexing’ also satisfies the ‘IsReindexing’.

```

lemma comp {n n1 n2 : ℕ} {c : Fin n → C} {c1 : Fin n1 → C}
  {c2 : Fin n2 → C} σ{ : Fin n1 → Fin n} σ2{ : Fin n2 → Fin n1}
  (h : IsReindexing c c1 σ) (h2 : IsReindexing c1 c2 σ2) : IsReindexing c c2 (σ ∘ σ2)

```

lemma TensorSpecies.Tensor.IsReindexing.append_congr_left[\[source\]](#)

```

lemma append_congr_left {n n' n2 : ℕ} {c : Fin n → C} {c' : Fin n' → C}
  σ{ : Fin n' → Fin n} (c2 : Fin n2 → C) (h : IsReindexing c c' σ) :
  IsReindexing (Fin.append c c2) (Fin.append c' c2)
  (Fin.append (Fin.castAdd n2 ∘ σ) (Fin.natAdd n))

```

lemma TensorSpecies.Tensor.IsReindexing.append_congr_right[\[source\]](#)

```

lemma append_congr_right {n n' n2 : ℕ} {c : Fin n → C} {c' : Fin n' → C}
  σ{ : Fin n' → Fin n} (c2 : Fin n2 → C) (h : IsReindexing c c' σ) :
  IsReindexing (Fin.append c2 c) (Fin.append c2 c')
  (Fin.append (Fin.castAdd n) (Fin.natAdd n2 ∘ σ))

```

lemma TensorSpecies.Tensor.IsReindexing.append_zero_right[\[source\]](#)

```

lemma append_zero_right {n} {c : Fin n → C}
  {c1 : Fin 0 → C} : IsReindexing c (Fin.append c c1) id

```

lemma TensorSpecies.Tensor.IsReindexing.append_swap[\[source\]](#)

```

lemma append_swap {n n2 : ℕ} {c : Fin n → C} {c2 : Fin n2 → C} :
  IsReindexing (Fin.append c c2) (Fin.append c2 c)
  (Fin.append (Fin.natAdd n) (Fin.castAdd n2))

```

lemma TensorSpecies.Tensor.IsReindexing.append_assoc_right[\[source\]](#)

```

lemma append_assoc_right {n1 n2 n3 : ℕ} {c : Fin n1 → C} {c2 : Fin n2 → C} {c3 : Fin n3 → C}
  :
  IsReindexing (Fin.append c (Fin.append c2 c3)) (Fin.append (Fin.append c c2) c3)
    (Fin.cast (by grind))

```

lemma TensorSpecies.Tensor.IsReindexing.append_assoc_left[\[source\]](#)

```

lemma append_assoc_left {n1 n2 n3 : ℕ} {c : Fin n1 → C} {c2 : Fin n2 → C} {c3 : Fin n3 → C} :
  IsReindexing (Fin.append (Fin.append c c2) c3) (Fin.append c (Fin.append c2 c3))
    (Fin.cast (by grind))

```

lemma TensorSpecies.Tensor.IsReindexing.append_succAbove_natAdd[\[source\]](#)

```

lemma append_succAbove_natAdd {n n1 : ℕ} {c : Fin n → C} {c1 : Fin (n1 + 1) → C}
  (i : Fin (n1 + 1)) :
  IsReindexing (Fin.append c c1 ∘ (Fin.natAdd n i).succAbove)
    (Fin.append c (c1 ∘ i.succAbove)) id

```

lemma TensorSpecies.Tensor.IsReindexing.append_succAbove_castAdd[\[source\]](#)

```

lemma append_succAbove_castAdd {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  (i : Fin (n + 1)) :
  IsReindexing (Fin.append c c1 ∘ (Fin.castAdd (n1 + 1) i).succAbove)
    (Fin.append (c ∘ i.succAbove) c1) (Fin.cast (by grind))

```

lemma TensorSpecies.Tensor.IsReindexing.append_succSuccAbove_natAdd[\[source\]](#)

*Removing two entries from the right component of ‘Fin.append c1 c’ commutes with the append: removing the ‘i’-th and ‘j’-th entries of ‘c’ and then appending ‘c1’ matches removing the corresponding entries of ‘Fin.append c1 c’, via the identity permutation. This is used for the commutation of taking a *product* of tensors with *contraction* of indices.*

```

lemma append_succSuccAbove_natAdd {n n1 : ℕ} {c : Fin (n + 1 + 1) → C}
  {c1 : Fin n1 → C} (i j : Fin (n + 1 + 1)) :
  IsReindexing (Fin.append c1 c ∘ (Fin.natAdd n1 i).succSuccAbove (Fin.natAdd n1 j))
    (Fin.append c1 (c ∘ i.succSuccAbove j)) id

```

lemma TensorSpecies.Tensor.IsReindexing.succAbove_of_eq_zero[\[source\]](#)

Given a reindexing of ‘c’ by ‘c1’ via ‘σ’ for which the index ‘i’ is sent to ‘0’, removing the ‘i’-th entry of ‘c1’ and the first entry of ‘c’ yields a reindexing of ‘c ∘ Fin.succ’ by ‘c1 ∘ i.succAbove’ via the map sending ‘j’ to the predecessor of ‘σ (i.succAbove j)’.

```

lemma succAbove_of_eq_zero {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  σ{ : Fin (n1 + 1) → Fin (n + 1)} (i : Fin (n1 + 1))
  (h : IsReindexing c c1 σ) (hi : σ i = 0) :
  IsReindexing (c ∘ Fin.succ) (c1 ∘ i.succAbove)
    (fun j => σ (i.succAbove j)).pred (by simp <[ hi, h.injective.eq_iff ]))

```

lemma TensorSpecies.Tensor.IsReindexing.succAbove_of_neq_zero[\[source\]](#)

Given a reindexing of c by $c1$ via σ for which the index i is not sent to 0 , removing the i -th entry of $c1$ and the (σi) -th entry of c yields a reindexing of $c \circ (\sigma i).succAbove$ by $c1 \circ i.succAbove$ via the map $(\sigma i).pred.predAbove \circ \sigma \circ i.succAbove$.

```
lemma succAbove_of_neq_zero {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  σ{ : Fin (n1 + 1) → Fin (n + 1)} (i : Fin (n1 + 1))
  (h : IsReindexing c c1 σ) (hi : σ i ≠ 0) :
  IsReindexing (c ∘ σ(i).succAbove) (c1 ∘ i.succAbove)
    ((Fin.pred σ(i) hi).predAbove ∘ σ ∘ i.succAbove)
```

lemma TensorSpecies.Tensor.IsReindexing.succAbove[\[source\]](#)

Given a reindexing of c by $c1$ via σ , removing the i -th entry of $c1$ and the (σi) -th entry of c yields a reindexing of $c \circ (\sigma i).succAbove$ by $c1 \circ i.succAbove$. This unifies `succAbove_of_eq_zero` and `succAbove_of_neq_zero` via a case split on whether $\sigma i = 0$. This is used for the commutation of **permutation** of indices with **evaluation** of indices.

```
lemma succAbove {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  σ{ : Fin (n1 + 1) → Fin (n + 1)} (i : Fin (n1 + 1))
  (h : IsReindexing c c1 σ) :
  IsReindexing (c ∘ σ(i).succAbove) (c1 ∘ i.succAbove)
    (if hi : σ i = 0 then fun j => σ(i.succAbove j)).pred (by simp <| hi, h.injective.
eq_iff))
    else (Fin.pred σ(i) hi).predAbove ∘ σ ∘ i.succAbove)
```

lemma TensorSpecies.Tensor.IsReindexing.succSuccAbove[\[source\]](#)

Given a reindexing of c by $c1$ via σ and two distinct indices $i \neq j$, removing the i -th and j -th entries of $c1$ and the (σi) -th and (σj) -th entries of c yields a reindexing of $c \circ (\sigma i).succSuccAbove (\sigma j)$ by $c1 \circ i.succSuccAbove j$. This is used for the commutation of **permutation** of indices with **contraction** of indices.

```
lemma succSuccAbove {n n1 : ℕ} {c : Fin (n + 1 + 1) → C}
  {c1 : Fin (n1 + 1 + 1) → C}
  (i j : Fin (n1 + 1 + 1)) (hij : i ≠ j)
  σ{ : Fin (n1 + 1 + 1) → Fin (n + 1 + 1)} (σh : IsReindexing c c1 σ) :
  IsReindexing (c ∘ σ(i).succSuccAbove σ(j))
    (c1 ∘ i.succSuccAbove j) (i.funPredPredAbove j hij σ σh.1)
```

lemma TensorSpecies.Tensor.IsReindexing.succSuccAbove_comm[\[source\]](#)

Removing two pairs of entries from c in either order gives the same colour list: removing the $i1$ -th and $j1$ -th entries and then the (shifted) $i2$ -th and $j2$ -th entries matches removing the $i2$ -th and $j2$ -th entries first and then the (shifted) $i1$ -th and $j1$ -th entries, via the identity permutation. This is used for the commutation of two **contractions**.

```
lemma succSuccAbove_comm {n : ℕ} {c : Fin (n + 1 + 1 + 1 + 1) → C}
  (i1 j1 : Fin (n + 1 + 1 + 1 + 1)) (i2 j2 : Fin (n + 1 + 1))
  (hij1 : i1 ≠ j1) (hij2 : i2 ≠ j2) :
  let i2'
```

lemma TensorSpecies.Tensor.IsReindexing.succAbove_succSuccAbove_comm[\[source\]](#)

Removing one entry and then a pair of entries from ‘ c ’ in either order gives the same colour list: removing the ‘ k ’-th entry and then the (shifted) ‘ i ’-th and ‘ j ’-th entries matches removing the ‘ i ’-th and ‘ j ’-th entries first and then the (shifted) ‘ k ’-th entry, via the identity permutation. This is used for the commutation of a **contraction** with an **evaluation**.

```
lemma succAbove_succSuccAbove_comm {n : ℕ} {c : Fin (n + 1 + 1 + 1) → C}
  (k : Fin (n + 1 + 1 + 1)) (i j : Fin (n + 1 + 1)) (hij : i ≠ j) :
  let i'
```

lemma TensorSpecies.Tensor.IsReindexing.succAbove_succAbove_comm[\[source\]](#)

Removing two single entries from ‘ c ’ in either order gives the same colour list: removing the ‘ $k1$ ’-th entry and then the (shifted) ‘ $k2$ ’-th entry matches removing the ‘ $k2$ ’-th entry first and then the (shifted) ‘ $k1$ ’-th entry, via the identity permutation. This is used for the commutation of two **evaluations** of indices.

```
lemma succAbove_succAbove_comm {n : ℕ} {c : Fin (n + 1 + 1) → C}
  (k1 : Fin (n + 1 + 1)) (k2 : Fin (n + 1)) :
  let k2'
```

lemma TensorSpecies.Tensor.IsReindexing.append_succ_last[\[source\]](#)

Splitting a list of colours ‘ $c : \text{Fin } (n + 1) \rightarrow C$ ’ into its first ‘ n ’ entries and its last entry recovers ‘ c ’: the identity permutation matches ‘ $\text{Fin.append } (c \circ (\text{Fin.last } n).succAbove) \text{ } [c \text{ (Fin.last } n)]$ ’ with ‘ c ’.

```
lemma append_succ_last {n : ℕ} (c : Fin (n + 1) → C) :
  IsReindexing (Fin.append (c ∘ (Fin.last n).succAbove) ![c (Fin.last n)]) c id
```

lemma TensorSpecies.Tensor.IsReindexing.append_of_first[\[source\]](#)

Splitting a list of colours ‘ $c : \text{Fin } (n + 1) \rightarrow C$ ’ into its first entry and its remaining ‘ n ’ entries recovers ‘ c ’: the canonical reindexing ‘ $\text{Fin } (1 + n) \simeq \text{Fin } (n + 1)$ ’ matches ‘ $\text{Fin.append } ![c \ 0] \ (c \circ \text{Fin.succAbove } 0)$ ’ with ‘ c ’.

```
lemma append_of_first {n : ℕ} (c : Fin (n + 1) → C) :
  IsReindexing (Fin.append ![c 0] (c ∘ Fin.succAbove 0)) c (Fin.cast (by grind))
```

lemma TensorSpecies.Tensor.IsReindexing.fin_cast_isReindexing[\[source\]](#)

Casting the domain along an equality ‘ $n1 = n$ ’ of lengths is a reindexing of ‘ c ’ by ‘ $c \circ \text{Fin.cast } h$ ’.

```
lemma fin_cast_isReindexing (n n1 : ℕ) {c : Fin n → C} (h : n1 = n) :
  IsReindexing c (c ∘ Fin.cast h) (Fin.cast h)
```

2.66 Physlib.Relativity.Tensors.Tensorial

[Physlib/Relativity/Tensors/Tensorial.lean](#) on GitHub.

lemma TensorSpecies.Tensorial.prod_eq_sum_eval[\[source\]](#)

Double basis expansion of an element of a tensor product $M \otimes [k] M_2$ of two ‘Tensorial’ one-index spaces. Given bases ‘b’, ‘b2’ of ‘M’, ‘M₂’ coming from the single-index tensor bases, every $x : M \otimes [k] M_2$ is the double sum over ‘i, j’ of the iterated evaluation coefficient $\text{toField } (\text{evalT } 0 \ j \ (\text{evalT } 0 \ i \ (\text{toTensor } x)))$ times $b \ i \otimes_t b2 \ j$:

```
lemma prod_eq_sum_eval {c c2 : C} {zM : Type}
  [Tensorial S ![c] M] [AddCommMonoid zM] [Module k zM]
  [Tensorial S ![c2] zM] {b : Module.Basis (basisIdx c) k M}
  {b2 : Module.Basis (basisIdx c2) k zM}
  (h : b = ((Tensor.basis (S := S) ![c]).map toTensor.symm).reindex ComponentIdx.single)
  (h2 : b2 = ((Tensor.basis (S := S) ![c2]).map toTensor.symm).reindex ComponentIdx.single)
  (x : M ⊗ [k] zM) : x = ∑ i : (basisIdx c), ∑ j : (basisIdx c2),
    toField (evalT 0 (basisIdxCongr (by rfl) j)
      (evalT 0 (basisIdxCongr (by rfl) i) (toTensor x))) • b i ⊗_t [k] b2 j
```

2.67 Physlib.Relativity.Tensors.TensorSpecies.Basic

[Physlib/Relativity/Tensors/TensorSpecies/Basic.lean](#) on GitHub.

lemma TensorSpecies.basisIdxCongr_heq_arg[\[source\]](#)

‘basisIdxCongr’ only depends on its endpoints up to ‘HEq’ of the arguments: equal target colours and heterogeneously equal labels give equal casts.

```
lemma basisIdxCongr_heq_arg {c1 c2 d : C} (h1 : c1 = d) (h2 : c2 = d)
  {x : basisIdx c1} {y : basisIdx c2} (hxy : HEq x y) :
  basisIdxCongr h1 x = basisIdxCongr h2 y
```

2.68 Physlib.SpaceAndTime.GalileanGroup.Basic

[Physlib/SpaceAndTime/GalileanGroup/Basic.lean](#) on GitHub.

structure GalileanGroup[\[source\]](#)

A Galilean transformation in ‘d’ spatial dimensions. The fields are, in order, the spatial orthogonal part, boost velocity, spatial translation, and time translation.

```
@[ext]
structure GalileanGroup (d : ℕ := 3) where
  rotation : Matrix.orthogonalGroup (Fin d) ℝ
  velocity : EuclideanSpace ℝ (Fin d)
  spaceTranslation : EuclideanSpace ℝ (Fin d)
  timeTranslation : Time
```

lemma GalileanGroup.orthogonal_smul_smul[\[source\]](#)

```
lemma orthogonal_smul_smul (R : Matrix.orthogonalGroup (Fin d) ℝ) (c : ℝ)
  (v : EuclideanSpace ℝ (Fin d)) :
  R • (c • v) = c • (R • v)
```

instance GalileanGroup.«instance@f34d5bc7»

[\[source\]](#)

The identity Galilean transformation.

instance : One (GalileanGroup d) **where**

lemma GalileanGroup.one_rotation

[\[source\]](#)

@[simp]

lemma one_rotation : (1 : GalileanGroup d).rotation = 1

lemma GalileanGroup.one_velocity

[\[source\]](#)

@[simp]

lemma one_velocity : (1 : GalileanGroup d).velocity = 0

lemma GalileanGroup.one_spaceTranslation

[\[source\]](#)

@[simp]

lemma one_spaceTranslation : (1 : GalileanGroup d).spaceTranslation = 0

lemma GalileanGroup.one_timeTranslation

[\[source\]](#)

@[simp]

lemma one_timeTranslation : (1 : GalileanGroup d).timeTranslation = 0

instance GalileanGroup.«instance@2d436bb6»

[\[source\]](#)

*The product whose action is composition: $(g * h) \bullet tx = g \bullet h \bullet tx$.*

instance : Mul (GalileanGroup d) **where**

lemma GalileanGroup.mul_rotation

[\[source\]](#)

@[simp]

lemma mul_rotation (g h : GalileanGroup d) :
 (g * h).rotation = g.rotation * h.rotation

lemma GalileanGroup.mul_velocity

[\[source\]](#)

@[simp]

lemma mul_velocity (g h : GalileanGroup d) :
 (g * h).velocity = g.rotation • h.velocity + g.velocity

lemma GalileanGroup.mul_spaceTranslation

[\[source\]](#)

@[simp]

lemma mul_spaceTranslation (g h : GalileanGroup d) :
 (g * h).spaceTranslation =
 g.spaceTranslation + g.rotation • h.spaceTranslation + h.timeTranslation.val • g.velocity

lemma GalileanGroup.mul_timeTranslation[\[source\]](#)

```
@[simp]
lemma mul_timeTranslation (g h : GalileanGroup d) :
  (g * h).timeTranslation = g.timeTranslation + h.timeTranslation
```

instance GalileanGroup.«instance@3107b840»[\[source\]](#)

The inverse Galilean transformation.

```
instance : Inv (GalileanGroup d) where
```

lemma GalileanGroup.inv_rotation[\[source\]](#)

```
@[simp]
lemma inv_rotation (g : GalileanGroup d) :
  -g1.rotation = g.-rotation1
```

lemma GalileanGroup.inv_velocity[\[source\]](#)

The inverse boost velocity formula.

```
@[simp]
lemma inv_velocity (g : GalileanGroup d) :
  -g1.velocity = -(g.-rotation1 • g.velocity)
```

lemma GalileanGroup.inv_spaceTranslation[\[source\]](#)

```
@[simp]
lemma inv_spaceTranslation (g : GalileanGroup d) :
  -g1.spaceTranslation =
    -(g.-rotation1 • g.spaceTranslation) +
    g.timeTranslation.val • (g.-rotation1 • g.velocity)
```

lemma GalileanGroup.inv_timeTranslation[\[source\]](#)

```
@[simp]
lemma inv_timeTranslation (g : GalileanGroup d) :
  -g1.timeTranslation = -g.timeTranslation
```

instance GalileanGroup.«instance@33c7352c»[\[source\]](#)

The Galilean transformations form a group under composition.

```
instance : Group (GalileanGroup d) where
```

instance GalileanGroup.«instance@5e5b6b3a»[\[source\]](#)

```
instance : Inhabited (GalileanGroup d) where
```

def GalileanGroup.actSpace[\[source\]](#)*The Galilean action on space at a fixed time.*

```
noncomputable def actSpace (g : GalileanGroup d) (t : Time) (x : Space d) : Space d
```

def GalileanGroup.act[\[source\]](#)*The Galilean action on $\text{Time} \times \text{Space } d$.*

```
noncomputable def act (g : GalileanGroup d) (tx : Time × Space d) : Time × Space d
```

instance GalileanGroup.«instance@c761f5d5»[\[source\]](#)

```
noncomputable instance : MulAction (GalileanGroup d) (Time × Space d) where
```

lemma GalileanGroup.smul_fst[\[source\]](#)

```
@[simp]
lemma smul_fst (g : GalileanGroup d) (tx : Time × Space d) :
  (g • tx).1 = tx.1 + g.timeTranslation
```

lemma GalileanGroup.smul_snd[\[source\]](#)

```
@[simp]
lemma smul_snd (g : GalileanGroup d) (tx : Time × Space d) :
  (g • tx).2 = g.actSpace tx.1 tx.2
```

lemma GalileanGroup.smul_mk[\[source\]](#)

```
@[simp]
lemma smul_mk (g : GalileanGroup d) (t : Time) (x : Space d) :
  g • ((t, x) : Time × Space d) = (t + g.timeTranslation, g.actSpace t x)
```

lemma GalileanGroup.actSpace_apply[\[source\]](#)

```
@[simp]
lemma actSpace_apply (g : GalileanGroup d) (t : Time) (x : Space d) (i : Fin d) :
  g.actSpace t x i =
    (g.rotation • (x v- (0 : Space d))) i + t.val * g.velocity i + g.spaceTranslation i
```

def GalileanGroup.ofEuclidean[\[source\]](#)*A Euclidean spatial transformation as a Galilean transformation with zero boost and no time translation.*

```
def ofEuclidean (g : EuclideanGroup d) : GalileanGroup d
```

lemma GalileanGroup.ofEuclidean_rotation[\[source\]](#)

```
@[simp]
lemma ofEuclidean_rotation (g : EuclideanGroup d) :
  (ofEuclidean g).rotation = g.linear
```

lemma GalileanGroup.ofEuclidean_velocity[\[source\]](#)

```
@[simp]
lemma ofEuclidean_velocity (g : EuclideanGroup d) :
  (ofEuclidean g).velocity = 0
```

lemma GalileanGroup.ofEuclidean_spaceTranslation[\[source\]](#)

```
@[simp]
lemma ofEuclidean_spaceTranslation (g : EuclideanGroup d) :
  (ofEuclidean g).spaceTranslation = g.translation
```

lemma GalileanGroup.ofEuclidean_timeTranslation[\[source\]](#)

```
@[simp]
lemma ofEuclidean_timeTranslation (g : EuclideanGroup d) :
  (ofEuclidean g).timeTranslation = 0
```

def GalileanGroup.euclidean.incl[\[source\]](#)

Inclusion of the Euclidean group into the Galilean group.

```
def euclidean.incl : EuclideanGroup d →* GalileanGroup d where
```

def GalileanGroup.ofOrthogonal[\[source\]](#)

A pure orthogonal spatial transformation as a Galilean transformation.

```
def ofOrthogonal (R : Matrix.orthogonalGroup (Fin d) ℝ) : GalileanGroup d
```

lemma GalileanGroup.ofOrthogonal_rotation[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_rotation (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).rotation = R
```

lemma GalileanGroup.ofOrthogonal_velocity[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_velocity (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).velocity = 0
```

lemma GalileanGroup.ofOrthogonal_spaceTranslation[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_spaceTranslation (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).spaceTranslation = 0
```

lemma GalileanGroup.ofOrthogonal_timeTranslation[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_timeTranslation (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).timeTranslation = 0
```

def GalileanGroup.orthogonal.incl[\[source\]](#)*Inclusion of the orthogonal group into the Galilean group.*

```
def orthogonal.incl : Matrix.orthogonalGroup (Fin d) ℝ →* GalileanGroup d where
```

def GalileanGroup.rotation.incl[\[source\]](#)*Inclusion of the existing spatial rotation subgroup into the Galilean group.*

```
noncomputable def rotation.incl : EuclideanGroup.RotationGroup d →* GalileanGroup d where
```

def GalileanGroup.ofSpaceTranslation[\[source\]](#)*A pure spatial translation as a Galilean transformation.*

```
def ofSpaceTranslation (a : EuclideanSpace ℝ (Fin d)) : GalileanGroup d
```

lemma GalileanGroup.ofSpaceTranslation_rotation[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_rotation (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).rotation = 1
```

lemma GalileanGroup.ofSpaceTranslation_velocity[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_velocity (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).velocity = 0
```

lemma GalileanGroup.ofSpaceTranslation_spaceTranslation[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_spaceTranslation (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).spaceTranslation = a
```

lemma GalileanGroup.ofSpaceTranslation_timeTranslation[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_timeTranslation (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).timeTranslation = 0
```

def GalileanGroup.spaceTranslation.incl[\[source\]](#)*Inclusion of spatial translations into the Galilean group.*

```
def spaceTranslation.incl :  
  Multiplicative (EuclideanSpace ℝ (Fin d)) →* GalileanGroup d where
```

def GalileanGroup.spaceTranslationGroup.incl[\[source\]](#)*Inclusion of the existing spatial translation subgroup into the Galilean group.*

```
noncomputable def spaceTranslationGroup.incl :
  EuclideanGroup.TranslationGroup d →* GalileanGroup d
```

def GalileanGroup.ofTimeTranslation

[\[source\]](#)

A pure time translation as a Galilean transformation.

```
def ofTimeTranslation (b : Time) : GalileanGroup d
```

lemma GalileanGroup.ofTimeTranslation_rotation

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_rotation (b : Time) :
  (ofTimeTranslation (d := d) b).rotation = 1
```

lemma GalileanGroup.ofTimeTranslation_velocity

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_velocity (b : Time) :
  (ofTimeTranslation (d := d) b).velocity = 0
```

lemma GalileanGroup.ofTimeTranslation_spaceTranslation

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_spaceTranslation (b : Time) :
  (ofTimeTranslation (d := d) b).spaceTranslation = 0
```

lemma GalileanGroup.ofTimeTranslation_timeTranslation

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_timeTranslation (b : Time) :
  (ofTimeTranslation (d := d) b).timeTranslation = b
```

def GalileanGroup.timeTranslation.incl

[\[source\]](#)

Inclusion of time translations into the Galilean group.

```
def timeTranslation.incl : Multiplicative Time →* GalileanGroup d where
```

2.69 Physlib.SpaceAndTime.Space.Basic

[Physlib/SpaceAndTime/Space/Basic.lean](#) on GitHub.

instance Space.instNontrivial

[\[source\]](#)

```
instance instNontrivial {d : ℕ} [NeZero d] : Nontrivial (Space d) where
```

def Space.homEuclideanSpaceSpace

[\[source\]](#)

'Space d' is homoemorphic to d-dimensional euclidean space. This is used to define the 'IsManifold' instance on 'Space d'.

```
noncomputable def homEuclideanSpaceSpace (d : ℕ) : EuclideanSpace ℝ (Fin d) ≈ₜ Space d where
```

```
instance Space.«instance@b9d497fa» \[source\]
```

```
noncomputable instance (d : ℕ) : ChartedSpace (EuclideanSpace ℝ (Fin d)) (Space d)
```

```
instance Space.«instance@d376417d» \[source\]
```

```
instance (d : ℕ) : IsManifold  $\square$ (d)  $\top$  (Space d)
```

```
lemma Space.chartAt_eq \[source\]
```

```
lemma chartAt_eq (d : ℕ) (p : Space d) :  
  chartAt (EuclideanSpace ℝ (Fin d)) p =  
  (homEuclideanSpaceSpace d).symm.toOpenPartialHomeomorph
```

2.70 Physlib.SpaceAndTime.Space.Derivatives.Basic

[Physlib/SpaceAndTime/Space/Derivatives/Basic.lean](#) on GitHub.

```
lemma Space.deriv_sub \[source\]
```

Derivatives on space distribute over subtraction.

```
@[to_fun]  
lemma deriv_sub [NormedAddCommGroup M] [NormedSpace ℝ M]  
  (f1 f2 : Space d → M) (hf1 : Differentiable ℝ f1) (hf2 : Differentiable ℝ f2) :  
   $\partial[u]$  (f1 - f2) =  $\partial[u]$  f1 -  $\partial[u]$  f2
```

2.71 Physlib.SpaceAndTime.Space.Derivatives.Curl

[Physlib/SpaceAndTime/Space/Derivatives/Curl.lean](#) on GitHub.

```
lemma Space.distCurl_coord_apply \[source\]
```

```
lemma distCurl_coord_apply {i : Fin 3}  
  (f : Space  $\rightarrow$  dℝ[] (EuclideanSpace ℝ (Fin 3)))  $\eta$  ( :  $\square$ (Space, ℝ)) :  
   $\nabla^d$ ( $\times$  f)  $\eta$  i = - f (SchwartzMap.evalCLM ℝ Space ℝ (basis (i+2)) (fderivCLM ℝ Space ℝ  $\eta$ )) (i  
+1)  
+ f (SchwartzMap.evalCLM ℝ Space ℝ (basis (i+1)) (fderivCLM ℝ Space ℝ  $\eta$ )) (i+2)
```

2.72 Physlib.SpaceAndTime.Space.Derivatives.Grad

[Physlib/SpaceAndTime/Space/Derivatives/Grad.lean](#) on GitHub.

```
lemma Space.grad_eq_gradient \[source\]
```

```
lemma grad_eq_gradient {d} (f : Space d → ℝ) :  
   $\nabla$  f = basis.repr  $\circ$  gradient f
```

lemma Space.distGrad_const[\[source\]](#)

```
@[simp]
lemma distGrad_const {d} (c : ℝ) :
  ∇d (Distribution.const ℝ (Space d) c) = 0
```

2.73 Physlib.SpaceAndTime.Space.Derivatives.Laplacian

[Physlib/SpaceAndTime/Space/Derivatives/Laplacian.lean](#) on GitHub.

lemma Space.laplacian_eq_sum_snd_deriv[\[source\]](#)

```
lemma laplacian_eq_sum_snd_deriv {d} (f : Space d → ℝ) :
  Δ f = fun x => ∑ i, ∂[i] ∂([i] f) x
```

def Space.distLaplacian[\[source\]](#)

The distributional ‘distLaplacian’ operator.

```
noncomputable def distLaplacian {d} :
  ((Space d) → ℝ) → ℝ → (Space d) → ℝ
```

lemma Space.distLaplacian_const[\[source\]](#)

```
@[simp]
lemma distLaplacian_const {d : ℕ} (c : ℝ) :
  Δd (Distribution.const ℝ (Space d) c) = 0
```

2.74 Physlib.SpaceAndTime.Space.DistOfFunction

[Physlib/SpaceAndTime/Space/DistOfFunction.lean](#) on GitHub.

lemma Space.distDeriv_distOfFunction[\[source\]](#)

The distributional spatial derivative of the distribution associated to a differentiable function is the distribution associated to the corresponding pointwise spatial derivative.

```
lemma distDeriv_distOfFunction {d : ℕ} μ( : Fin d) (f : Space d → F)
  (hf : IsDistBounded f) (hf_deriv : IsDistBounded ∂μ([ ] f))
  (hf_diff : Differentiable ℝ f) :
  ∂dμ([ ] (distOfFunction f hf)) = distOfFunction ∂μ([ ] f) hf_deriv
```

2.75 Physlib.SpaceAndTime.Space.EuclideanGroup.Action

[Physlib/SpaceAndTime/Space/EuclideanGroup/Action.lean](#) on GitHub.

instance EuclideanGroup.«instance@6372eb27»[\[source\]](#)

The action of the Euclidean group on the affine space of points ‘Space d’: ‘g = ⟨t, Q⟩’ rotates ‘p’ about the coordinate origin by the orthogonal part ‘Q’ and then translates by ‘t’.

```
noncomputable instance : MulAction (EuclideanGroup d) (Space d) where
```

lemma EuclideanGroup.dist_smul[\[source\]](#)

The Euclidean group acts on ‘Space d’ *by isometries*: every rigid motion preserves distance.

```
lemma dist_smul (g : EuclideanGroup d) (p q : Space d) :
  dist (g • p) (g • q) = dist p q
```

lemma EuclideanGroup.rotation_smul_vsub_origin[\[source\]](#)

A rotation acts on the displacement from the origin by its orthogonal part, for every ‘p’: ‘(r • p) -_v origin = Q • (p -_v origin)’.

```
lemma rotation_smul_vsub_origin (r : RotationGroup d) (p : Space d) :
  (r • p) - (0 : Space d) = (r : EuclideanGroup d).linear • (p - (0 : Space d))
```

lemma EuclideanGroup.rotation_dist_smul[\[source\]](#)

The rotation group acts on ‘Space d’ *by isometries* (inherited from ‘dist_smul’).

```
lemma rotation_dist_smul (r : RotationGroup d) (p q : Space d) :
  dist (r • p) (r • q) = dist p q
```

lemma EuclideanGroup.chartEuclidean_smul[\[source\]](#)

Under the standard chart, the Euclidean action on ‘Space d’ is the transport of ‘toAffineIsometryMulEquiv’ acting on ‘EuclideanSpace’: ‘chart (g • p) = (toAffineIsometryMulEquiv g) (chart p)’.

```
lemma chartEuclidean_smul (g : EuclideanGroup d) (p : Space d) :
  Space.chartEuclidean d (g • p) = toAffineIsometryMulEquiv g (Space.chartEuclidean d p)
```

2.76 Physlib.SpaceAndTime.Space.EuclideanGroup.AffineGroup

[Physlib/SpaceAndTime/Space/EuclideanGroup/AffineGroup.lean](#) on GitHub.

def EuclideanGroup.orthogonalToLinearIsometryEquiv[\[source\]](#)

An orthogonal matrix viewed as a linear isometry equivalence of ‘EuclideanSpace $\mathbb{R}$ (Fin n)’; the linear ingredient of the first leg ‘toAffineIsometryHom’.

```
noncomputable def orthogonalToLinearIsometryEquiv
  (Q : Matrix.orthogonalGroup (Fin n)  $\mathbb{R}$ ) :
  EuclideanSpace  $\mathbb{R}$  (Fin n)  $\approx_{\text{LI}}$   $\mathbb{R}[]$  EuclideanSpace  $\mathbb{R}$  (Fin n)
```

def EuclideanGroup.toAffineIsometryHom[\[source\]](#)

The first leg of the inclusion: ‘(t, Q) $\mapsto$ (x $\mapsto$ Q x + t)’, bundled as a monoid homomorphism from the Euclidean group into the affine isometry group of ‘EuclideanSpace $\mathbb{R}$ (Fin n)’.

```
noncomputable def toAffineIsometryHom :
  EuclideanGroup n  $\rightarrow^*$ 
  AffineIsometryEquiv  $\mathbb{R}$  (EuclideanSpace  $\mathbb{R}$  (Fin n)) (EuclideanSpace  $\mathbb{R}$  (Fin n)) where
```


def AffineIsometryEquiv.toAffineEquivHom[\[source\]](#)

The second leg of the inclusion: ‘AffineIsometryEquiv.toAffineEquiv’ bundled as a monoid homomorphism into the affine automorphism group.

```
noncomputable def _root_.AffineIsometryEquiv.toAffineEquivHom :
  AffineIsometryEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n)) →*
  AffineEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n)) where
```

def EuclideanGroup.toAffineEquiv[\[source\]](#)

The inclusion of the Euclidean group into Mathlib’s affine automorphism group: the composite of the two legs ‘toAffineIsometryHom’ and ‘AffineIsometryEquiv.toAffineEquivHom’.

```
noncomputable def toAffineEquiv :
  EuclideanGroup n →* AffineEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n))
```

def EuclideanGroup.linearIsometryEquivToOrthogonal[\[source\]](#)

The ‘invFun’ ingredient of ‘toAffineIsometryMulEquiv’: a linear isometry equivalence read back as an orthogonal matrix, inverse to the linear bridge ‘orthogonalToLinearIsometryEquiv’ (see the round-trip ‘@[simp]’ lemmas below).

```
noncomputable def linearIsometryEquivToOrthogonal
  (L : EuclideanSpace ℝ (Fin n) ≈ℝ EuclideanSpace ℝ (Fin n)) :
  Matrix.orthogonalGroup (Fin n) ℝ
```

def EuclideanGroup.toAffineIsometryMulEquiv[\[source\]](#)

‘EuclideanGroup n ≈ AffineIsometryEquiv ℝ (EuclideanSpace ℝ (Fin n)) _’: the Euclidean group is the full group of affine isometries of Euclidean space. This upgrades ‘toAffineIsometryHom’ to a group isomorphism with the same underlying map.*

```
noncomputable def toAffineIsometryMulEquiv :
  EuclideanGroup n ≈*
  AffineIsometryEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n)) where
```

2.77 Physlib.SpaceAndTime.Space.EuclideanGroup.Basic

[Physlib/SpaceAndTime/Space/EuclideanGroup/Basic.lean](#) on GitHub.

structure EuclideanGroup[\[source\]](#)

An n -dimensional ‘Euclidean group’ is a group of rotations, reflections, and translations.

```
@[ext]
structure EuclideanGroup (n : ℕ) where
  translation : EuclideanSpace ℝ (Fin n)
  linear : Matrix.orthogonalGroup (Fin n) ℝ
```

instance EuclideanGroup.«instance@134bed7b»

[\[source\]](#)

*Group structure on $E(n) = \mathbb{R}^n \rtimes O(n)$. The orthogonal component acts on translations by the inherited matrix-vector action on $EuclideanSpace \mathbb{R} (Fin n)$, transported through $WithLp$ from the coordinate action $Q.val *_v v.ofLp$.*

noncomputable instance : Group (EuclideanGroup n) **where**

lemma EuclideanGroup.det_coe_inv

[\[source\]](#)

private lemma det_coe_inv {n : ℕ} (Q : Matrix.orthogonalGroup (Fin n) ℝ) :
 $(-Q^1).val.det = (Q.val.det)^{-1}$

lemma EuclideanGroup.coe_inv

[\[source\]](#)

private lemma coe_inv {n : ℕ} (Q : Matrix.orthogonalGroup (Fin n) ℝ) :
 $(-Q^1).val = (Q.val)^{-1}$

def EuclideanGroup.SpecialEuclideanGroup

[\[source\]](#)

Special Euclidean Group is the subgroup with $\det(Q) = 1$ where $Q \in O(n)$

def SpecialEuclideanGroup (n : ℕ) : Subgroup (EuclideanGroup n) **where**

def EuclideanGroup.SpecialEuclideanGroup.incl

[\[source\]](#)

The inclusion of the special Euclidean group into the Euclidean group.

noncomputable def SpecialEuclideanGroup.incl (n : ℕ) :
 SpecialEuclideanGroup n →* EuclideanGroup n

def EuclideanGroup.TranslationGroup

[\[source\]](#)

Translation is the subgroup with $Q = 1$.

def TranslationGroup (n : ℕ) : Subgroup (EuclideanGroup n) **where**

def EuclideanGroup.TranslationGroup.incl

[\[source\]](#)

The inclusion of the translation subgroup into the Euclidean group.

noncomputable def TranslationGroup.incl (n : ℕ) :
 TranslationGroup n →* EuclideanGroup n

def EuclideanGroup.translationVector.incl

[\[source\]](#)

MonoidHom including a translation vector into the Euclidean Group.

def translationVector.incl (n : ℕ) :
 Multiplicative (EuclideanSpace ℝ (Fin n)) →* EuclideanGroup n **where**

lemma EuclideanGroup.translationVector.incl_range[\[source\]](#)

An API feature: the translation vector inclusion image is the ‘TranslationGroup’ carrier.

```
lemma translationVector.incl_range :
  Set.range (@translationVector.incl n) = (TranslationGroup n : Set (EuclideanGroup n))
```

lemma EuclideanGroup.translation_zero[\[source\]](#)

The translation by the zero vector is the identity of the Euclidean group.

```
lemma translation_zero : translationVector.incl n
  (Multiplicative.ofAdd (0 : EuclideanSpace ℝ (Fin n))) = 1
```

def EuclideanGroup.OriginStabilizer[\[source\]](#)

The subgroup of ‘EuclideanGroup n’ whose elements fix the origin (translation = 0). This is the copy of ‘O(n)’ sitting inside ‘E(n)’.

```
def OriginStabilizer (n : ℕ) : Subgroup (EuclideanGroup n) where
```

def EuclideanGroup.RotationGroup[\[source\]](#)

Rotation Group is the subgroup of $E(n)$ consisting of rotations about the origin: elements with ‘det = 1’ (orientation-preserving) and ‘translation = 0’ (origin-fixing).

```
noncomputable def RotationGroup (n : ℕ) : Subgroup (EuclideanGroup n)
```

def EuclideanGroup.RotationGroup.incl[\[source\]](#)

The inclusion of the rotation subgroup into the Euclidean group.

```
noncomputable def RotationGroup.incl (n : ℕ) :
  RotationGroup n →* EuclideanGroup n
```

def EuclideanGroup.RotationsAbout[\[source\]](#)

*The subgroup of rotation about a spatial point ‘p : EuclideanSpace ℝ (Fin n)’ consists of elements of the form $T(p) * r * T(-p)$ with $T(\cdot) : \text{translationVector.incl } n$ (Multiplicative.ofAdd $\cdot$) and $r : \text{RotationGroup}$ where r is viewed as a rotation about the origin. Note $T(-p) = T(p)^{-1}$.*

```
noncomputable def RotationsAbout : Subgroup (EuclideanGroup n) where
```

def EuclideanGroup.RotationsAbout.incl[\[source\]](#)

The inclusion of rotations about ‘p’ into the Euclidean group.

```
noncomputable def RotationsAbout.incl : RotationsAbout p →* EuclideanGroup n
```

def EuclideanGroup.RotationsAbout.toOrigin[\[source\]](#)

Conjugate a rotation about 'p' back to a rotation about the origin.

```
noncomputable def RotationsAbout.toOrigin :
  RotationsAbout p →* RotationGroup n where
```

def EuclideanGroup.RotationsAbout.fromOrigin[\[source\]](#)

Conjugate a rotation about the origin to a rotation about 'p'.

```
noncomputable def RotationsAbout.fromOrigin :
  RotationGroup n →* RotationsAbout p where
```

lemma EuclideanGroup.RotationsAbout.fromOrigin_comp_toOrigin[\[source\]](#)

*'RotationsAbout.toOrigin p' followed by 'RotationsAbout.fromOrigin p' is the identity;
the forward leg of the isomorphism 'RotationsAboutEquiv'.*

```
lemma RotationsAbout.fromOrigin_comp_toOrigin :
  (RotationsAbout.fromOrigin p).comp (RotationsAbout.toOrigin p) =
  MonoidHom.id (RotationsAbout p)
```

lemma EuclideanGroup.RotationsAbout.toOrigin_comp_fromOrigin[\[source\]](#)

*'RotationsAbout.fromOrigin p' followed by 'RotationsAbout.toOrigin p' is the identity;
the backward leg of the isomorphism 'RotationsAboutEquiv'.*

```
lemma RotationsAbout.toOrigin_comp_fromOrigin :
  (RotationsAbout.toOrigin p).comp (RotationsAbout.fromOrigin p) =
  MonoidHom.id (RotationGroup n)
```

def EuclideanGroup.RotationsAboutEquiv[\[source\]](#)

*API feature: conjugation by the translation 'T(p)' exhibits the rotations about 'p' as
isomorphic to the rotations about the origin 'RotationGroup n'.*

```
noncomputable def RotationsAboutEquiv : RotationsAbout p ≈* RotationGroup n
```

lemma EuclideanGroup.RotationsAbout_zero[\[source\]](#)

API feature: the degenerate identity that 'RotationsAbout 0 = RotationGroup n'

```
lemma RotationsAbout_zero : RotationsAbout (0 : EuclideanSpace ℝ (Fin n)) = RotationGroup n
```

def EuclideanGroup.specialOrthogonal.incl[\[source\]](#)

*Rotations are members of special orthogonal groups and can be viewed as members
of orthogonal groups.*

```
def specialOrthogonal.incl (n : ℕ) :
  Matrix.specialOrthogonalGroup (Fin n) ℝ →* Matrix.orthogonalGroup (Fin n) ℝ
```

def EuclideanGroup.ofRotation[\[source\]](#)

The Euclidean group element given by a rotation about the origin (zero translation).

```
def ofRotation (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ) :
  EuclideanGroup n
```

def EuclideanGroup.ofRotationTranslation[\[source\]](#)

Specialization to a group element from a rotation and a translation.

```
def ofRotationTranslation (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) : EuclideanGroup n
```

lemma EuclideanGroup.ofRotationTranslation_translation[\[source\]](#)

The specialization projects back to the translation component.

```
@[simp]
lemma ofRotationTranslation_translation (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) :
  (ofRotationTranslation Q t).translation = t
```

lemma EuclideanGroup.ofRotationTranslation_linear[\[source\]](#)

The specialization projects back to the linear (rotation) component.

```
@[simp]
lemma ofRotationTranslation_linear (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) :
  (ofRotationTranslation Q t).linear = specialOrthogonal.incl n Q
```

lemma EuclideanGroup.ofRotationTranslation_decompose[\[source\]](#)

API feature: the inclusion image decomposes as group product.

```
@[simp]
lemma ofRotationTranslation_decompose (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) :
  (ofRotationTranslation Q t) =
  (translationVector.incl n (Multiplicative.ofAdd t)) * (ofRotation (Q))
```

def EuclideanGroup.specialOrthogonal.toRotation[\[source\]](#)

The isomorphism's forward map: a special orthogonal matrix as a rotation about the origin.

```
noncomputable def specialOrthogonal.toRotation (n : ℕ) :
  Matrix.specialOrthogonalGroup (Fin n) ℝ →* RotationGroup n where
```

def EuclideanGroup.specialOrthogonal.fromRotation[\[source\]](#)

The isomorphism's inverse map: the linear part of a rotation about the origin, as a special orthogonal matrix.

```
noncomputable def specialOrthogonal.fromRotation (n : ℕ) :
  RotationGroup n →* Matrix.specialOrthogonalGroup (Fin n) ℝ where
```

lemma EuclideanGroup.specialOrthogonal.fromRotation_comp_toRotation

[\[source\]](#)

‘specialOrthogonal.toRotation n’ followed by ‘specialOrthogonal.fromRotation n’ is the identity; the forward leg of the isomorphism ‘specialOrthogonalEquiv’.

```
lemma specialOrthogonal.fromRotation_comp_toRotation :
  (specialOrthogonal.fromRotation n).comp (specialOrthogonal.toRotation n) =
  MonoidHom.id (Matrix.specialOrthogonalGroup (Fin n) ℝ)
```

lemma EuclideanGroup.specialOrthogonal.toRotation_comp_fromRotation

[\[source\]](#)

‘specialOrthogonal.fromRotation n’ followed by ‘specialOrthogonal.toRotation n’ is the identity; the backward leg of the isomorphism ‘specialOrthogonalEquiv’.

```
lemma specialOrthogonal.toRotation_comp_fromRotation :
  (specialOrthogonal.toRotation n).comp (specialOrthogonal.fromRotation n) =
  MonoidHom.id (RotationGroup n)
```

def EuclideanGroup.specialOrthogonalEquiv

[\[source\]](#)

API feature: $SO(n) \simeq^ \text{RotationGroup } n$*

```
noncomputable def specialOrthogonalEquiv :
  Matrix.specialOrthogonalGroup (Fin n) ℝ ≈* RotationGroup n
```

2.78 Physlib.SpaceAndTime.Space.Integrals.Basic

[Physlib/SpaceAndTime/Space/Integrals/Basic.lean](#) on GitHub.

lemma Space.integrable_spherical_of_integrable

[\[source\]](#)

```
lemma integrable_spherical_of_integrable {d : ℕ} {F : Type*}
  [NormedAddCommGroup F] [NormedSpace ℝ F] {f : Space d → F}
  (hf : Integrable f volume) :
  Integrable
    (fun x : ↑(Metric.sphere (0 : Space d) 1) × Set.Ioi (0 : ℝ) =>
      f (x.2.1 • x.1.1))
  (volume α( := Space d).toSphere.prod
    (Measure.volumeIoiPow (Module.finrank ℝ (Space d) - 1)))
```

lemma Space.integral_volume_eq_spherical_iterated

[\[source\]](#)

```
lemma integral_volume_eq_spherical_iterated {d : ℕ} [NeZero d] {F : Type*}
  [NormedAddCommGroup F] [NormedSpace ℝ F] (f : Space d → F)
  (hf : Integrable f volume) :
  ∫ x : Space d, f x =
  ∫ n : ↑(Metric.sphere (0 : Space d) 1),
  ∫ r : Set.Ioi (0 : ℝ), f (r.1 • n.1)
    ∂(Measure.volumeIoiPow (Module.finrank ℝ (Space d) - 1))
    ∂(volume α( := Space d).toSphere)
```

lemma Space.integral_volume_eq_spherical_integral[\[source\]](#)

```

lemma integral_volume_eq_spherical_integral {d : ℕ} [NeZero d] {F : Type*}
  [NormedAddCommGroup F] [NormedSpace ℝ F] (f : Space d → F)
  (hf : Integrable f volume) :
  ∫ x : Space d, f x =
  ∫ n : ↑(Metric.sphere (0 : Space d) 1),
  ∫ r : Set.Ioi (0 : ℝ), (r.1 ^ (d - 1)) • f (r.1 • n.1)
    ∂(.comap Subtype.val volume)
  ∂(volume α( := Space d).toSphere)

```

2.79 Physlib.SpaceAndTime.Space.Integrals.NormPow

[Physlib/SpaceAndTime/Space/Integrals/NormPow.lean](#) on GitHub.

lemma Space.radial_jacobian_zpow_mul_self[\[source\]](#)

```

lemma radial_jacobian_zpow_mul_self
  {d p : ℕ} [NeZero d] {q : ℤ} (hp_int : (p : ℤ) = q + (d : ℤ))
  {r : ℝ} (hr : 0 < r) :
  r ^ (d - 1) * (r ^ q * r) = r ^ p

```

lemma Space.radial_jacobian_zpow[\[source\]](#)

```

private lemma radial_jacobian_zpow
  {d p : ℕ} [NeZero d] {q : ℤ} (hp_int : (p : ℤ) = q + (d : ℤ))
  (hp_pos : 0 < p) {r : ℝ} (hr : 0 < r) :
  r ^ (d - 1) * r ^ q = r ^ (p - 1)

```

lemma Space.radial_norm_power_spherical_integral_eq_space_integral[\[source\]](#)

```

lemma radial_norm_power_spherical_integral_eq_space_integral
  {d p : ℕ} [NeZero d] {q : ℤ} (hp_int : (p : ℤ) = q + (d : ℤ))
  (hp_pos : 0 < p) η( : □(Space d, ℝ)) :
  ∫ x : Space d, η x * ||x ^ q =
  ∫ (n : ↑(Metric.sphere (0 : Space d) 1)),
  ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ (p - 1) * η (r.1 • n.1)
    ∂(.comap Subtype.val volume)
  ∂(volume α( := Space d).toSphere)

```

2.80 Physlib.SpaceAndTime.Space.Module

[Physlib/SpaceAndTime/Space/Module.lean](#) on GitHub.

lemma Space.norm_smul_sphere[\[source\]](#)

```

lemma norm_smul_sphere {d : ℕ} (n : ↑(Metric.sphere (0 : Space d) 1))
  {r : ℝ} (hr : 0 ≤ r) :
  ||(r • (n : Space d))|| = r

```

2.81 Physlib.SpaceAndTime.Space.Norm.Basic

[Physlib/SpaceAndTime/Space/Norm/Basic.lean](#) on GitHub.

lemma Space.integrable_real_pow_mul_schwartz[\[source\]](#)

```
private lemma integrable_real_pow_mul_schwartz
  (ψ : ℝ(, ℝ)) (k : ℕ) :
  Integrable (fun x : ℝ => x ^ k * ψ x) volume
```

lemma Space.radial_power_deriv_integral_by_parts[\[source\]](#)

```
private lemma radial_power_deriv_integral_by_parts
  {d : ℕ} η( : ℝ(Space d, ℝ))
  (n : ↑(Metric.sphere (0 : Space d) 1))
  (p : ℕ) (hp : 0 < p) :
  - ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ p * (_root_.deriv (fun a => η (a • n.1)) r.1)
    ∂(.comap Subtype.val volume)
  =
  (p : ℝ) * ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ (p - 1) * η (r.1 • n.1)
    ∂(.comap Subtype.val volume)
```

lemma Space.distDiv_norm_zpow_smul_repr_self_apply_eq_radial_deriv[\[source\]](#)

```
private lemma distDiv_norm_zpow_smul_repr_self_apply_eq_radial_deriv
  {d p : ℕ} [NeZero d] (q : ℤ) (hq : 0 < q + (d : ℤ))
  (hp_int : (p : ℤ) = q + (d : ℤ))
  η( : ℝ(Space d, ℝ)) :
  ∇d( ∫ (distOfFunction (fun x : Space d => |||x ^ q • basis.repr x)
    (IsDistBounded.zpow_smul_repr_self q (by omega)))) η =
  - ∫ n, ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ p * (_root_.deriv (fun a => η (a • n.1)) r.1)
    ∂(.comap Subtype.val volume)
  ∂(volume α( := Space d).toSphere)
```

lemma Space.distDiv_norm_zpow_smul_repr_self_eq_smul[\[source\]](#)

```
lemma distDiv_norm_zpow_smul_repr_self_eq_smul
  {d : ℕ} [NeZero d] (q : ℤ) (hq : 0 < q + (d : ℤ)) :
  ∇d ∫ (distOfFunction (fun x : Space d => |||x ^ q • basis.repr x)
    (IsDistBounded.zpow_smul_repr_self q (by omega))) =
  (((q + d : ℤ) : ℝ) •
    distOfFunction (fun x : Space d => |||x ^ q)
    (IsDistBounded.pow q (by omega)))
```

lemma Space.distLaplacian_distOfFunction_norm_zpow[\[source\]](#)

```
lemma distLaplacian_distOfFunction_norm_zpow {d : ℕ} [NeZero d] (m : ℤ)
  (hdiv : 0 < m - 2 + (d : ℤ)) :
  Δd (distOfFunction (fun x : Space d => |||x ^ m)
    (IsDistBounded.pow m (by omega))) =
  (((m : ℝ) * ((m - 2 + d : ℤ) : ℝ))) •
    distOfFunction (fun x : Space d => |||x ^ (m - 2))
    (IsDistBounded.pow (m - 2) (by omega)))
```

lemma Space.distLaplacian_fundamentalSolution_norm_zpow[\[source\]](#)

The distributional Laplacian of $\|x\|^{-(2-d)}$ is $(2-d) * d * \text{volume}(\text{Metric.ball } 0 \ 1)$ times the Dirac delta at the origin. For $d \geq 3$ this $\|x\|^{-(2-d)}$ is the (singular) fundamental solution of the Laplacian, and for $d = 1$ it is $\|x\|$. When $d = 2$ the exponent vanishes, so the identity collapses to the trivial $\Delta^d 1 = 0$; the genuine two-dimensional fundamental solution is the logarithm, proved in `distLaplacian_fundamentalSolution_log_norm`. The statement also holds vacuously for $d = 0$, where the space is trivial.

```
lemma distLaplacian_fundamentalSolution_norm_zpow {d : ℕ} :
  Δd (distOfFunction (fun x : Space d => |||x ^ (- ((d : ℤ) - 2)))
    (IsDistBounded.pow _ (by omega))) =
    ((- ((d : ℝ) - 2)) * d *
      (volume α( := Space d)).real (Metric.ball 0 1)) • diracDelta ℝ 0
```

lemma Space.distLaplacian_fundamentalSolution_log_norm

[\[source\]](#)

In dimension two the fundamental solution of the Laplacian is the logarithm: the distributional Laplacian of $\text{Real.log } \|x\|$ is $2 * \text{volume}(\text{Metric.ball } 0 \ 1)$ times the Dirac delta at the origin.

```
lemma distLaplacian_fundamentalSolution_log_norm :
  Δd (distOfFunction (fun x : Space 2 => Real.log |||x) IsDistBounded.log_norm) =
    (2 * (volume α( := Space 2)).real (Metric.ball 0 1)) • diracDelta ℝ 0
```

2.82 Physlib.SpaceAndTime.Space.Norm.IteratedLaplacian

[Physlib/SpaceAndTime/Space/Norm/IteratedLaplacian.lean](#) on GitHub.

def Space.oddNormIteratedLaplacianCoeff

[\[source\]](#)

The scalar factor in the odd-dimensional iterated Laplacian of the norm.

```
noncomputable def oddNormIteratedLaplacianCoeff (m : ℕ) : ℝ
```

lemma Space.oddNormIteratedLaplacianCoeff_factor_ne_zero

[\[source\]](#)

```
private lemma oddNormIteratedLaplacianCoeff_factor_ne_zero {m k : ℕ} (hk : k < m) :
  (((1 : ℤ) - 2 * (k : ℤ) : ℤ) : ℝ) *
  (((1 : ℤ) - 2 * (k : ℤ) - 2 + (2 * m + 1 : ℤ) : ℤ) : ℝ)) ≠ 0
```

lemma Space.oddNormIteratedLaplacianCoeff_ne_zero

[\[source\]](#)

The scalar factor in the odd-dimensional iterated Laplacian of the norm is nonzero.

```
lemma oddNormIteratedLaplacianCoeff_ne_zero (m : ℕ) :
  oddNormIteratedLaplacianCoeff m ≠ 0
```

lemma Space.distLaplacian_norm_zpow_odd_boundary

[\[source\]](#)

```
private lemma distLaplacian_norm_zpow_odd_boundary (m : ℕ) :
  Δd (distOfFunction (fun x : Space (2 * m + 1) =>
    |||x ^ ((1 : ℤ) - 2 * (m : ℤ)))
    (IsDistBounded.pow _ (by omega))) =
    (((1 : ℤ) - 2 * (m : ℤ) : ℝ) * (2 * m + 1 : ℝ) *
```

```
(volume α( := Space (2 * m + 1))).real (Metric.ball 0 1)) •
  diracDelta ℝ 0
```

lemma Space.iterated_distLaplacian_norm_zpow_odd_until_boundary

[\[source\]](#)

```
private lemma iterated_distLaplacian_norm_zpow_odd_until_boundary
  (m k : ℕ) (hk : k ≤ m) :
  ((distLaplacian (d := 2 * m + 1))^[k])
  (distOfFunction (fun x : Space (2 * m + 1) => |||x ^ (1 : ℤ))
    (IsDistBounded.pow 1 (by omega))) =
  ∏( j ∈ Finset.range k,
    (((1 : ℤ) - 2 * (j : ℤ) : ℤ) : ℝ) *
    (((1 : ℤ) - 2 * (j : ℤ) - 2 + (2 * m + 1 : ℤ) : ℤ) : ℝ))) •
  distOfFunction (fun x : Space (2 * m + 1) => |||x ^ ((1 : ℤ) - 2 * (k : ℤ)))
    (IsDistBounded.pow _ (by omega)))
```

lemma Space.iterated_distLaplacian_norm_zpow_odd_eq_smul_diracDelta

[\[source\]](#)

*In dimension ‘ $2 * m + 1$ ’, the ‘ $(m + 1)$ ’-fold distributional Laplacian of the distribution induced by the norm is a nonzero multiple of the Dirac delta at the origin.*

```
lemma iterated_distLaplacian_norm_zpow_odd_eq_smul_diracDelta (m : ℕ) :
  ((distLaplacian (d := 2 * m + 1))^[m + 1])
  (distOfFunction (fun x : Space (2 * m + 1) => |||x ^ (1 : ℤ))
    (IsDistBounded.pow 1 (by omega))) =
  oddNormIteratedLaplacianCoeff m • diracDelta ℝ 0
```

2.83 Physlib.SpaceAndTime.Space.Norm.Regularized

[Physlib/SpaceAndTime/Space/Norm/Regularized.lean](#) on GitHub.

def Space.normRegularizedPow

[\[source\]](#)

Power of regularized norm, ‘ $(\|x\|^2 + \varepsilon^2)^{(s/2)}$ ’.

```
def normRegularizedPow (d : ℕ) ε( s : ℝ) : Space d → ℝ
```

lemma Space.normRegularizedPow_eq

[\[source\]](#)

```
lemma normRegularizedPow_eq (d : ℕ) ε( s : ℝ) :
  normRegularizedPow d ε s = fun x => ||(|x ^ 2 + ε ^ 2) ^ (s / 2)
```

lemma Space.norm_sq_add_unit_sq_pos

[\[source\]](#)

For a nonzero regularization parameter, ‘ $\|x\|^2 + \varepsilon^2$ ’ is positive.

```
lemma norm_sq_add_unit_sq_pos {d : ℕ} ε( : ℝ×) (x : Space d) : 0 < |||x ^ 2 + ε ^ 2
```

lemma Space.normRegularizedPow_pos

[\[source\]](#)

The regularized norm power is positive for nonzero regularization parameter.

```
lemma normRegularizedPow_pos (d : ℕ) ε( : ℝ×) (s : ℝ) (x : Space d) :
  0 < normRegularizedPow d ε s x
```

lemma Space.normRegularizedPow_hasTemperateGrowth[\[source\]](#)*The regularized norm power has temperate growth.*

```
lemma normRegularizedPow_hasTemperateGrowth (d : ℕ) ε( : ℝ×) (s : ℝ) :
  HasTemperateGrowth (normRegularizedPow d ε s)
```

lemma Space.normRegularizedPow_measurable[\[source\]](#)

```
@[fun_prop]
lemma normRegularizedPow_measurable (d : ℕ) ε( s : ℝ) :
  Measurable (normRegularizedPow d ε s)
```

2.84 Physlib.SpaceAndTime.Space.Origin[Physlib/SpaceAndTime/Space/Origin.lean](#) on GitHub.**instance Space.«instance@570dcd05»**[\[source\]](#)

```
instance {d} : Zero (Space d) where
```

lemma Space.zero_val[\[source\]](#)

```
@[simp]
lemma zero_val {d : ℕ} : (0 : Space d).val = fun _ => 0
```

lemma Space.zero_apply[\[source\]](#)

```
@[simp]
lemma zero_apply {d : ℕ} (i : Fin d) :
  (0 : Space d) i = 0
```

def Space.vectorToSpace[\[source\]](#)*A Euclidean vector, based at the chosen origin, viewed as a point of ‘Space d’.*

```
noncomputable def vectorToSpace {d : ℕ} (v : EuclideanSpace ℝ (Fin d)) : Space d
```

lemma Space.vectorToSpace_apply[\[source\]](#)

```
@[simp]
lemma vectorToSpace_apply {d : ℕ} (v : EuclideanSpace ℝ (Fin d)) (i : Fin d) :
  vectorToSpace v i = v i
```

lemma Space.vectorToSpace_vsub_zero[\[source\]](#)

```
@[simp]
lemma vectorToSpace_vsub_zero {d : ℕ} (v : EuclideanSpace ℝ (Fin d)) :
  vectorToSpace v v - (0 : Space d) = v
```

def Space.chartEuclidean[\[source\]](#)

The standard chart ‘Space $d \simeq^{\text{ai}} \mathbb{R}$ EuclideanSpace $\mathbb{R}$ (Fin d)’ , ‘ $p \mapsto p -_v 0$ ’, identifying a point with its coordinate vector relative to the origin (the vector-space zero ‘ $0 : \text{Space } d$ ’).

```
noncomputable def chartEuclidean (d : ℕ) :
  Space d  $\simeq^{\text{ai}} \mathbb{R}$  EuclideanSpace  $\mathbb{R}$  (Fin d)
```

2.85 Physlib.SpaceAndTime.SpaceTime.Derivatives

[Physlib/SpaceAndTime/SpaceTime/Derivatives.lean](#) on GitHub.

def SpaceTime.manifoldDeriv[\[source\]](#)

The derivative of a function from ‘SpaceTime d’ to a manifold along the ‘ μ ’ coordinate, as a tangent vector at the value of the function.

```
noncomputable def manifoldDeriv {E H N : Type} [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
  [TopologicalSpace H] (I : ModelWithCorners  $\mathbb{R}$  E H) [TopologicalSpace N]
  [ChartedSpace H N] {d : ℕ}  $\mu$  ( : Fin 1  $\otimes$  Fin d) (f : SpaceTime d  $\rightarrow$  N) :
  (y : SpaceTime d)  $\rightarrow$  TangentSpace I (f y)
```

lemma SpaceTime.manifoldDeriv_eq[\[source\]](#)

```
lemma manifoldDeriv_eq {E H N : Type} [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
  [TopologicalSpace H] (I : ModelWithCorners  $\mathbb{R}$  E H) [TopologicalSpace N]
  [ChartedSpace H N] {d : ℕ}  $\mu$  ( : Fin 1  $\otimes$  Fin d) (f : SpaceTime d  $\rightarrow$  N)
  (y : SpaceTime d) :
  manifoldDeriv I  $\mu$  f y =
    mfderiv  $\square \mathbb{R}$ (, SpaceTime d) I f y
    ((Lorentz.Vector.basis  $\mu$  : SpaceTime d) : TangentSpace  $\square \mathbb{R}$ (, SpaceTime d) y)
```

lemma SpaceTime.deriv_eq_mfderiv[\[source\]](#)

The spacetime derivative is the manifold derivative for functions into normed spaces.

```
lemma deriv_eq_mfderiv {M : Type} [NormedAddCommGroup M] [NormedSpace  $\mathbb{R}$  M]
  {d : ℕ}  $\mu$  ( : Fin 1  $\otimes$  Fin d) (f : SpaceTime d  $\rightarrow$  M) (y : SpaceTime d) :
  deriv  $\mu$  f y =
    mfderiv  $\square \mathbb{R}$ (, SpaceTime d)  $\square \mathbb{R}$ (, M) f y
    ((Lorentz.Vector.basis  $\mu$  : SpaceTime d) : TangentSpace  $\square \mathbb{R}$ (, SpaceTime d) y)
```

lemma SpaceTime.deriv_eq_manifoldDeriv[\[source\]](#)

```
lemma deriv_eq_manifoldDeriv {M : Type} [NormedAddCommGroup M] [NormedSpace  $\mathbb{R}$  M]
  {d : ℕ}  $\mu$  ( : Fin 1  $\otimes$  Fin d) (f : SpaceTime d  $\rightarrow$  M) (y : SpaceTime d) :
  deriv  $\mu$  f y = manifoldDeriv  $\square \mathbb{R}$ (, M)  $\mu$  f y
```

lemma SpaceTime.manifoldDeriv_const[\[source\]](#)

```
@[simp]
lemma manifoldDeriv_const {E H N : Type} [NormedAddCommGroup E] [NormedSpace  $\mathbb{R}$  E]
  [TopologicalSpace H] (I : ModelWithCorners  $\mathbb{R}$  E H) [TopologicalSpace N]
  [ChartedSpace H N] {d : ℕ}  $\mu$  ( : Fin 1  $\otimes$  Fin d) (n : N) (y : SpaceTime d) :
```

```
manifoldDeriv I μ (fun _ : SpaceTime d => n) y = 0
```

lemma SpaceTime.contDiff_deriv

[\[source\]](#)

If 'f' is ' C^{n+1} ' then ' $\partial_{\mu} f$ ' is ' C^n '.

@[fun_prop]

```
lemma contDiff_deriv {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M] {d : ℕ}
  {n : WithTop ℕ} μ( : Fin 1 ⊗ Fin d) (f : SpaceTime d → M) (hf : ContDiff ℝ (n + 1) f) :
  ContDiff ℝ n ∂(_ μ f)
```

lemma SpaceTime.differentiable_deriv

[\[source\]](#)

If 'f' is ' C^2 ' then ' $\partial_{\mu} f$ ' is differentiable.

@[fun_prop]

```
lemma differentiable_deriv {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M] {d : ℕ}
  μ( : Fin 1 ⊗ Fin d) (f : SpaceTime d → M) (hf : ContDiff ℝ 2 f) :
  Differentiable ℝ ∂(_ μ f)
```

lemma SpaceTime.deriv_commute

[\[source\]](#)

Derivatives on spacetime commute with one another (Clairaut's theorem).

```
lemma deriv_commute {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M] {d : ℕ}
  μ( v : Fin 1 ⊗ Fin d) (f : SpaceTime d → M) (hf : ContDiff ℝ 2 f) :
  ∂_ μ ∂(_ v f) = ∂_ v ∂(_ μ f)
```

2.86 Physlib.SpaceAndTime.Time.Derivatives

[Physlib/SpaceAndTime/Time/Derivatives.lean](#) on GitHub.

def Time.manifoldDeriv

[\[source\]](#)

The time derivative of a function from 'Time' to a manifold, as a tangent vector at the value of the function.

```
noncomputable def manifoldDeriv {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] (f : Time → N) : (t : Time) → TangentSpace I (f t)
```

lemma Time.manifoldDeriv_eq

[\[source\]](#)

```
lemma manifoldDeriv_eq {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] (f : Time → N) (t : Time) :
  manifoldDeriv I f t =
  mfderiv (I (f t)) (t) (t) (t)
```

lemma Time.deriv_eq_mfderiv

[\[source\]](#)

The time derivative is the manifold derivative for functions into normed spaces.

```

lemma deriv_eq_mfderiv [NormedAddCommGroup M] [NormedSpace ℝ M]
  (f : Time → M) (t : Time) :
  deriv f t =
    mfderiv ℝ(, Time) ℝ(, M) f t
    ((1 : Time) : TangentSpace ℝ(, Time) t)

```

lemma Time.deriv_eq_manifoldDeriv

[\[source\]](#)

```

lemma deriv_eq_manifoldDeriv [NormedAddCommGroup M] [NormedSpace ℝ M]
  (f : Time → M) (t : Time) :
  deriv f t = manifoldDeriv ℝ(, M) f t

```

lemma Time.manifoldDeriv_const

[\[source\]](#)

```

@[simp]
lemma manifoldDeriv_const {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] (n : N) :
  manifoldDeriv I (fun _ : Time => n) t = 0

```

2.87 Physlib.SpaceAndTime.TimeAndSpace.Basic

[Physlib/SpaceAndTime/TimeAndSpace/Basic.lean](#) on GitHub.

abbrev TimeAndSpace

[\[source\]](#)

Euclidean spacetime as the product of time and ‘d’-dimensional space.

```

abbrev TimeAndSpace (d : ℕ := 3)

```

def TimeAndSpace.time

[\[source\]](#)

The time-coordinate projection from ‘TimeAndSpace d’.

```

noncomputable def time {d : ℕ} : TimeAndSpace d → ℝ[] Time

```

def TimeAndSpace.space

[\[source\]](#)

The spatial-coordinate projection from ‘TimeAndSpace d’.

```

noncomputable def space {d : ℕ} : TimeAndSpace d → ℝ[] Space d

```

lemma TimeAndSpace.time_apply

[\[source\]](#)

```

@[simp]
lemma time_apply (tx : TimeAndSpace d) :
  time tx = tx.1

```

lemma TimeAndSpace.space_apply

[\[source\]](#)

```

@[simp]
lemma space_apply (tx : TimeAndSpace d) :
  space tx = tx.2

```

lemma TimeAndSpace.dist_time_le[\[source\]](#)

The time projection is nonexpanding for the product metric.

```
lemma dist_time_le (tx ty : TimeAndSpace d) :
  dist (time tx) (time ty) ≤ dist tx ty
```

lemma TimeAndSpace.dist_space_le[\[source\]](#)

The spatial projection is nonexpanding for the product metric.

```
lemma dist_space_le (tx ty : TimeAndSpace d) :
  dist (space tx) (space ty) ≤ dist tx ty
```

2.88 Physlib.SpaceAndTime.TimeAndSpace.EuclideanGroup.Action

[Physlib/SpaceAndTime/TimeAndSpace/EuclideanGroup/Action.lean](#) on GitHub.

instance TimeAndSpace.instMulActionEuclideanGroup[\[source\]](#)

The Euclidean group acts on ‘TimeAndSpace d’ by fixing time and acting on space.

```
noncomputable instance instMulActionEuclideanGroup :
  MulAction (EuclideanGroup d) (TimeAndSpace d) where
```

lemma TimeAndSpace.smul_mk[\[source\]](#)

Coordinate formula for the Euclidean-group action on ‘TimeAndSpace d’.

```
@[simp]
lemma smul_mk (g : EuclideanGroup d) (t : Time) (x : Space d) :
  g • ((t, x) : TimeAndSpace d) = (t, g • x)
```

lemma TimeAndSpace.fst_smul[\[source\]](#)

The Euclidean-group action fixes the first coordinate.

```
@[simp]
lemma fst_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  (g • tx).1 = tx.1
```

lemma TimeAndSpace.snd_smul[\[source\]](#)

The Euclidean-group action applies to the second coordinate.

```
@[simp]
lemma snd_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  (g • tx).2 = g • tx.2
```

lemma TimeAndSpace.time_smul[\[source\]](#)

The Euclidean-group action fixes the time projection.

```
lemma time_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  time (g • tx) = time tx
```

lemma TimeAndSpace.space_smul[\[source\]](#)

The Euclidean-group action applies to the space projection.

```
lemma space_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  space (g • tx) = g • space tx
```

lemma TimeAndSpace.dist_smul[\[source\]](#)

The Euclidean-group action on 'TimeAndSpace d' preserves product distance.

```
lemma dist_smul (g : EuclideanGroup d) (tx ty : TimeAndSpace d) :
  dist (g • tx) (g • ty) = dist tx ty
```

def TimeAndSpace.actionLinearMap[\[source\]](#)

The linear part of the Euclidean-group action on 'TimeAndSpace d'.

```
noncomputable def actionLinearMap (g : EuclideanGroup d) :
  TimeAndSpace d → LR[] TimeAndSpace d
```

def TimeAndSpace.actionTranslation[\[source\]](#)

The translation part of the Euclidean-group action on 'TimeAndSpace d'.

```
noncomputable def actionTranslation (g : EuclideanGroup d) : TimeAndSpace d
```

lemma TimeAndSpace.smul_eq_actionLinearMap_add[\[source\]](#)

```
lemma smul_eq_actionLinearMap_add (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  g • tx = actionLinearMap g tx + actionTranslation g
```

lemma TimeAndSpace.smul_hasTemperateGrowth[\[source\]](#)

The Euclidean-group action on 'TimeAndSpace d' has temperate growth.

```
lemma smul_hasTemperateGrowth (g : EuclideanGroup d) :
  Function.HasTemperateGrowth (fun tx : TimeAndSpace d => g • tx)
```

lemma TimeAndSpace.isometry_smul[\[source\]](#)

The Euclidean-group action on 'TimeAndSpace d' is an isometry.

```
lemma isometry_smul (g : EuclideanGroup d) :
  Isometry (fun tx : TimeAndSpace d => g • tx)
```

lemma TimeAndSpace.antilipschitz_smul[\[source\]](#)

The Euclidean-group action on 'TimeAndSpace d' is antilipschitz.

```
lemma antilipschitz_smul (g : EuclideanGroup d) :
  AntilipschitzWith 1 (fun tx : TimeAndSpace d => g • tx)
```


2.89 Physlib.SpaceAndTime.TimeAndSpace.EuclideanGroup.SchwartzAction

[Physlib/SpaceAndTime/TimeAndSpace/EuclideanGroup/SchwartzAction.lean](#) on GitHub.

def TimeAndSpace.schwartzEuclideanAction [\[source\]](#)

The Euclidean-group pullback action on Schwartz maps over ‘TimeAndSpace d’, as a continuous representation ‘ContRepresentation $\mathbb{R}$ (EuclideanGroup d) $\mathcal{S}(\text{TimeAndSpace } d, F)$ ’:

‘ContRepresentation $R \ G \ V$ ’ unfolds to the monoid homomorphism ‘ $G \rightarrow^ V \rightarrow L[R] V$ ’ into the **continuous**-linear maps: the builder ‘SchwartzMap.compCLMOfAntilipschitz’ already yields a continuous-linear map for free, and the ‘ $\rightarrow L$ ’ codomain lets the action compose under ‘ $\circ L$ ’. The plain mathlib ‘Representation’ ($G \rightarrow^* V \rightarrow_l[\mathbb{R}] V$) is recovered via ‘ContRepresentation.toRepresentation’.*

```
noncomputable def schwartzEuclideanAction {d : ℕ} :
  ContRepresentation ℝ (EuclideanGroup d)  $\square$ (TimeAndSpace d, F) where
```

lemma TimeAndSpace.schwartzEuclideanAction_apply [\[source\]](#)

Pointwise formula for the monoid-homomorphism form of the Schwartz-map pullback action.

```
@[simp]
lemma schwartzEuclideanAction_apply {d : ℕ} (g : EuclideanGroup d)  $\eta$ ( :  $\square$ (TimeAndSpace d, F))
  (tx : TimeAndSpace d) :
  (schwartzEuclideanAction g  $\eta$ ) tx =  $\eta$  (⁻g¹ • tx)
```

instance TimeAndSpace.instMulActionSchwartzMap [\[source\]](#)

The Euclidean group acts on Schwartz maps over ‘TimeAndSpace d’ by pullback.

```
noncomputable instance instMulActionSchwartzMap {d : ℕ} :
  MulAction (EuclideanGroup d)  $\square$ (TimeAndSpace d, F) where
```

lemma TimeAndSpace.smul_schwartzMap_apply [\[source\]](#)

Pointwise formula for the ‘MulAction’ instance on Schwartz maps.

```
@[simp]
lemma smul_schwartzMap_apply {d : ℕ} (g : EuclideanGroup d)  $\eta$ ( :  $\square$ (TimeAndSpace d, F))
  (tx : TimeAndSpace d) :
  (g •  $\eta$ ) tx =  $\eta$  (⁻g¹ • tx)
```

lemma TimeAndSpace.schwartzEuclideanAction_mul_apply [\[source\]](#)

*Applying ‘g’ and then ‘h’ to a Schwartz map is the pullback action of ‘h * g’.*

```
lemma schwartzEuclideanAction_mul_apply {d : ℕ} (g h : EuclideanGroup d)
   $\eta$ ( :  $\square$ (TimeAndSpace d, F)) :
  schwartzEuclideanAction h (schwartzEuclideanAction g  $\eta$ ) =
  schwartzEuclideanAction (h * g)  $\eta$ 
```

lemma TimeAndSpace.schwartzEuclideanAction_injective[\[source\]](#)

Each Euclidean-group pullback action on Schwartz maps is injective.

```
lemma schwartzEuclideanAction_injective {d : ℕ} (g : EuclideanGroup d) :
  Function.Injective (schwartzEuclideanAction (F := F) g)
```

lemma TimeAndSpace.schwartzEuclideanAction_surjective[\[source\]](#)

Each Euclidean-group pullback action on Schwartz maps is surjective.

```
lemma schwartzEuclideanAction_surjective {d : ℕ} (g : EuclideanGroup d) :
  Function.Surjective (schwartzEuclideanAction (F := F) g)
```

2.90 Physlib.Thermodynamics.Temperature.Basic

[Physlib/Thermodynamics/Temperature/Basic.lean](#) on GitHub.

lemma Temperature.β_toReal[\[source\]](#)

```
lemma β_toReal (T : Temperature) : β( T : ℝ) = 1 / (kB * (T : ℝ))
```

lemma Temperature.ofβ_toReal[\[source\]](#)

```
lemma βof_toReal β( : ℝ≥0) : (βof β).toReal = 1 / (kB * β( : ℝ))
```

2.91 Physlib.Units.WithDim.Basic

[Physlib/Units/WithDim/Basic.lean](#) on GitHub.

instance WithDim.«instance@ec98a162»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Inhabited M] : Inhabited (WithDim d M) where
```

instance WithDim.«instance@c39a3b54»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Zero M] : Zero (WithDim d M) where
```

lemma WithDim.val_zero[\[source\]](#)

```
@[simp]
lemma val_zero {d : Dimension} {M : Type} [Zero M] :
  (0 : WithDim d M).val = 0
```

instance WithDim.«instance@ccb84cd6»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Add M] : Add (WithDim d M) where
```

lemma WithDim.val_add[\[source\]](#)

```
@[simp]
lemma val_add {d : Dimension} {M : Type} [Add M] (m1 m2 : WithDim d M) :
  (m1 + m2).val = m1.val + m2.val
```

instance WithDim.«instance@61de96eb»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Neg M] : Neg (WithDim d M) where
```

lemma WithDim.val_neg[\[source\]](#)

```
@[simp]
lemma val_neg {d : Dimension} {M : Type} [Neg M] (m : WithDim d M) :
  (-m).val = -m.val
```

instance WithDim.«instance@9b66d0ce»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Sub M] : Sub (WithDim d M) where
```

lemma WithDim.val_sub[\[source\]](#)

```
@[simp]
lemma val_sub {d : Dimension} {M : Type} [Sub M] (m1 m2 : WithDim d M) :
  (m1 - m2).val = m1.val - m2.val
```

instance WithDim.«instance@a15a8cbf»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddSemigroup M] :
  AddSemigroup (WithDim d M) where
```

instance WithDim.«instance@e53aef60»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddCommSemigroup M] :
  AddCommSemigroup (WithDim d M) where
```

instance WithDim.«instance@a79c96d0»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddMonoid M] :
  AddMonoid (WithDim d M) where
```

instance WithDim.«instance@2b3671f3»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddCommMonoid M] :
  AddCommMonoid (WithDim d M) where
```

instance WithDim.«instance@17f8c5c8»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddGroup M] :
  AddGroup (WithDim d M) where
```

instance WithDim.«instance@060f3ddb»[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddCommGroup M] :
  AddCommGroup (WithDim d M) where
```

instance WithDim.«instance@84bf74c6»

[\[source\]](#)

```
instance (d : Dimension) (M : Type) [LE M] : LE (WithDim d M) where
```

lemma WithDim.le_def

[\[source\]](#)

```
@[simp]
lemma le_def {d : Dimension} {M : Type} [LE M] (m1 m2 : WithDim d M) :
  m1 ≤ m2 ↔ m1.val ≤ m2.val
```

instance WithDim.«instance@fac005f7»

[\[source\]](#)

```
instance (d : Dimension) (M : Type) [LT M] : LT (WithDim d M) where
```

lemma WithDim.lt_def

[\[source\]](#)

```
@[simp]
lemma lt_def {d : Dimension} {M : Type} [LT M] (m1 m2 : WithDim d M) :
  m1 < m2 ↔ m1.val < m2.val
```

instance WithDim.«instance@0a2efc58»

[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Preorder M] :
  Preorder (WithDim d M) where
```

instance WithDim.«instance@86c24b45»

[\[source\]](#)

```
instance (d : Dimension) (M : Type) [PartialOrder M] :
  PartialOrder (WithDim d M) where
```

2.92 PhyslibAlpha.ClassicalFieldTheory.Local.Action

[PhyslibAlpha/ClassicalFieldTheory/Local/Action.lean](#) on GitHub.

def ClassicalFieldTheory.Local.actionDensity

[\[source\]](#)

The action density determined by a local Lagrangian and a field.

```
noncomputable def actionDensity (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

def ClassicalFieldTheory.Local.HasFiniteAction

[\[source\]](#)

The integrability condition for the action density of a field.

```
def HasFiniteAction (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ClassicalFieldTheory.Local.IsAdmissibleForAction

[\[source\]](#)

Symmetric admissibility predicate for the action functional: the pair ‘(L, f)’ is admissible when the field is smooth and the corresponding action density is integrable.

```
def IsAdmissibleForAction (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ClassicalFieldTheory.Local.action

[\[source\]](#)

The action associated with a local Lagrangian.

```
noncomputable def action (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : ℝ
```

lemma ClassicalFieldTheory.Local.actionDensity_apply

[\[source\]](#)

```
@[simp]
lemma actionDensity_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  actionDensity L f x = L (jetAt k f x)
```

lemma ClassicalFieldTheory.Local.action_eq_integral

[\[source\]](#)

```
@[simp]
lemma action_eq_integral (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) :
  action L f = ∫ x, actionDensity L f x
```

def ClassicalFieldTheory.Local.variedField

[\[source\]](#)

The field obtained by varying ‘f’ in the direction ‘η’ with parameter ‘s’.

```
noncomputable def variedField (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s : ℝ) :
  Space d → EuclideanSpace ℝ (Fin m)
```

lemma ClassicalFieldTheory.Local.variedField_contDiff

[\[source\]](#)

Smoothness of the varied field for smooth ‘f’ and admissible ‘η’.

```
lemma variedField_contDiff (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s : ℝ)
  (hf : ContDiff ℝ ∞ f) :
  ContDiff ℝ ∞ (variedField f η s)
```

def ClassicalFieldTheory.Local.actionVariation

[\[source\]](#)

The action of a field under an admissible variation.

```
noncomputable def actionVariation (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : ℝ → ℝ
```

def ClassicalFieldTheory.Local.HasFiniteActionVariation

[\[source\]](#)

Explicit integrability condition for the family of varied action densities.

```
def HasFiniteActionVariation (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Prop
```

def ClassicalFieldTheory.Local.HasCompactlySupportedActionVariationDifference [\[source\]](#)

Locality package for the action density under compactly supported variations: for every variation parameter 's', the change in the action density is a continuous compactly supported function. Combined with finiteness of the base action, this implies finiteness of the varied action.

```
def HasCompactlySupportedActionVariationDifference (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Prop
```

lemma ClassicalFieldTheory.Local.hasFiniteActionVariation_of_hasFiniteAction_of_compactlySupportedDiff [\[source\]](#)

If the base action is finite and every varied action density differs from the base density by a continuous compactly supported function, then the whole varied family has finite action.

```
lemma hasFiniteActionVariation_of_hasFiniteAction_of_compactlySupportedDifference
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hbase : HasFiniteAction L f)
  (hdiff : HasCompactlySupportedActionVariationDifference L f η) :
  HasFiniteActionVariation L f η
```

lemma ClassicalFieldTheory.Local.variedField_apply [\[source\]](#)

```
@[simp]
lemma variedField_apply (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (s : ℝ)
  (x : Space d) :
  variedField f η s x = f x + s • η x
```

lemma ClassicalFieldTheory.Local.actionVariation_apply [\[source\]](#)

```
@[simp]
lemma actionVariation_apply (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (s : ℝ) :
  actionVariation L f η s = action L (variedField f η s)
```

lemma ClassicalFieldTheory.Local.jetAt_variedField_eq_of_notMem_tsupport [\[source\]](#)

```
lemma jetAt_variedField_eq_of_notMem_tsupport
  (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (s : ℝ)
  {x : Space d}
  (hf : ContDiff ℝ ∞ f) (hx : x ∉ tsupport η.toFun) :
  jetAt k (variedField f η s) x = jetAt k f x
```

lemma ClassicalFieldTheory.Local.actionDensity_variedField_eq_of_notMem_tsupport [\[source\]](#)

```
lemma actionDensity_variedField_eq_of_notMem_tsupport
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s : ℝ)
  {x : Space d} (hf : ContDiff ℝ ∞ f) (hx : x ∉ tsupport η.toFun) :
  actionDensity L (variedField f η s) x = actionDensity L f x
```

lemma ClassicalFieldTheory.Local.hasCompactlySupportedActionVariationDifference_of_continuousInCoordinates [\[source\]](#)

```
lemma hasCompactlySupportedActionVariationDifference_of_continuousInCoordinates
  (L : Lagrangian d m k) (hcontL : Lagrangian.ContinuousInCoordinates L)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) :
  HasCompactlySupportedActionVariationDifference L f η
```

def ClassicalFieldTheory.Local.IsCritical [\[source\]](#)

A field is critical for a local action if every admissible compactly supported variation has vanishing first derivative at $s = 0$, under the explicit finite-action hypothesis for the varied family.

```
def IsCritical (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

2.93 PhyslibAlpha.ClassicalFieldTheory.Local.EulerLagrange

[PhyslibAlpha/ClassicalFieldTheory/Local/EulerLagrange.lean](#) on GitHub.

def ClassicalFieldTheory.Local.eulerLagrangeTerm [\[source\]](#)

The summand $(-1)^{|I|} D_I (\partial L / \partial u^a_I)$ in the local Euler-Lagrange formula.

```
noncomputable def eulerLagrangeTerm (L : Lagrangian d m k) (I : DerivativeIndex d k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

lemma ClassicalFieldTheory.Local.eulerLagrangeTerm_apply [\[source\]](#)

```
@[simp]
lemma eulerLagrangeTerm_apply (L : Lagrangian d m k) (I : DerivativeIndex d k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  eulerLagrangeTerm L I a f x =
  (-1 : ℝ) ^ I.1.order * iteratedTotalDerivative k I.1 (L.coordDeriv I a) f x
```

def ClassicalFieldTheory.Local.eulerLagrangeComponent [\[source\]](#)

The a -th component of the local Euler-Lagrange operator.

```
noncomputable def eulerLagrangeComponent (L : Lagrangian d m k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

lemma ClassicalFieldTheory.Local.eulerLagrangeComponent_apply[\[source\]](#)

```
@[simp]
lemma eulerLagrangeComponent_apply (L : Lagrangian d m k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  eulerLagrangeComponent L a f x =  $\sum$  I : DerivativeIndex d k, eulerLagrangeTerm L I a f x
```

def ClassicalFieldTheory.Local.eulerLagrangeOp[\[source\]](#)

The local Euler-Lagrange operator attached to a local Lagrangian.

```
noncomputable def eulerLagrangeOp (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) :
  Space d → EuclideanSpace ℝ (Fin m)
```

lemma ClassicalFieldTheory.Local.eulerLagrangeOp_apply[\[source\]](#)

```
lemma eulerLagrangeOp_apply (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (a : Fin m) :
  eulerLagrangeOp L f x a = eulerLagrangeComponent L a f x
```

lemma ClassicalFieldTheory.Local.eulerLagrangeOp_eq[\[source\]](#)

```
@[simp]
lemma eulerLagrangeOp_eq (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp L f = fun x => WithLp.toLp 2 fun a => eulerLagrangeComponent L a f x
```

2.94 PhyslibAlpha.ClassicalFieldTheory.Local.FirstOrder

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstOrder.lean](#) on GitHub.

abbrev ClassicalFieldTheory.Local.FirstOrderJetCoordinates[\[source\]](#)

Coordinate data for first-order local jet points.

```
abbrev FirstOrderJetCoordinates (d m : ℕ)
```

abbrev ClassicalFieldTheory.Local.FirstOrderJetPoint[\[source\]](#)

Coordinate-level first-order jet points.

```
abbrev FirstOrderJetPoint (d m : ℕ)
```

abbrev ClassicalFieldTheory.Local.FirstOrderJetFiberData[\[source\]](#)

Fiber-direction data for first-order jet points.

```
abbrev FirstOrderJetFiberData (d m : ℕ)
```

abbrev ClassicalFieldTheory.Local.FirstOrderLagrangian[\[source\]](#)

First-order local lagrangians.

```
abbrev FirstOrderLagrangian (d m : ℕ)
```


def ClassicalFieldTheory.Local.firstDerivativeIndex[\[source\]](#)

The first-derivative coordinate index in the source direction ‘i’.

```
def firstDerivativeIndex (d : ℕ) (i : Fin d) : DerivativeIndex d 1
```

lemma ClassicalFieldTheory.Local.firstDerivativeIndex_coe[\[source\]](#)

@[simp]

```
lemma firstDerivativeIndex_coe (d : ℕ) (i : Fin d) :  
  ((firstDerivativeIndex d i : DerivativeIndex d 1) : MultiIndex d) =  
    Physlib.MultiIndex.increment 0 i
```

lemma ClassicalFieldTheory.Local.firstDerivativeIndex_order[\[source\]](#)

```
lemma firstDerivativeIndex_order (d : ℕ) (i : Fin d) :  
  ((firstDerivativeIndex d i : DerivativeIndex d 1) : MultiIndex d).order = 1
```

def ClassicalFieldTheory.Local.JetPoint.firstDerivCoord[\[source\]](#)

The first derivative coordinate ‘ u^a_i ’ of a first-order jet point.

```
def firstDerivCoord (J : FirstOrderJetPoint d m) (i : Fin d) (a : Fin m) : ℝ
```

def ClassicalFieldTheory.Local.JetPoint.firstDerivVector[\[source\]](#)

The bundled target vector of first derivative coordinates in the source direction ‘i’.

```
def firstDerivVector (J : FirstOrderJetPoint d m) (i : Fin d) : EuclideanSpace ℝ (Fin m)
```

lemma ClassicalFieldTheory.Local.JetPoint.firstDerivVector_apply[\[source\]](#)

@[simp]

```
lemma firstDerivVector_apply (J : FirstOrderJetPoint d m) (i : Fin d) (a : Fin m) :  
  J.firstDerivVector i a = J.firstDerivCoord i a
```

lemma ClassicalFieldTheory.Local.JetPoint.firstDerivCoord_ofBaseCoordinates[\[source\]](#)

@[simp]

```
lemma firstDerivCoord_ofBaseCoordinates (x : Space d) (u : FirstOrderJetCoordinates d m)  
  (i : Fin d) (a : Fin m) :  
  (JetPoint.ofBaseCoordinates x u).firstDerivCoord i a = u (firstDerivativeIndex d i) a
```

abbrev ClassicalFieldTheory.Local.firstOrderJetAt[\[source\]](#)

The first-order coordinate-level jet point of a field at a point.

```
noncomputable abbrev firstOrderJetAt (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :  
  FirstOrderJetPoint d m
```

lemma ClassicalFieldTheory.Local.firstOrderJetAt_base[\[source\]](#)

```
@[simp]
lemma firstOrderJetAt_base (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  (firstOrderJetAt f x).base = x
```

lemma ClassicalFieldTheory.Local.firstOrderJetAt_value[\[source\]](#)

```
lemma firstOrderJetAt_value (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  (firstOrderJetAt f x).value = f x
```

lemma ClassicalFieldTheory.Local.firstOrderJetAt_firstDerivCoord[\[source\]](#)

```
@[simp]
lemma firstOrderJetAt_firstDerivCoord (f : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (i : Fin d) (a : Fin m) :
  (firstOrderJetAt f x).firstDerivCoord i a =
    ∂^[Physlib.MultiIndex.increment 0 i] (fun y => (f y) a) x
```

abbrev ClassicalFieldTheory.Local.firstOrderActionDensity[\[source\]](#)

Evaluate a first-order local lagrangian along the first jets of a field.

```
noncomputable abbrev firstOrderActionDensity (L : FirstOrderLagrangian d m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

abbrev ClassicalFieldTheory.Local.firstOrderAction[\[source\]](#)

The action of a first-order local lagrangian.

```
noncomputable abbrev firstOrderAction (L : FirstOrderLagrangian d m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : ℝ
```

abbrev ClassicalFieldTheory.Local.firstOrderEulerLagrangeOp[\[source\]](#)

The Euler-Lagrange operator of a first-order local lagrangian.

```
noncomputable abbrev firstOrderEulerLagrangeOp (L : FirstOrderLagrangian d m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → EuclideanSpace ℝ (Fin m)
```

theorem ClassicalFieldTheory.Local.firstOrder_isCritical_iff_eulerLagrange_zero[\[source\]](#)

```
theorem firstOrder_isCritical_iff_eulerLagrange_zero
  (L : FirstOrderLagrangian d m) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hLf : IsAdmissibleForAction L f)
  (hL : Lagrangian.SmoothInCoordinates L) :
  IsCritical L f ↔ firstOrderEulerLagrangeOp L f = 0
```

2.95 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation.lean](#) on GitHub.

theorem ClassicalFieldTheory.Local.hasSmoothEulerLagrangeRegularity_of_smoothInCoordinates
[\[source\]](#)

```
theorem hasSmoothEulerLagrangeRegularity_of_smoothInCoordinates
  (L : Lagrangian d m k) (hL : Lagrangian.SmoothInCoordinates L) :
  HasSmoothEulerLagrangeRegularity L
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothInCoordinates
[\[source\]](#)

```
theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothInCoordinates
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hL : Lagrangian.SmoothInCoordinates L)
  (hbase : HasFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_admissibleForAction_and_smoothInCoordinates
[\[source\]](#)

```
theorem isCritical_iff_eulerLagrange_zero_of_admissibleForAction_and_smoothInCoordinates
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hLf : IsAdmissibleForAction L f)
  (hL : Lagrangian.SmoothInCoordinates L) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

2.96 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Basic

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Basic.lean](#) on GitHub.

def ClassicalFieldTheory.Local.firstVariationDensityTerm [\[source\]](#)

A single term in the linearized first-variation density before integration by parts.

```
noncomputable def firstVariationDensityTerm (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (I : DerivativeIndex d k)
  (a : Fin m) :
  Space d → ℝ
```

def ClassicalFieldTheory.Local.firstVariationDensity [\[source\]](#)

The pointwise linearized first-variation density before integration by parts.

```
noncomputable def firstVariationDensity (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Space d → ℝ
```

def ClassicalFieldTheory.Local.firstVariationValue [\[source\]](#)

The value predicted by the first-variation formula for an admissible variation.

```
noncomputable def firstVariationValue (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : ℝ
```

lemma ClassicalFieldTheory.Local.firstVariationDensityTerm_apply[\[source\]](#)

```
@[simp]
lemma firstVariationDensityTerm_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (I : DerivativeIndex d k)
  (a : Fin m) (x : Space d) :
  firstVariationDensityTerm L f η I a x =
    L.coordDeriv I a (jetAt k f x) * ∂^[I.1] (fun y => η(y) a) x
```

lemma ClassicalFieldTheory.Local.firstVariationDensity_apply[\[source\]](#)

```
@[simp]
lemma firstVariationDensity_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (x : Space d) :
  firstVariationDensity L f η x =
    ∑ I : DerivativeIndex d k, ∑ a : Fin m, firstVariationDensityTerm L f η I a x
```

lemma ClassicalFieldTheory.Local.firstVariationValue_eq_integral[\[source\]](#)

```
@[simp]
lemma firstVariationValue_eq_integral (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) :
  firstVariationValue L f η = ∫ x, ⌊eulerLagrangeOp L f x, η ⌋ ℝ x_
```

lemma ClassicalFieldTheory.Local.firstVariationValue_eq_zero_of_eulerLagrange_zero[\[source\]](#)

```
lemma firstVariationValue_eq_zero_of_eulerLagrange_zero
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (hEuler : eulerLagrangeOp L f = 0) :
  firstVariationValue L f η = 0
```

2.97 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Criterion

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Criterion.lean](#) on GitHub.

def ClassicalFieldTheory.Local.AllVariationsHaveFiniteAction[\[source\]](#)

Explicit global finiteness hypothesis for all admissible variations.

```
def AllVariationsHaveFiniteAction (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ClassicalFieldTheory.Local.HasFirstVariationFormula[\[source\]](#)

The first-variation formula for the local action, packaged as a reusable hypothesis.

```
def HasFirstVariationFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

lemma ClassicalFieldTheory.Local.isCritical_of_eulerLagrange_zero[\[source\]](#)

```

lemma isCritical_of_eulerLagrange_zero (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hfirst : HasFirstVariationFormula L f) (hEuler : eulerLagrangeOp L f = 0) :
  IsCritical L f

```

lemma ClassicalFieldTheory.Local.eulerLagrange_zero_of_isCritical[\[source\]](#)

```

lemma eulerLagrange_zero_of_isCritical (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hfirst : HasFirstVariationFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f))
  (hcrit : IsCritical L f) :
  eulerLagrangeOp L f = 0

```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero[\[source\]](#)

```

theorem isCritical_iff_eulerLagrange_zero (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hfirst : HasFirstVariationFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0

```

lemma ClassicalFieldTheory.Local.hasFirstVariationFormula_of_underIntegral_linearized_and_parts[\[source\]](#)

```

private lemma hasFirstVariationFormula_of_underIntegral_linearized_and_parts
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hibp : HasIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f

```

lemma ClassicalFieldTheory.Local.hasFirstVariationFormula_of_contDiff_underIntegral_and_parts[\[source\]](#)

```

private lemma hasFirstVariationFormula_of_contDiff_underIntegral_and_parts
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hibp : HasIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f

```

lemma ClassicalFieldTheory.Local.hasFirstVariationFormula_of_underIntegral_linearized_and_termwise[\[source\]](#)

```

private lemma hasFirstVariationFormula_of_underIntegral_linearized_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f

```

lemma ClassicalFieldTheory.Local.hasFirstVariationFormula_of_contDiff_underIntegral_and_termwise
[\[source\]](#)

```
private lemma hasFirstVariationFormula_of_contDiff_underIntegral_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_parts
[\[source\]](#)

The local Euler-Lagrange criterion obtained from the packaged analytic ingredients of the first-variation formula. This is the current formalized form of Theorem 5.2.

```
private theorem isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_parts
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hibp : HasIntegratedByPartsFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_termwise
[\[source\]](#)

Variant of the local Euler-Lagrange criterion where the integration-by-parts input is reduced to a termwise hypothesis.

```
private theorem isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_underIntegral_and_termwise
[\[source\]](#)

```
private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_underIntegral_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_continuous_coordDeriv_and_termwise
[\[source\]](#)

```
private theorem
  isCritical_iff_eulerLagrange_zero_of_contDiff_continuous_coordDeriv_and_termwise
```

```

(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hcontVar : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
  ∀ I : DerivativeIndex d k, ∀ a : Fin m,
  Continuous (fun p : ℝ × Space d =>
    L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)))
(hterm : HasTermwiseIntegratedByPartsFormula L f)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_regular
[\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regular
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hcontVar : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
  Lagrangian.ContinuousCoordDerivAlongFamily L (fun s : ℝ => variedField f η s))
(hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt
[\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hreg : HasEulerLagrangeRegularityAt L f)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity
[\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hreg : HasSmoothEulerLagrangeRegularity L)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity
[\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt'

[\[source\]](#)

```
private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt'
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hreg : HasEulerLagrangeRegularityAt L f)
  (hfin : AllVariationsHaveFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity'

[\[source\]](#)

```
private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity'
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hreg : HasSmoothEulerLagrangeRegularity L)
  (hfin : AllVariationsHaveFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity'

[\[source\]](#)

```
private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity'
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
  (hfin : AllVariationsHaveFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.«theorem@4a640fdf»

[\[source\]](#)

```
private theorem
  isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity_of_hasFiniteAction
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
  (hbase : HasFiniteAction L f)
  (hlocal :
    ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
      HasCompactlySupportedActionVariationDifference L f η) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem ClassicalFieldTheory.Local.isCritical_iff_eulerLagrange_zero_of_hasFiniteAction_and_continuousInCoordinates'

[\[source\]](#)

```
theorem isCritical_iff_eulerLagrange_zero_of_hasFiniteAction_and_continuousInCoordinates
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
  (hcontL : Lagrangian.ContinuousInCoordinates L)
  (hbase : HasFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

2.98 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Density

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Density.lean](#) on GitHub.

def ClassicalFieldTheory.Local.HasActionVariationDerivativeUnderIntegral [\[source\]](#)

Differentiation of the varied action under the integral sign, packaged as a hypothesis.

```
def HasActionVariationDerivativeUnderIntegral (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  Prop
```

def ClassicalFieldTheory.Local.HasPointwiseLinearizedDensityFormula [\[source\]](#)

Pointwise linearization of the action density before integration by parts.

```
def HasPointwiseLinearizedDensityFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

lemma ClassicalFieldTheory.Local.actionVariation_hasDerivAt_of_dominated_loc_of_deriv_le [\[source\]](#)

A direct bridge from Mathlib's dominated differentiation-under-the-integral theorem to the local action variation. This isolates the measure-theoretic input needed to prove 'HasActionVariationDerivativeUnderIntegral'.

```
lemma actionVariation_hasDerivAt_of_dominated_loc_of_deriv_le
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  {F' : ℝ → Space d → ℝ} {bound : Space d → ℝ} ε{ : ℝ} (εh : 0 < ε)
  (hmeas :
    ∀f s in nhds (0 : ℝ),
      AESTronglyMeasurable (actionDensity L (variedField f η s)) volume)
  (hfinite0 : Integrable (actionDensity L (variedField f η 0)))
  (hF'_meas : AESTronglyMeasurable (F' 0) volume)
  (hbound :
    ∀m x ∂volume, ∀ s ∈ Metric.ball (0 : ℝ) ε, ||F' s ||x ≤ bound x)
  (hbound_int : Integrable bound volume)
  (hderiv :
    ∀m x ∂volume, ∀ s ∈ Metric.ball (0 : ℝ) ε,
      HasDerivAt (fun r : ℝ => actionDensity L (variedField f η r) x) (F' s x) s) :
  HasDerivAt (actionVariation L f η) ∫( x, F' 0 x) 0
```

def ClassicalFieldTheory.Local.HasDominatedVariationDerivativeNear [\[source\]](#)

A dominated bound for the pointwise first variation along the varied-field family near 's = 0'. This packages the analytic domination needed to differentiate the action under the integral sign.

```
def HasDominatedVariationDerivativeNear (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Prop
```

lemma ClassicalFieldTheory.Local.continuous_coordDeriv_at_zero [\[source\]](#)

```
private lemma continuous_coordDeriv_at_zero
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hcont : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    Continuous (fun p : ℝ × Space d =>
```

```

    L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)))
(I : DerivativeIndex d k) (a : Fin m) :
Continuous (fun x : Space d => L.coordDeriv I a (jetAt k f x))

```

lemma ClassicalFieldTheory.Local.firstVariationDensityTerm_norm_le_of_coeff_bound [\[source\]](#)

```

private lemma firstVariationDensityTerm_norm_le_of_coeff_bound
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
ψ( : DerivativeIndex d k → Fin m → Space d → ℝ)
(ψh : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x : Space d,
  ψ I a x = ∂^[I.1] (fun y => η( y) a) x)
(C : DerivativeIndex d k → Fin m → ℝ)
(K : DerivativeIndex d k → Fin m → Set (Space d))
(hKzero : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x, x ∉ K I a → ψ I a x = 0)
(hC : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
  ∀ p ∈ Metric.closedBall (0 : ℝ) 1 s× K I a,
    ||L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)|| ≤ C I a)
{x : Space d} {s : ℝ} (hs : s ∈ Metric.ball (0 : ℝ) 1)
(I : DerivativeIndex d k) (a : Fin m) :
||firstVariationDensityTerm L (variedField f η s) η I a ||x ≤ C I a * ||ψ I a ||x

```

lemma ClassicalFieldTheory.Local.norm_firstVariationDensity_le_bound

[\[source\]](#)

```

private lemma norm_firstVariationDensity_le_bound
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
ψ( : DerivativeIndex d k → Fin m → Space d → ℝ)
(ψh : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x : Space d,
  ψ I a x = ∂^[I.1] (fun y => η( y) a) x)
(C : DerivativeIndex d k → Fin m → ℝ)
(K : DerivativeIndex d k → Fin m → Set (Space d))
(hKzero : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x, x ∉ K I a → ψ I a x = 0)
(hC : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
  ∀ p ∈ Metric.closedBall (0 : ℝ) 1 s× K I a,
    ||L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)|| ≤ C I a)
{x : Space d} {s : ℝ} (hs : s ∈ Metric.ball (0 : ℝ) 1) :
||firstVariationDensity L (variedField f η s) η ||x
  ≤ ∑ I : DerivativeIndex d k, ∑ a : Fin m, C I a * ||ψ I a ||x

```

lemma ClassicalFieldTheory.Local.hasDominatedVariationDerivativeNear_of_continuous_coordDeriv

[\[source\]](#)

A concrete dominated bound for the varied first-variation density, obtained from continuity of the jet-coordinate coefficient family in ‘(s, x)’. The compact support of the iterated derivatives of the admissible variation supplies the integrable majorant.

```

lemma hasDominatedVariationDerivativeNear_of_continuous_coordDeriv
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
(hcont : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
  Continuous (fun p : ℝ × Space d =>
    L.coordDeriv I a (jetAt k (variedField f η p.1) p.2))) :
HasDominatedVariationDerivativeNear L f η

```

lemma ClassicalFieldTheory.Local.hasDerivAt_actionDensity_variedField_zero_of_contDiff [\[source\]](#)

```
lemma hasDerivAt_actionDensity_variedField_zero_of_contDiff
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f) :
  ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)), ∀ x : Space d,
  HasDerivAt (fun s : ℝ => actionDensity L (variedField f η s) x)
    (firstVariationDensity L f η x) 0
```

lemma ClassicalFieldTheory.Local.hasDerivAt_actionDensity_variedField_of_contDiff [\[source\]](#)

```
lemma hasDerivAt_actionDensity_variedField_of_contDiff
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (x : Space d) (s : ℝ) :
  HasDerivAt (fun r : ℝ => actionDensity L (variedField f η r) x)
    (firstVariationDensity L (variedField f η s) η x) s
```

lemma ClassicalFieldTheory.Local.hasPointwiseLinearizedDensityFormula_of_contDiff [\[source\]](#)

```
lemma hasPointwiseLinearizedDensityFormula_of_contDiff
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f) :
  HasPointwiseLinearizedDensityFormula L f
```

lemma ClassicalFieldTheory.Local.hasActionVariationDerivativeUnderIntegral_of_contDiff_of_dominatedVariation [\[source\]](#)

```
lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_dominatedVariation
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hdom : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    HasFiniteActionVariation L f η →
    HasDominatedVariationDerivativeNear L f η) :
  HasActionVariationDerivativeUnderIntegral L f
```

lemma ClassicalFieldTheory.Local.hasActionVariationDerivativeUnderIntegral_of_contDiff_of_continuous_coordDeriv [\[source\]](#)

```
lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_continuous_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcontVar : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    Continuous (fun p : ℝ × Space d =>
      L.coordDeriv I a (jetAt k (variedField f η p.1) p.2))) :
  HasActionVariationDerivativeUnderIntegral L f
```

lemma ClassicalFieldTheory.Local.hasActionVariationDerivativeUnderIntegral_of_contDiff_of_regular [\[source\]](#)

```
lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcontVar : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    Lagrangian.ContinuousCoordDerivAlongFamily L (fun s : ℝ => variedField f η s)) :
  HasActionVariationDerivativeUnderIntegral L f
```

2.99 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.IntegrationByParts

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/IntegrationByParts.lean](#) on GitHub.

def ClassicalFieldTheory.Local.HasIntegratedByPartsFormula [\[source\]](#)

The integration-by-parts step sending the linearized density to the Euler-Lagrange pairing.

```
def HasIntegratedByPartsFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ClassicalFieldTheory.Local.HasTermwiseIntegratedByPartsFormula [\[source\]](#)

Termwise integration-by-parts data for the linearized first-variation density.

```
def HasTermwiseIntegratedByPartsFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

lemma ClassicalFieldTheory.Local.integral_mul_space_deriv_eq_neg_deriv_mul [\[source\]](#)

```
private lemma integral_mul_space_deriv_eq_neg_deriv_mul (i : Fin d) {g h : Space d → ℝ}
  (hg : ContDiff ℝ ∞ g) (hh : IsTestFunction h) :
  ∫ x, g x * ∂[i] h x = ∫ x, ∂(-[i] g x) * h x
```

lemma ClassicalFieldTheory.Local.integral_mul_iteratedDerivList_eq_sign [\[source\]](#)

```
private lemma integral_mul_iteratedDerivList_eq_sign (L : List (Fin d)) {g h : Space d → ℝ}
  (hg : ContDiff ℝ ∞ g) (hh : IsTestFunction h) :
  ∫ x, g x * L.foldr (fun i f => ∂[i] f) h x =
  ∫ x, (((-1 : ℝ) ^ L.length) * L.foldr (fun i f => ∂[i] f) g x) * h x
```

lemma ClassicalFieldTheory.Local.integral_mul_iteratedDeriv_eq_sign [\[source\]](#)

Repeated integration by parts for iterated coordinate derivatives against a test function.

```
lemma integral_mul_iteratedDeriv_eq_sign {g h : Space d → ℝ} (I : MultiIndex d)
  (hg : ContDiff ℝ ∞ g) (hh : IsTestFunction h) :
  ∫ x, g x * ∂^[I] h x =
  ∫ x, (((-1 : ℝ) ^ I.order) * ∂^[I] g x) * h x
```

lemma ClassicalFieldTheory.Local.hasTermwiseIntegratedByPartsFormula_of_contDiff_coordDeriv [\[source\]](#)

Concrete termwise integration by parts for the first-variation density, assuming the jet coordinate coefficient functions are smooth along the field.

```
lemma hasTermwiseIntegratedByPartsFormula_of_contDiff_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    ContDiff ℝ ∞ (fun x => L.coordDeriv I a (jetAt k f x))) :
  HasTermwiseIntegratedByPartsFormula L f
```

lemma ClassicalFieldTheory.Local.hasTermwiseIntegratedByPartsFormula_of_regular [\[source\]](#)

```
lemma hasTermwiseIntegratedByPartsFormula_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f) :
  HasTermwiseIntegratedByPartsFormula L f
```

lemma ClassicalFieldTheory.Local.integral_firstVariationDensity_eq_termwise_sum [\[source\]](#)

```
private lemma integral_firstVariationDensity_eq_termwise_sum
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  ∫ x, firstVariationDensity L f η x =
  ∑ I : DerivativeIndex d k, ∑ a : Fin m,
  ∫ x, firstVariationDensityTerm L f η I a x
```

lemma ClassicalFieldTheory.Local.integral_eulerLagrange_termwise_sum_eq_pairing [\[source\]](#)

```
private lemma integral_eulerLagrange_termwise_sum_eq_pairing
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  ∑ I : DerivativeIndex d k, ∑ a : Fin m, ∫ x, eulerLagrangeTerm L I a f x * η( x) a
  = firstVariationValue L f η
```

lemma ClassicalFieldTheory.Local.hasIntegratedByPartsFormula_of_termwise [\[source\]](#)

```
lemma hasIntegratedByPartsFormula_of_termwise (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  HasIntegratedByPartsFormula L f
```

2.100 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Regularity

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Regularity.lean](#) on GitHub.

def ClassicalFieldTheory.Local.HasEulerLagrangeRegularityAt [\[source\]](#)

The combined regularity package currently needed to derive the local Euler-Lagrange criterion for a field ‘f’: smooth coefficient functions along ‘f’, and continuity of the corresponding coefficient families along every admissible varied-field family based at ‘f’.

```
def HasEulerLagrangeRegularityAt (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ClassicalFieldTheory.Local.HasSmoothEulerLagrangeRegularity [\[source\]](#)

A local Lagrangian has smooth Euler-Lagrange regularity if every smooth field satisfies the regularity package currently needed by the local first-variation proof.

```
def HasSmoothEulerLagrangeRegularity (L : Lagrangian d m k) : Prop
```

lemma ClassicalFieldTheory.Local.continuous_eulerLagrangeTerm_of_regular[\[source\]](#)

```

lemma continuous_eulerLagrangeTerm_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f)
  (I : DerivativeIndex d k) (a : Fin m) :
  Continuous (eulerLagrangeTerm L I a f)

```

lemma ClassicalFieldTheory.Local.continuous_eulerLagrangeComponent_of_regular[\[source\]](#)

```

lemma continuous_eulerLagrangeComponent_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f)
  (a : Fin m) :
  Continuous (eulerLagrangeComponent L a f)

```

lemma ClassicalFieldTheory.Local.continuous_eulerLagrangeOp_of_regular[\[source\]](#)

```

lemma continuous_eulerLagrangeOp_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f) :
  Continuous (eulerLagrangeOp L f)

```

theorem ClassicalFieldTheory.Local.hasSmoothEulerLagrangeRegularity_of_contDiffCoordDerivInCoordinates[\[source\]](#)

```

theorem hasSmoothEulerLagrangeRegularity_of_contDiffCoordDerivInCoordinates
  (L : Lagrangian d m k) (hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L) :
  HasSmoothEulerLagrangeRegularity L

```

2.101 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Support

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Support.lean](#) on GitHub.

lemma ClassicalFieldTheory.Local.variedField_zero[\[source\]](#)

```

@[simp]
lemma variedField_zero (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) :
  variedField f η 0 = f

```

lemma ClassicalFieldTheory.Local.variedField_variedField[\[source\]](#)

```

@[simp]
lemma variedField_variedField (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s t : ℝ) :
  variedField (variedField f η s) η t = variedField f η (s + t)

```

lemma ClassicalFieldTheory.Local.contDiff_space_deriv[\[source\]](#)

```

lemma contDiff_space_deriv {g : Space d → ℝ} (hg : ContDiff ℝ ∞ g)
  (i : Fin d) : ContDiff ℝ ∞ ∂([i] g)

```

lemma ClassicalFieldTheory.Local.isTestFunction_space_deriv[\[source\]](#)

```
lemma isTestFunction_space_deriv {g : Space d → ℝ} (hg : IsTestFunction g) (i : Fin d) :
  IsTestFunction ∂([i] g)
```

lemma ClassicalFieldTheory.Local.iteratedDerivList_contDiff[\[source\]](#)

```
lemma iteratedDerivList_contDiff (L : List (Fin d)) {g : Space d → ℝ}
  (hg : ContDiff ℝ ∞ g) :
  ContDiff ℝ ∞ (L.foldr (fun i h => ∂[i] h) g)
```

lemma ClassicalFieldTheory.Local.iteratedDerivList_isTestFunction[\[source\]](#)

```
lemma iteratedDerivList_isTestFunction (L : List (Fin d)) {g : Space d → ℝ}
  (hg : IsTestFunction g) :
  IsTestFunction (L.foldr (fun i h => ∂[i] h) g)
```

lemma ClassicalFieldTheory.Local.iteratedDerivList_commute_deriv[\[source\]](#)

```
lemma iteratedDerivList_commute_deriv (L : List (Fin d)) (i : Fin d)
  {g : Space d → ℝ} (hg : ContDiff ℝ ∞ g) :
  L.foldr (fun j h => ∂[j] h) ∂([i] g) = ∂[i] (L.foldr (fun j h => ∂[j] h) g)
```

lemma ClassicalFieldTheory.Local.iteratedDeriv_coord_isTestFunction[\[source\]](#)

```
lemma iteratedDeriv_coord_isTestFunction η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (I : DerivativeIndex d k) (a : Fin m) :
  IsTestFunction (fun x => ∂^[I.1] (fun y => η( y) a) x)
```

lemma ClassicalFieldTheory.Local.firstVariationDensityTerm_integrable_of_continuous_coordDeriv[\[source\]](#)

```
lemma firstVariationDensityTerm_integrable_of_continuous_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (I : DerivativeIndex d k) (a : Fin m)
  (hcont : Continuous (fun x : Space d =>
    L.coordDeriv I a (jetAt k f x))) :
  Integrable (firstVariationDensityTerm L f η I a)
```

lemma ClassicalFieldTheory.Local.continuous_jetBaseCoordinates_variedField[\[source\]](#)

```
lemma continuous_jetBaseCoordinates_variedField
  (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m))
  η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hf : ContDiff ℝ ∞ f) :
  Continuous
    (fun p : ℝ × Space d => (p.2, jetCoordinatesAt k (variedField f η p.1) p.2))
```

2.102 PhyslibAlpha.ClassicalFieldTheory.Local.JetPoint

[PhyslibAlpha/ClassicalFieldTheory/Local/JetPoint.lean](#) on GitHub.

abbrev ClassicalFieldTheory.Local.JetCoordinates[\[source\]](#)

Coordinate data $\hat{u}_a^I$ for ' k '-th order jet points.

For each derivative index ' I ', these are the scalar components of the corresponding 'EuclideanSpace $\mathbb{R}$ (Fin m)' fiber coordinate. We keep this componentwise form because the local Euler-Lagrange formulas are indexed by both ' I ' and ' a '.

```
abbrev JetCoordinates (d m k : ℕ)
```

structure ClassicalFieldTheory.Local.JetPoint[\[source\]](#)

The coordinate-level points of the locally trivialized ' k '-jet bundle for fields 'Space $d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ '.

```
structure JetPoint (d m k : ℕ) where
  base : Space d
  fiber : JetCoordinates d m k
```

def ClassicalFieldTheory.Local.JetPoint.coord[\[source\]](#)

The coordinate $\hat{u}_a^I$ of a jet point.

```
def coord (J : JetPoint d m k) (I : DerivativeIndex d k) (a : Fin m) : ℝ
```

def ClassicalFieldTheory.Local.JetPoint.value[\[source\]](#)

The zero-th order value encoded by a jet point.

```
def value (J : JetPoint d m k) : EuclideanSpace ℝ (Fin m)
```

lemma ClassicalFieldTheory.Local.JetPoint.value_apply[\[source\]](#)

```
@[simp]
lemma value_apply (J : JetPoint d m k) (a : Fin m) :
  J.value a = J.coord 0 a
```

lemma ClassicalFieldTheory.Local.JetPoint.ext[\[source\]](#)

```
@[ext]
lemma ext (J K : JetPoint d m k) (hbase : J.base = K.base)
  (hcoord : ∀ I a, J.coord I a = K.coord I a) :
  J = K
```

def ClassicalFieldTheory.Local.JetPoint.ofBaseCoordinates[\[source\]](#)

Build a jet point from a base point and its jet coordinates.

```
def ofBaseCoordinates (x : Space d) (u : JetCoordinates d m k) : JetPoint d m k where
```

def ClassicalFieldTheory.Local.JetPoint.toBaseCoordinates[\[source\]](#)

The base point and jet coordinates of a jet point.

```
def toBaseCoordinates (J : JetPoint d m k) : Space d × JetCoordinates d m k
```


lemma ClassicalFieldTheory.Local.JetPoint.ofBaseCoordinates_base[\[source\]](#)

@[simp]

```
lemma ofBaseCoordinates_base (x : Space d) (u : JetCoordinates d m k) :
  (ofBaseCoordinates x u).base = x
```

lemma ClassicalFieldTheory.Local.JetPoint.ofBaseCoordinates_fiber[\[source\]](#)

@[simp]

```
lemma ofBaseCoordinates_fiber (x : Space d) (u : JetCoordinates d m k) :
  (ofBaseCoordinates x u).fiber = u
```

lemma ClassicalFieldTheory.Local.JetPoint.ofBaseCoordinates_coord[\[source\]](#)

@[simp]

```
lemma ofBaseCoordinates_coord (x : Space d) (u : JetCoordinates d m k)
  (I : DerivativeIndex d k) (a : Fin m) :
  (ofBaseCoordinates x u).coord I a = u I a
```

lemma ClassicalFieldTheory.Local.JetPoint.ofBaseCoordinates_base_fiber[\[source\]](#)

@[simp]

```
lemma ofBaseCoordinates_base_fiber (J : JetPoint d m k) :
  ofBaseCoordinates J.base J.fiber = J
```

lemma ClassicalFieldTheory.Local.JetPoint.toBaseCoordinates_ofBaseCoordinates[\[source\]](#)

@[simp]

```
lemma toBaseCoordinates_ofBaseCoordinates (x : Space d) (u : JetCoordinates d m k) :
  (ofBaseCoordinates x u).toBaseCoordinates = (x, u)
```

def ClassicalFieldTheory.Local.jetCoordinatesAt[\[source\]](#)

The jet-coordinate function determined by a field ‘g’.

```
noncomputable def jetCoordinatesAt (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) :
  JetCoordinates d m k
```

def ClassicalFieldTheory.Local.jetAt[\[source\]](#)

The coordinate-level ‘k’-jet point of a field ‘f’ at the point ‘x’.

```
noncomputable def jetAt (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  JetPoint d m k where
```

lemma ClassicalFieldTheory.Local.jetAt_base[\[source\]](#)

@[simp]

```
lemma jetAt_base (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  (jetAt k f x).base = x
```

lemma ClassicalFieldTheory.Local.jetAt_value[\[source\]](#)

```
@[simp]
lemma jetAt_value (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  (jetAt k f x).value = f x
```

lemma ClassicalFieldTheory.Local.jetAt_coord[\[source\]](#)

```
@[simp]
lemma jetAt_coord (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (I : DerivativeIndex d k) (a : Fin m) :
  (jetAt k f x).coord I a = ∂^[I.1] (fun y => (f y) a) x
```

lemma ClassicalFieldTheory.Local.jetAt_coord_zero[\[source\]](#)

```
lemma jetAt_coord_zero (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (a : Fin m) :
  (jetAt k f x).coord 0 a = (f x) a
```

lemma ClassicalFieldTheory.Local.jetCoordinatesAt_eq[\[source\]](#)

```
@[simp]
lemma jetCoordinatesAt_eq (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (I : DerivativeIndex d k) (a : Fin m) :
  jetCoordinatesAt k g x I a = ∂^[I.1] (fun y => (g y) a) x
```

lemma ClassicalFieldTheory.Local.jetCoordinatesAt_zero[\[source\]](#)

```
lemma jetCoordinatesAt_zero (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (a : Fin m) :
  jetCoordinatesAt k g x 0 a = (g x) a
```

lemma ClassicalFieldTheory.Local.jetAt_eq_ofBaseCoordinates[\[source\]](#)

```
lemma jetAt_eq_ofBaseCoordinates (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) :
  jetAt k f x = JetPoint.ofBaseCoordinates x (jetCoordinatesAt k f x)
```

2.103 PhyslibAlpha.ClassicalFieldTheory.Local.JetPointFiber

[PhyslibAlpha/ClassicalFieldTheory/Local/JetPointFiber.lean](#) on GitHub.

structure ClassicalFieldTheory.Local.JetFiberData[\[source\]](#)

Fiber-coordinate data for jet points of order ‘k’. This records only the coordinates ‘ u^a_I ’, not the base point in ‘Space d’.

```
structure JetFiberData (d m k : ℕ) where
  coord : JetCoordinates d m k
```

instance ClassicalFieldTheory.Local.JetFiberData.«instance@24e9455d»[\[source\]](#)

```
instance : CoeFun (JetFiberData d m k) (fun _ => DerivativeIndex d k → Fin m → ℝ) where
```

lemma ClassicalFieldTheory.Local.JetFiberData.ext[\[source\]](#)`@[ext]``lemma ext (V W : JetFiberData d m k) (hcoord : $\forall I a, V.coord I a = W.coord I a$) : V = W`**def ClassicalFieldTheory.Local.JetFiberData.value**[\[source\]](#)*The zero-th order component of a jet-fiber direction, corresponding to the field value.*`def value (V : JetFiberData d m k) : EuclideanSpace $\mathbb{R}$ (Fin m)`**lemma ClassicalFieldTheory.Local.JetFiberData.value\_apply**[\[source\]](#)`@[simp]``lemma value_apply (V : JetFiberData d m k) (a : Fin m) :
 V.value a = V.coord 0 a`**instance ClassicalFieldTheory.Local.JetFiberData.«instance@14f5f97d»**[\[source\]](#)`instance : Zero (JetFiberData d m k) where`**instance ClassicalFieldTheory.Local.JetFiberData.«instance@47407776»**[\[source\]](#)`instance : Add (JetFiberData d m k) where`**instance ClassicalFieldTheory.Local.JetFiberData.«instance@cddd31c6»**[\[source\]](#)`instance : SMul $\mathbb{R}$ (JetFiberData d m k) where`**lemma ClassicalFieldTheory.Local.JetFiberData.zero\_coord**[\[source\]](#)`@[simp]``lemma zero_coord (I : DerivativeIndex d k) (a : Fin m) :
 (0 : JetFiberData d m k).coord I a = 0`**lemma ClassicalFieldTheory.Local.JetFiberData.add\_coord**[\[source\]](#)`@[simp]``lemma add_coord (V W : JetFiberData d m k) (I : DerivativeIndex d k) (a : Fin m) :
 (V + W).coord I a = V.coord I a + W.coord I a`**lemma ClassicalFieldTheory.Local.JetFiberData.smul\_coord**[\[source\]](#)`@[simp]``lemma smul_coord (c : $\mathbb{R}$) (V : JetFiberData d m k) (I : DerivativeIndex d k) (a : Fin m) :
 (c • V).coord I a = c * V.coord I a`**def ClassicalFieldTheory.Local.JetPoint.addFiber**[\[source\]](#)*Translate a jet point by a fiber-direction increment.*`def addFiber (J : JetPoint d m k) (V : JetFiberData d m k) : JetPoint d m k where`

def ClassicalFieldTheory.Local.JetPoint.lineMap[\[source\]](#)

The affine line in jet space through ‘J’ in the fiber direction ‘V’.

```
def lineMap (J : JetPoint d m k) (V : JetFiberData d m k) (s : ℝ) : JetPoint d m k
```

lemma ClassicalFieldTheory.Local.JetPoint.addFiber_base[\[source\]](#)

```
@[simp]
```

```
lemma addFiber_base (J : JetPoint d m k) (V : JetFiberData d m k) :  
  (J.addFiber V).base = J.base
```

lemma ClassicalFieldTheory.Local.JetPoint.addFiber_value[\[source\]](#)

```
@[simp]
```

```
lemma addFiber_value (J : JetPoint d m k) (V : JetFiberData d m k) :  
  (J.addFiber V).value = J.value + V.value
```

lemma ClassicalFieldTheory.Local.JetPoint.addFiber_coord[\[source\]](#)

```
@[simp]
```

```
lemma addFiber_coord (J : JetPoint d m k) (V : JetFiberData d m k)  
  (I : DerivativeIndex d k) (a : Fin m) :  
  (J.addFiber V).coord I a = J.coord I a + V.coord I a
```

lemma ClassicalFieldTheory.Local.JetPoint.lineMap_base[\[source\]](#)

```
@[simp]
```

```
lemma lineMap_base (J : JetPoint d m k) (V : JetFiberData d m k) (s : ℝ) :  
  (J.lineMap V s).base = J.base
```

lemma ClassicalFieldTheory.Local.JetPoint.lineMap_coord[\[source\]](#)

```
@[simp]
```

```
lemma lineMap_coord (J : JetPoint d m k) (V : JetFiberData d m k) (s : ℝ)  
  (I : DerivativeIndex d k) (a : Fin m) :  
  (J.lineMap V s).coord I a = J.coord I a + s * V.coord I a
```

def ClassicalFieldTheory.Local.jetDirectionAt[\[source\]](#)

The jet-fiber direction determined by a field ‘g’ at a point ‘x’.

```
noncomputable def jetDirectionAt (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))  
  (x : Space d) :  
  JetFiberData d m k where
```

lemma ClassicalFieldTheory.Local.jetDirectionAt_coord[\[source\]](#)

```
@[simp]
```

```
lemma jetDirectionAt_coord (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)  
  (I : DerivativeIndex d k) (a : Fin m) :  
  (jetDirectionAt k g x).coord I a = ∂^[I.1] (fun y => (g y) a) x
```

lemma ClassicalFieldTheory.Local.jetDirectionAt_coord_zero[\[source\]](#)

```
lemma jetDirectionAt_coord_zero (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (a : Fin m) :
  (jetDirectionAt k g x).coord 0 a = (g x) a
```

lemma ClassicalFieldTheory.Local.jetCoordinatesAt_add_smul[\[source\]](#)

```
lemma jetCoordinatesAt_add_smul (k : ℕ) (f g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (s : ℝ)
  (hf : ContDiff ℝ ∞ f) (hg : ContDiff ℝ ∞ g) :
  jetCoordinatesAt k (fun y => f y + s • g y) x =
    jetCoordinatesAt k f x + s • jetCoordinatesAt k g x
```

lemma ClassicalFieldTheory.Local.jetAt_add_smul[\[source\]](#)

```
lemma jetAt_add_smul (k : ℕ) (f g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (s : ℝ)
  (hf : ContDiff ℝ ∞ f) (hg : ContDiff ℝ ∞ g) :
  jetAt k (fun y => f y + s • g y) x = (jetAt k f x).lineMap (jetDirectionAt k g x) s
```

2.104 PhyslibAlpha.ClassicalFieldTheory.Local.JetPointRegularity

[PhyslibAlpha/ClassicalFieldTheory/Local/JetPointRegularity.lean](#) on GitHub.

lemma ClassicalFieldTheory.Local.jetCoordinatesAt_contDiff[\[source\]](#)

Smoothness of the jet-coordinate map of a smooth field.

```
lemma jetCoordinatesAt_contDiff (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (hg : ContDiff ℝ ∞ g) :
  ContDiff ℝ ∞ (jetCoordinatesAt k g)
```

lemma ClassicalFieldTheory.Local.jetBaseCoordinates_contDiff[\[source\]](#)

Smoothness of the base-plus-coordinate jet map of a smooth field.

```
lemma jetBaseCoordinates_contDiff (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (hg : ContDiff ℝ ∞ g) :
  ContDiff ℝ ∞ (fun x : Space d => (x, jetCoordinatesAt k g x))
```

lemma ClassicalFieldTheory.Local.notMem_tsupport_coord[\[source\]](#)

```
private lemma notMem_tsupport_coord {g : Space d → EuclideanSpace ℝ (Fin m)} {x : Space d}
  (hx : x ∉ tsupport g) (a : Fin m) :
  x ∉ tsupport (fun y => (g y) a)
```

lemma ClassicalFieldTheory.Local.jetCoordinatesAt_eq_zero_of_notMem_tsupport[\[source\]](#)

Outside the topological support of a field, all of its jet coordinates vanish.

```
lemma jetCoordinatesAt_eq_zero_of_notMem_tsupport (k : ℕ)
  (g : Space d → EuclideanSpace ℝ (Fin m)) {x : Space d}
  (hx : x ∉ tsupport g) :
  jetCoordinatesAt k g x = 0
```

lemma ClassicalFieldTheory.Local.jetDirectionAt_eq_zero_of_notMem_tsupport [\[source\]](#)

Outside the topological support of a field, the corresponding jet-fiber direction vanishes.

```
lemma jetDirectionAt_eq_zero_of_notMem_tsupport (k : ℕ)
  (g : Space d → EuclideanSpace ℝ (Fin m)) {x : Space d}
  (hx : x ∉ tsupport g) :
  jetDirectionAt k g x = 0
```

2.105 PhyslibAlpha.ClassicalFieldTheory.Local.Lagrangian

[PhyslibAlpha/ClassicalFieldTheory/Local/Lagrangian.lean](#) on GitHub.

structure ClassicalFieldTheory.Local.Lagrangian [\[source\]](#)

A local ‘k’-th order Lagrangian for fields ‘Space d → EuclideanSpace ℝ (Fin m)’:

```
structure Lagrangian (d m k : ℕ) where
  toFun : JetPoint d m k → ℝ
  coordDeriv : DerivativeIndex d k → Fin m → JetPoint d m k → ℝ
  hasDerivAlongFiber :
    ∀ J : JetPoint d m k, ∀ V : JetFiberData d m k,
      HasDerivAt (fun s : ℝ => toFun (J.lineMap V s))
        (Σ (I : DerivativeIndex d k, Σ a : Fin m, coordDeriv I a J * V.coord I a) 0
```

instance ClassicalFieldTheory.Local.Lagrangian.«instance@43bf9c65» [\[source\]](#)

```
instance : CoeFun (Lagrangian d m k) (fun _ => JetPoint d m k → ℝ) where
```

def ClassicalFieldTheory.Local.Lagrangian.fiberDerivative [\[source\]](#)

The derivative of ‘L’ along the fiber-jet direction ‘V’ based at ‘J’.

```
noncomputable def fiberDerivative (L : Lagrangian d m k) (J : JetPoint d m k)
  (V : JetFiberData d m k) : ℝ
```

lemma ClassicalFieldTheory.Local.Lagrangian.hasDerivAt_lineMap [\[source\]](#)

```
lemma hasDerivAt_lineMap (L : Lagrangian d m k) (J : JetPoint d m k) (V : JetFiberData d m k) :
  HasDerivAt (fun s : ℝ => L (J.lineMap V s)) (L.fiberDerivative J V) 0
```

def ClassicalFieldTheory.Local.Lagrangian.ContinuousInCoordinates [\[source\]](#)

Continuity of a local lagrangian in the explicit local jet coordinates ‘(x, u^a_I)’.

```
def ContinuousInCoordinates (L : Lagrangian d m k) : Prop
```

def ClassicalFieldTheory.Local.Lagrangian.ContDiffCoordDerivInCoordinates [\[source\]](#)

Intrinsic smoothness of the jet-coordinate derivatives of a local Lagrangian in the explicit local jet coordinates ‘(x, u^a_I)’.

```
def ContDiffCoordDerivInCoordinates (L : Lagrangian d m k) : Prop
```

def ClassicalFieldTheory.Local.Lagrangian.SmoothInCoordinates[\[source\]](#)

Public regularity package for a smooth local lagrangian in explicit local jet coordinates: continuity of the lagrangian itself together with smoothness of all jet-coordinate derivatives.

```
def SmoothInCoordinates (L : Lagrangian d m k) : Prop
```

def ClassicalFieldTheory.Local.Lagrangian.ContDiffCoordDerivAlongField[\[source\]](#)

Smoothness of all coefficient functions ' $x \mapsto \partial L / \partial u_I^a (j^k f(x))$ ' along a field 'f'.

```
def ContDiffCoordDerivAlongField (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ClassicalFieldTheory.Local.Lagrangian.ContinuousCoordDerivAlongFamily[\[source\]](#)

Continuity of the coefficient family ' $(s, x) \mapsto \partial L / \partial u_I^a (j^k (F s)(x))$ ' along a one-parameter family of fields ' $F : \mathbb{R} \rightarrow \text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} (\text{Fin } m)$ '.

```
def ContinuousCoordDerivAlongFamily (L : Lagrangian d m k)
  (F : ℝ → Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

lemma ClassicalFieldTheory.Local.Lagrangian.continuousAlongField_of_inCoordinates[\[source\]](#)

```
lemma continuousAlongField_of_inCoordinates (L : Lagrangian d m k)
  (hcont : ContinuousInCoordinates L) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hf : ContDiff ℝ ∞ f) :
  Continuous (fun x : Space d => L (jetAt k f x))
```

lemma ClassicalFieldTheory.Local.Lagrangian.contDiffCoordDerivAlongField_of_inCoordinates[\[source\]](#)

```
lemma contDiffCoordDerivAlongField_of_inCoordinates (L : Lagrangian d m k)
  (hcoord : ContDiffCoordDerivInCoordinates L) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hf : ContDiff ℝ ∞ f) :
  ContDiffCoordDerivAlongField L f
```

lemma ClassicalFieldTheory.Local.Lagrangian.continuousCoordDerivAlongFamily_of_inCoordinates[\[source\]](#)

```
lemma continuousCoordDerivAlongFamily_of_inCoordinates (L : Lagrangian d m k)
  (hcoord : ContDiffCoordDerivInCoordinates L)
  (F : ℝ → Space d → EuclideanSpace ℝ (Fin m))
  (hjet : Continuous (fun p : ℝ × Space d => (p.2, jetCoordinatesAt k (F p.1) p.2))) :
  ContinuousCoordDerivAlongFamily L F
```

def ClassicalFieldTheory.Local.Lagrangian.alongField[\[source\]](#)

Evaluate a local Lagrangian along the 'k'-jets of a field.

```
noncomputable def alongField (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

lemma ClassicalFieldTheory.Local.Lagrangian.alongField_apply[\[source\]](#)

```
@[simp]
lemma alongField_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  L.alongField f x = L (jetAt k f x)
```

2.106 PhyslibAlpha.ClassicalFieldTheory.Local.TotalDerivative

[PhyslibAlpha/ClassicalFieldTheory/Local/TotalDerivative.lean](#) on GitHub.

def ClassicalFieldTheory.Local.evalOnJet[\[source\]](#)

Evaluate a local jet-dependent function along the ‘k’-jet of a field.

```
noncomputable def evalOnJet (k : ℕ) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  Space d → ℝ
```

def ClassicalFieldTheory.Local.totalDerivative[\[source\]](#)

The total derivative of a jet-dependent function in the coordinate direction ‘i’.

```
noncomputable def totalDerivative (k : ℕ) (i : Fin d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

def ClassicalFieldTheory.Local.iteratedTotalDerivative[\[source\]](#)

The iterated total derivative indexed by a multi-index.

```
noncomputable def iteratedTotalDerivative (k : ℕ) (I : MultiIndex d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

lemma ClassicalFieldTheory.Local.evalOnJet_apply[\[source\]](#)

```
@[simp]
lemma evalOnJet_apply (k : ℕ) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  evalOnJet k F f x = F (jetAt k f x)
```

lemma ClassicalFieldTheory.Local.totalDerivative_eq[\[source\]](#)

```
@[simp]
lemma totalDerivative_eq (k : ℕ) (i : Fin d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  totalDerivative k i F f = ∂[i] (fun x => F (jetAt k f x))
```

lemma ClassicalFieldTheory.Local.iteratedTotalDerivative_eq[\[source\]](#)

```
@[simp]
lemma iteratedTotalDerivative_eq (k : ℕ) (I : MultiIndex d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  iteratedTotalDerivative k I F f = ∂^[I] (fun x => F (jetAt k f x))
```


lemma ClassicalFieldTheory.Local.iteratedTotalDerivative_zero[\[source\]](#)

```
lemma iteratedTotalDerivative_zero (k : ℕ) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  iteratedTotalDerivative k (0 : MultiIndex d) F f = evalOnJet k F f
```

lemma ClassicalFieldTheory.Local.iteratedTotalDerivative_single[\[source\]](#)

```
lemma iteratedTotalDerivative_single (k : ℕ) (i : Fin d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  iteratedTotalDerivative k (Physlib.MultiIndex.increment 0 i) F f =
  totalDerivative k i F f
```

2.107 PhyslibAlpha.ClassicalFieldTheory.Local.TotalDivergence

[PhyslibAlpha/ClassicalFieldTheory/Local/TotalDivergence.lean](#) on GitHub.

def ClassicalFieldTheory.Local.HasTotalDivergenceDensity[\[source\]](#)

A lagrangian density has total-divergence form if, along every field, it is the sum of the total derivatives of a jet-dependent current. A current of order ‘k’ gives a density of order ‘k + 1’.

```
def HasTotalDivergenceDensity (L : Lagrangian d m (k + 1))
  (current : Fin d → JetPoint d m k → ℝ) : Prop
```

def ClassicalFieldTheory.Local.IsEulerLagrangeTrivial[\[source\]](#)

A local lagrangian is Euler-Lagrange-trivial if its Euler-Lagrange operator vanishes along every field.

```
def IsEulerLagrangeTrivial (L : Lagrangian d m k) : Prop
```

structure ClassicalFieldTheory.Local.TotalDivergence[\[source\]](#)

A packaged local total divergence.

The current has order ‘k’, while the associated lagrangian has order ‘k + 1’, matching the coordinate expression ‘ $L = \sum_i D_i B_i$ ’. The Euler-Lagrange-triviality field records the variationally trivial direction needed by later equivalence results.

```
structure TotalDivergence (d m k : ℕ) where
  lagrangian : Lagrangian d m (k + 1)
  current : Fin d → JetPoint d m k → ℝ
  hasTotalDivergenceDensity : HasTotalDivergenceDensity lagrangian current
  isEulerLagrangeTrivial : IsEulerLagrangeTrivial lagrangian
```

instance ClassicalFieldTheory.Local.TotalDivergence.«instance@5c9b98ce»[\[source\]](#)

```
instance : CoeFun (TotalDivergence d m k) (fun _ => Lagrangian d m (k + 1)) where
```

lemma ClassicalFieldTheory.Local.TotalDivergence.actionDensity_eq[\[source\]](#)

```
@[simp]
lemma actionDensity_eq (T : TotalDivergence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  actionDensity T.lagrangian f =
    fun x => ∑ i : Fin d, totalDerivative k i (T.current i) f x
```

lemma ClassicalFieldTheory.Local.TotalDivergence.eulerLagrangeOp_eq_zero[\[source\]](#)

```
lemma eulerLagrangeOp_eq_zero (T : TotalDivergence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp T.lagrangian f = 0
```

lemma ClassicalFieldTheory.Local.TotalDivergence.isCritical[\[source\]](#)

```
lemma isCritical (T : TotalDivergence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hTf : IsAdmissibleForAction T.lagrangian f)
  (hT : Lagrangian.SmoothInCoordinates T.lagrangian) :
  IsCritical T.lagrangian f
```

2.108 PhyslibAlpha.ClassicalFieldTheory.Local.TotalDivergenceEquivalence

[PhyslibAlpha/ClassicalFieldTheory/Local/TotalDivergenceEquivalence.lean](#) on GitHub.

def ClassicalFieldTheory.Local.HasTotalDivergenceDifference[\[source\]](#)

A lagrangian ‘target’ differs from ‘source’ by the density of a packaged total divergence.

The total divergence has current order ‘k’, hence its associated lagrangian and both compared lagrangians have order ‘k + 1’.

```
def HasTotalDivergenceDifference (source target : Lagrangian d m (k + 1))
  (T : TotalDivergence d m k) : Prop
```

def ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent[\[source\]](#)

Two local lagrangians are Euler-Lagrange equivalent when they determine the same local Euler-Lagrange operator along every field.

```
def IsEulerLagrangeEquivalent (source target : Lagrangian d m k) : Prop
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.refl[\[source\]](#)

```
lemma refl (L : Lagrangian d m k) : IsEulerLagrangeEquivalent L L
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.symm[\[source\]](#)

```
lemma symm {L1 L2 : Lagrangian d m k} (h : IsEulerLagrangeEquivalent L1 L2) :
  IsEulerLagrangeEquivalent L2 L1
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.trans[\[source\]](#)

```
lemma trans {L1 L2 L3 : Lagrangian d m k}
  (h12 : IsEulerLagrangeEquivalent L1 L2) (h23 : IsEulerLagrangeEquivalent L2 L3) :
  IsEulerLagrangeEquivalent L1 L3
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.eulerLagrange0p_eq[\[source\]](#)

```
lemma eulerLagrange0p_eq {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrange0p target f = eulerLagrange0p source f
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.eulerLagrange0p_eq_zero_iff[\[source\]](#)

```
lemma eulerLagrange0p_eq_zero_iff {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrange0p target f = 0 ↔ eulerLagrange0p source f = 0
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.isCritical_iff[\[source\]](#)

```
lemma isCritical_iff {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction source f)
  (htargetf : IsAdmissibleForAction target f)
  (hsource : Lagrangian.SmoothInCoordinates source)
  (htarget : Lagrangian.SmoothInCoordinates target) :
  IsCritical target f ↔ IsCritical source f
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.isCritical_target_of_source[\[source\]](#)

```
lemma isCritical_target_of_source {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction source f)
  (htargetf : IsAdmissibleForAction target f)
  (hsource : Lagrangian.SmoothInCoordinates source)
  (htarget : Lagrangian.SmoothInCoordinates target)
  (hcrit : IsCritical source f) :
  IsCritical target f
```

lemma ClassicalFieldTheory.Local.IsEulerLagrangeEquivalent.isCritical_source_of_target[\[source\]](#)

```
lemma isCritical_source_of_target {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction source f)
  (htargetf : IsAdmissibleForAction target f)
  (hsource : Lagrangian.SmoothInCoordinates source)
  (htarget : Lagrangian.SmoothInCoordinates target)
  (hcrit : IsCritical target f) :
  IsCritical source f
```

structure ClassicalFieldTheory.Local.TotalDivergenceEquivalence[\[source\]](#)

A packaged equivalence between two local lagrangians whose densities differ by a total divergence.

The field ‘sameEulerLagrangeOp’ records the variational consequence in the current Alpha API. It will become a theorem once the stack has enough symbolic calculus for coordinate derivatives of total divergences.

```
structure TotalDivergenceEquivalence (d m k : ℕ) where
  source : Lagrangian d m (k + 1)
  target : Lagrangian d m (k + 1)
  divergence : TotalDivergence d m k
  hasTotalDivergenceDifference :
    HasTotalDivergenceDifference source target divergence
  sameEulerLagrangeOp : IsEulerLagrangeEquivalent source target
```

lemma ClassicalFieldTheory.Local.TotalDivergenceEquivalence.actionDensity_eq[\[source\]](#)

```
@[simp]
lemma actionDensity_eq (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  actionDensity E.target f =
    fun x => actionDensity E.source f x + actionDensity E.divergence.lagrangian f x
```

lemma ClassicalFieldTheory.Local.TotalDivergenceEquivalence.eulerLagrangeOp_eq[\[source\]](#)

```
lemma eulerLagrangeOp_eq (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp E.target f = eulerLagrangeOp E.source f
```

lemma ClassicalFieldTheory.Local.TotalDivergenceEquivalence.eulerLagrangeOp_eq_zero_iff[\[source\]](#)

```
lemma eulerLagrangeOp_eq_zero_iff (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp E.target f = 0 ↔ eulerLagrangeOp E.source f = 0
```

lemma ClassicalFieldTheory.Local.TotalDivergenceEquivalence.isCritical_iff[\[source\]](#)

```
lemma isCritical_iff (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction E.source f)
  (htargetf : IsAdmissibleForAction E.target f)
  (hsource : Lagrangian.SmoothInCoordinates E.source)
  (htarget : Lagrangian.SmoothInCoordinates E.target) :
  IsCritical E.target f ↔ IsCritical E.source f
```

lemma ClassicalFieldTheory.Local.TotalDivergenceEquivalence.isCritical_target_of_source[\[source\]](#)

```
lemma isCritical_target_of_source (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction E.source f)
  (htargetf : IsAdmissibleForAction E.target f)
```

```
(hsource : Lagrangian.SmoothInCoordinates E.source)
(htarget : Lagrangian.SmoothInCoordinates E.target)
(hcrit : IsCritical E.source f) :
IsCritical E.target f
```

lemma ClassicalFieldTheory.Local.TotalDivergenceEquivalence.isCritical_source_of_target
[\[source\]](#)

```
lemma isCritical_source_of_target (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction E.source f)
  (htargetf : IsAdmissibleForAction E.target f)
  (hsource : Lagrangian.SmoothInCoordinates E.source)
  (htarget : Lagrangian.SmoothInCoordinates E.target)
  (hcrit : IsCritical E.target f) :
IsCritical E.source f
```

2.109 PhyslibAlpha.ClassicalFieldTheory.Local.Variation

[PhyslibAlpha/ClassicalFieldTheory/Local/Variation.lean](#) on GitHub.

lemma ClassicalFieldTheory.Local.AdmissibleVariation.coord_euclidean [\[source\]](#)

A Euclidean component of an admissible variation is again a test function.

```
@[fun_prop]
lemma coord_euclidean η( : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (a : Fin m) :
IsTestFunction (fun x => η(.toFun x) a)
```

2.110 PhyslibAlpha.ClassicalMechanics.CoupledSpringPotential

[PhyslibAlpha/ClassicalMechanics/CoupledSpringPotential.lean](#) on GitHub.

def couplingPotential [\[source\]](#)

The potential energy of a pair of coupled springs.

```
noncomputable def couplingPotential (x : EuclideanSpace ℝ (Fin 2)) : ℝ
```

lemma couplingPotential_analyticAt [\[source\]](#)

```
lemma couplingPotential_analyticAt (z : EuclideanSpace ℝ (Fin 2)) :
AnalyticAt ℝ couplingPotential z
```

lemma couplingPotential_hasFDerivAt [\[source\]](#)

```
lemma couplingPotential_hasFDerivAt (x : EuclideanSpace ℝ (Fin 2)) :
HasFDerivAt couplingPotential
  ((2 * x 0 + x 1) • (EuclideanSpace.proj []( := ℝ) 0)
   + (x 0 + 2 * x 1) • (EuclideanSpace.proj []( := ℝ) 1)) x
```

lemma couplingPotential_gradient_zero[\[source\]](#)

```
lemma couplingPotential_gradient_zero :
  gradient couplingPotential z (![0, 0] : EuclideanSpace ℝ (Fin 2)) = 0
```

lemma couplingPotential_iteratedFDeriv_two[\[source\]](#)

```
lemma couplingPotential_iteratedFDeriv_two (z : EuclideanSpace ℝ (Fin 2))
  (a b : EuclideanSpace ℝ (Fin 2)) :
  iteratedFDeriv ℝ 2 couplingPotential z ![a, b]
    = 2 * a 0 * b 0 + a 0 * b 1 + a 1 * b 0 + 2 * a 1 * b 1
```

lemma couplingPotential_posDef[\[source\]](#)

```
lemma couplingPotential_posDef :
  (iteratedFDerivQuadraticMap couplingPotential
    z (![0, 0] : EuclideanSpace ℝ (Fin 2))).PosDef
```

lemma coupled_spring_potential[\[source\]](#)

```
lemma coupled_spring_potential :
  let U
```

2.111 PhyslibAlpha.Mathematics.PartialDerivativeTest

[PhyslibAlpha/Mathematics/PartialDerivativeTest.lean](#) on GitHub.

lemma Function.update₀[\[source\]](#)

Update a vector of length 2 in coordinate 0.

```
@[simp]
lemma Function.update α {α : Type*} {a b c : α} : Function.update ![a,b] 0 c = ![c,b]
```

lemma Function.update₁[\[source\]](#)

Update a vector of length 2 in coordinate 1.

```
@[simp]
lemma Function.update α {α : Type*} {a b c : α} : Function.update ![a,b] 1 c = ![a,c]
```

def hessianBilinearCompanion[\[source\]](#)

The Hessian companion as a bilinear map.

```
noncomputable def hessianBilinearCompanion {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] (f : V → ℝ) (x : V) : V →1ℝ[] V →1ℝ[] ℝ
```

def iteratedFDerivQuadraticMap[\[source\]](#)

The second iterated Frechét derivative as a quadratic map.

```
noncomputable def iteratedFDerivQuadraticMap {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] (f : V → ℝ) (x : V) : QuadraticMap ℝ V ℝ
```

def continuousBilinearMapOfContinuousMultilinearMap[\[source\]](#)

A continuous multilinear map is bilinear.

```
noncomputable def continuousBilinearMapOfContinuousMultilinearMap
  {α : Type*} [NontriviallyNormedField α] [CompleteSpace α]
  {V : Type*}
  [NormedAddCommGroup V] [NormedSpace α V] [FiniteDimensional α V]
  (g : ContinuousMultilinearMap α (fun _ : Fin 2 => V) α) : V →L[α] V →L[α] α
```

def QuadraticMap.toMultilinearMap[\[source\]](#)

```
def QuadraticMap.toMultilinearMap {V : Type*} [AddCommGroup V] [Module ℝ V]
  (Q : QuadraticMap ℝ V ℝ) : MultilinearMap ℝ (fun _ : Fin 2 => V) ℝ
```

def QuadraticMap.toMultilinearMapHalfPolarBilin[\[source\]](#)

```
noncomputable def QuadraticMap.toMultilinearMapHalfPolarBilin
  {V : Type*} [AddCommGroup V] [Module ℝ V]
  (Q : QuadraticMap ℝ V ℝ) :
  MultilinearMap ℝ (fun _ : Fin 2 => V) ℝ
```

theorem QuadraticMap.toMultilinearMap_continuous[\[source\]](#)

The multilinear map associated to a quadratic map is continuous.

```
theorem QuadraticMap.toMultilinearMap_continuous {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) : Continuous Q.toMultilinearMap
```

theorem QuadraticMap.toMultilinearMapHalfPolarBilin_continuous[\[source\]](#)

```
theorem QuadraticMap.toMultilinearMapHalfPolarBilin_continuous {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) :
  Continuous Q.toMultilinearMapHalfPolarBilin
```

def QuadraticMap.toContinuousMultilinearMap[\[source\]](#)

Construct a continuous multilinear map from a quadratic map on a finite dimensional space.

```
def QuadraticMap.toContinuousMultilinearMap {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) :
  ContinuousMultilinearMap ℝ (fun _ : Fin 2 → V) ℝ
```

def QuadraticMap.toContinuousMultilinearMapHalfPolarBilin

[\[source\]](#)

```

noncomputable def QuadraticMap.toContinuousMultilinearMapHalfPolarBilin {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V] [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) :
  ContinuousMultilinearMap ℝ (fun _ : Fin 2 → V) ℝ

```

theorem QuadraticMap.toContinuousMultilinearMap_apply

[\[source\]](#)

The constructed continuous multilinear map agrees with the polar bilinear form.

```

theorem QuadraticMap.toContinuousMultilinearMap_apply {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) (x y : V) :
  Q.toContinuousMultilinearMap ![x, y] = Q.polarBilin x y

```

theorem QuadraticMap.toContinuousMultilinearMap_applyHalf

[\[source\]](#)

```

theorem QuadraticMap.toContinuousMultilinearMap_applyHalf {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) (x y : V) :
  Q.toContinuousMultilinearMapHalfPolarBilin ![x, y] = (1/2) * Q.polarBilin x y

```

lemma coercive_of_posdefHalf

[\[source\]](#)

```

lemma coercive_of_posdefHalf {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] {F : QuadraticMap ℝ V ℝ}
  (hf' : F.PosDef) :
  IsCoercive (continuousBilinearMapOfContinuousMultilinearMap
    F.toContinuousMultilinearMapHalfPolarBilin)

```

lemma polarBilin_iteratedFDerivQuadraticMap

[\[source\]](#)

The polar bilinear form of the second-derivative quadratic map is the symmetrized Hessian.

```

lemma polarBilin_iteratedFDerivQuadraticMap {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  ]
  (f : V → ℝ) (ox : V) (x y : V) :
  (iteratedFDerivQuadraticMap f ox).polarBilin x y
  = iteratedFDeriv ℝ 2 f ox ![x, y] + iteratedFDeriv ℝ 2 f ox ![y, x]

```

lemma iteratedFDeriv_two_swap

[\[source\]](#)

Symmetry of the second iterated Fréchet derivative (Schwarz's theorem).

```

lemma iteratedFDeriv_two_swap {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  {f : V → ℝ} {ox : V} (hf : ContDiffAt ℝ 1 f ox) (x y : V) :
  iteratedFDeriv ℝ 2 f ox ![y, x] = iteratedFDeriv ℝ 2 f ox ![x, y]

```


lemma coercive_of_posdef_of_contdiff[\[source\]](#)

Positive definiteness implies coercivity. The proof uses the general fact ‘coercive_of_posdefHalf’ but requires a differentiability assumption on ‘f’.

```
lemma coercive_of_posdef_of_contdiff {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] {f : V → ℝ} {x : V}
  (hf : ContDiffAt ℝ τ f x)
  (hf' : (iteratedFDerivQuadraticMap f x).PosDef) :
  IsCoercive (continuousBilinearMapOfContinuousMultilinearMap
    (iteratedFDeriv ℝ 2 f x))
```

lemma coercive_of_posdef[\[source\]](#)

Positive definiteness implies coercivity.

```
lemma coercive_of_posdef {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] {f : V → ℝ} {x : V}
  (hf' : (iteratedFDerivQuadraticMap f x).PosDef) :
  IsCoercive (continuousBilinearMapOfContinuousMultilinearMap
    (iteratedFDeriv ℝ 2 f x))
```

theorem le_of_little0[\[source\]](#)

```
theorem le_of_little0 {V : Type*}
  [NormedAddCommGroup V] [InnerProductSpace ℝ V]
  {f : V → ℝ} {x x : V} {C : ℝ}
  (hx : C * ||x - x||^2 ≤ iteratedFDeriv ℝ 2 f x fun _ => x - x)
  (hx : fderiv ℝ f x x = fderiv ℝ f x x)
  (hx : ||f x - ∑ i ∈ range 3, 1 / i! * iteratedFDeriv ℝ i f x fun _ => x - x||
    ≤ C / 2 * ||x - x||^2) :
  f x ≤ f x
```

theorem isLocalMin_of_posDef_of_little0[\[source\]](#)

Second partial derivative test, "little oh" form.

```
theorem isLocalMin_of_posDef_of_little0 {V : Type*} [NormedAddCommGroup V]
  [InnerProductSpace ℝ V] [FiniteDimensional ℝ V] {f : V → ℝ} {x : V}
  (h : (fun x => f x - ∑ i ∈ range 3, 1 / i! * iteratedFDeriv ℝ i f x fun _ => x - x)
    =o[nhds x] fun x => ||x - x||^2) (hx : gradient f x = 0) (hcd : ContDiffAt ℝ τ f x)
  (hf : (iteratedFDerivQuadraticMap f x).PosDef) : IsLocalMin f x
```

lemma little0_of_powerseries[\[source\]](#)

Having a power series implies quadratic approximation.

```
lemma little0_of_powerseries {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  {f : V → ℝ} {x : V} {p : FormalMultilinearSeries ℝ V ℝ}
  {r : NNReal} (h : HasFPowerSeriesOnBall f p x r) :
  (fun x => f x - p.partialSum 3 (x - x)) =o[nhds x] fun x => ||x - x||^2
```

lemma isLocalMin.of_subsingleton[\[source\]](#)

```
@[nontriviality]
lemma isLocalMin.of_subsingleton {V : Type*} [TopologicalSpace V]
  [Subsingleton V] {f : V → ℝ} {x : V} : IsLocalMin f x
```

theorem second_derivative_test[\[source\]](#)

The second partial derivative test.

```
theorem second_derivative_test {V : Type*} [NormedAddCommGroup V] [InnerProductSpace ℝ V]
  [FiniteDimensional ℝ V] {f : V → ℝ} {x : V} {p : FormalMultilinearSeries ℝ V ℝ}
  (oh : gradient f x = 0) {r : NNReal}
  (ih : HasFPowerSeriesOnBall f p x r)
  (hf : (iteratedFDerivQuadraticMap f x).PosDef) : IsLocalMin f x
```

theorem second_derivative_test_analyticAt[\[source\]](#)

A more convenient form of the second partial derivative test: it suffices that ‘f’ is analytic at ‘x₀’ (so that an explicit power series need not be produced by hand).

```
theorem second_derivative_test_analyticAt {V : Type*} [NormedAddCommGroup V]
  [InnerProductSpace ℝ V] [FiniteDimensional ℝ V] {f : V → ℝ} {x : V}
  (oh : gradient f x = 0) (han : AnalyticAt ℝ f x)
  (hf : (iteratedFDerivQuadraticMap f x).PosDef) : IsLocalMin f x
```

2.112 PhyslibAlpha.QuantumMechanics.QuantumHarmonicOscillator

[PhyslibAlpha/QuantumMechanics/QuantumHarmonicOscillator.lean](#) on GitHub.

def a[\[source\]](#)

Annihilation operator.

```
def a (x : ℕ → ℂ) : ℕ → ℂ
```

def aLin[\[source\]](#)

```
def aLin : ℕ( → ℂ) →1ℂ[] ℕ( → ℂ)
```

def a_dag[\[source\]](#)

Creation operator.

```
def a_dag (x : ℕ → ℂ) : ℕ → ℂ
```

def a_dagLin[\[source\]](#)

```
def a_dagLin : ℕ( → ℂ) →1ℂ[] ℕ( → ℂ)
```

def ε[\[source\]](#)

```
def ε (n : ℕ) (c : ℂ) : ℕ → ℂ
```

lemma a_dag_eq[\[source\]](#)

Verify that a_dag really is the transpose of a .

```
lemma a_dag_eq (i j : ℕ) : a_dag ε( i 1) j = star (a ε( j 1) i)
```

lemma verify_a[\[source\]](#)

Verify that $a |n + 1\rangle = \sqrt{(n + 1)} |n\rangle$

```
lemma verify_a (n : ℕ) :
  a ε( (n + 1) 1) =
    ε n √((n + 1))
```

lemma verify_a_dag[\[source\]](#)

Verify that $a^\dagger |n\rangle = \sqrt{(n+1)} |n+1\rangle$.

```
lemma verify_a_dag (n : ℕ) :
  a_dag ε( n 1) = ε (n + 1) √((n + 1))
```

lemma verify_a_dag_a[\[source\]](#)

```
lemma verify_a_dag_a (n : ℕ) (x : ℕ → ℂ) :
  a_dag (a x) n = n * x n
```

lemma verify_a_a_dag[\[source\]](#)

```
lemma verify_a_a_dag (n : ℕ) (x : ℕ → ℂ) :
  a (a_dag x) n = (n + 1) * x n
```

lemma commutation_relation[\[source\]](#)

```
lemma commutation_relation :
  a ∘ a_dag - a_dag ∘ a = id
```

def coherentState[\[source\]](#)

```
def coherentState α( : ℂ) : ℕ → ℂ
```

def probabilityOf[\[source\]](#)

```
def probabilityOf (n : ℕ) α( : ℂ) : NNReal
```

lemma probabilityOf_eq_poisson_C[\[source\]](#)

Coherent state has a Poisson distribution.

```
lemma probabilityOf_eq_poisson_C (n : ℕ) α( : ℂ) :
  let Λ
```

lemma coherentState_only_eigenvector[\[source\]](#)

The only eigenvectors of 'a' are the coherent states.

```
lemma coherentState_only_eigenvector α( : ℂ) (v : ℕ → ℂ) :
  a v = α • v ↔
  v = (Complex.exp ↑||α|| ( ^ 2 / 2) * v 0) • coherentState α
```

lemma eigenvector_coherentState[\[source\]](#)

```
lemma eigenvector_coherentState α( : ℂ) :
  a (coherentState α) = α • coherentState α
```

lemma distinct_eigenvectors_a[\[source\]](#)

None of the 'coherentState' eigenvectors are proportional.

```
lemma distinct_eigenvectors_a α( β c : ℂ)
  (hc : coherentState α = c • coherentState β) : α = β
```

lemma a_dagalmost_eigenvector[\[source\]](#)

Formal eigenvectors for 'a_dag' (not in ' $\ell^2(\mathbb{C})$ ') (fails at $n=0$) .

```
lemma a_dagalmost_eigenvector α{ : ℂ} (αh : α ≠ 0) {n : ℕ} (hn : n ≠ 0) :
  a_dag (fun n => √(n.factorial) / α^n) n =
  α • (fun n => √(n.factorial) / α^n) n
```

lemma a_dag_nullspace[\[source\]](#)

```
lemma a_dag_nullspace
  {v : ℕ → ℂ} (hv : a_dag v = 0) : v = 0
```

lemma no_a_dag_eigenvector[\[source\]](#)

```
lemma no_a_dag_eigenvector α( : ℂ) (v : ℕ → ℂ) :
  a_dag v = α • v ↔ v = 0
```

lemma commutationRelation[\[source\]](#)

```
lemma commutationRelation : {aLin, }a_dagLin = 1
```

def coherentState_ℓ2[\[source\]](#)

The coherent state belongs to ' $\ell^2(\mathbb{C})$ '.

```
def coherentState_2 α( : ℂ) : lp (fun _ : ℕ => ℂ) 2
```

2.113 PhyslibAlpha.QuantumMechanics.StinespringDilation

[PhyslibAlpha/QuantumMechanics/StinespringDilation.lean](#) on GitHub.

def krausApply[\[source\]](#)

Completely positive map given by a (not necessarily minimal) Kraus family.

```
def krausApply {R : Type*} [Mul R] [Star R] [AddCommMonoid R]
  {q r : Type*} [Fintype q] [Fintype r]
  (K : r → Matrix q q R) (ρ : Matrix q q R) : Matrix q q R
```

lemma krausApply.posSemidef[\[source\]](#)

Kraus operator preserves PSD property.

```
lemma krausApply.posSemidef {R : Type*} [Ring R] [PartialOrder R] [StarRing R]
  [AddLeftMono R]
  {q r : Type*} [Fintype q] [Fintype r]
  (K : r → Matrix q q R)
  (ρ : Matrix q q R) (ph : ρ.PosSemidef) :
  (krausApply K ρ).PosSemidef
```

def QuantumChannel[\[source\]](#)

Quantum channel.

```
def QuantumChannel {R : Type*} [Mul R] [One R] [Star R] [AddCommMonoid R]
  {q r : Type*} [Fintype q] [Fintype r] [DecidableEq q]
  (K : r → Matrix q q R)
```

def QuantumOperation[\[source\]](#)

Quantum operation.

```
def QuantumOperation {R : Type*} [RCLike R]
  {q r : Type*} [Fintype q] [Fintype r] [DecidableEq q]
  (K : r → Matrix q q R)
```

def densityMatrix[\[source\]](#)

Density matrix.

```
def densityMatrix {R : Type*} [Ring R] [PartialOrder R] [StarRing R] (d : Type*) [Fintype d]
```

def densityMatrix.convex_comb[\[source\]](#)

Density matrices are closed under real convex combinations.

```
def densityMatrix.convex_comb {R : Type*} [RCLike R]
  {d : ℕ} (ρ₀ (ρ₁ : densityMatrix (Fin d) (R := R)) {t : R}
  (h₀ : 0 ≤ t) (h₁ : 0 ≤ 1 - t) : densityMatrix (Fin d) (R := R)
```

def tr₂[\[source\]](#)

Also known as ‘partialTraceRight’.

```
def ztr {R : Type*} [Ring R] {m n m' : Type*} [Fintype n]
  (ρ : Matrix (m × n) (m' × n) R) : Matrix m m' R
```

def stinespringOp[\[source\]](#)

‘stinespringOp’ is often written as ‘V’.

```
def stinespringOp {R : Type*} [Ring R]
  {m r : Type*} [Fintype r] [DecidableEq r]
  (K : r → Matrix m m R) : Matrix (m × r) m R
```

def stinespringDilation[\[source\]](#)

The Stinespring dilation.

```
def stinespringDilation {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R)
  ρ( : Matrix m m R)
```

def stinespringForm[\[source\]](#)

The partial trace of the Stinespring dilation.

```
def stinespringForm {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R)
```

lemma stinespringOp_adjoint_mul_self[\[source\]](#)

A useful identity for Stinespring dilations.

```
lemma stinespringOp_adjoint_mul_self {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R) :
  ∑ i, star K i * K i = (stinespringOp K)h * stinespringOp K
```

lemma stinespringForm_CPTNI[\[source\]](#)

A useful identity for completely positive, trace non-increasing maps.

```
lemma stinespringForm_CPTNI {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  (K : r → Matrix m m R)
  (hK : ∑ i, (K i)h * K i ≤ 1) :
  (stinespringOp K)h * (stinespringOp K) ≤ 1
```

lemma stinespringForm_CPTP_isometry[\[source\]](#)

The Stinespring operator of a completely positive, trace-preserving maps is an isometry. (Note that ‘stinespringOp K’ is not a square matrix in general.)

```
lemma stinespringForm_CPTP_isometry {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (hK : ∑ i, (K i)h * K i = 1) :
  (stinespringOp K)h * (stinespringOp K) = 1
```

lemma stinespringOrtho[\[source\]](#)

Proving the columns of ‘ $V = \text{stinespringOp } K$ ’ are independent is a step on the way to constructing the unitary dilation.

```
lemma stinespringOrtho {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) :
  Orthonormal  $\square$  ( := R)
  fun j : m => WithLp.toLp 2 fun i : m × r => stinespringOp K i j
```

lemma basisCard[\[source\]](#)

‘m’ will of course be finite and bounded by ‘n’ here, but no need to assume or prove that.

```
lemma basisCard {R : Type*} [RCLike R] {n m : Type*} [Fintype n] {s : Matrix n m R}
  (ho : Orthonormal R fun j ↦ WithLp.toLp 2 fun i ↦ s i j) :
  Fintype.card n =
  ho.toSubtypeRange.exists_orthonormalBasis_extension.choose.card
```

lemma stinespringCard[\[source\]](#)

Calculating the cardinality of the orthonormal basis obtained by extending the Stinespring orthonormal set, the columns of ‘ $\text{stinespringOp } K$ ’.

```
lemma stinespringCard {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) :
  Fintype.card (m × r) = (stinespringOrtho
  hK).toSubtypeRange.exists_orthonormalBasis_extension.choose.card
```

theorem complCard[\[source\]](#)

We need the 1 matrix, which we don’t seem to have for an arbitrary ‘ $[Fintype m]$ ’. Since we are comparing ‘ $\text{Fin } r$ ’ and ‘ $\text{Fin } (r-1)$ ’ we also cannot too easily use an arbitrary ‘ $[Fintype r]$ ’ $[Zero r]$ ’.

```
theorem complCard {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  let  $\square$ 
```

def Fin.predAbove_of_ne[\[source\]](#)

See discussion at <https://leanprover.zulipchat.com/#narrow/channel/217875-Is-there-code-for-X.3F/topic/succAbove.20and.20predAbove.20lemmas/with/584270574>

```
def Fin.predAbove_of_ne {n : ℕ} {k i : Fin n}
  (h : i ≠ k) : Fin (n - 1)
```

lemma Fin.predAbove_of_ne_injective[\[source\]](#)

A "missing lemma" for 'Fin' types.

```
lemma Fin.predAbove_of_ne_injective (n : ℕ) (k x y : Fin n)
  (hx : x ≠ k) (hy : y ≠ k)
  (heq : Fin.predAbove_of_ne hx = Fin.predAbove_of_ne hy) : x = y
```

def onbPart

[\[source\]](#)

The way this is written, 'Fin r' and 'Fin (r-1)' both occur so it is tricky to go to a general 'Fintype'.

```
def onbPart {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (x : Fin m × Fin r) {z : Fin r} (hx : ¬x.2 = z) :
```

lemma onbPart_inner

[\[source\]](#)

```
lemma onbPart_inner {R : Type*} [RCLike R] {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) {z : Fin r}
  {y : Fin m × Fin r} (hy : ¬y.2 = z)
  {x : Fin m × Fin r} (hx : ¬x.2 = z)
  (h : y ≠ x) :
  inner R (WithLp.toLp 2 <| onbPart hK y hy)
    (WithLp.toLp 2 <| onbPart hK x hx) = 0
```

lemma onbPart_norm

[\[source\]](#)

The vectors in the Stinespring orthonormal basis have norm 1.

```
lemma onbPart_norm {R : Type*} [RCLike R] {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (x : Fin m × Fin r)
  {z : Fin r} (hx : ¬x.2 = z) :
  ||WithLp.toLp 2 <| onbPart hK x ||hx = 1
```

def Ud

[\[source\]](#)

Also known as 'unitaryDilation'. Respects x,y order.

```
def Ud {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  Matrix (Fin m × Fin r) (Fin m × Fin r) R
```

def general_dilation

[\[source\]](#)

This generalization of Stinespring dilation has the right "shape" but otherwise nothing specific to it.

```
def general_dilation {R : Type*}
  {m r : Type*} [DecidableEq r]
  (z : r)
  (S : Matrix (m × r) m R)
  (M : Matrix (m × r) (m × r) R) :
  Matrix (m × r) (m × r) R
```


def dilation[\[source\]](#)

A general, not necessarily unitary, dilation.

```
def dilation {R : Type*} [Ring R]
  {m r : Type*} [Fintype r] [DecidableEq r]
  (K : r → Matrix m m R) (z : r) (M : Matrix (m × r) (m × r) R) :
  Matrix (m × r) (m × r) R
```

theorem Ud_orthonormal₁[\[source\]](#)

One version of orthonormality of ‘stinespringOp’.

```
theorem 1Ud_orthonormal {R : Type*} [RCLike R] {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  Orthonormal R fun y ↦ if hy : y.2 = z then WithLp.toLp 2 fun i ↦ stinespringOp K i y.1
  else WithLp.toLp 2 fun i ↦ onbPart hK y hy i
```

theorem Ud_orthonormal₂[\[source\]](#)

The Stinespring dilation columns form an orthonormal basis.

```
theorem 2Ud_orthonormal {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  Orthonormal R fun y ↦
    WithLp.toLp 2 fun i ↦ if hy : y.2 = z then stinespringOp K i y.1 else onbPart hK y hy i
```

lemma smul_self_one_of_norm_one[\[source\]](#)

If ‘β’ has length 1 then the dot product of ‘β’ with itself is 1.

```
lemma smul_self_one_of_norm_one {R : Type*} [RCLike R]
  {t : Type*} [Fintype t] {β : t → R} (hj : ||WithLp.toLp 2 β|| = 1) :
  ∑ x, (starRingEnd R) β( x) * β x = 1
```

theorem unitary_of_orthonormal[\[source\]](#)

A matrix whose columns are orthonormal is unitary.

```
theorem unitary_of_orthonormal {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  α( : Matrix (m × r) (m × r) R)
  (oh : Orthonormal R fun i ↦ WithLp.toLp 2 α( i)) : α * star α = 1
```

lemma Ud_unitaryT[\[source\]](#)

The transpose of the unitary dilation is unitary.

```
lemma Ud_unitaryT {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  (Ud hK z)T ∈ unitary _
```

lemma Ud_unitary[\[source\]](#)

The unitary dilation ‘Ud’ is in fact unitary.

```
lemma Ud_unitary {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  (Ud hK z) ∈ unitary _
```

lemma tr₂_e₀Xe₀[\[source\]](#)

Taking the partial trace of a tensor product with a matrix of trace 1 is the identity map.

```
lemma ztr_eXe {R : Type*} [RCLike R]
  {m w : Type*} [Fintype w] [Zero w]
  (e : Matrix w w R) (htr : e.trace = 1)
  (ρ : Matrix m m R) :
  ztr ρ( ⊗ e) = ρ
```

def stinespringUnitaryForm[\[source\]](#)

The Stinespring unitary form.

```
def stinespringUnitaryForm {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r)
  (ρ : Matrix (Fin m) (Fin m) R) :
  (Matrix (Fin m) (Fin m) R)
```

def stinespringUnitaryForm_e[\[source\]](#)

The Stinespring unitary form, general version.

```
def stinespringUnitaryForm_e {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) (e : Matrix (Fin r) (Fin r) R)
  (ρ : Matrix (Fin m) (Fin m) R) :
  (Matrix (Fin m) (Fin m) R)
```

theorem tracefree_version[\[source\]](#)

Trace-free version of the Stinespring Dilation Theorem.

```
theorem tracefree_version {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Zero r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (ρ : Matrix m m R) :
  let K'
```

theorem heisenberg_schrödinger[\[source\]](#)

A Heisberg picture / Schrödinger picture view of the Stinespring dilation.

```
theorem heisenberg_schrödinger {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Zero r] [Fintype m] [DecidableEq m]
```

```

{K : r → Matrix m m R}
p( : Matrix m m R) :
let K'

```

def generalForm[\[source\]](#)

A further generalization of ‘stinespringGeneralForm’.

```

def generalForm {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (z : r)
  (S : Matrix (m × r) m R)
  (M : Matrix (m × r) (m × r) R)

```

def stinespringGeneralForm[\[source\]](#)

General form of the Stinespring dilation.

```

def stinespringGeneralForm {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R) (z : r)
  (M : Matrix (m × r) (m × r) R)

```

def stinespringGeneralForm_e[\[source\]](#)

Even more general form of the Stinespring dilation.

```

def stinespringGeneralForm_e {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R) (z : r) (e : Matrix r r R)
  (M : Matrix (m × r) (m × r) R)

```

theorem unitaryForm_of_general[\[source\]](#)

When we plug in ‘M = Ud hK’ into the general ‘stinespringGeneralForm’, then we do get ‘stinespringUnitaryForm hK’.

```

theorem unitaryForm_of_general {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  stinespringGeneralForm K z (Ud hK z) =
  stinespringUnitaryForm hK z

```

theorem unitaryForm_of_general_e[\[source\]](#)

The Stinespring unitary form as a general form applied to the unitary dilation.

```

theorem unitaryForm_of_general_e {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) (e : Matrix (Fin r) (Fin r) R) :
  stinespringGeneralForm_e K z e (Ud hK z) =
  stinespringUnitaryForm_e hK z e

```

lemma stinespringGeneralForm_works[\[source\]](#)

Note we don't need any special properties of M , and we don't need K to be CPTP.

Uses 'Fin' types because of the use of 'Fin.sum_univ_succAbove' in the proof.

```
lemma stinespringGeneralForm_works {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R) (z : Fin r)
  (M : Matrix (Fin m × Fin r) (Fin m × Fin r) R) :
  stinespringGeneralForm K z M = krausApply K
```

lemma stinespringUnitaryForm_works[\[source\]](#)

Notice that unitarity is a side property, it is not why the Stinespring form works.

Here 'z' is the coordinate used for the ancilla.

```
lemma stinespringUnitaryForm_works {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  stinespringUnitaryForm hK z = krausApply K
```

def krausCompletion[\[source\]](#)

The "orthogonal" CPTP completion of a CPTNI map. 'Vtilde' is an alternative name for 'krausCompletion'.

```
def krausCompletion {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R) :
  Matrix (Fin m × Fin (r+1)) (Fin m) R
```

theorem stinespringOp_apply[\[source\]](#)

Entrywise formula for the Stinespring isometry: its ' $((x_1, x_2), y)$ ' entry is ' $K x_2 x_1 y$ '.

```
theorem stinespringOp_apply {R : Type*} [Ring R] {m r : Type*} [Fintype r] [DecidableEq r]
  [Fintype m] [DecidableEq m] (K : r → Matrix m m R) (x : m × r) (y : m) :
  stinespringOp K x y = K x.2 x.1 y
```

theorem stinespringOp_gram[\[source\]](#)

*The Gram matrix of the Stinespring isometry is ' $\sum i, (K i)^H * K i$ '.*

```
theorem stinespringOp_gram {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R) :
  (stinespringOp K)H * stinespringOp K = ∑ i, star (K i) * K i
```

lemma krausCompletion_isometry_of_TNI[\[source\]](#)

Mar 14, 2026 by Bjørn for 4.27 June 13, 2026 by Aristotle for 4.31 including 'stinespringOp_gram' and 'stinespringOp_apply'.

```
lemma krausCompletion_isometry_of_TNI {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, star K i * K i ≤ 1) :
  (krausCompletion K)H * krausCompletion K = 1
```

def unital[\[source\]](#)*A unital operator.*

```
def unital {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R)
```

def subunital[\[source\]](#)*A subunital operator.*

```
def subunital {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R)
```

lemma partialTrace_tensor[\[source\]](#)*The identity $\text{Tr}_B (A \otimes B) = \text{Tr}(B) \cdot A$*

```
lemma partialTrace_tensor {R : Type*} [RCLike R] {m n : ℕ}
  (A : Matrix (Fin m) (Fin m) R) (B : Matrix (Fin n) (Fin n) R) :
  ztr (A ⊗k B) = (trace B) • A
```

lemma krausApply_of_tensor[\[source\]](#)*A unitary dilation view of the application of a Kraus operator.*

```
lemma krausApply_of_tensor {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r)
  ρ( α : Matrix (Fin m) (Fin m) R) β( : Matrix (Fin r) (Fin r) R)
  (βh : β.trace = 1) (h : (Ud hK z) * ρ( ⊗k (single z z 1)) * (Ud hK z)H = α ⊗k β) :
  krausApply K ρ = α
```

lemma trace_tr₂[\[source\]](#)*Trace of partial trace equals trace.*

```
lemma ztrace_tr {R : Type*} [RCLike R] {m n : ℕ}
  ρ( : Matrix (Fin m × Fin n) (Fin m × Fin n) R) :
  trace ρ = trace (ztr ρ)
```

def krausCompletionChannelMap[\[source\]](#)*The Kraus completion as a map from operations to channels.*

```
def krausCompletionChannelMap {R : Type*} [RCLike R] {q r : ℕ}
  {K : Fin r → Matrix (Fin q) (Fin q) R} (hK : QuantumOperation K) :
  {K : Fin (r+1) → Matrix (Fin q) (Fin q) R | QuantumChannel K}
```

lemma CPTP_of_CPTNI[\[source\]](#)*The "not orthogonal" CPTP completion of a CPTNI map.*

```
lemma CPTP_of_CPTNI {R : Type*} [RCLike R]
  {q r : ℕ}
  {K : Fin r → Matrix (Fin q) (Fin q) R}
```

```

(hq : QuantumOperation K) :
  ∃ K' : Fin (r+1) → Matrix (Fin q) (Fin q) R,
  QuantumChannel K' ∧
  ∀ i, ∀ H : i ≠ Fin.last r, K' i = K (i.1, Fin.val_lt_last )H

```

def partialTraceLeft[\[source\]](#)

Partial trace on the left of a tensor product.

```

def partialTraceLeft {R : Type*} [RCLike R]
  {m n : Type*} [Fintype m]
  ρ( : Matrix (m × n)
      (m × n) R) : Matrix (n) (n) R

```

theorem stinespringForm_eq[\[source\]](#)

A version of the Stinespring Dilation Theorem.

```

theorem stinespringForm_eq {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R)
  ρ( : Matrix (Fin m) (Fin m) R) :
  ⌊tr (stinespringDilation K ρ) = krausApply K ρ

```

2.114 PhyslibAlpha.SpaceAndTime.Space.Surfaces.HalfPlane

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/HalfPlane.lean](#) on GitHub.

def Space.halfPlaneDomain[\[source\]](#)

The domain of the half-plane inside ‘Space 2’.

```

def halfPlaneDomain : Set (Space 2)

```

def Space.halfPlane[\[source\]](#)

The coordinate plane embedding used for the half-plane surface in ‘Space 3’.

```

def halfPlane : Space 2 → Space 3

```

lemma Space.halfPlane_eq[\[source\]](#)

```

lemma halfPlane_eq : halfPlane = (slice (2 : Fin 3)).symm ∘ (fun x : Space 2 => (0, x))

```

lemma Space.halfPlane_injective[\[source\]](#)

```

lemma halfPlane_injective : Function.Injective halfPlane

```

lemma Space.halfPlane_continuous[\[source\]](#)

```

@[fun_prop]
lemma halfPlane_continuous : Continuous halfPlane

```

lemma Space.halfPlane_measurableEmbedding[\[source\]](#)

```
lemma halfPlane_measurableEmbedding : MeasurableEmbedding halfPlane
```

lemma Space.norm_halfPlane[\[source\]](#)

```
@[simp]
```

```
lemma norm_halfPlane (x : Space 2) : ||halfPlane ||x = |||x
```

def Space.halfPlaneMeasure[\[source\]](#)

The measure on ‘Space 3’ corresponding to integration over a half-plane.

```
def halfPlaneMeasure : Measure (Space 3)
```

instance Space.halfPlaneMeasure_hasTemperateGrowth[\[source\]](#)

```
instance halfPlaneMeasure_hasTemperateGrowth :  
  halfPlaneMeasure.HasTemperateGrowth
```

instance Space.halfPlaneMeasure_sFinite[\[source\]](#)

```
instance halfPlaneMeasure_sFinite : SFinite halfPlaneMeasure
```

def Space.halfPlaneDist[\[source\]](#)

The distribution on ‘Space 3’ corresponding to integration over a half-plane.

```
def halfPlaneDist : (Space 3) → dℝ[] ℝ
```

lemma Space.halfPlaneDist_apply_eq_integral_halfPlaneMeasure[\[source\]](#)

```
lemma halfPlaneDist_apply_eq_integral_halfPlaneMeasure (f : ℝ(Space 3, ℝ)) :  
  halfPlaneDist f = ∫ x, f x ∂halfPlaneMeasure
```

lemma Space.halfPlaneDist_apply_eq_integral_volume[\[source\]](#)

```
lemma halfPlaneDist_apply_eq_integral_volume (f : ℝ(Space 3, ℝ)) :  
  halfPlaneDist f = ∫ x, f (halfPlane x) ∂(volume.restrict halfPlaneDomain)
```

def Space.halfPlaneSubmodule[\[source\]](#)

The coordinate plane containing the half-plane in ‘Space 3’.

```
def halfPlaneSubmodule : Submodule ℝ (Space 3) where
```

lemma Space.halfPlane_mem_halfPlaneSubmodule[\[source\]](#)

```
lemma halfPlane_mem_halfPlaneSubmodule (x : Space 2) : halfPlane x ∈ halfPlaneSubmodule
```

lemma Space.halfPlane_image_domain_subset_halfPlaneSubmodule[\[source\]](#)

```
lemma halfPlane_image_domain_subset_halfPlaneSubmodule :
  halfPlane '' halfPlaneDomain  $\subseteq$  (halfPlaneSubmodule : Set (Space 3))
```

lemma Space.halfPlaneSubmodule_ne_top[\[source\]](#)

```
lemma halfPlaneSubmodule_ne_top : halfPlaneSubmodule  $\neq$   $\top$ 
```

lemma Space.volume_halfPlane_image_domain[\[source\]](#)

```
lemma volume_halfPlane_image_domain :
  volume (halfPlane '' halfPlaneDomain : Set (Space 3)) = 0
```

2.115 PhyslibAlpha.SpaceAndTime.Space.Surfaces.Line

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/Line.lean](#) on GitHub.

def Space.line[\[source\]](#)

The coordinate line embedded in ‘Space d’.

```
def line (d :  $\mathbb{N}$ ) [NeZero d] :  $\mathbb{R} \rightarrow$  Space d
```

lemma Space.line_eq_smul_basis[\[source\]](#)

```
lemma line_eq_smul_basis (d :  $\mathbb{N}$ ) [NeZero d] :
  line d = fun r => r • basis (0 : Fin d)
```

lemma Space.line_injective[\[source\]](#)

```
lemma line_injective (d :  $\mathbb{N}$ ) [NeZero d] : Function.Injective (line d)
```

lemma Space.line_continuous[\[source\]](#)

```
@[fun_prop]
lemma line_continuous (d :  $\mathbb{N}$ ) [NeZero d] : Continuous (line d)
```

lemma Space.line_measurableEmbedding[\[source\]](#)

```
lemma line_measurableEmbedding (d :  $\mathbb{N}$ ) [NeZero d] : MeasurableEmbedding (line d)
```

lemma Space.norm_line[\[source\]](#)

```
@[simp]
lemma norm_line (d :  $\mathbb{N}$ ) [NeZero d] (r :  $\mathbb{R}$ ) : ||line d ||r = |||r
```

def Space.lineMeasure[\[source\]](#)

The measure on ‘Space d’ corresponding to integration along a coordinate line.

```
def lineMeasure (d :  $\mathbb{N}$ ) [NeZero d] : Measure (Space d)
```


instance Space.lineMeasure_hasTemperateGrowth[\[source\]](#)

```
instance lineMeasure_hasTemperateGrowth (d : ℕ) [NeZero d] :
  (lineMeasure d).HasTemperateGrowth
```

def Space.lineDist[\[source\]](#)

*The distribution on ‘Space d’ corresponding to integration along a coordinate line.
One can roughly think of this distribution as taking a test function ‘f’ to its integral
against a mass, charge or current density concentrated on a line.*

```
def lineDist (d : ℕ) [NeZero d] : (Space d) → dℝ[] ℝ
```

lemma Space.lineDist_apply_eq_integral_lineMeasure[\[source\]](#)

```
lemma lineDist_apply_eq_integral_lineMeasure (d : ℕ) [NeZero d] (f : ℝ(Space d, ℝ)) :
  lineDist d f = ∫ x, f x ∂lineMeasure d
```

lemma Space.lineDist_apply_eq_integral_volume[\[source\]](#)

```
lemma lineDist_apply_eq_integral_volume (d : ℕ) [NeZero d] (f : ℝ(Space d, ℝ)) :
  lineDist d f = ∫ r : ℝ, f (line d r)
```

def Space.lineSubmodule[\[source\]](#)

The linear subspace spanned by the coordinate line in ‘Space d’.

```
def lineSubmodule (d : ℕ) [NeZero d] : Submodule ℝ (Space d)
```

lemma Space.line_mem_lineSubmodule[\[source\]](#)

```
lemma line_mem_lineSubmodule (d : ℕ) [NeZero d] (r : ℝ) : line d r ∈ lineSubmodule d
```

lemma Space.range_line_subset_lineSubmodule[\[source\]](#)

```
lemma range_line_subset_lineSubmodule (d : ℕ) [NeZero d] :
  Set.range (line d) ⊆ (lineSubmodule d : Set (Space d))
```

lemma Space.lineSubmodule_ne_top[\[source\]](#)

```
lemma lineSubmodule_ne_top (d : ℕ) [NeZero d] (hd : 2 ≤ d) : lineSubmodule d ≠ ⊤
```

lemma Space.volume_line_range[\[source\]](#)

```
lemma volume_line_range (d : ℕ) [NeZero d] (hd : 2 ≤ d) :
  volume (Set.range (line d) : Set (Space d)) = 0
```

2.116 PhyslibAlpha.SpaceAndTime.Space.Surfaces.Ring

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/Ring.lean](#) on GitHub.

def Space.ring[\[source\]](#)

The embedding of the unit ring (a circle ' S^1 ' in 'Space 2') into 'Space 3'.

```
def ring : Metric.sphere (0 : Space 2) 1 → Space 3
```

lemma Space.ring_eq[\[source\]](#)

```
lemma ring_eq : ring = (slice 2).symm ∘ (fun x => (0, sphericalShell 2 x))
```

lemma Space.ring_injective[\[source\]](#)

```
lemma ring_injective : Function.Injective ring
```

lemma Space.ring_continuous[\[source\]](#)

```
@[fun_prop]
lemma ring_continuous : Continuous ring
```

lemma Space.ring_measurableEmbedding[\[source\]](#)

```
lemma ring_measurableEmbedding : MeasurableEmbedding ring
```

def Space.ringMeasure[\[source\]](#)

The measure on 'Space 3' corresponding to integration around a ring.

```
def ringMeasure : Measure (Space 3)
```

instance Space.ringMeasure_hasTemperateGrowth[\[source\]](#)

```
instance ringMeasure_hasTemperateGrowth :
  ringMeasure.HasTemperateGrowth
```

instance Space.ringMeasure_prod_volume_hasTemperateGrowth[\[source\]](#)

```
instance ringMeasure_prod_volume_hasTemperateGrowth :
  (ringMeasure.prod (volume α( := Space))).HasTemperateGrowth
```

instance Space.ringMeasure_sFinite[\[source\]](#)

```
instance ringMeasure_sFinite: SFinite ringMeasure
```

instance Space.ringMeasure_finite[\[source\]](#)

```
instance ringMeasure_finite: IsFiniteMeasure ringMeasure
```

lemma Space.integrable_ringMeasure_of_continuous[\[source\]](#)

```
lemma integrable_ringMeasure_of_continuous (f : Space → ℝ) (hf : Continuous (f ∘ ring)) :
  Integrable f ringMeasure
```

lemma Space.integrable_ringMeasure_of_continuous_euclid[\[source\]](#)

```
lemma integrable_ringMeasure_of_continuous_euclid (f : Space → EuclideanSpace ℝ (Fin n))
  (hf : Continuous (f ∘ ring)) :
  Integrable f ringMeasure
```

lemma Space.ringMeasure_prod_volume_map[\[source\]](#)

```
lemma ringMeasure_prod_volume_map :
  (ringMeasure.prod (volume α( := Space))).map (fun x : Space × Space => (x.1, x.2 + x.1))
  = (ringMeasure.prod (volume α( := Space)))
```

lemma Space.ringMeasure_univ[\[source\]](#)

```
@[simp]
lemma ringMeasure_univ : ringMeasure Set.univ = ENNReal.ofReal ((2 : ℝ) * π)
```

def Space.ringDist[\[source\]](#)

The distribution on ‘Space 3’ corresponding to integration around a ring.

```
def ringDist : (Space 3) → dℝ[] ℝ
```

lemma Space.ringDist_apply_eq_integral_ringMeasure[\[source\]](#)

```
lemma ringDist_apply_eq_integral_ringMeasure (f : ℝ(Space 3, ℝ)) :
  ringDist f = ∫ x, f x ∂ringMeasure
```

lemma Space.ringDist_eq_integral_delta[\[source\]](#)

```
lemma ringDist_eq_integral_delta (f : ℝ(Space 3, ℝ)) :
  ringDist f = ∫ z, diracDelta ℝ z f ∂ringMeasure
```

2.117 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SolidCylinder

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SolidCylinder.lean](#) on GitHub.

def Space.solidCylinder[\[source\]](#)

The map embedding a cross-sectional disk extruded along the axis into ‘Space 3’. The disk is cut out by the solid-cylinder measure, which is supported on the closed unit disk.

```
def solidCylinder : Space 2 × ℝ → Space 3
```

lemma Space.solidCylinder_eq[\[source\]](#)

```
lemma solidCylinder_eq :
  solidCylinder = (slice 2).symm ∘ (fun x : Space 2 × ℝ => (x.2, x.1))
```

lemma Space.solidCylinder_injective[\[source\]](#)

```
lemma solidCylinder_injective : Function.Injective solidCylinder
```

lemma Space.solidCylinder_continuous[\[source\]](#)

```
@[fun_prop]
```

```
lemma solidCylinder_continuous : Continuous solidCylinder
```

lemma Space.solidCylinder_measurableEmbedding[\[source\]](#)

```
lemma solidCylinder_measurableEmbedding : MeasurableEmbedding solidCylinder
```

lemma Space.norm_solidCylinder[\[source\]](#)

```
@[simp]
```

```
lemma norm_solidCylinder (x : Space 2 × ℝ) :  
  ||solidCylinder ||x = √(||x||.2 ^ 2 + ||x||.1 ^ 2)
```

def Space.solidCylinderMeasure[\[source\]](#)

The measure on ‘Space 3’ corresponding to integration over a solid cylinder, i.e. the pushforward of the product of the solid-sphere (closed unit disk) measure on ‘Space 2’ with the line measure along the axis.

```
def solidCylinderMeasure : Measure (Space 3)
```

instance Space.solidCylinderMeasure_hasTemperateGrowth[\[source\]](#)

```
instance solidCylinderMeasure_hasTemperateGrowth :  
  solidCylinderMeasure.HasTemperateGrowth
```

instance Space.solidCylinderMeasure_sFinite[\[source\]](#)

```
instance solidCylinderMeasure_sFinite : SFinite solidCylinderMeasure
```

def Space.solidCylinderDist[\[source\]](#)

The distribution on ‘Space 3’ corresponding to integration over a solid cylinder. One can roughly think of this distribution as taking a test function ‘f’ to its integral against a mass, charge or current density spread over a solid cylinder.

```
def solidCylinderDist : (Space 3) → dℝ[] ℝ
```

lemma Space.solidCylinderDist_apply_eq_integral_solidCylinderMeasure[\[source\]](#)

```
lemma solidCylinderDist_apply_eq_integral_solidCylinderMeasure (f : ℝ → ℝ) :  
  solidCylinderDist f = ∫ x, f x ∂solidCylinderMeasure
```

lemma Space.solidCylinderDist_apply_eq_integral_disk_volume[\[source\]](#)

```
lemma solidCylinderDist_apply_eq_integral_disk_volume (f :  $\mathbb{R}$  →  $\mathbb{R}$ ) :
  solidCylinderDist f =
   $\int x, f (\text{solidCylinder } x) \partial((\text{solidSphereMeasure } 2).prod (\text{volume } \alpha( := \mathbb{R})))$ 
```

lemma Space.solidCylinderMeasure_univ_pos[\[source\]](#)

```
lemma solidCylinderMeasure_univ_pos : 0 < solidCylinderMeasure Set.univ
```

2.118 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SolidSphere

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SolidSphere.lean](#) on GitHub.

def Space.solidSphere[\[source\]](#)

The inclusion into ‘Space d’ of its closed unit ball ‘ B^d ’ (the subtype coercion ‘ $x \mapsto x.1$ ’).

```
def solidSphere (d :  $\mathbb{N}$ ) : Metric.closedBall (0 : Space d) 1 → Space d
```

lemma Space.solidSphere_injective[\[source\]](#)

```
lemma solidSphere_injective (d :  $\mathbb{N}$ ) : Function.Injective (solidSphere d)
```

lemma Space.solidSphere_continuous[\[source\]](#)

```
lemma solidSphere_continuous (d :  $\mathbb{N}$ ) : Continuous (solidSphere d)
```

lemma Space.solidSphere_measurableEmbedding[\[source\]](#)

```
lemma solidSphere_measurableEmbedding (d :  $\mathbb{N}$ ) : MeasurableEmbedding (solidSphere d)
```

lemma Space.norm_solidSphere_le[\[source\]](#)

```
@[simp]
lemma norm_solidSphere_le (d :  $\mathbb{N}$ ) (x : Metric.closedBall (0 : Space d) 1) :
  ||solidSphere d ||x ≤ 1
```

def Space.solidSphereMeasure[\[source\]](#)

The measure on ‘Space d’ corresponding to integration over a solid sphere, i.e. the ambient volume measure restricted to the closed unit ball.

```
def solidSphereMeasure (d :  $\mathbb{N}$ ) : Measure (Space d)
```

instance Space.solidSphereMeasure_isFiniteMeasure[\[source\]](#)

```
instance solidSphereMeasure_isFiniteMeasure (d :  $\mathbb{N}$ ) :
  IsFiniteMeasure (solidSphereMeasure d)
```

instance Space.solidSphereMeasure_hasTemperateGrowth[\[source\]](#)

```
instance solidSphereMeasure_hasTemperateGrowth (d : ℕ) :
  (solidSphereMeasure d).HasTemperateGrowth
```

def Space.solidSphereDist[\[source\]](#)

The distribution on ‘Space d’ corresponding to integration over a solid sphere. One can roughly think of this distribution as taking a test function ‘f’ to its integral against a mass, charge or current density spread over a solid ball.

```
def solidSphereDist (d : ℕ) : (Space d) → ℝ → ℝ
```

lemma Space.solidSphereDist_apply_eq_integral_solidSphereMeasure[\[source\]](#)

```
lemma solidSphereDist_apply_eq_integral_solidSphereMeasure (d : ℕ) (f : ℝ → ℝ) :
  solidSphereDist d f = ∫ x, f x ∂solidSphereMeasure d
```

lemma Space.solidSphereDist_apply_eq_integral_closedBall[\[source\]](#)

```
lemma solidSphereDist_apply_eq_integral_closedBall (d : ℕ) (f : ℝ → ℝ) :
  solidSphereDist d f = ∫ x in Metric.closedBall (0 : Space d) 1, f x
```

lemma Space.solidSphere_volume_pos[\[source\]](#)

```
lemma solidSphere_volume_pos (d : ℕ) :
  0 < volume (Metric.closedBall (0 : Space d) 1)
```

lemma Space.solidSphereMeasure_univ_pos[\[source\]](#)

```
lemma solidSphereMeasure_univ_pos (d : ℕ) : 0 < solidSphereMeasure d Set.univ
```

2.119 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SphericalCylinder

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SphericalCylinder.lean](#) on GitHub.

def Space.sphericalCylinder[\[source\]](#)

The map embedding the unit circular shell extruded along the axis into ‘Space 3’.

```
def sphericalCylinder : Metric.sphere (0 : Space 2) 1 × ℝ → Space 3
```

lemma Space.sphericalCylinder_eq[\[source\]](#)

```
lemma sphericalCylinder_eq :
  sphericalCylinder = (slice 2).symm ∘ (fun x => (x.2, sphericalShell 2 x.1))
```

lemma Space.sphericalCylinder_injective[\[source\]](#)

```
lemma sphericalCylinder_injective : Function.Injective sphericalCylinder
```

lemma Space.sphericalCylinder_continuous[\[source\]](#)

```
@[fun_prop]
lemma sphericalCylinder_continuous : Continuous sphericalCylinder
```

lemma Space.sphericalCylinder_measurableEmbedding[\[source\]](#)

```
lemma sphericalCylinder_measurableEmbedding : MeasurableEmbedding sphericalCylinder
```

lemma Space.norm_sphericalCylinder[\[source\]](#)

```
@[simp]
lemma norm_sphericalCylinder (x : Metric.sphere (0 : Space 2) 1 × ℝ) :
  ||sphericalCylinder ||x = √(||x||.2 ^ 2 + 1)
```

def Space.sphericalCylinderMeasure[\[source\]](#)

The measure on ‘Space 3’ corresponding to integration over a spherical cylinder.

```
def sphericalCylinderMeasure : Measure (Space 3)
```

instance Space.sphericalCylinderMeasure_hasTemperateGrowth[\[source\]](#)

```
instance sphericalCylinderMeasure_hasTemperateGrowth :
  sphericalCylinderMeasure.HasTemperateGrowth
```

instance Space.sphericalCylinderMeasure_sFinite[\[source\]](#)

```
instance sphericalCylinderMeasure_sFinite : SFinite sphericalCylinderMeasure
```

def Space.sphericalCylinderDist[\[source\]](#)

The distribution on ‘Space 3’ corresponding to integration over a spherical cylinder.

```
def sphericalCylinderDist : (Space 3) → dℝ[] ℝ
```

lemma Space.sphericalCylinderDist_apply_eq_integral_sphericalCylinderMeasure[\[source\]](#)

```
lemma sphericalCylinderDist_apply_eq_integral_sphericalCylinderMeasure (f : ℝ → ℝ) :
  sphericalCylinderDist f = ∫ x, f x ∂sphericalCylinderMeasure
```

lemma Space.sphericalCylinderDist_apply_eq_integral_sphere_volume[\[source\]](#)

```
lemma sphericalCylinderDist_apply_eq_integral_sphere_volume (f : ℝ → ℝ) :
  sphericalCylinderDist f =
    ∫ x, f (sphericalCylinder x)
    ∂((MeasureTheory.Measure.toSphere volume).prod (volume α( := ℝ)))
```

2.120 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SphericalShell

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SphericalShell.lean](#) on GitHub.

def Space.sphericalShell[\[source\]](#)

The inclusion into ‘Space d ’ of its unit sphere ‘ $S^{\{d-1\}}$ ’ (the subtype coercion ‘ $x \mapsto x.1$ ’).

```
def sphericalShell (d : ℕ) : Metric.sphere (0 : Space d) 1 → Space d
```

lemma Space.sphericalShell_injective[\[source\]](#)

```
lemma sphericalShell_injective (d : ℕ) : Function.Injective (sphericalShell d)
```

lemma Space.sphericalShell_continuous[\[source\]](#)

```
lemma sphericalShell_continuous (d : ℕ) : Continuous (sphericalShell d)
```

lemma Space.sphericalShell_measurableEmbedding[\[source\]](#)

```
lemma sphericalShell_measurableEmbedding (d : ℕ) : MeasurableEmbedding (sphericalShell d)
```

lemma Space.norm_sphericalShell[\[source\]](#)

```
@[simp]
lemma norm_sphericalShell (d : ℕ) (x : Metric.sphere (0 : Space d) 1) :
  ||sphericalShell d ||x = 1
```

def Space.sphericalShellMeasure[\[source\]](#)

The measure on ‘Space d ’ corresponding to integration around a spherical shell.

```
def sphericalShellMeasure (d : ℕ) : Measure (Space d)
```

instance Space.sphericalShellMeasure_hasTemperateGrowth[\[source\]](#)

```
instance sphericalShellMeasure_hasTemperateGrowth (d : ℕ) :
  (sphericalShellMeasure d).HasTemperateGrowth
```

def Space.sphericalShellDist[\[source\]](#)

The distribution associated with a spherical shell. One can roughly think of this distribution as the distribution which takes test functions ‘ $f(r)$ ’ to ‘ $\int d^3r f(r) \rho(r)$ ’ where ‘ $\rho(r)$ ’ is the mass, charge or current etc. distribution.

```
def sphericalShellDist (d : ℕ) : (Space d) → dℝ[] ℝ
```

lemma Space.sphericalShellDist_apply_eq_integral_sphericalShellMeasure[\[source\]](#)

```
lemma sphericalShellDist_apply_eq_integral_sphericalShellMeasure (d : ℕ) (f : ℝ(Space d, ℝ)) :
  sphericalShellDist d f = ∫ x, f x ∂sphericalShellMeasure d
```


lemma Space.sphericalShellDist_apply_eq_integral_sphere_volume[\[source\]](#)

```

lemma sphericalShellDist_apply_eq_integral_sphere_volume (d : ℕ) (f : ℝ → ℝ) :
  sphericalShellDist d f =
  ∫ x, f (sphericalShell d x) ∂(MeasureTheory.Measure.toSphere volume)

```

2.121 QuantumInfo.States.Mixed.MState

[QuantumInfo/States/Mixed/MState.lean](#) on GitHub.

theorem MState.traceNorm_eq_one[\[source\]](#)

```

@[simp]
theorem traceNorm_eq_one (ρ : MState d) : ρ.m.traceNorm = 1

```

2.122 scripts.MetaPrograms.module_doc_lint

[scripts/MetaPrograms/module_doc_lint.lean](#) on GitHub.

def linterExemptions[\[source\]](#)

*The array of modules exempt from all linters, read from 'scripts/LinterExemption.txt'.
This is used to lint 'QuantumInfo' file-by-file.*

```

def linterExemptions : IO (Array FilePath)

```

def importedFilePaths[\[source\]](#)

The file paths of the modules imported into the module 'mods' (e.g. 'Physlib').

```

def importedFilePaths (mods : Name) : IO (Array FilePath)

```

2.123 scripts.MetaPrograms.runPhyslibLinters

[scripts/MetaPrograms/runPhyslibLinters.lean](#) on GitHub.

def linterExemptions[\[source\]](#)

*The file paths of modules exempt from all linters, read from 'scripts/LinterExemption.txt'.
This is used to lint 'QuantumInfo' file-by-file.*

```

def linterExemptions : IO (Array String)

```

def moduleToFilePathString[\[source\]](#)

The file path (as a string e.g. 'QuantumInfo/Foo/Bar.lean') of the module 'mod'.

```

def moduleToFilePathString (mod : Name) : String

```

2.124 scripts.MetaPrograms.style_lint

[scripts/MetaPrograms/style_lint.lean](#) on GitHub.

def importedFilePaths[\[source\]](#)

The file paths of the modules imported into the module ‘mods’ (e.g. ‘Physlib’), found by reading the module’s ‘.olean’ file.

```
def importedFilePaths (mods : Name) : IO (Array FilePath)
```

def linterExemptions[\[source\]](#)

The file paths of modules which should be skipped by the linters, read from ‘scripts/LinterExemption.txt’. This is used to lint ‘QuantumInfo’ file-by-file.

```
def linterExemptions : IO (Array String)
```

2.125 scripts.PhyslibAlpha.runPhyslibAlphaLinters

[scripts/PhyslibAlpha/runPhyslibAlphaLinters.lean](#) on GitHub.

def runLinterOnModule[\[source\]](#)

Run the Batteries linter on a given module and update the linter if ‘update’ is ‘true’.

```
unsafe def runLinterOnModule (module : Name): IO Unit
```

def main[\[source\]](#)

```
unsafe def main (_ : List String) : IO Unit
```